

STATISTICS PAPER II · AUTHENTIC PYQS 2018-2026

Statistical Inference and Hypothesis Testing

436 authentic UPSC ISS questions with exam shortcut, tips & tricks and step-by-step solution

436

Authentic PYQs

7

Topics Covered

9

Years 2018-2026

AI-DERIVED EXPLANATIONS

The source booklets carry no official answer key. Every Exam Shortcut, Tips & Tricks and Step-by-Step Solution here is independently derived and verified by the builder of this platform, not copied from an official source.

Contents

Methods of Estimation	77 questions
Maximum Likelihood Estimation	38
Method of Moments	24
Optimal Properties of MLE	15
Point Estimation	78 questions
Characteristics of a Good Estimator (Unbiasedness, Consistency, Efficiency, MSE)	72
Cross-over Distribution Theory Items	6
Sufficiency and Completeness	62 questions
Sufficient Estimator and Factorization Theorem	43
Completeness, Minimal Sufficiency, Ancillarity, Basu's Theorem	19
Optimal Estimation	72 questions
Cramer-Rao Inequality, MVB and Fisher Information	38
Rao-Blackwell Theorem and UMVUE Construction	32
Bhattacharya Bounds	2
Interval Estimation	29 questions
Confidence Intervals, Pivots and Optimum Bounds	27
Resampling: Bootstrap and Jackknife	2
Hypothesis Testing	91 questions
Randomized Tests	11
Simple and Composite Hypotheses, Two Kinds of Error, Size and Power	28
Neyman-Pearson Lemma and Best Critical Region	17
Likelihood Ratio Test	14
UMP, MLR, Unbiased and Similar Tests	21
Sequential and Decision Theory	27 questions
Wald's SPRT, OC and ASN Functions	17
Decision Theory, Loss, Risk and Bayes Estimators	10

Methods of Estimation

77 authentic PYQs in this topic (2018-2026)

Maximum Likelihood Estimation

Q1.

For a sample of unit size from a population with p.d.f. $f(x, \alpha) = \frac{2}{\alpha^2}(\alpha - x)$; $0 < x < \alpha$, consider:

1. $2x$ is the maximum likelihood estimator of α .
2. $2x$ is the unbiased estimator of α .

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\log L = \log(\alpha - x) - 2 \log \alpha$ gives $\frac{1}{\alpha - x} = \frac{2}{\alpha} \Rightarrow \hat{\alpha} = 2x$. But $E(X) = \alpha/3$, so $E(2X) = 2\alpha/3 \neq \alpha$: MLE yes, unbiased no.

TIPS & TRICKS

- For a triangular density on $(0, \alpha)$, $E(X) = \alpha/3$ - integrate $\frac{2}{\alpha^2} \int_0^\alpha x(\alpha - x) dx$ once and remember it.
- Being the MLE never implies unbiasedness; invariance transformations routinely destroy it.
- Here $3X$ is the unbiased estimator while $2X$ is the MLE - a neat illustration that the two criteria disagree.

STEP-BY-STEP SOLUTION

1. With $n=1$, $L(\alpha) = \frac{2}{\alpha^2}(\alpha - x)$ for $\alpha > x$, so $\log L = \log 2 - 2 \log \alpha + \log(\alpha - x)$.

2. $\frac{d \log L}{d\alpha} = -\frac{2}{\alpha} + \frac{1}{\alpha - x} = 0 \Rightarrow \alpha = 2(\alpha - x) \Rightarrow \hat{\alpha} = 2x$. Statement 1 is true.

3. $E(X) = \int_0^\alpha x \cdot \frac{2}{\alpha^2}(\alpha - x) dx = \frac{2}{\alpha^2} \left[\frac{\alpha^3}{2} - \frac{\alpha^3}{3} \right] = \frac{\alpha}{3}$.

4. Hence $E(2X) = \frac{2\alpha}{3} \neq \alpha$: $2X$ is biased. Statement 2 is false.

5. Only statement 1 holds - option (a).

Q2.

Suppose a random sample of size 2 was selected from $f_{\theta}(x) = \frac{2}{\theta^2}(\theta - x)$; $0 < x < \theta$. The MLE of θ will be

- A. $\frac{(x_1 + x_2) + \sqrt{(x_1 - x_2)^2 + 4x_1x_2}}{4}$
- B. $\frac{(x_1 + x_2) + \sqrt{9(x_1 - x_2)^2 + 4x_1x_2}}{4}$
- C. $\frac{3(x_1 + x_2) + \sqrt{(x_1 - x_2)^2 + 4x_1x_2}}{4}$
- D. $\frac{\sqrt{3(x_1 + x_2) + \sqrt{9(x_1 - x_2)^2 + 4x_1x_2}}}{4}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$$\frac{d}{d\theta} \log L = -\frac{4}{\theta} + \frac{1}{\theta - x_1} + \frac{1}{\theta - x_2} = 0 \text{ reduces to } 2\theta^2 - 3(x_1 + x_2)\theta + 4x_1x_2 = 0; \text{ the admissible root is}$$

$$\frac{3(x_1 + x_2) + \sqrt{9(x_1 - x_2)^2 + 4x_1x_2}}{4}.$$

TIPS & TRICKS

- Take logs before differentiating: $\log L = \text{const} - 4 \log \theta + \log(\theta - x_1) + \log(\theta - x_2)$.
- Rewrite the discriminant using $9(x_1 + x_2)^2 - 32x_1x_2 = 9(x_1 - x_2)^2 + 4x_1x_2$ - that identity is what matches the printed options.
- Only the + root can exceed $\max(x_1, x_2)$, which the support $0 < x < \theta$ demands.

STEP-BY-STEP SOLUTION

$$1. L(\theta) = \frac{4}{\theta^4}(\theta - x_1)(\theta - x_2) \text{ for } \theta > \max(x_1, x_2).$$

$$2. \frac{d \log L}{d\theta} = -\frac{4}{\theta} + \frac{1}{\theta - x_1} + \frac{1}{\theta - x_2} = 0. \text{ Multiplying through by } \theta(\theta - x_1)(\theta - x_2) \text{ gives}$$

$$-4(\theta - x_1)(\theta - x_2) + \theta(\theta - x_2) + \theta(\theta - x_1) = 0.$$

$$3. \text{ Expanding: } -2\theta^2 + 3(x_1 + x_2)\theta - 4x_1x_2 = 0, \text{ i.e. } 2\theta^2 - 3(x_1 + x_2)\theta + 4x_1x_2 = 0.$$

$$4. \theta = \frac{3(x_1 + x_2) \pm \sqrt{9(x_1 + x_2)^2 - 32x_1x_2}}{4}, \text{ and } 9(x_1 + x_2)^2 - 32x_1x_2 = 9(x_1 - x_2)^2 + 4x_1x_2.$$

5. Taking the + root (the only one exceeding $\max(x_1, x_2)$) gives option (d).

SI & HT

2018 Q25

Numerical

Medium

Q3.

If one observation from Bernoulli with $p \in [\frac{1}{3}, \frac{2}{3}]$ is obtained, the MLE of p is

- A. x
- B. $(2x+1)/3$
- C. $(x-1)/3$
- D. $(x+1)/3$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

The unrestricted MLE is $x \in \{0, 1\}$; projecting onto $[\frac{1}{3}, \frac{2}{3}]$ gives $\frac{1}{3}$ when $x=0$ and $\frac{2}{3}$ when $x=1$, i.e. $\frac{x+1}{3}$.

TIPS & TRICKS

- With a restricted parameter space, the MLE is the admissible point closest to the unrestricted maximiser (likelihood is monotone on either side).
- Test the candidate formulae at $x=0$ and $x=1$ - only $(x+1)/3$ returns $1/3$ and $2/3$.
- $L(p) = p^x(1-p)^{1-x}$ is increasing in p when $x=1$ and decreasing when $x=0$, which is why the answer sits at an endpoint.

STEP-BY-STEP SOLUTION

- For one Bernoulli observation, $L(p) = p^x(1-p)^{1-x}$.
- If $x=1$, $L(p)=p$ is increasing, so the maximum over $[\frac{1}{3}, \frac{2}{3}]$ is at $p = \frac{2}{3}$.
- If $x=0$, $L(p)=1-p$ is decreasing, so the maximum is at $p = \frac{1}{3}$.
- Both values are produced by $\hat{p} = \frac{x+1}{3}$ - option (d).

Q4.

Let $X_i \sim N(\mu, \sigma^2)$, $i=1, \dots, n$, μ known. The MLE of σ^2 is

A. $\frac{\sum_{i=1}^n (X_i - \mu)^2}{n}$

B. $\frac{\sum_{i=1}^{n-1} (X_i - \mu)^2}{n-1}$

C. $\frac{n-1}{n^2} \sum_{i=1}^n (X_i - \mu)^2$

D. $\frac{n-1}{n} \sum_{i=1}^n (X_i - \mu)^2$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

With μ known, $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \mu)^2$ - and this one is unbiased as well as the MLE.

TIPS & TRICKS

- μ known $\Rightarrow$ no degree of freedom is spent estimating it, so the divisor is n and the estimator is unbiased.
- Contrast μ unknown: $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ is the MLE but is biased; $\frac{1}{n-1} \sum (X_i - \bar{X})^2$ is unbiased.
- $\sum (X_i - \mu)^2 / \sigma^2 \sim \chi^2_n$ here, versus χ^2_{n-1} when $\bar{X}$ replaces μ .

STEP-BY-STEP SOLUTION

1. $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (X_i - \mu)^2$.

2. $\frac{\partial \log L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (X_i - \mu)^2 = 0$.

3. Solving: $\hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \mu)^2}{n}$.

4. Option (a).

Q5.

Let $(X_1, \dots, X_n)$ be i.i.d. with $f(x, \mu) = e^{-(x-\mu)}$, $x \geq \mu$. The MLE of μ is

A. $X_{(1)}$

B. $X_{(n)}$

C. $\frac{X_{(1)} + X_{(n)}}{2}$

D. None of the above

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Support is $x \geq \mu$, so the likelihood $e^{-\sum (x_i - \mu)} = e^{n\mu - \sum x_i}$ increases in μ up to the constraint $\mu \leq X_{(1)}$: $\hat{\mu} = X_{(1)}$.

TIPS & TRICKS

- When the support depends on the parameter, never differentiate - push the parameter to the boundary allowed by the data.
- $X_{(1)}$ is also the complete sufficient statistic here, and $X_{(1)} - \frac{1}{n}$ is the UMVUE of μ .
- $n(X_{(1)} - \mu) \sim \text{Exp}(1)$ - the fact used to build exact confidence intervals in later papers.

STEP-BY-STEP SOLUTION

1. $L(\mu) = \prod e^{-(x_i - \mu)} = e^{n\mu - \sum x_i}$ for $\mu \leq \min_i x_i$, and $L=0$ otherwise.
2. $e^{n\mu}$ is strictly increasing in μ , so the likelihood is maximised by taking μ as large as the constraint permits.
3. The constraint is $\mu \leq X_{(1)}$, so $\hat{\mu} = X_{(1)}$.
4. Option (a).

Q6.

Let $(X_1, \dots, X_n)$ be from $f(x, \theta) = \sqrt{\theta} x^{\sqrt{\theta}-1}$, $0 \leq x \leq 1$. The MLE of θ is

- A. $\frac{\sum_{i=1}^n \log x_i}{n}$
- B. $\left(\frac{\sum_{i=1}^n \log x_i}{n} \right)^2$
- C. $\frac{n}{\sum_{i=1}^n \log x_i}$
- D. ✓ $\frac{n^2}{\left(\sum_{i=1}^n \log x_i \right)^2}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Put $\varphi = \sqrt{\theta}$; then $\hat{\varphi} = -n / \sum \log x_i$, and by invariance $\hat{\theta} = \hat{\varphi}^2 = n^2 / \left(\sum \log x_i \right)^2$.

TIPS & TRICKS

- Reparametrise to the familiar $\theta x^{\theta-1}$ form first, then apply invariance - far faster than differentiating in θ .
- $\sum \log x_i < 0$ since $0 < x_i < 1$, so $-n / \sum \log x_i > 0$ as required; squaring removes the sign issue.
- Options (a) and (b) are negative or wrongly scaled - a sign check disposes of them instantly.

STEP-BY-STEP SOLUTION

1. Write $\varphi = \sqrt{\theta}$, so $f(x) = \varphi x^{\varphi-1}$ on $(0, 1)$.

2. $\log L = n \log \varphi + (\varphi - 1) \sum \log x_i$, and $\frac{\partial \log L}{\partial \varphi} = \frac{n}{\varphi} + \sum \log x_i = 0$ gives $\hat{\varphi} = \frac{-n}{\sum \log x_i}$.

3. By the invariance property of the MLE, $\hat{\theta} = \hat{\varphi}^2$.

4. $\hat{\theta} = \frac{n^2}{\left(\sum_{i=1}^n \log x_i \right)^2}$ - option (d).

Q7.

The mle's of α and λ (λ large) of

$$f(x) = \frac{1}{\Gamma(\lambda)} \left(\frac{\lambda}{\alpha} \right)^\lambda e^{-\lambda x/\alpha} x^{\lambda-1}; \quad 0 < x < \infty, \lambda > 0$$

are respectively [given $\frac{\partial}{\partial \lambda} \log \Gamma(\lambda) = \log \lambda - \frac{1}{2\lambda}$]

where G is the geometric mean.

A. G and $\frac{1}{2 \log(\bar{x}/G)}$

B. $\bar{x}$ and $\frac{1}{2 \log(G/\bar{x})}$

C. $\bar{x}$ and $\frac{1}{2 \log(\bar{x}/G)}$

D. G and $\frac{1}{2 \log(G/\bar{x})}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$\partial/\partial \alpha$ gives $\hat{\alpha} = \bar{x}$; substituting into $\partial/\partial \lambda = 0$ and using $\psi(\lambda) \approx \log \lambda - \frac{1}{2\lambda}$ gives $\frac{1}{2\lambda} = \log(\bar{x}/G)$, i.e.

$$\hat{\lambda} = \frac{1}{2 \log(\bar{x}/G)}.$$

TIPS & TRICKS

- This is the gamma density re-parametrised so that $E(X) = \alpha$; hence the MLE of α is the arithmetic mean.
- $\sum \log x_i = n \log G$ where G is the geometric mean - this is what makes $\log(\bar{x}/G)$ appear.
- By the AM-GM inequality $\bar{x} \geq G$, so $\log(\bar{x}/G) \geq 0$ and $\hat{\lambda} > 0$: options with $\log(G/\bar{x})$ give a negative λ and are impossible.

STEP-BY-STEP SOLUTION

$$1. \log L = -n \log \Gamma(\lambda) + n\lambda \log \lambda - n\lambda \log \alpha - \frac{\lambda}{\alpha} \sum x_i + (\lambda - 1) \sum \log x_i.$$

$$2. \frac{\partial \log L}{\partial \alpha} = -\frac{n\lambda}{\alpha} + \frac{\lambda}{\alpha^2} \sum x_i = 0 \Rightarrow \hat{\alpha} = \bar{x}.$$

$$3. \frac{\partial \log L}{\partial \lambda} = -n\psi(\lambda) + n \log \lambda + n - \frac{\sum x_i}{\alpha} + \sum \log x_i. \text{ At } \alpha = \bar{x} \text{ the terms } n \text{ and } \sum x_i/\bar{x} = n \text{ cancel.}$$

4. Using $\psi(\lambda) = \log \lambda - \frac{1}{2\lambda}$ and $\sum \log x_i = n \log \bar{x}$: $\frac{n}{2\lambda} - n \log \bar{x} + n \log G = 0$, so $\frac{1}{2\lambda} = \log \frac{\bar{x}}{G}$.

5. Hence $\hat{\lambda} = \frac{1}{2 \log(\bar{x}/G)}$ and the pair is $(\bar{x}, \frac{1}{2 \log(\bar{x}/G)})$ - option (c).

SI & HT

2019 Q10

Conceptual

Hard

Q8.

Let $X_1, \dots, X_n \sim U(\theta, \theta+1)$. Which is **not** correct?

- A. $(\bar{X} - \frac{1}{2})$ is an unbiased estimate of θ
- B. UMVUE will not exist for θ
- C. Any value of θ in $[X_{(n)} - 1, X_{(1)}]$ is an mle
- D. $X_{(1)}$ is an mle which is sufficient for θ

Correct answer: D

EXAM SHORTCUT AI-derived explanation

For $U(\theta, \theta+1)$ the minimal sufficient statistic is the PAIR $(X_{(1)}, X_{(n)})$, not $X_{(1)}$ alone. So (d) is the incorrect statement.

TIPS & TRICKS

- Both endpoints constrain θ : $X_{(n)} - 1 \leq \theta \leq X_{(1)}$, so both order statistics are needed.
- The MLE is any point of that interval - non-unique, which makes (c) correct.
- $(X_{(1)}, X_{(n)})$ is minimal sufficient but NOT complete (the range is ancillary), which is why no UMVUE exists and (b) is correct.

STEP-BY-STEP SOLUTION

- The likelihood is $L(\theta) = 1$ if $\theta \leq X_{(1)}$ and $X_{(n)} \leq \theta + 1$, i.e. $X_{(n)} - 1 \leq \theta \leq X_{(1)}$, and 0 otherwise.
- (a) $E(X) = \theta + \frac{1}{2}$, so $\bar{X} - \frac{1}{2}$ is unbiased for θ . Correct.
- (c) The likelihood is flat on $[X_{(n)} - 1, X_{(1)}]$, so every point there is an MLE. Correct.
- (b) $(X_{(1)}, X_{(n)})$ is minimal sufficient but not complete (the range $X_{(n)} - X_{(1)}$ is ancillary), so Lehmann-Scheffe does not apply and no UMVUE exists. Correct.
- (d) $X_{(1)}$ alone is NOT sufficient - the upper constraint $\theta \geq X_{(n)} - 1$ is lost. This is the incorrect statement, option (d).

Q9.

L differentiable at least twice, $\theta = \hat{\theta}$ solves $\frac{\partial(\log L)}{\partial \theta} = 0$. If $l = \left[\frac{\partial^2 \log L}{\partial \theta^2} \right]_{\theta = \hat{\theta}}$, $\hat{\theta}$ is a maximum likelihood estimate if

- A. $l < 0$
- B. $l > 0$
- C. $l = 0$
- D. $l \geq 0$

Correct answer: A**EXAM SHORTCUT** AI-derived explanation

A stationary point is a MAXIMUM when the second derivative is negative: $l < 0$.

TIPS & TRICKS

- $\frac{\partial \log L}{\partial \theta} = 0$ only locates a stationary point; the sign of the second derivative classifies it.
- $-\frac{\partial^2 \log L}{\partial \theta^2}$ evaluated at $\hat{\theta}$ is the observed information, which must be positive.
- If $l = 0$ the test is inconclusive and higher derivatives must be examined.

STEP-BY-STEP SOLUTION

1. $\hat{\theta}$ solves the likelihood equation $\frac{\partial \log L}{\partial \theta} = 0$.
2. By the second-derivative test, this stationary point is a local maximum of $\log L$ (and hence of L) if $\left[\frac{\partial^2 \log L}{\partial \theta^2} \right]_{\hat{\theta}} < 0$.
3. So the condition is $l < 0$ - option (a).

Q10.

Let X_1, X_2, X_3 be iid $U(\theta, \theta^2)$, $\theta > 1$. The mle of θ is

- A. $X_{(1)}^2$
 B. $\sqrt{X_{(3)}}$
 C. $\sqrt{X_{(1)}}$
 D. $\alpha X_{(1)} + (1-\alpha)X_{(3)}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$L = (\theta^2 - \theta)^{-3}$ subject to $\theta \leq X_{(1)}$ and $\theta^2 \geq X_{(3)}$. For $\theta > 1$, L decreases in θ , so take θ as small as allowed:
 $\hat{\theta} = \sqrt{X_{(3)}}$.

TIPS & TRICKS

- Support-dependent parameter: read off the two constraints, then see which direction the likelihood pushes.
- $\theta^2 - \theta$ is increasing for $\theta > 1$, so $1/(\theta^2 - \theta)^3$ is decreasing - the maximum is at the SMALLEST admissible θ .
- The binding lower constraint is $\theta \geq \sqrt{X_{(3)}}$, coming from the upper endpoint of the support.

STEP-BY-STEP SOLUTION

1. The density is $\frac{1}{\theta^2 - \theta}$ on (θ, θ^2) , so $L(\theta) = \frac{1}{(\theta^2 - \theta)^3}$ provided $\theta \leq X_{(1)}$ and $X_{(3)} \leq \theta^2$.
2. The constraints give $\sqrt{X_{(3)}} \leq \theta \leq X_{(1)}$.
3. For $\theta > 1$, $\theta^2 - \theta$ is increasing, so L is strictly decreasing in θ ; the maximum is at the smallest admissible value.
4. Hence $\hat{\theta} = \sqrt{X_{(3)}}$ - option (b).

Q11.

Let $X_1, \dots, X_n$ be from $U(0, 5\theta)$. Define $X_{(1)} = \min(\dots)$, $X_{(n)} = \max(\dots)$. The MLE of θ is

- A. $\frac{1}{5} X_{(1)}$
- B. $X_{(1)}$
- C. $X_{(n)}$
- D. $\frac{1}{5} X_{(n)}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Need $5\theta \geq X_{(n)}$, i.e. $\theta \geq X_{(n)}/5$, and $L = (5\theta)^{-n}$ decreases in θ : $\hat{\theta} = X_{(n)}/5$.

TIPS & TRICKS

- Rescale: if $Y = X/5 \sim U(0, \theta)$ then $\hat{\theta} = Y_{(n)} = X_{(n)}/5$ - invariance does the work.
- Support-dependent parameter $\Rightarrow$ push θ to the smallest admissible value, since L is decreasing.
- $X_{(1)}$ carries no information about the upper endpoint - options (a) and (b) are structurally wrong.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \frac{1}{(5\theta)^n}$ provided every $x_i \leq 5\theta$, i.e. $\theta \geq \frac{X_{(n)}}{5}$.

2. L is strictly decreasing in θ , so it is maximised at the smallest admissible value.

3. $\hat{\theta} = \frac{X_{(n)}}{5}$ - option (d).

Q12.

Let $X_1, \dots, X_n$ be iid $\text{Poisson}(\lambda)$. The MLE of λ is

A. $\bar{X}$

B. $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$

C. $\bar{X} - X_{(1)}$

D. $\sum_{i=1}^n X_i$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$$\frac{\partial \log L}{\partial \lambda} = -n + \frac{\sum x_i}{\lambda} = 0 \text{ gives } \hat{\lambda} = \bar{X}.$$

TIPS & TRICKS

- For Poisson the MLE, the moment estimator and the UMVUE of λ all coincide with $\bar{X}$.
- $\sum X_i$ is the sufficient statistic but is not itself an estimator of λ (its expectation is $n\lambda$).
- Since mean = variance for Poisson, the sample variance is also consistent - but it is not the MLE.

STEP-BY-STEP SOLUTION

1. $L(\lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!}$, so $\log L = -n\lambda + \left(\sum x_i\right) \log \lambda - \log \prod x_i!$.

2. $\frac{\partial \log L}{\partial \lambda} = -n + \frac{\sum x_i}{\lambda} = 0 \Rightarrow \hat{\lambda} = \frac{\sum x_i}{n} = \bar{X}.$

3. Second derivative $-\sum x_i / \lambda^2 < 0$ confirms a maximum. Option (a).

Q13.

Consider a random sample $\{8, 4, \frac{1}{2}, 1\}$ from the distribution with the most general pmf form

$$f(x, \theta) = \left(\frac{x}{\theta}\right)^{\theta A'(\theta)} \exp[A(\theta) + C(x)]$$

where $A'(\theta)$ is the derivative of $A(\theta)$ w.r.t. θ . The MLE of θ is

A. $\frac{27}{8}$

B. $\sqrt{2}$

C. $\frac{32}{27}$

D. $\frac{5}{2}$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Differentiating $\log L$ collapses to $\sum \log x_i = n \log \theta$, so $\hat{\theta}$ is the GEOMETRIC MEAN. For $\{8, 4, \frac{1}{2}, 1\}$ the product is 16 and $16^{1/4} = 2$.

TIPS & TRICKS

- $\log f = \theta A'(\theta) [\log x - \log \theta] + A(\theta) + C(x)$; the A -terms cancel when you differentiate and simplify.
- The likelihood equation reduces to $\frac{1}{n} \sum \log x_i = \log \theta$, i.e. $\hat{\theta} = \left(\prod x_i\right)^{1/n}$.
- $8 \times 4 \times \frac{1}{2} \times 1 = 16$ and $16^{1/4} = 2$ - clean by design.

STEP-BY-STEP SOLUTION

1. $\log L = \theta A'(\theta) \left[\sum \log x_i - n \log \theta \right] + nA(\theta) + \sum C(x_i)$.

2. $\frac{\partial \log L}{\partial \theta} = [A'(\theta) + \theta A''(\theta)] \left[\sum \log x_i - n \log \theta \right] - \theta A'(\theta) \frac{n}{\theta} + nA'(\theta)$.

3. The last two terms cancel, leaving $[A' + \theta A''] \left[\sum \log x_i - n \log \theta \right] = 0$, hence $\log \theta = \frac{1}{n} \sum \log x_i$.

4. So $\hat{\theta} = \left(\prod x_i\right)^{1/n} = \left(8 \cdot 4 \cdot \frac{1}{2} \cdot 1\right)^{1/4} = 16^{1/4} = 2$.

5. Option (b).

SI & HT

2020 Q24

Numerical

Medium

Q14.

Let $\{7.05, 6.89, 6.62, 7.32, 7.48, 6.93, 7.17, 6.78\}$ be a random sample from $U\left(\theta - \frac{1}{2}, \theta + \frac{1}{2}\right)$. Which would be MLE for θ ?

A. 6.99 as well as 7.10

B. 6.87 as well as 7.10

C. 7.54 as well as 6.92

D. 6.92 as well as 6.87

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Every $\theta \in \left[X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}\right] = [6.98, 7.12]$ maximises the likelihood. Only 6.99 and 7.10 both lie there.

TIPS & TRICKS

- $U\left(\theta - \frac{1}{2}, \theta + \frac{1}{2}\right)$ has a FLAT likelihood over an interval - the MLE is not unique.
- The interval is $\left[\max x_i - \frac{1}{2}, \min x_i + \frac{1}{2}\right]$; here $\max = 7.48$, $\min = 6.62$.
- Check both numbers in each option against $[6.98, 7.12]$: (b), (c), (d) each contain a value outside.

STEP-BY-STEP SOLUTION

1. The likelihood is 1 whenever $\theta - \frac{1}{2} \leq x_i \leq \theta + \frac{1}{2}$ for all i , i.e. $X_{(n)} - \frac{1}{2} \leq \theta \leq X_{(1)} + \frac{1}{2}$.
2. Here $X_{(n)} = 7.48$ and $X_{(1)} = 6.62$, giving $6.98 \leq \theta \leq 7.12$.
3. Both 6.99 and 7.10 lie in $[6.98, 7.12]$, so both are maximum likelihood estimates.
4. 6.87, 7.54 and 6.92 all fall outside. Option (a).

Q15.

If $X_1, \dots, X_m$ be iid from $B(n, P)$, the MLE of P is

- A. $\frac{\sum \bar{X}}{n}$
- B. $\bar{X}$
- C. $\frac{\bar{X}}{m}$
- D. $\frac{\bar{X}}{mn}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$E(X_i) = nP$, so the MLE (and moment estimator) is $\hat{P} = \bar{X}/n$.

TIPS & TRICKS

- There are m observations each on n trials, so the total number of trials is mn and $\hat{P} = \frac{\sum X_i}{mn} = \frac{\bar{X}}{n}$.
- Divide by the NUMBER OF TRIALS n , not by the number of observations m - the two are easy to swap.
- Check with $n=1$ (Bernoulli): the formula returns $\bar{X}$, as it must.

STEP-BY-STEP SOLUTION

$$1. L(P) = \prod_{i=1}^m \binom{n}{x_i} P^{x_i} (1-P)^{n-x_i}, \text{ so } \log L = \text{const} + \left(\sum x_i \right) \log P + \left(mn - \sum x_i \right) \log(1-P).$$

$$2. \frac{\partial \log L}{\partial P} = \frac{\sum x_i}{P} - \frac{mn - \sum x_i}{1-P} = 0.$$

$$3. \text{ Solving: } \hat{P} = \frac{\sum x_i}{mn} = \frac{\bar{X}}{n}.$$

4. Option (a).

Q16.

Let $x_1, \dots, x_n \sim U(0, \theta)$, order statistics $x_{(1)} \leq \dots \leq x_{(n)}$:

1. $x_{(n)}$ is the MLE for θ .
2. $x_{(n)}$ is the consistent estimator of θ .

Codes:

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$x_{(n)}$ is the MLE (the likelihood θ^{-n} decreases in θ) and it is consistent, since $P(\theta - X_{(n)} > \varepsilon) = (1 - \varepsilon/\theta)^n \rightarrow 0$.

TIPS & TRICKS

- $X_{(n)}$ is biased ($E = \frac{n\theta}{n+1}$) but consistent - bias and consistency are independent properties.
- Convergence is at rate $1/n$, faster than the usual $1/\sqrt{n}$ - a super-efficient estimator.
- $P(X_{(n)} \leq \theta - \varepsilon) = \left(\frac{\theta - \varepsilon}{\theta}\right)^n \rightarrow 0$ for every fixed $\varepsilon > 0$.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \theta^{-n}$ for $\theta \geq X_{(n)}$, which is decreasing, so $\hat{\theta} = X_{(n)}$. Statement 1 true.
2. $P(|X_{(n)} - \theta| > \varepsilon) = P(X_{(n)} < \theta - \varepsilon) = \left(\frac{\theta - \varepsilon}{\theta}\right)^n \rightarrow 0$ as $n \rightarrow \infty$.
3. So $X_{(n)} \rightarrow P\theta$: statement 2 is true.
4. Option (c).

Q17.

For $x_1, \dots, x_n$ from $\text{Poisson}(\theta)$:

1. $\frac{1}{n} \sum_{i=1}^n x_i$ is the MLE of θ .
2. $\frac{1}{n} \sum_{i=1}^n x_i$ is sufficient for estimating θ .

Codes:

- A. 1 only
B. 2 only
C. Both 1 and 2
D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\bar{x}$ is both the MLE and (being a one-to-one function of $\sum x_i$) sufficient for θ .

TIPS & TRICKS

- $\sum X_i$ is sufficient; $\bar{X} = \sum X_i/n$ is a one-to-one function of it, hence equally sufficient.
- MLEs are always functions of the sufficient statistic when one exists - the two statements reinforce each other.
- For Poisson, $\bar{X}$ is simultaneously MLE, MoM estimator, UMVUE and MVB estimator.

STEP-BY-STEP SOLUTION

1. $\log L = -n\theta + \left(\sum x_i\right) \log \theta - \log \prod x_i!$, so $\hat{\theta} = \bar{x}$. Statement 1 true.
2. By factorisation, $\sum X_i$ is sufficient; $\bar{X}$ is a one-to-one function of it, so $\bar{X}$ is sufficient too. Statement 2 true.
3. Option (c).

Q18.

Let $X \sim b(1, p)$, $p \in \left[\frac{a}{a+b}, \frac{b}{a+b} \right]$; $a, b > 0$. The MLE of p is

- A. $\frac{X+b}{a+b}$
- B. $\frac{(b-a)X-1}{a+b}$
- C. $\frac{(b-a)X+a}{a+b}$
- D. $\frac{(b-a)X+b}{a+b}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

The unrestricted MLE is X ; projecting onto $\left[\frac{a}{a+b}, \frac{b}{a+b} \right]$ gives $\frac{a}{a+b}$ at $X=0$ and $\frac{b}{a+b}$ at $X=1$ - i.e. $\frac{(b-a)X+a}{a+b}$.

TIPS & TRICKS

- Test each candidate formula at $X=0$ and $X=1$; only one returns the two endpoints.
- $X=0$: $L(p)=1-p$ decreasing $\Rightarrow$ take the smallest admissible p . $X=1$: $L(p)=p$ increasing $\Rightarrow$ take the largest.
- Restricted parameter space $\Rightarrow$ the MLE is the projection of the unrestricted maximiser.

STEP-BY-STEP SOLUTION

1. For one Bernoulli observation, $L(p)=p^X(1-p)^{1-X}$.
2. If $X=0$, $L(p)=1-p$ is decreasing, so $\hat{p}=\frac{a}{a+b}$ (the left endpoint).
3. If $X=1$, $L(p)=p$ is increasing, so $\hat{p}=\frac{b}{a+b}$ (the right endpoint).
4. Both are produced by $\hat{p}=\frac{(b-a)X+a}{a+b}$: at $X=0$ it gives $\frac{a}{a+b}$, at $X=1$ it gives $\frac{b-a+a}{a+b}=\frac{b}{a+b}$.
5. Option (c).

Q19.

Let $x_1, \dots, x_n$ be from $U(\alpha, \beta)$. The MLE of α and β are respectively

- A. $\sum x_i$ and $\sum x_i^2$
- B. $\prod x_i$ and $\prod x_i^2$
- C. $\min(x_1, \dots, x_n)$ and $\max(x_1, \dots, x_n)$
- D. $\frac{x_{(n)} - x_{(1)}}{2}$ and $\frac{x_{(n)} + x_{(1)}}{2}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Support-dependent endpoints: $\hat{\alpha} = X_{(1)} = \min x_i$ and $\hat{\beta} = X_{(n)} = \max x_i$.

TIPS & TRICKS

- $L = (\beta - \alpha)^{-n}$ is maximised by making $\beta - \alpha$ as SMALL as the data permit.
- The smallest admissible interval containing all observations is $[\min x_i, \max x_i]$.
- Option (d) gives the half-range and midrange, which are the MLEs in the $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ parametrisation, not here.

STEP-BY-STEP SOLUTION

1. $L(\alpha, \beta) = \frac{1}{(\beta - \alpha)^n}$ subject to $\alpha \leq X_{(1)}$ and $\beta \geq X_{(n)}$.
2. The likelihood increases as $\beta - \alpha$ shrinks, so it is maximised by taking α as large and β as small as the constraints allow.
3. Hence $\hat{\alpha} = X_{(1)} = \min(x_1, \dots, x_n)$ and $\hat{\beta} = X_{(n)} = \max(x_1, \dots, x_n)$.
4. Option (c).

SHARED CONTEXT

Let X have uniform density $f(x; \mu, \sigma) = \frac{1}{2\sqrt{3}\sigma}$, $\mu - \sqrt{3}\sigma < x < \mu + \sqrt{3}\sigma$; $-\infty < \mu < \infty$, $\sigma > 0$.

Q20.

What is the MLE estimator of μ ?

- A. $x_{(1)}$
- B. $x_{(1)} + x_{(n)}$
- C. $[x_{(1)} + x_{(n)}]/2$
- D. $2[x_{(1)} + x_{(n)}]$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

The support endpoints are $\mu \pm \sqrt{3}\sigma$, matched to $x_{(1)}$ and $x_{(n)}$; solving gives $\hat{\mu} = \frac{x_{(1)} + x_{(n)}}{2}$, the midrange.

TIPS & TRICKS

- This is $U(a, b)$ in disguise with $a = \mu - \sqrt{3}\sigma$, $b = \mu + \sqrt{3}\sigma$; the MLEs of a, b are $x_{(1)}, x_{(n)}$.
- Then invert: $\mu = \frac{a+b}{2}$ and $\sigma = \frac{b-a}{2\sqrt{3}}$, and use invariance.
- The parametrisation is chosen so that $E(X) = \mu$ and $V(X) = \sigma^2$ - check: $\frac{(2\sqrt{3}\sigma)^2}{12} = \sigma^2$.

STEP-BY-STEP SOLUTION

- The density is uniform on $(\mu - \sqrt{3}\sigma, \mu + \sqrt{3}\sigma)$, so the MLEs of the two endpoints are $x_{(1)}$ and $x_{(n)}$.
- Setting $\hat{\mu} - \sqrt{3}\hat{\sigma} = x_{(1)}$ and $\hat{\mu} + \sqrt{3}\hat{\sigma} = x_{(n)}$ and adding:
- $2\hat{\mu} = x_{(1)} + x_{(n)}$, so $\hat{\mu} = \frac{x_{(1)} + x_{(n)}}{2}$.
- Option (c).

SHARED CONTEXT

Let X have uniform density $f(x; \mu, \sigma) = \frac{1}{2\sqrt{3}\sigma}$, $\mu - \sqrt{3}\sigma < x < \mu + \sqrt{3}\sigma$; $-\infty < \mu < \infty$, $\sigma > 0$.

Q21.

What is the MLE estimator of σ ?

- A. $x_{(n)}$
- B. $x_{(n)} - x_{(1)}$
- C. $[x_{(n)} - x_{(1)}] / (2\sqrt{3})$
- D. $[x_{(n)} - x_{(1)}] / 2$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Subtracting the two endpoint equations gives $2\sqrt{3}\hat{\sigma} = x_{(n)} - x_{(1)}$, so $\hat{\sigma} = \frac{x_{(n)} - x_{(1)}}{2\sqrt{3}}$.

TIPS & TRICKS

- The range of the support is $2\sqrt{3}\sigma$, so divide the observed range by $2\sqrt{3}$.
- Dividing by 2 only (option (d)) would be right for $U(\mu - \sigma, \mu + \sigma)$, a different parametrisation.
- $2\sqrt{3} \approx 3.46$, so $\hat{\sigma}$ is roughly $0.29 \times$ the sample range.

STEP-BY-STEP SOLUTION

- From $\mu - \sqrt{3}\hat{\sigma} = x_{(1)}$ and $\mu + \sqrt{3}\hat{\sigma} = x_{(n)}$, subtract:
- $2\sqrt{3}\hat{\sigma} = x_{(n)} - x_{(1)}$.
- Hence $\hat{\sigma} = \frac{x_{(n)} - x_{(1)}}{2\sqrt{3}}$.
- Option (c).

SHARED CONTEXT

Let $X_1, \dots, X_n$ be iid $N(\mu, \mu)$ — i.e. mean = variance = μ ($\mu > 0$).

Q22.

Consider:

1. $\sum x_i^2$ is sufficient for μ .

2. MLE of μ is $\frac{-1 + \sqrt{1 + 4m_2}}{2}$, where $m_2 = \frac{\sum x_i^2}{n}$.

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\log L$ contains $\sum x_i$ only with coefficient 1 (free of μ), so $\sum x_i^2$ is sufficient; the likelihood equation reduces to $\mu^2 + \mu - m_2 = 0$, whose positive root is $\frac{-1 + \sqrt{1 + 4m_2}}{2}$.

TIPS & TRICKS

- Expand $\sum (x_i - \mu)^2 / (2\mu)$: the cross term $-2\mu \sum x_i / (2\mu) = -\sum x_i$ loses its μ - that is why $\sum x_i$ drops out.
- Hence $\sum x_i^2$ alone is sufficient even though the model has both mean and variance equal to μ .
- Take the POSITIVE root of the quadratic since $\mu > 0$.

STEP-BY-STEP SOLUTION

$$1. \log L = -\frac{n}{2} \log(2\pi\mu) - \frac{1}{2\mu} \sum (x_i - \mu)^2 = -\frac{n}{2} \log(2\pi\mu) - \frac{\sum x_i^2}{2\mu} + \sum x_i - \frac{n\mu}{2}.$$

2. The term $\sum x_i$ carries no μ , so the likelihood factorises through $\sum x_i^2$; statement 1 is true.

$$3. \frac{\partial \log L}{\partial \mu} = -\frac{n}{2\mu} + \frac{\sum x_i^2}{2\mu^2} - \frac{n}{2} = 0. \text{ Multiplying by } \frac{2\mu^2}{n}: -\mu + m_2 - \mu^2 = 0 \text{ with } m_2 = \frac{\sum x_i^2}{n}.$$

4. So $\mu^2 + \mu - m_2 = 0$ and the admissible root is $\hat{\mu} = \frac{-1 + \sqrt{1 + 4m_2}}{2}$; statement 2 is true.

5. Option (c).

SI & HT

2023 Q5

Numerical

Medium

Q23.

Let $\{6, 11, 4, 13, 5\}$ be a random sample from the exponential distribution $f(x, \theta) = e^{-(x-\theta)}$, $x \geq \theta$, $\theta \in (-\infty, 3]$ is unknown. Then the Maximum Likelihood Estimate of e^θ is:

A. e^3

B. e^4

C. $e^{7.8}$

D. e^6

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The unconstrained MLE is $X_{(1)} = 4$, but $\theta \leq 3$; since L increases in θ , the constrained maximum is at $\theta = 3$, so $\hat{e^\theta} = e^3$.

TIPS & TRICKS

- Restricted parameter space: the MLE is the admissible value closest to the unrestricted maximiser along the increasing direction.
- $L(\theta) = e^{n\theta} \sum x_i$ is increasing in θ up to $\theta \leq X_{(1)}$; with the extra cap $\theta \leq 3$ the binding constraint is 3.
- Invariance then gives $\hat{e^\theta} = e^{\hat{\theta}} = e^3$ - no further work needed.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \prod e^{-(x_i - \theta)} = e^{5\theta} \sum x_i$ for $\theta \leq X_{(1)} = \min\{6, 11, 4, 13, 5\} = 4$.
2. L is increasing in θ , so without restriction $\hat{\theta} = 4$.
3. The parameter space is $(-\infty, 3]$, so the maximum over the admissible set is attained at $\hat{\theta} = 3$.
4. By invariance, the MLE of e^θ is e^3 - option (a).

Q24.

Let $\{4.5, 3.8, 2.5, 5.2\}$ be a random sample from $N(\mu, 1)$ distribution. If the parametric space for parameter μ is $\Omega = \{-2, -1, 1, 2\}$, then the MLE of μ is:

- A. 5.2
- B. 4
- C. -2
- D. 2

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$\bar{x} = 4$; over $\Omega = \{-2, -1, 1, 2\}$ the likelihood is maximised by the admissible value closest to $\bar{x}$, namely $\mu = 2$.

TIPS & TRICKS

- $\log L = \text{const} - \frac{1}{2} \sum (x_i - \mu)^2$, and $\sum (x_i - \mu)^2 = \sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$.
- So maximising L over a restricted set means minimising $|\mu - \bar{x}|$ over that set.
- $\bar{x} = \frac{4.5 + 3.8 + 2.5 + 5.2}{4} = \frac{16}{4} = 4$, and 2 is the nearest point of Ω .

STEP-BY-STEP SOLUTION

$$1. \bar{x} = \frac{4.5 + 3.8 + 2.5 + 5.2}{4} = \frac{16}{4} = 4.$$

$$2. \log L(\mu) = \text{const} - \frac{1}{2} \sum (x_i - \mu)^2 = \text{const} - \frac{1}{2} \left[\sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 \right].$$

3. Maximising over $\mu \in \Omega$ is therefore minimising $(\bar{x} - \mu)^2$, i.e. choosing the element of Ω closest to 4.

4. The distances are 6, 5, 3, 2 for $\mu = -2, -1, 1, 2$, so $\hat{\mu} = 2$ - option (d).

Q25.

In which one of the following cases, MLE for a parameter θ need not be unique based on a random sample of size n ?

- A. $U(0, \theta)$
- B. $U(\theta - 0.5, \theta + 0.5)$
- C. $f(x, \theta) = e^{-(x-\theta)}, x \geq \theta$
- D. $f(x, \theta) = (1+\theta)x^\theta, 0 < x < 1$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

For $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ the likelihood is flat on $[X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}]$, so every point there is an MLE.

TIPS & TRICKS

- Non-uniqueness arises when BOTH support endpoints move with θ , trapping θ in an interval.
- $U(0, \theta)$ and the shifted exponential each have ONE moving endpoint, giving a unique boundary solution.
- $(1+\theta)x^\theta$ is a smooth regular family with a unique interior maximum.

STEP-BY-STEP SOLUTION

1. (a) $U(0, \theta)$: $L = \theta^{-n}$ for $\theta \geq X_{(n)}$, decreasing, so $\hat{\theta} = X_{(n)}$ - unique.
2. (c) $e^{-(x-\theta)}, x \geq \theta$: L increases in θ up to $X_{(1)}$, so $\hat{\theta} = X_{(1)}$ - unique.
3. (d) $(1+\theta)x^\theta$: a regular family with $\hat{\theta} = -\frac{n}{\sum \ln x_i} - 1$ - unique.
4. (b) $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$: $L = 1$ for every $\theta \in [X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}]$, an interval of maximisers - NOT unique.
5. Option (b).

Q26.

Let

$$X_i \sim f(x) = \begin{cases} \frac{1}{N}, & x=1,2,\dots,N \\ 0, & \text{otherwise} \end{cases}; \quad i=1,2,3,\dots,n$$

The Maximum Likelihood Estimator (MLE) of N has which one of the following properties?

A. Consistent, sufficient and complete

B. Unbiased, sufficient and complete

C. Unbiased and sufficient only

D. Unbiased, complete and consistent

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\hat{N} = X_{(n)}$ is sufficient, complete and consistent, but BIASED ($E[X_{(n)}] < N$).

TIPS & TRICKS

- $X_{(n)} \leq N$ always, so its expectation is strictly below N - unbiasedness is impossible.
- The UMVUE is $\frac{X_{(n)}^{n+1} - (X_{(n)} - 1)^{n+1}}{X_{(n)}^n - (X_{(n)} - 1)^n}$ - a bias correction of the MLE.
- Three options assert unbiasedness; eliminating that word leaves only one choice.

STEP-BY-STEP SOLUTION

- $L(N) = N^{-n} I(X_{(n)} \leq N)$ is decreasing in N , so $\hat{N} = X_{(n)}$.
- $X_{(n)}$ is sufficient (factorisation) and complete for the discrete uniform family.
- $P(X_{(n)} \leq N-1) = \left(\frac{N-1}{N}\right)^n \rightarrow 0$, so $X_{(n)} \rightarrow PN$: consistent.
- But $X_{(n)} \leq N$ with positive probability of being strictly less, so $E[X_{(n)}] < N$: biased.
- Hence consistent, sufficient and complete - option (a).

Q27.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from the distribution having pdf $f(x, \theta) = \frac{1}{\theta} e^{-(x-\theta)/\theta}$; $x \geq \theta$, $\theta > 0$. The maximum likelihood estimator of $\ln \theta$ is

A. $\ln \left[\frac{\sum_{i=1}^n x_i}{n} + x_{(1)} \right]$

B. $\ln \left[\frac{\sum_{i=1}^n x_{(i)}}{n} - x_{(1)} \right]$

C. $\ln [x_{(1)}]$

D. $\ln \left[\sum_{i=1}^n x_{(i)} - n \right]$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\log L = -n \log \theta - \frac{\sum x_i}{\theta} + n$ increases in θ up to $\bar{x}$, but the support forces $\theta \leq X_{(1)} < \bar{x}$; hence $\hat{\theta} = X_{(1)}$ and $\hat{\ln \theta} = \ln X_{(1)}$.

TIPS & TRICKS

- Two constraints compete: the smooth maximum at $\bar{x}$ and the support constraint $\theta \leq X_{(1)}$.
- Since $X_{(1)} \leq \bar{x}$ always, the support constraint binds and the MLE sits on the boundary.
- Invariance then gives the MLE of $\ln \theta$ directly as $\ln(X_{(1)})$.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \theta^{-n} \exp \left[-\frac{\sum (x_i - \theta)}{\theta} \right] = \theta^{-n} e^n \exp \left(-\frac{\sum x_i}{\theta} \right)$, valid for $\theta \leq X_{(1)}$.

2. $\frac{\partial \log L}{\partial \theta} = -\frac{n}{\theta} + \frac{\sum x_i}{\theta^2} = \frac{n(\bar{x} - \theta)}{\theta^2} > 0$ for $\theta < \bar{x}$.

3. Since $X_{(1)} \leq \bar{x}$, the likelihood is increasing over the whole admissible range, so $\hat{\theta} = X_{(1)}$.

4. By invariance, the MLE of $\ln \theta$ is $\ln [x_{(1)}]$ - option (c).

Q28.

Let $x_1, x_2, x_3, \dots, x_{10}$ be a random sample from a uniform $U(-\theta, \theta)$, $\theta > 0$ distribution. It is given that the values of the largest and smallest observations in the sample are 7 and -12 respectively. Then MLE of θ is

A. -12

B. $-\frac{5}{2}$

C. 7

D. 12

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$U(-\theta, \theta)$ requires $\theta \geq \max_i |x_i| = \max(7, 12) = 12$; the likelihood $(2\theta)^{-n}$ decreases, so $\hat{\theta} = 12$.

TIPS & TRICKS

- The binding constraint is on the ABSOLUTE values - the sufficient statistic is $\max |X_i|$.
- $|-12| = 12 > |7|$, so the smallest observation drives the estimate here.
- $\theta > 0$ by assumption, so negative options are impossible on sight.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \frac{1}{(2\theta)^n}$ provided $-\theta \leq x_i \leq \theta$ for all i , i.e. $\theta \geq \max_i |x_i|$.

2. $\max_i |x_i| = \max(|7|, |-12|) = 12$.

3. L is decreasing in θ , so the maximum is at the smallest admissible value $\hat{\theta} = 12$.

4. Option (d).

Q29.

A biased coin with probability of head p is tossed m times independently. It is known that $p \in \{1/4, 3/4\}$ and $m \in \{3, 5\}$. If 3 heads are observed in these m tosses, then which one of the following is correct?

- A. $(5, 3/4)$ is MLE of m and p
- B. $(5, 1/4)$ is MLE of m and p
- C. **$(3, 3/4)$ is MLE of m and p**
- D. MLE of m and p is not unique

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Evaluate $\binom{m}{3} p^3 (1-p)^{m-3}$ at all four (m, p) pairs; the largest is $\left(\frac{3}{4}\right)^3 = 0.4219$ at $(3, \frac{3}{4})$.

TIPS & TRICKS

- With a finite parameter set, just tabulate the likelihood at every admissible point.
- $(3, \frac{3}{4})$: $\binom{3}{3} \left(\frac{3}{4}\right)^3 = \frac{27}{64} = 0.4219$ - the clear maximum.
- $(5, \frac{3}{4})$: $\binom{5}{3} \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^2 = 10 \cdot \frac{27}{64} \cdot \frac{1}{16} = 0.2637$.

STEP-BY-STEP SOLUTION

1. $L(m, p) = \binom{m}{3} p^3 (1-p)^{m-3}$.
2. $(3, \frac{1}{4})$: $1 \cdot \left(\frac{1}{4}\right)^3 = 0.0156$. $(3, \frac{3}{4})$: $1 \cdot \left(\frac{3}{4}\right)^3 = 0.4219$.
3. $(5, \frac{1}{4})$: $10 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2 = 0.0879$. $(5, \frac{3}{4})$: $10 \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^2 = 0.2637$.
4. The maximum is at $(m, p) = (3, \frac{3}{4})$ - option (c).

Q30.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from a distribution with probability density function $f(x; \theta) = \frac{2x}{\theta} e^{-x^2/\theta}$; $x > 0, \theta > 0$. Which of the following statements are correct in respect of parameter θ of the distribution?

I. The maximum likelihood estimate of θ is given by $\hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i^2$.

II. $\hat{\theta}_{MLE}$ is an unbiased estimator of θ .

III. $\hat{\theta}_{MLE}$ is sufficient estimator of θ .

Select the answer using the code given below.

- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II and III**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$\hat{\theta} = \frac{1}{n} \sum x_i^2$ (I), $X^2 \sim \text{Exp}(\text{mean } \theta)$ so $E(\hat{\theta}) = \theta$ (II), and $\sum X_i^2$ is the sufficient statistic (III). All three hold.

TIPS & TRICKS

- The transformation $Y = X^2$ turns this Rayleigh-type density into an exponential with mean θ - everything follows.
- $\log L = -n \log \theta - \frac{\sum x_i^2}{\theta} + \text{const}$, so the MLE is the mean of the squares.
- Unbiasedness here is unusual for an MLE - it happens because the model is exponential in $\sum X_i^2$.

STEP-BY-STEP SOLUTION

1. $\log L = n \log 2 + \sum \log x_i - n \log \theta - \frac{\sum x_i^2}{\theta}$, so $\frac{\partial \log L}{\partial \theta} = -\frac{n}{\theta} + \frac{\sum x_i^2}{\theta^2} = 0$ gives $\hat{\theta} = \frac{1}{n} \sum x_i^2$. Statement I true.

2. Setting $Y = X^2$, $f_Y(y) = \frac{1}{\theta} e^{-y/\theta}$, so $E(X^2) = \theta$ and $E(\hat{\theta}) = \theta$. Statement II true.

3. By factorisation the likelihood depends on the data only through $\sum x_i^2$, so $\hat{\theta}$ is sufficient. Statement III true.
4. Option (d).
-

Q31.

Let $X_i \sim U(0, \theta)$, $\theta \in (0, \infty)$. Then which of the following statements are correct?

I. $X_{(n)}$ is maximum likelihood estimate of θ .

II. $X_{(n)}$ is sufficient for θ .

III. Family of pdfs of X is complete.

Select the answer using the code given below:

A. I and II only

B. II and III only

C. I and III only

D. I, II and III

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

For $U(0, \theta)$: $X_{(n)}$ is both the MLE and sufficient, and the family $\{U(0, \theta)\}$ is complete. All three statements hold.

TIPS & TRICKS

- $L = \theta^{-n} I(X_{(n)} \leq \theta)$ is decreasing, so the MLE sits at the boundary $X_{(n)}$.
- Completeness: $\int_0^\theta g(x) dx = 0$ for all θ forces $g \equiv 0$ by differentiation.
- Complete + sufficient makes $\frac{n+1}{n} X_{(n)}$ the UMVUE of θ .

STEP-BY-STEP SOLUTION

1. I: $L(\theta) = \theta^{-n}$ for $\theta \geq X_{(n)}$, decreasing, so $\hat{\theta} = X_{(n)}$. True.

2. II: the same factorisation shows $X_{(n)}$ is sufficient. True.

3. III: if $E_\theta[g(X)] = \frac{1}{\theta} \int_0^\theta g(x) dx = 0$ for all $\theta > 0$, then $\int_0^\theta g = 0$ for all θ ; differentiating gives $g(\theta) = 0$ a.e.

So the family is complete. True.

4. Option (d).

SHARED CONTEXT

Let the probability density function $f_{\theta}(x) = \frac{1}{\theta} e^{-x/\theta}$, $0 < x < \infty$.

Q32.

What is the maximum likelihood estimator of θ ?

A. $\frac{1}{2} \bar{x}$

B. $\bar{x}$

C. $2\bar{x}$

D. $4\bar{x}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\log L = -n \log \theta - \frac{\sum x_i}{\theta} \text{ gives } \hat{\theta} = \bar{x}.$$

TIPS & TRICKS

- Here θ is the MEAN of the exponential, so the sample mean estimates it directly.
- If the density were $\theta e^{-\theta x}$ (rate parametrisation) the MLE would be $1/\bar{x}$ - read the density carefully.
- $\bar{x}$ is simultaneously MLE, MoM estimator, UMVUE and MVB estimator here.

STEP-BY-STEP SOLUTION

$$1. L(\theta) = \theta^{-n} \exp\left(-\frac{\sum x_i}{\theta}\right), \text{ so } \log L = -n \log \theta - \frac{\sum x_i}{\theta}.$$

$$2. \frac{\partial \log L}{\partial \theta} = -\frac{n}{\theta} + \frac{\sum x_i}{\theta^2} = 0 \Rightarrow \hat{\theta} = \frac{\sum x_i}{n} = \bar{x}.$$

3. Option (b).

SHARED CONTEXT

Consider a random sample from a normal $N(\mu, \sigma^2)$.

Q33.

What is the maximum likelihood estimator (MLE) of $\hat{\mu}$?

A. $\bar{x}/2$

B. $\bar{x}$

C. $3\bar{x}/2$

D. None of the above

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\frac{\partial \log L}{\partial \mu} = \frac{\sum (x_i - \mu)}{\sigma^2} = 0 \text{ gives } \hat{\mu} = \bar{x}.$$

TIPS & TRICKS

- The MLE of a normal mean is the sample mean, whether or not σ^2 is known.
- It is simultaneously unbiased, sufficient, consistent, UMVUE and MVB.
- No scaling factor is involved - options (a) and (c) have no derivation behind them.

STEP-BY-STEP SOLUTION

$$1. \log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2.$$

$$2. \frac{\partial \log L}{\partial \mu} = \frac{1}{\sigma^2} \sum (x_i - \mu) = 0 \Rightarrow \sum x_i = n\mu.$$

$$3. \hat{\mu} = \bar{x} - \text{option (b).}$$

SHARED CONTEXT

Consider a random sample from a normal $N(\mu, \sigma^2)$.

Q34.

What is the maximum likelihood estimator (MLE) of σ^2 when μ is not known?

A. $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

B. $s^2 = \frac{1}{n+1} \sum_{i=1}^n (x_i - \bar{x})^2$

C. $\check{s}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

D. None of the above

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

The MLE of σ^2 uses divisor n : $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ (biased).

TIPS & TRICKS

- MLE divisor is n ; the UNBIASED divisor is $n-1$. The question asks for the MLE.
- $E(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2$ - biased downwards.
- Both estimators are consistent; only one is unbiased and only one is the MLE.

STEP-BY-STEP SOLUTION

1. $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{\sum (x_i - \mu)^2}{2\sigma^2}$; substituting $\hat{\mu} = \bar{x}$ and differentiating in σ^2 :

2. $-\frac{n}{2\sigma^2} + \frac{\sum (x_i - \bar{x})^2}{2\sigma^4} = 0$.

3. $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ - option (c).

Q35.

If $X \sim \text{Exp}(\lambda)$, then based on a random sample of size n from this population, consider the following statements:

I. MLE and MME estimators of λ are equal to $\hat{\lambda} = 1/\bar{x}$.

II. MLE and MME estimators of λ are equal to $\hat{\lambda} = \bar{x}$.

III. MLE and MME of λ are unequal.

Which of the statements given above is correct?

A. I

B. II

C. III

D. Neither I nor II nor III

Correct answer: A

EXAM SHORTCUT AI-derived explanation

For $\text{Exp}(\lambda)$ (rate parametrisation), $E(X) = 1/\lambda$, so both the MLE and the moment estimator are $1/\bar{x}$.

TIPS & TRICKS

- Rate parametrisation: $f(x) = \lambda e^{-\lambda x}$, mean $1/\lambda$ - so the estimator is the **RECIPROCAL** of $\bar{x}$.
- If the parameter were the **MEAN** instead, both estimators would be $\bar{x}$ - read the parametrisation.
- MoM and MLE agree here because λ is a one-to-one function of the first moment.

STEP-BY-STEP SOLUTION

1. $\log L = n \log \lambda - \lambda \sum x_i$, so $\frac{\partial \log L}{\partial \lambda} = \frac{n}{\lambda} - \sum x_i = 0$ gives $\hat{\lambda}_{MLE} = \frac{1}{\bar{x}}$.

2. $E(X) = \frac{1}{\lambda}$, so the moment equation $\frac{1}{\lambda} = \bar{x}$ gives $\hat{\lambda}_{MME} = \frac{1}{\bar{x}}$.

3. The two coincide at $1/\bar{x}$ - statement I is correct, option (a).

Q36.

Let $x_1, x_2, \dots, x_n$ be a random sample from a population with probability density function $f(x; \theta) = \theta x^{\theta-1}$; $0 < x < 1, \theta > 0$. Which of the following statements are correct?

- I. $\prod_{i=1}^n x_i$ is a sufficient estimator of θ .
- II. $\sum_{i=1}^n x_i$ is a sufficient estimator of θ .
- III. $-\frac{n}{\sum_{i=1}^n \log x_i}$ is a maximum likelihood estimator of θ .

Select the answer using the code given below.

- A. I and II only
- B. I and III only**
- C. II and III only
- D. I, II and III

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$\prod x_i$ is sufficient (I true), $\sum x_i$ is not (II false), and the MLE is $-n/\sum \log x_i$ (III true).

TIPS & TRICKS

- The likelihood involves $(\prod x_i)^{\theta-1}$ - the product, never the sum.
- $\log L = n \log \theta + (\theta-1) \sum \log x_i$, so $\hat{\theta} = -n / \sum \log x_i$.
- $\sum \log x_i < 0$, so the MLE is positive as required.

STEP-BY-STEP SOLUTION

- $L(\theta) = \theta^n (\prod x_i)^{\theta-1}$, so $\prod x_i$ is sufficient: I true, and $\sum x_i$ plays no role: II false.
- $\log L = n \log \theta + (\theta-1) \sum \log x_i$, and $\frac{\partial \log L}{\partial \theta} = \frac{n}{\theta} + \sum \log x_i = 0$ gives $\hat{\theta} = -\frac{n}{\sum \log x_i}$: III true.
- Hence I and III only - option (b).

Q37.

Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$; $-\infty < \mu < \infty, \sigma^2 > 0$, where both μ and σ^2 are unknown. Which of the following statements are correct?

I. $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ maximize the likelihood function $L(\mu, \sigma^2; x)$.

II. $\hat{\mu} = \bar{x}$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ maximize the likelihood function $L(\mu, \sigma^2; x)$.

III. $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimator of σ^2 .

IV. $S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimator of σ^2 .

Select the answer using the code given below.

A. II and IV

B. II and III

C. I and III

D. I and IV

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

The likelihood is maximised by $(\bar{x}, \frac{1}{n} \sum (x_i - \bar{x})^2)$ (I true, II false); and unbiasedness belongs to the $n-1$ divisor (IV true, III false).

TIPS & TRICKS

- The MLE and the unbiased estimator of σ^2 are different estimators - the question tests exactly that split.
- Maximising $\Rightarrow$ divisor n ; unbiased $\Rightarrow$ divisor $n-1$.
- $E\left[\frac{1}{n} \sum (x_i - \bar{x})^2\right] = \frac{n-1}{n} \sigma^2 < \sigma^2$.

STEP-BY-STEP SOLUTION

1. Maximising $\log L$ gives $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$: statement I true, statement II false.

2. $E\left[\sum (x_i - \bar{x})^2\right] = (n-1)\sigma^2$, so $\hat{\sigma}^2$ is biased (III false) while S^2 with divisor $n-1$ is unbiased (IV true).

3. Hence I and IV - option (d).

SI & HT

2026 Q38

Conceptual

Easy

Q38.

Let $x_1, x_2, \dots, x_n$ be a random sample from $f(x; \alpha, \beta) = \frac{1}{\beta - \alpha}$, $\alpha < x < \beta$; 0 otherwise. Then the MLEs of α and β are respectively given by

- A. $\hat{\alpha} = \bar{x}$ and $\hat{\beta} = \bar{x} - 1$
- B. $\hat{\alpha} = 1/\bar{x}$ and $\hat{\beta} = 2/\bar{x}$
- C. $\hat{\alpha} = x_{(1)}$ and $\hat{\beta} = x_{(n)}$
- D. $\hat{\alpha} = x_1$ and $\hat{\beta} = x_n$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$L = (\beta - \alpha)^{-n}$ is maximised by making the interval as short as the data allow: $\hat{\alpha} = x_{(1)}$, $\hat{\beta} = x_{(n)}$.

TIPS & TRICKS

- Support-dependent parameters: never differentiate, push to the boundary.
- Note the ORDER statistics, not the first and last observations - option (d) is the typographic trap.
- Both MLEs are biased inward; the UMVUEs extrapolate outward.

STEP-BY-STEP SOLUTION

1. $L(\alpha, \beta) = \frac{1}{(\beta - \alpha)^n}$ provided $\alpha \leq X_{(1)}$ and $\beta \geq X_{(n)}$.

2. The likelihood increases as $\beta - \alpha$ decreases, so the maximum occurs at the tightest admissible interval.

3. $\hat{\alpha} = x_{(1)}$ and $\hat{\beta} = x_{(n)}$ - option (c).

Method of Moments

Q1.

Let $(X_1, \dots, X_n)$ be a random sample of size n from $f(x) = \frac{1}{\theta_2 - \theta_1}$, $\theta_1 < x < \theta_2$. Let T_1, T_2 be the method-of-moments estimates of θ_1, θ_2 . Which is correct?

- A. $T_1 = \bar{X} + \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$ and $T_2 = \bar{X} - \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$
- B. ✓ $T_1 = \bar{X} - \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$ and $T_2 = \bar{X} + \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$
- C. $T_1 = \bar{X} + \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$ and $T_2 = \bar{X} + \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$
- D. $T_1 = \bar{X} - \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$ and $T_2 = \bar{X} - \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

For $U(\theta_1, \theta_2)$: mean $= \frac{\theta_1 + \theta_2}{2}$, variance $= \frac{(\theta_2 - \theta_1)^2}{12}$. Equating gives $\hat{\theta}_{1,2} = \bar{X} \mp \sqrt{3m_2}$.

TIPS & TRICKS

- The half-width is $\sqrt{3}$ times the sample standard deviation (with divisor n) - the factor $\sqrt{3}$ is the whole question.
- Order matters: $\theta_1 < \theta_2$, so the minus sign belongs to T_1 . Options (c) and (d) give equal estimates, which is impossible.
- Derivation: $\frac{(\theta_2 - \theta_1)^2}{12} = m_2 \Rightarrow \theta_2 - \theta_1 = 2\sqrt{3m_2}$, so each endpoint is $\sqrt{3m_2}$ from the mean.

STEP-BY-STEP SOLUTION

- For $U(\theta_1, \theta_2)$: $\mu_1' = \frac{\theta_1 + \theta_2}{2}$ and $\mu_2 = \frac{(\theta_2 - \theta_1)^2}{12}$.
- Method of moments: $\frac{\hat{\theta}_1 + \hat{\theta}_2}{2} = \bar{X}$ and $\frac{(\hat{\theta}_2 - \hat{\theta}_1)^2}{12} = m_2 = \frac{1}{n} \sum (X_i - \bar{X})^2$.
- The second gives $\hat{\theta}_2 - \hat{\theta}_1 = 2\sqrt{3m_2}$, so each endpoint is $\sqrt{3m_2}$ away from $\bar{X}$.
- Hence $T_1 = \bar{X} - \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$ and $T_2 = \bar{X} + \sqrt{\frac{3}{n} \sum (X_i - \bar{X})^2}$ - option (b).

Q2.

If $X_1, \dots, X_n$ is a random sample from $f(x, \theta) = \frac{2(\theta-x)}{\theta^2}$; $0 < x < \theta$, the estimator of θ by the method of moments is

- A. $\bar{X}$
 B. $3\bar{X}$
 C. $\bar{X}/3$
 D. $\bar{X}+3$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$E(X) = \theta/3$ for this decreasing triangular density, so the moment estimator is $\hat{\theta} = 3\bar{X}$.

TIPS & TRICKS

- $\int_0^\theta x \cdot \frac{2(\theta-x)}{\theta^2} dx = \frac{2}{\theta^2} \left[\frac{\theta^3}{2} - \frac{\theta^3}{3} \right] = \frac{\theta}{3}$.
- For the mirror-image density $\frac{2x}{\theta^2}$ the mean would be $\frac{2\theta}{3}$ - check which way the triangle leans.
- Method of moments: set $E(X) = \bar{X}$ and solve; no likelihood needed.

STEP-BY-STEP SOLUTION

1. $E(X) = \int_0^\theta x \cdot \frac{2(\theta-x)}{\theta^2} dx = \frac{2}{\theta^2} \left[\frac{\theta \cdot \theta^2}{2} - \frac{\theta^3}{3} \right] = \frac{2}{\theta^2} \cdot \frac{\theta^3}{6} = \frac{\theta}{3}$.
2. Equating the population and sample first moments: $\frac{\hat{\theta}}{3} = \bar{X}$.
3. Hence $\hat{\theta} = 3\bar{X}$ - option (b).

Q3.

Let 4,3,6,3,6,2,3,4,4,3,5,5 be a sample of size 12 from a geometric population, $f(x)=\theta(1-\theta)^{x-1}$, $x=1,2,3,\dots$. The method-of-moments estimator of θ

A. is 0.25.

B. is 4.

C. is 48.

D. does not exist.

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Geometric on $\{1,2,\dots\}$ has $E(X)=1/\theta$; $\bar{x}=48/12=4$, so $\hat{\theta}=1/4=0.25$.

TIPS & TRICKS

- For $f(x)=\theta(1-\theta)^{x-1}$ (support starting at 1), $E(X)=1/\theta$.
- If the support started at 0 the mean would be $(1-\theta)/\theta$ - check the support before writing the mean.
- Sum the data once: $4+3+6+3+6+2+3+4+4+3+5+5=48$.

STEP-BY-STEP SOLUTION

1. For the geometric distribution $f(x)=\theta(1-\theta)^{x-1}$, $x=1,2,\dots$, the mean is $E(X)=\frac{1}{\theta}$.
2. $\sum x_i=4+3+6+3+6+2+3+4+4+3+5+5=48$, so $\bar{x}=\frac{48}{12}=4$.
3. Method of moments: $\frac{1}{\theta}=\bar{x}=4$, so $\hat{\theta}=0.25$.
4. Option (a).

Q4.

Let $x_1, \dots, x_n$ be from $f(x, \theta) = \frac{x}{\theta} \exp\left(-\frac{x^2}{2\theta}\right)$; $x > 0, \theta > 0$. The method-of-moments estimator of θ is

- A. $\bar{x}$
 B. $\bar{x}^2$
 C. $\frac{\bar{2x}}{\pi}$
 D. $\frac{\sqrt{2\bar{x}^2}}{\pi}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

For this Rayleigh form $E(X) = \sqrt{\pi\theta/2}$; equating to $\bar{x}$ gives $\hat{\theta} = \frac{2\bar{x}^2}{\pi}$.

TIPS & TRICKS

- Rayleigh with $f(x) = \frac{x}{\theta} e^{-x^2/(2\theta)}$ has $E(X) = \sqrt{\frac{\pi\theta}{2}}$ and $E(X^2) = 2\theta$.
- Since $E(X) \propto \sqrt{\theta}$, the moment estimator must involve $\bar{x}^2$ - options linear in $\bar{x}$ are impossible.
- Using the SECOND moment instead gives the simpler $\hat{\theta} = \frac{1}{2n} \sum x_i^2$, which is also the MLE.

STEP-BY-STEP SOLUTION

1. $E(X) = \int_0^\infty x \cdot \frac{x}{\theta} e^{-x^2/(2\theta)} dx$. Substituting $x = \sqrt{2\theta} u$ gives $E(X) = \sqrt{2\theta} \int_0^\infty 2u^2 e^{-u^2} du = \sqrt{2\theta} \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\frac{\pi\theta}{2}}$.

2. Method of moments: $\sqrt{\frac{\pi\hat{\theta}}{2}} = \bar{x}$.

3. Squaring: $\frac{\pi\hat{\theta}}{2} = \bar{x}^2$, so $\hat{\theta} = \frac{2\bar{x}^2}{\pi}$.

4. Option (d).

SHARED CONTEXT

Let $X_1, \dots, X_n$ be iid $N(\mu, \mu)$ — i.e. mean = variance = μ ($\mu > 0$).

Q5.

Consider:

1. $(\sum x_i, x_n)$ is sufficient for μ .
2. Moment estimator of μ is $\bar{x}$.

Codes:

- A. 1 only
- B. 2 only**
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$\sum x_i^2$ (not $\sum x_i$) is the sufficient statistic here, so statement 1 fails; but $E(X) = \mu$ makes $\bar{x}$ the moment estimator, so statement 2 holds.

TIPS & TRICKS

- $(\sum x_i, x_n)$ cannot reproduce $\sum x_i^2$, so it is not sufficient.
- Method of moments uses the FIRST moment by default: $E(X) = \mu = \bar{x}$.
- Note the MoM estimator $\bar{x}$ differs from the MLE $\frac{-1 + \sqrt{1 + 4m_2}}{2}$ - a good illustration that the two methods disagree.

STEP-BY-STEP SOLUTION

1. From the previous item, $\log L$ depends on the data only through $\sum x_i^2$; the pair $(\sum x_i, x_n)$ does not determine $\sum x_i^2$, so it is not sufficient. Statement 1 false.
2. The first population moment is $E(X) = \mu$. Equating to the first sample moment gives $\hat{\mu} = \bar{x}$. Statement 2 true.
3. Hence 2 only - option (b).

SHARED CONTEXT

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from uniform distribution $U(0, \theta)$. Let $T_1 = \frac{2}{n} \sum_{i=1}^n X_i$ and $T_2 = \max(X_1, X_2, X_3, \dots, X_n)$.

Q6.

Consider the following statements:

1. T_1 is not sufficient but T_2 is a sufficient estimator of θ .
2. T_1 is the method of moments estimator of θ .

Which of the statements given above is/are correct?

- A. 1 only
 B. 2 only
 C. Both 1 and 2
 D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$T_2 = X_{(n)}$ is sufficient while $T_1 = 2\bar{X}$ is not; and T_1 is exactly the moment estimator since $E(X) = \theta/2$. Both statements hold.

TIPS & TRICKS

- $U(0, \theta)$: sufficiency lives entirely in the maximum; the mean discards information about the boundary.
- Method of moments: $\bar{X} = \hat{\theta}/2 \Rightarrow \hat{\theta} = 2\bar{X}$.
- A moment estimator is rarely a function of the sufficient statistic - this pair is the textbook illustration.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \theta^{-n} I(X_{(n)} \leq \theta)$, so $T_2 = X_{(n)}$ is sufficient; $T_1 = 2\bar{X}$ does not determine $X_{(n)}$, so it is not sufficient.
Statement 1 true.
2. $E(X) = \frac{\theta}{2}$, so equating moments gives $\hat{\theta} = 2\bar{X} = T_1$. Statement 2 true.
3. Option (c).

SHARED CONTEXT

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

Q7.

The estimator for 'a' by the method of moments is:

- A. $\bar{X} - \sqrt{\frac{\sum (X_i - \bar{X})^2}{n}}$
- B. $\bar{X} - \sqrt{\frac{2 \sum (X_i - \bar{X})^2}{n}}$
- C. ✓ $\bar{X} - \sqrt{\frac{3 \sum (X_i - \bar{X})^2}{n}}$
- D. $\bar{X} - 2\sqrt{\frac{\sum (X_i - \bar{X})^2}{n}}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

For $U(a, b)$, $\hat{a} = \bar{X} - \sqrt{3m_2}$ where $m_2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ - the factor $\sqrt{3}$ is the point.

TIPS & TRICKS

- $V(X) = \frac{(b-a)^2}{12}$ gives half-range $= \sqrt{3} \times$ standard deviation.
- Use the sample second central moment with divisor n (method of moments convention).
- Check numerically: for $U(0,1)$, $m_2 \approx 1/12$ and $\sqrt{3/12} = 0.5$ - exactly the half-range.

STEP-BY-STEP SOLUTION

1. For $U(a, b)$: $\mu_1' = \frac{a+b}{2}$ and $\mu_2' = \frac{(b-a)^2}{12}$.
2. Method of moments: $\frac{\hat{a} + \hat{b}}{2} = \bar{X}$ and $\frac{(\hat{b} - \hat{a})^2}{12} = m_2 = \frac{1}{n} \sum (X_i - \bar{X})^2$.
3. The second equation gives $\hat{b} - \hat{a} = 2\sqrt{3m_2}$, so each endpoint lies $\sqrt{3m_2}$ from $\bar{X}$.

4. Hence $\hat{\sigma}^2 = \frac{3 \sum (X_i - \bar{X})^2}{n}$ - option (c).

SHARED CONTEXT

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

Q8.

The estimator for 'b' by the method of moments is:

A. $\frac{\overline{\sum (X_i - \bar{X})^2}}{\bar{X} + \sqrt{\frac{n}{n}}}$

B. $\frac{2 \sum (X_i - \bar{X})^2}{\bar{X} + \sqrt{\frac{n}{n}}}$

C. ✓ $\frac{3 \sum (X_i - \bar{X})^2}{\bar{X} + \sqrt{\frac{n}{n}}}$

D. $\frac{\sum (X_i - \bar{X})^2}{\bar{X} + 2\sqrt{\frac{n}{n}}}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

By symmetry with the previous item, $\hat{b} = \bar{X} + \sqrt{3m_2}$.

TIPS & TRICKS

- $\hat{a}$ and $\hat{b}$ are equidistant from $\bar{X}$ - the only difference is the sign.
- Consistency check: $\hat{b} - \hat{a} = 2\sqrt{3m_2}$, matching the moment equation for the variance.
- The same $\sqrt{3}$ factor was tested in 2018 Q27 and 2025 Q39-40.

STEP-BY-STEP SOLUTION

1. From $\frac{\hat{a} + \hat{b}}{2} = \bar{X}$ and $\hat{b} - \hat{a} = 2\sqrt{3m_2}$:

2. $\hat{b} = \bar{X} + \sqrt{3m_2}$ where $m_2 = \frac{1}{n} \sum (X_i - \bar{X})^2$.

3. Hence $\hat{b} = \bar{X} + \sqrt{\frac{3 \sum (X_i - \bar{X})^2}{n}}$ - option (c).

Q9.

Let 2.5, -2.0, 1.5, 3.5, 0.5 be the observations of a random sample of size 5 from the continuous distribution with pdf

$$f(x) = \frac{1}{8} e^{-|x-2|} + \frac{3}{4\sqrt{2\pi}} e^{-\frac{1}{2}(x-\theta)^2}; \quad x, \theta \in \mathbb{R}, \theta \text{ unknown.}$$

Then the method of moment estimators of θ belongs to the interval:

A. (0.60, 0.70)

B. (0.70, 0.80)

C. (0.80, 0.90)

D. (0.90, 0.99)

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

The density is a mixture with weights $\frac{1}{4}$ (Laplace at 2) and $\frac{3}{4}$ (normal at θ), so $E(X) = \frac{1}{2} + \frac{3}{4}\theta$. With

$$\bar{x} = 1.2: \hat{\theta} = \frac{0.7}{0.75} = 0.933.$$

TIPS & TRICKS

- Read the mixture weights by integrating each piece: $\int \frac{1}{8} e^{-|x-2|} dx = \frac{1}{4}$ and the normal piece carries $\frac{3}{4}$.
- Mixture mean = weighted average of component means: $\frac{1}{4}(2) + \frac{3}{4}\theta$.
- $\sum x_i = 2.5 - 2.0 + 1.5 + 3.5 + 0.5 = 6.0$, so $\bar{x} = 1.2$.

STEP-BY-STEP SOLUTION

1. The Laplace component $\frac{1}{8} e^{-|x-2|}$ integrates to $\frac{1}{4}$ and has mean 2; the normal component carries weight $\frac{3}{4}$ and mean θ .

2. Hence $E(X) = \frac{1}{4}(2) + \frac{3}{4}\theta = \frac{1}{2} + \frac{3}{4}\theta$.

3. $\bar{x} = \frac{2.5 - 2.0 + 1.5 + 3.5 + 0.5}{5} = \frac{6.0}{5} = 1.2$.

4. Method of moments: $\frac{1}{2} + \frac{3}{4}\hat{\theta} = 1.2 \Rightarrow \hat{\theta} = \frac{0.7}{0.75} = 0.9333$.

5. This lies in (0.90,0.99) - option (d).

SI & HT

2023 Q3

Conceptual

Easy

Q10.

Consider the following statements:

Statement-I: The method of moments provides consistent estimators of population moments.

Statement-II: From the Weak Law of Large Numbers, it follows that $\frac{1}{n} \sum_{i=1}^n x_i^j \rightarrow P E(X_j)$ provided $E(X_j)$ exists.

Which one of the following is correct in respect of the above statements?

A. Statement-I and Statement-II are individually correct and Statement-II is the correct explanation of Statement-I.

B. Statement-I and Statement-II are individually correct but Statement-II is **not** the correct explanation of Statement-I.

C. Statement-I is correct but Statement-II is incorrect.

D. Statement-I is incorrect but Statement-II is correct.

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Both statements are true, and the WLLN is precisely why sample moments converge in probability to population moments - so II explains I.

TIPS & TRICKS

- Method of moments works because sample moments are consistent for population moments (WLLN) and the solving map is continuous.
- Consistency of MoM ESTIMATORS then follows by the continuous mapping theorem.
- MoM estimators are consistent but generally neither unbiased nor efficient.

STEP-BY-STEP SOLUTION

1. Statement II: by the weak law of large numbers, $\frac{1}{n} \sum x_i^j \rightarrow P E(X^j)$ whenever that expectation exists.
True.
2. Statement I: the method of moments equates sample to population moments, so it inherits this consistency. True.
3. The WLLN is exactly the mechanism that makes the method of moments consistent, so II explains I.
4. Option (a).

SHARED CONTEXT

Let $x_1 = \frac{17}{3}$, $x_2 = \frac{10}{3}$, $x_3 = \frac{13}{3}$, $x_4 = 4$ be the observed values of a random sample of size 4 from the distribution having pdf

$$f(x, \theta) = \frac{1}{3} \left[\frac{1}{2\theta} e^{-\frac{x}{2\theta}} + \frac{1}{\theta^2} e^{-\frac{x}{\theta^2}} + e^{-x} \right]; \quad x > 0, \theta > 0$$

Q11.

What is $E(X)$ equal to?

A. $\frac{\theta^2 + \theta + 1}{3}$

B. $\frac{\theta^2 + 2\theta + 1}{3}$

C. $\frac{\theta^2 + 2\theta + 2}{3}$

D. $\frac{\theta^2 + 3\theta + 1}{3}$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

An equal-weight mixture of three exponentials with means 2θ , θ^2 and 1: $E(X) = \frac{2\theta + \theta^2 + 1}{3} = \frac{(\theta + 1)^2}{3}$.

TIPS & TRICKS

- Each bracketed term $\frac{1}{m} e^{-x/m}$ is an exponential density with mean m - read off 2θ , θ^2 and 1.
- Mixture mean = weighted average of component means, with weights $\frac{1}{3}$ each.
- $\theta^2 + 2\theta + 1 = (\theta + 1)^2$ - recognising the square makes the companion item immediate.

STEP-BY-STEP SOLUTION

- The density is $\frac{1}{3} \left[\frac{1}{2\theta} e^{-x/(2\theta)} + \frac{1}{\theta^2} e^{-x/\theta^2} + e^{-x} \right]$, an equal mixture of exponentials with means 2θ , θ^2 and 1.
- $E(X) = \frac{1}{3} (2\theta + \theta^2 + 1)$.

$$3. = \frac{\theta^2 + 2\theta + 1}{3} - \text{option (b).}$$

SHARED CONTEXT

Let $x_1 = \frac{17}{3}$, $x_2 = \frac{10}{3}$, $x_3 = \frac{13}{3}$, $x_4 = 4$ be the observed values of a random sample of size 4 from the distribution having pdf

$$f(x, \theta) = \frac{1}{3} \left[\frac{1}{2\theta} e^{-\frac{x}{2\theta}} + \frac{1}{\theta^2} e^{-\frac{x}{\theta^2}} + e^{-x} \right]; \quad x > 0, \theta > 0$$

Q12.

The method of moment estimate of θ is:

A. $\sqrt[3]{13-1}$

B. $\sqrt{13}+1$

C. 1

D. 3

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\bar{x} = \frac{13}{3} \text{ and } \frac{(\theta+1)^2}{3} = \frac{13}{3} \text{ gives } (\theta+1)^2 = 13, \text{ so } \hat{\theta} = \sqrt{13}-1.$$

TIPS & TRICKS

- $\sum x_i = \frac{17}{3} + \frac{10}{3} + \frac{13}{3} + 4 = \frac{52}{3}$, so $\bar{x} = \frac{13}{3}$.
- Using $E(X) = \frac{(\theta+1)^2}{3}$ turns the moment equation into a perfect square.
- Take the positive root since $\theta > 0$: $\theta = \sqrt{13}-1 \approx 2.606$.

STEP-BY-STEP SOLUTION

$$1. \bar{x} = \frac{1}{4} \left(\frac{17}{3} + \frac{10}{3} + \frac{13}{3} + 4 \right) = \frac{1}{4} \cdot \frac{52}{3} = \frac{13}{3}.$$

$$2. \text{Method of moments: } \frac{(\hat{\theta}+1)^2}{3} = \frac{13}{3}.$$

$$3. (\hat{\theta}+1)^2 = 13 \Rightarrow \hat{\theta}+1 = \sqrt{13} \text{ (positive root).}$$

$$4. \hat{\theta} = \sqrt{13}-1 \text{ - option (a).}$$

SHARED CONTEXT

Let $(X_1, X_2, X_3, \dots, X_n)$ be a random sample from a distribution with pdf $f(x/\alpha, \beta) = \frac{e^{-x/\beta} x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)}$; $x > 0, \alpha > 1, \beta > 0$.

Q13.

What is the estimator of α obtained by method of moments?

- A. $\checkmark \quad \frac{n\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$
- B. $\frac{\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$
- C. $\frac{\bar{X}}{\sum_{i=1}^n (X_i - \bar{X})^2}$
- D. $\frac{n\bar{X}}{\sum_{i=1}^n (X_i - \bar{X})^2}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Gamma has mean $\alpha\beta$ and variance $\alpha\beta^2$, so $\hat{\alpha} = \frac{\bar{X}^2}{m_2} = \frac{n\bar{X}^2}{\sum (X_i - \bar{X})^2}$.

TIPS & TRICKS

- $\frac{(\text{mean})^2}{\text{variance}} = \frac{\alpha^2 \beta^2}{\alpha \beta^2} = \alpha$ - the shape is the squared coefficient of variation inverted.
- $m_2 = \frac{1}{n} \sum (X_i - \bar{X})^2$, so writing it out introduces the factor n in the numerator.
- Check dimensions: α is dimensionless, and $\bar{X}^2/m_2$ indeed is.

STEP-BY-STEP SOLUTION

1. For Gamma(α, β) in this parametrisation: $E(X) = \alpha\beta$ and $V(X) = \alpha\beta^2$.

2. Method of moments: $\hat{\alpha}\hat{\beta} = \bar{X}$ and $\hat{\alpha}\hat{\beta}^2 = m_2 = \frac{1}{n} \sum (X_i - \bar{X})^2$.

3. Dividing: $\hat{\beta} = \frac{m_2}{\bar{X}}$, hence $\hat{\alpha} = \frac{\bar{X}}{\hat{\beta}} = \frac{\bar{X}^2}{m_2}$.

$$4. = \frac{n\bar{X}^2}{\sum (X_i - \bar{X})^2} - \text{option (a).}$$

SHARED CONTEXT

Let $(X_1, X_2, X_3, \dots, X_n)$ be a random sample from a distribution with pdf $f(x/\alpha, \beta) = \frac{e^{-x/\beta} x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)}$; $x > 0, \alpha > 1, \beta > 0$.

Q14.

What is the estimator of β obtained by method of moments?

A. $\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n\bar{X}}$

B. $\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\bar{X}}$

C. $\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\bar{X}^2}$

D. $\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n\bar{X}^2}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\hat{\beta} = \frac{m_2}{\bar{X}} = \frac{\sum (X_i - \bar{X})^2}{n\bar{X}}.$$

TIPS & TRICKS

- $\frac{\text{variance}}{\text{mean}} = \frac{\alpha\beta^2}{\alpha\beta} = \beta$ - the scale is the variance-to-mean ratio.
- Remember m_2 carries the divisor n .
- Cross-check: $\hat{\alpha}\hat{\beta} = \frac{n\bar{X}^2}{\sum (X_i - \bar{X})^2} \cdot \frac{\sum (X_i - \bar{X})^2}{n\bar{X}} = \bar{X}$ - consistent.

STEP-BY-STEP SOLUTION

1. From $\hat{\alpha}\hat{\beta} = \bar{X}$ and $\hat{\alpha}\hat{\beta}^2 = m_2$, dividing gives $\hat{\beta} = \frac{m_2}{\bar{X}}$.

2. With $m_2 = \frac{1}{n} \sum (X_i - \bar{X})^2$:

3. $\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n\bar{X}}$ - option (a).

SHARED CONTEXT

Let $(X_1, X_2, X_3, \dots, X_n)$ be independently and identically distributed random sample having pdf $f(x) = \theta x^{\theta-1}$; $0 \leq x \leq 1$, $0 < \theta < \infty$.

Q15.

What is $E(X)$ equal to?

A. $\frac{\theta}{\theta+1}$

B. $\frac{\theta}{\theta-1}$

C. $\frac{2\theta}{\theta+1}$

D. $\frac{2\theta}{\theta-1}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$E(X) = \int_0^1 x \cdot \theta x^{\theta-1} dx = \frac{\theta}{\theta+1}.$$

TIPS & TRICKS

- $\int_0^1 \theta x^{\theta} dx = \frac{\theta}{\theta+1}$ - one line.
- Sanity check: as $\theta \rightarrow \infty$ the density piles up near 1 and $E(X) \rightarrow 1$; as $\theta \rightarrow 0$, $E(X) \rightarrow 0$.
- $E(X) < 1$ always, which rules out options with $\theta-1$ in the denominator.

STEP-BY-STEP SOLUTION

1. $E(X) = \int_0^1 x \cdot \theta x^{\theta-1} dx = \theta \int_0^1 x^{\theta} dx.$

2. $= \theta \left[\frac{x^{\theta+1}}{\theta+1} \right]_0^1 = \frac{\theta}{\theta+1}.$

3. Option (a).

SHARED CONTEXT

Let $(X_1, X_2, X_3, \dots, X_n)$ be independently and identically distributed random sample having pdf $f(x) = \theta x^{\theta-1}$; $0 \leq x \leq 1$, $0 < \theta < \infty$.

Q16.

What is the moment estimator of θ ?

- A. $\frac{\sum_{i=1}^n X_i}{n}$
- B. ✓ $\frac{\sum_{i=1}^n X_i}{n - \sum_{i=1}^n X_i}$
- C. $\frac{\sum_{i=1}^n X_i}{1 + \sum_{i=1}^n X_i}$
- D. $\frac{\sum_{i=1}^n X_i}{n + \sum_{i=1}^n X_i}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\frac{\hat{\theta}}{\hat{\theta}+1} = \bar{X} \text{ gives } \hat{\theta} = \frac{\bar{X}}{1-\bar{X}} = \frac{\sum X_i}{n - \sum X_i}.$$

TIPS & TRICKS

- Solve $\frac{\theta}{\theta+1} = m$ for θ : $\theta = \frac{m}{1-m}$.
- Substituting $m = \frac{\sum X_i}{n}$ and clearing the fraction gives the printed form.
- Sanity: as $\bar{X} \rightarrow 1$ the estimate $\rightarrow \infty$, matching the shape of the density.

STEP-BY-STEP SOLUTION

1. Method of moments equates $E(X) = \frac{\theta}{\theta+1}$ to $\bar{X}$.

$$2. \frac{\hat{\theta}}{\hat{\theta}+1} = \bar{X} \Rightarrow \hat{\theta} = \bar{X}\hat{\theta} + \bar{X} \Rightarrow \hat{\theta}(1-\bar{X}) = \bar{X}.$$

$$3. \hat{\theta} = \frac{\bar{X}}{1-\bar{X}} = \frac{\sum X_i/n}{1 - \sum X_i/n} = \frac{\sum_{i=1}^n X_i}{n - \sum_{i=1}^n X_i}.$$

4. Option (b).

SI & HT

2025 Q15

Numerical

Easy

Q17.

Let $(1.25, 0.85, 1.35)$ be a random sample from $U(\theta, 2\theta)$, $\theta > 0$. What is the value of moment estimate of θ ?

A. 0.77

B. 1.25

C. 2.30

D. 3.45

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$E(X) = \frac{\theta + 2\theta}{2} = \frac{3\theta}{2}; \text{ with } \bar{x} = 1.15, \hat{\theta} = \frac{2\bar{x}}{3} = 0.767 \approx 0.77.$$

TIPS & TRICKS

- $U(\theta, 2\theta)$ has midpoint $\frac{3\theta}{2}$ - the mean of the endpoints.
- $\bar{x} = \frac{1.25 + 0.85 + 1.35}{3} = \frac{3.45}{3} = 1.15$.
- $\hat{\theta} = \frac{2}{3}(1.15) = 0.7667$; option (c) is the raw total 3.45 and (b) is one observation.

STEP-BY-STEP SOLUTION

1. For $U(\theta, 2\theta)$, $E(X) = \frac{\theta + 2\theta}{2} = \frac{3\theta}{2}$.

2. $\bar{x} = \frac{1.25 + 0.85 + 1.35}{3} = \frac{3.45}{3} = 1.15$.

3. Method of moments: $\frac{3\theta}{2} = 1.15 \Rightarrow \hat{\theta} = \frac{2(1.15)}{3} = 0.7667$.

4. ≈ 0.77 - option (a).

Q18.

Let $x_1=3, x_2=4, x_3=3.5, x_4=2.5$ be four observations from a distribution with pdf

$$f(x/\theta) = \frac{1}{3} \left[\frac{1}{\theta} e^{-x/\theta} + \frac{1}{\theta^2} e^{-x/\theta^2} + e^{-x} \right]; \quad \theta > 0$$

The method of moment estimator of θ based on the above observations is:

- A. 1
- B. 2.5**
- C. 3.25
- D. 3.5

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Equal mixture of exponentials with means $\theta, \theta^2, 1$ gives $E(X) = \frac{\theta^2 + \theta + 1}{3}$; $\bar{x} = 3.25$ leads to $\theta^2 + \theta - 8.75 = 0$, whose positive root is 2.5.

TIPS & TRICKS

- Identify the three component means directly from $\frac{1}{m} e^{-x/m}$: here θ, θ^2 and 1.
- $\bar{x} = \frac{3+4+3.5+2.5}{4} = \frac{13}{4} = 3.25$.
- $\theta^2 + \theta + 1 = 9.75$ gives $\theta = \frac{-1 + \sqrt{36}}{2} = 2.5$ - the discriminant is a perfect square by design.

STEP-BY-STEP SOLUTION

1. The density is an equal-weight mixture of exponentials with means θ, θ^2 and 1, so $E(X) = \frac{\theta + \theta^2 + 1}{3}$.

2. $\bar{x} = \frac{3+4+3.5+2.5}{4} = 3.25$.

3. $\frac{\hat{\theta}^2 + \hat{\theta} + 1}{3} = 3.25 \Rightarrow \hat{\theta}^2 + \hat{\theta} - 8.75 = 0$.

4. $\hat{\theta} = \frac{-1 + \sqrt{1+35}}{2} = \frac{-1+6}{2} = 2.5$ - option (b).

SHARED CONTEXT

Let the probability density function $f_{\theta}(x) = \frac{1}{\theta} e^{-x/\theta}$, $0 < x < \infty$.

Q19.

Consider the following statements:

- I. The estimator of θ with method of moments is equal to $\bar{2x}$.
- II. Both maximum likelihood estimator and method of moments estimator of θ will be same.

Which of the statements given above is/are correct?

- A. I only
- B. II only**
- C. Both I and II
- D. Neither I nor II

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$E(X) = \theta$, so the moment estimator is $\bar{x}$ (not $\bar{2x}$) - and it coincides with the MLE.

TIPS & TRICKS

- $\bar{2x}$ would be the moment estimator for $U(0, \theta)$, not for the exponential - a deliberate cross-family trap.
- For the exponential with mean θ , $\text{MoM} = \text{MLE} = \bar{x}$: the two methods agree.
- Agreement of MoM and MLE happens whenever the parameter IS a moment of the distribution.

STEP-BY-STEP SOLUTION

- $E(X) = \theta$ for $f_{\theta}(x) = \frac{1}{\theta} e^{-x/\theta}$, so the method of moments gives $\hat{\theta} = \bar{x}$, not $\bar{2x}$. Statement I false.
- The MLE is also $\bar{x}$ (previous item), so the two estimators coincide. Statement II true.
- Hence II only - option (b).

SHARED CONTEXT

Let $x_1, x_2, x_3, \dots, x_n$ be a sample from $f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$.

Q20.

What is the estimator of 'a' obtained by using method of moments?

- A. $\bar{X} + \sqrt{\frac{4 \sum (X_i - \bar{X})^2}{n}}$
- B. ✓ $\bar{X} - \sqrt{\frac{3 \sum (X_i - \bar{X})^2}{n}}$
- C. $\bar{X} - \sqrt{\frac{2 \sum (X_i - \bar{X})^2}{n}}$
- D. $\bar{X} + \sqrt{\frac{\sum (X_i - \bar{X})^2}{n}}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\hat{a} = \bar{X} - \sqrt{3m_2} \text{ with } m_2 = \frac{1}{n} \sum (X_i - \bar{X})^2.$$

TIPS & TRICKS

- The $\sqrt{3}$ comes from $V(X) = \frac{(b-a)^2}{12}$: half-range = $\sqrt{3 \times \text{variance}}$.
- Lower endpoint takes the MINUS sign.
- Same item appeared as 2018 Q27 and 2022 Q74 - a three-time repeat.

STEP-BY-STEP SOLUTION

1. $E(X) = \frac{a+b}{2}$ and $V(X) = \frac{(b-a)^2}{12}$.

2. Method of moments: $\frac{\hat{a} + \hat{b}}{2} = \bar{X}$ and $\frac{(\hat{b} - \hat{a})^2}{12} = m_2$, so $\hat{b} - \hat{a} = 2\sqrt{3m_2}$.

3. Hence $\hat{a} = \bar{X} - \sqrt{3m_2} = \bar{X} - \sqrt{\frac{3 \sum (X_i - \bar{X})^2}{n}}$.

4. Option (b).

SHARED CONTEXT

Let $x_1, x_2, x_3, \dots, x_n$ be a sample from $f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$.

Q21.

What is the estimator of 'b' obtained by using method of moments?

A. $\bar{X} - \sqrt{\frac{4 \sum (X_i - \bar{X})^2}{n}}$

B. ✓ $\bar{X} + \sqrt{\frac{3 \sum (X_i - \bar{X})^2}{n}}$

C. $\bar{X} + \sqrt{\frac{2 \sum (X_i - \bar{X})^2}{n}}$

D. $\bar{X} - \sqrt{\frac{\sum (X_i - \bar{X})^2}{n}}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\hat{b} = \bar{X} + \sqrt{3m_2}.$$

TIPS & TRICKS

- Mirror image of the previous item - only the sign changes.
- Check: $\hat{b} - \hat{a} = 2\sqrt{3m_2}$ reproduces the moment equation for the variance.
- The factor $\sqrt{3}$, not $\sqrt{2}$ or 2, is the discriminating detail.

STEP-BY-STEP SOLUTION

1. From $\frac{\hat{a} + \hat{b}}{2} = \bar{X}$ and $\hat{b} - \hat{a} = 2\sqrt{3m_2}$:

2. $\hat{b} = \bar{X} + \sqrt{3m_2} = \bar{X} + \sqrt{\frac{3 \sum (X_i - \bar{X})^2}{n}}$.

3. Option (b).

Q22.

Let $x_1, x_2, \dots, x_n$ be a random sample from the following probability distribution:

$$P(X=1) = \frac{2(1-\theta)}{2-\theta}, \quad P(X=2) = \frac{\theta}{2-\theta}; \quad \theta \in (0,1)$$

The moment estimate of θ is

- A. $\bar{x}$
- B. $2(1-\bar{x})$
- C. $2\left(1 - \frac{1}{\bar{x}}\right)$
- D. $\frac{2}{2-\bar{x}}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$E(X) = \frac{2(1-\theta) + 2\theta}{2-\theta} = \frac{2}{2-\theta}; \text{ setting this to } \bar{x} \text{ gives } \hat{\theta} = 2\left(1 - \frac{1}{\bar{x}}\right).$$

TIPS & TRICKS

- Compute $E(X) = 1 \cdot P(X=1) + 2 \cdot P(X=2)$ and watch the numerator collapse to 2.
- $\frac{2}{2-\theta} = \bar{x} \Rightarrow 2-\theta = \frac{2}{\bar{x}} \Rightarrow \theta = 2 - \frac{2}{\bar{x}}$.
- Since $X \in \{1, 2\}$, $\bar{x} \in [1, 2]$, and the formula returns $\theta \in [0, 1]$ as required.

STEP-BY-STEP SOLUTION

$$1. E(X) = 1 \cdot \frac{2(1-\theta)}{2-\theta} + 2 \cdot \frac{\theta}{2-\theta} = \frac{2-2\theta+2\theta}{2-\theta} = \frac{2}{2-\theta}.$$

$$2. \text{ Method of moments: } \frac{2}{2-\hat{\theta}} = \bar{x}.$$

$$3. 2 - \hat{\theta} = \frac{2}{\bar{x}} \Rightarrow \hat{\theta} = 2 - \frac{2}{\bar{x}} = 2\left(1 - \frac{1}{\bar{x}}\right).$$

4. Option (c).

Q23.

Let $x_1, x_2, \dots, x_n$ constitute a random sample of size n from the probability density function $f(x; \theta) = \theta e^{-\theta x}$; $0 < x < \infty$, $\theta > 0$. Consider the following statements:

- I. The estimator of θ with method of moments is $\bar{x}$.
- II. Both maximum likelihood estimator and moment estimator of θ are same.

Which of the statements given above is/are correct?

- A. I only
- B. II only**
- C. Both I and II
- D. Neither I nor II

Correct answer: B

EXAM SHORTCUT AI-derived explanation

With the RATE parametrisation $E(X) = 1/\theta$, so the moment estimator is $1/\bar{x}$, not $\bar{x}$ - statement I fails; both methods do give the same $1/\bar{x}$, so statement II holds.

TIPS & TRICKS

- $f(x) = \theta e^{-\theta x}$ means θ is the RATE, and the mean is its reciprocal.
- Compare the 2025 version where the density was $\frac{1}{\theta} e^{-x/\theta}$ (mean parametrisation) and the answer was $\bar{x}$.
- MoM and MLE agree for both parametrisations - only the formula changes.

STEP-BY-STEP SOLUTION

- $E(X) = \frac{1}{\theta}$, so the moment equation $\frac{1}{\theta} = \bar{x}$ gives $\hat{\theta} = \frac{1}{\bar{x}}$, not $\bar{x}$: statement I false.
- $\log L = n \log \theta - \theta \sum x_i$ gives $\hat{\theta}_{MLE} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}$, the same value: statement II true.
- Hence II only - option (b).

Q24.

Let $x_1, x_2, \dots, x_n$ be a random sample from a distribution with probability density function $f(x; \alpha) = \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}$;

$0 \leq x < \infty$, $\alpha > 0$. Then the estimator of α obtained by method of moments is

A. $\frac{\bar{x}}{2}$

B. $\bar{x}$

C. $\frac{1}{\bar{x}}$

D. $2\bar{x}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

With unit scale, Gamma(α) has $E(X) = \alpha$, so the moment estimator is $\hat{\alpha} = \bar{x}$.

TIPS & TRICKS

- The density $\frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}$ has scale fixed at 1, so mean = shape.
- Equating $E(X) = \alpha$ to $\bar{x}$ gives the estimator immediately - no second moment needed.
- With a free scale parameter you would need both moments (as in the 2024 items).

STEP-BY-STEP SOLUTION

- For $f(x; \alpha) = \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}$, the scale is 1 and $E(X) = \alpha$.
- Method of moments: $\hat{\alpha} = \bar{x}$.
- Option (b).

Optimal Properties of MLE

Q1.

If T is an MLE of θ and $\varphi(\theta)$ is a one-to-one function of θ , then $\varphi(t)$ is the MLE of $\varphi(\theta)$. This property is known as

A. Invariance property of MLE

B. asymptotic property of MLE

C. consistency property of MLE

D. regularity conditions of MLE

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

This is the invariance property of maximum likelihood estimation.

TIPS & TRICKS

- Invariance holds for arbitrary g , not merely one-to-one g (extended invariance, via the induced likelihood).
- It is the reason $\hat{\sigma} = \sqrt{\overline{\hat{\sigma}^2}}$ and $\hat{e}^\theta = e^{\hat{\theta}}$.
- Invariance does not preserve unbiasedness - $\bar{e^X}$ is the MLE of e^μ but is biased upwards.

STEP-BY-STEP SOLUTION

1. If $\hat{\theta}$ maximises $L(\theta)$ and φ is one-to-one, then re-parametrising by $\eta = \varphi(\theta)$ gives a likelihood maximised at $\eta = \varphi(\hat{\theta})$.
2. So the MLE of $\varphi(\theta)$ is $\varphi(\hat{\theta})$.
3. This is called the invariance property of the MLE - option (a).

Q2.

Which of the following statements are correct?

1. MLEs are consistent.
2. MLEs are always unbiased.
3. MLEs follow asymptotically normal distribution.
4. MLE need not be a unique estimator.

Codes:

A. 1 and 3 only

B. 1, 3 and 4

C. 1, 2 and 3

D. 2, 3 and 4

Correct answer: B

EXAM SHORTCUT AI-derived explanation

MLEs are consistent and asymptotically normal and may be non-unique, but they are NOT always unbiased ($\hat{\sigma}^2$ with divisor n). So 1, 3, 4.

TIPS & TRICKS

- Standard counterexample to unbiasedness: $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ in the normal model.
- Standard counterexample to uniqueness: $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, where any θ in an interval maximises L .
- The correct slogan is 'asymptotically unbiased, consistent, asymptotically normal, asymptotically efficient' - all the good properties are asymptotic.

STEP-BY-STEP SOLUTION

1. Statement 1: under regularity conditions MLEs are consistent. True.
2. Statement 2: MLEs are often biased in finite samples (e.g. $\hat{\sigma}^2$ with divisor n). False.
3. Statement 3: under regularity, $\sqrt{n}(\hat{\theta} - \theta) \rightarrow dN(0, 1/I_1(\theta))$. True.
4. Statement 4: the MLE need not be unique - e.g. $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$. True.
5. Hence 1, 3 and 4 - option (b).

Q3.

Let $X_1, \dots, X_n$ be iid $N(\mu, \sigma^2)$, $\hat{\mu}, \hat{\sigma}^2$ the mle's:

1. $\hat{\mu}$ is unbiased for μ .
2. $\hat{\sigma}^2$ is unbiased for σ^2 .

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\hat{\mu} = \bar{X}$ is unbiased, but $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ has expectation $\frac{n-1}{n} \sigma^2$ - biased.

TIPS & TRICKS

- The MLE of σ^2 uses divisor n and is the classic example of a biased MLE.
- $E\left[\sum (X_i - \bar{X})^2\right] = (n-1)\sigma^2$ always - one df is lost estimating μ .
- The bias disappears asymptotically, consistent with MLEs being asymptotically unbiased.

STEP-BY-STEP SOLUTION

1. Maximising the normal likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$.
2. $E(\hat{\mu}) = \mu$: statement 1 is true.
3. $E\left[\sum (X_i - \bar{X})^2\right] = (n-1)\sigma^2$, so $E(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2 \neq \sigma^2$: statement 2 is false.
4. Hence 1 only - option (a).

Q4.

Let $X_1, \dots, X_n$ be iid $f(x, \mu) = e^{-(x-\mu)}$, $x \geq \mu$. The MLE of $P(X_1 \geq t)$, $t > \mu$, is

- A. $e^{-(t-\bar{x})}$
- B. $e^{-(t-X_{(1)})}$, $X_{(1)} = \min(X_1, \dots, X_n)$
- C. $e^{-(t-X_{(n)})}$, $X_{(n)} = \max(X_1, \dots, X_n)$
- D. $e^{-\left(t - \frac{X_{(1)} + X_{(n)}}{2}\right)}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$P(X_1 \geq t) = e^{-(t-\mu)}$ and $\hat{\mu} = X_{(1)}$; by invariance the MLE is $e^{-(t-X_{(1)})}$.

TIPS & TRICKS

- Invariance: MLE of $g(\mu)$ is $g(\hat{\mu})$ - no fresh maximisation required.
- $\hat{\mu} = X_{(1)}$ because the support is $x \geq \mu$ and the likelihood increases in μ .
- Compute the survival function first: $\int_t^\infty e^{-(x-\mu)} dx = e^{-(t-\mu)}$.

STEP-BY-STEP SOLUTION

1. $P(X_1 \geq t) = \int_t^\infty e^{-(x-\mu)} dx = e^{-(t-\mu)}$ for $t > \mu$.
2. The MLE of μ is $\hat{\mu} = X_{(1)}$ (the likelihood $e^{n\mu - \sum x_i}$ increases in μ up to $X_{(1)}$).
3. By the invariance property, the MLE of $e^{-(t-\mu)}$ is $e^{-(t-\hat{\mu})} = e^{-(t-X_{(1)})}$.
4. Option (b).

Q5.

Let X be Bernoulli(p), $h(p)=p(1-p)$. The mle of $h(p)$ from a sample of size n is

A. $\sum_{i=1}^n X_i \left(1 - \sum_{i=1}^n X_i\right)$

B. $\frac{\bar{X}^2}{n^2}$

C. $\bar{X}(1-\bar{X})$

D. $\frac{\bar{X}}{n} \left(1 - \frac{\bar{X}}{n}\right)$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\hat{p} = \bar{X}$, so by invariance the MLE of $p(1-p)$ is $\bar{X}(1-\bar{X})$.

TIPS & TRICKS

- Invariance applies to any function, not just one-to-one ones - $p(1-p)$ is not monotone on $(0,1)$ yet the rule still holds.
- The MLE $\bar{X}(1-\bar{X})$ is biased; the UMVUE is $\frac{Y(n-Y)}{n(n-1)}$ with $Y = \sum X_i$.
- Options dividing $\bar{X}$ by n confuse the mean with the total - check the scale of $\bar{X}$ first.

STEP-BY-STEP SOLUTION

1. For a Bernoulli sample, $\log L = \left(\sum x_i\right) \log p + \left(n - \sum x_i\right) \log(1-p)$, giving $\hat{p} = \bar{X}$.
2. $h(p) = p(1-p)$.
3. By the invariance property of maximum likelihood, $h(\hat{p}) = h(\bar{X}) = \bar{X}(1-\bar{X})$.
4. Option (c).

Q6.

For a random sample $x_1, \dots, x_n$ from $N(\theta, 4)$:

1. MLE of θ is $\frac{1}{n} \sum_{i=1}^n x_i$.
2. MLE of θ^2 is $\frac{1}{n} \sum_{i=1}^n x_i^2$.

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\hat{\theta} = \bar{x}$ is correct; by invariance the MLE of θ^2 is $\bar{x}^2$, not $\frac{1}{n} \sum x_i^2$.

TIPS & TRICKS

- Invariance: MLE of $g(\theta)$ is $g(\hat{\theta})$ - apply g to the estimate, never to the data separately.
- $\frac{1}{n} \sum x_i^2$ estimates $E(X^2) = \theta^2 + 4$, a different quantity.
- The variance 4 is known here, so it plays no role in estimating θ .

STEP-BY-STEP SOLUTION

1. For $N(\theta, 4)$, $\log L = \text{const} - \frac{1}{8} \sum (x_i - \theta)^2$, so $\hat{\theta} = \bar{x}$. Statement 1 is true.
2. By the invariance property, the MLE of θ^2 is $(\hat{\theta})^2 = \bar{x}^2$.
3. $\frac{1}{n} \sum x_i^2$ has expectation $\theta^2 + 4 \neq \theta^2$ and is not the MLE. Statement 2 is false.
4. Hence 1 only - option (a).

Q7.

Consider the following statements:

1. An MLE is always unique.
2. An MLE may not be consistent.
3. An MLE may not be unbiased.

Codes:

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

Correct answer: B

EXAM SHORTCUT AI-derived explanation

MLEs need not be unique (1 false), may be inconsistent in irregular models (2 true), and are frequently biased (3 true).

TIPS & TRICKS

- Non-uniqueness: $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ has a whole interval of MLEs.
- Inconsistency: the Neyman-Scott problem (many incidental parameters) gives an inconsistent MLE of σ^2 .
- Bias: $\hat{\sigma}^2$ with divisor n in the normal model.

STEP-BY-STEP SOLUTION

1. Statement 1: for $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ every θ in an interval maximises L , so the MLE is not unique. False.
2. Statement 2: when the regularity conditions fail (e.g. the number of parameters grows with n , as in the Neyman-Scott problem), the MLE can be inconsistent. True.
3. Statement 3: $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ is the MLE and is biased. True.
4. Hence 2 and 3 only - option (b).

Q8.

Which of the following assertions is/are correct?

1. Maximum Likelihood Estimate of Poisson mean λ is unbiased, sufficient and consistent estimate for λ .
2. For $X \sim U(0, \theta)$, maximum likelihood estimate of θ is a biased, consistent and sufficient estimate for θ .

Select the correct answer using the code given below:

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Poisson MLE $\bar{X}$ is unbiased, sufficient and consistent; $U(0, \theta)$ MLE $X_{(n)}$ is biased, sufficient and consistent. Both statements hold.

TIPS & TRICKS

- $E[X_{(n)}] = \frac{n\theta}{n+1} < \theta$ - biased downward but the bias vanishes, so consistency survives.
- Both MLEs are functions of the respective sufficient statistics, as theory guarantees.
- The contrast between the two is exactly the point: unbiasedness is not a property MLEs must have.

STEP-BY-STEP SOLUTION

1. Statement 1: for Poisson, $\hat{\lambda} = \bar{X}$ with $E(\bar{X}) = \lambda$ (unbiased), $\bar{X}$ a one-to-one function of the sufficient $\sum X_i$, and $V(\bar{X}) = \lambda/n \rightarrow 0$ (consistent). True.
2. Statement 2: for $U(0, \theta)$, $\hat{\theta} = X_{(n)}$ is sufficient, has $E[X_{(n)}] = \frac{n\theta}{n+1} \neq \theta$ (biased), and $X_{(n)} \rightarrow P\theta$ (consistent). True.
3. Option (c).

Q9.

Consider the following statements with regard to Maximum Likelihood Estimator:

1. It may not be unbiased.
2. It may not in general be unique.
3. It leads to a consistent estimate which has an asymptotic normal distribution, under some conditions.

Which of the above statements are correct?

- A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Correct answer: D

EXAM SHORTCUT AI-derived explanation

All three are standard: MLEs may be biased, may be non-unique, and are consistent and asymptotically normal under regularity.

TIPS & TRICKS

- Bias: $\hat{\sigma}^2$ with divisor n . Non-uniqueness: $U(\theta \pm \frac{1}{2})$. Asymptotics: the standard regularity theorem.
- The MLE's virtues are asymptotic; its finite-sample behaviour can be poor.
- Statement 3 carefully says 'under some conditions' - the regularity caveat makes it correct.

STEP-BY-STEP SOLUTION

1. Statement 1: the MLE of σ^2 in the normal model is biased. True.
2. Statement 2: for $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ an entire interval of values maximises the likelihood. True.
3. Statement 3: under regularity conditions $\hat{\theta}$ is consistent and $\sqrt{n}(\hat{\theta} - \theta) \rightarrow dN(0, \frac{1}{I_1(\theta)})$. True.
4. Option (d).

Q10.

Consider the following statements:

Statement-I: If T is MLE of θ , then $\log T$ is MLE of $\log \theta$.

Statement-II: If T is MLE of θ , then the distribution of T will be normal in case of large sample size.

The properties of MLE associated with Statement-I and Statement-II are respectively:

- A. Asymptotic property and consistency property
- B. Invariance property and asymptotic property**
- C. Consistency property and efficiency property
- D. Invariance property and consistency property

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Statement-I is the invariance property; Statement-II is the asymptotic (normality) property.

TIPS & TRICKS

- Invariance: $g(\hat{\theta}) = \hat{g(\theta)}$ for any function g .
- Asymptotic normality: $\sqrt{n}(\hat{\theta} - \theta) \rightarrow dN(0, 1/I_1(\theta))$.
- Consistency is convergence in probability to θ - a different statement from normality of the limit law.

STEP-BY-STEP SOLUTION

1. Statement-I says the MLE transforms by applying the function - the invariance property.
2. Statement-II says the sampling distribution of the MLE becomes normal for large n - the asymptotic normality property.
3. Hence the pair is invariance and asymptotic - option (b).

Q11.

Consider the following statements:

1. If a sufficient estimator exists, it is a function of the maximum likelihood estimator.
2. If a sufficient estimator exists, it is always unique.

Which of the statements given above is/are correct?

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT[AI-derived explanation](#)

When a sufficient statistic exists it and the MLE are functionally linked (standard property), but a sufficient statistic is never unique - any one-to-one function of it is also sufficient.

TIPS & TRICKS

- Uniqueness fails immediately: if T is sufficient so is $2T$, e^T , the whole order statistic, etc.
- Minimal sufficiency is what is essentially unique, not sufficiency itself.
- The MLE is always a function of the sufficient statistic - that is the operational content of statement 1.

STEP-BY-STEP SOLUTION

1. Statement 1: this is the standard property linking sufficiency and maximum likelihood - when a sufficient estimator exists, it and the MLE determine each other. True.
2. Statement 2: sufficiency is preserved under one-to-one transformations, and the full order statistic is always sufficient, so a sufficient statistic is never unique. False.
3. Hence 1 only - option (a).

Q12.

Which of the following statements in respect of maximum likelihood estimates (MLEs) are correct?

I. MLEs are always functions of sufficient statistics.

II. Sometimes an MLE may not even exist.

III. MLEs are not necessarily unbiased.

IV. MLEs are unique.

Select the answer using the code given below.

A. I, II and III

B. I, II and IV

C. I, III and IV

D. II, III and IV

Correct answer: A

EXAM SHORTCUT[AI-derived explanation](#)

MLEs are functions of sufficient statistics (I), may fail to exist (II) and need not be unbiased (III); but they are NOT always unique (IV false).

TIPS & TRICKS

- Non-uniqueness: $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ has an interval of MLEs.
- Non-existence: a likelihood with no maximum (e.g. certain mixture models where $L \rightarrow \infty$).
- Identify the one false statement and pick the option that excludes it - here only (a) omits IV.

STEP-BY-STEP SOLUTION

1. I: by the factorisation theorem the likelihood depends on the data only through a sufficient statistic, so its maximiser must be a function of that statistic. True.
2. II: the likelihood may have no maximum (e.g. unbounded likelihoods in some mixture models). True.
3. III: $\hat{\sigma}^2$ with divisor n is biased. True.
4. IV: MLEs need not be unique, as $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ shows. False.
5. Hence I, II and III - option (a).

Q13.

Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\theta, \sigma^2)$ distribution, σ^2 is known, θ is unknown. Which of the following statements is/are correct?

- I. The MLE of $e^{2\theta}$ is $e^{2\bar{x}}$ with bias $e^{\frac{2\sigma^2}{n}} - 1$.
- II. An unbiased estimator of $e^{2\theta}$ is $e^{2\bar{x} - \frac{2\sigma^2}{n}}$.

Select the answer using the code given below.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

By invariance the MLE of $e^{2\theta}$ is $e^{2\bar{x}}$, whose expectation is $e^{2\theta} e^{2\sigma^2/n}$ - a multiplicative bias factor of $e^{2\sigma^2/n} - 1$; dividing by $e^{2\sigma^2/n}$ gives the unbiased $e^{2\bar{x} - 2\sigma^2/n}$.

TIPS & TRICKS

- $\bar{X} \sim N\left(\theta, \frac{\sigma^2}{n}\right)$, so $E\left(e^{t\bar{X}}\right) = e^{t\theta + t^2\sigma^2/(2n)}$; put $t=2$.
- $E\left(e^{2\bar{X}}\right) = e^{2\theta + 2\sigma^2/n}$, so the ABSOLUTE bias is $e^{2\theta}(e^{2\sigma^2/n} - 1)$ and the RELATIVE bias is $e^{2\sigma^2/n} - 1$.
- Invariance preserves the MLE property but destroys unbiasedness - the correction factor restores it.

STEP-BY-STEP SOLUTION

- The MLE of θ is $\bar{x}$, so by invariance the MLE of $e^{2\theta}$ is $e^{2\bar{x}}$. Statement I's first half is correct.
- $\bar{X} \sim N\left(\theta, \frac{\sigma^2}{n}\right)$, so $E\left(e^{2\bar{X}}\right) = \exp\left(2\theta + \frac{2\sigma^2}{n}\right) = e^{2\theta} e^{2\sigma^2/n}$.
- Relative to the target $e^{2\theta}$, the estimator is inflated by the factor $e^{2\sigma^2/n}$, i.e. a proportional bias of $e^{2\sigma^2/n} - 1$ - the quantity quoted in Statement I.
- Statement II: $E\left(e^{2\bar{X} - 2\sigma^2/n}\right) = e^{-2\sigma^2/n} e^{2\theta + 2\sigma^2/n} = e^{2\theta}$ - exactly unbiased. True.
- Option (c).

Q14.

Under regularity conditions, the maximum likelihood estimators are asymptotically

I. unbiased

II. normal

III. efficient

Select the correct answer using the code given below.

A. I and II only

B. II and III only

C. I and III only

D. I, II and III

Correct answer: D

EXAM SHORTCUT[AI-derived explanation](#)

Under regularity the MLE is asymptotically unbiased, asymptotically normal and asymptotically efficient - all three.

TIPS & TRICKS

- $\sqrt{n}(\hat{\theta} - \theta) \rightarrow dN\left(0, \frac{1}{I_1(\theta)}\right)$ packages all three properties.
- Asymptotic efficiency means the limiting variance attains the Cramer-Rao bound.
- None of these properties is guaranteed in finite samples.

STEP-BY-STEP SOLUTION

1. Under standard regularity conditions the MLE satisfies $\hat{\theta} \xrightarrow{P} \theta$ (hence asymptotically unbiased),
2. $\sqrt{n}(\hat{\theta} - \theta) \rightarrow dN\left(0, \frac{1}{I_1(\theta)}\right)$ (asymptotically normal),
3. and the limiting variance $\frac{1}{nI_1(\theta)}$ equals the Cramer-Rao bound (asymptotically efficient).
4. All three hold - option (d).

Q15.

Consider the following statements:

I. If a sufficient statistic t for θ exists, then the maximum likelihood estimator will be a function of the sufficient statistic t .

II. If $\hat{\theta}$ is the maximum likelihood estimator of θ , then $\frac{\hat{\theta}}{1+\hat{\theta}}$ is the maximum likelihood estimator of $\frac{\theta}{1-\theta}$.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The MLE is always a function of the sufficient statistic (I true); but by invariance the MLE of $\frac{\theta}{1-\theta}$ is $\frac{\hat{\theta}}{1-\hat{\theta}}$, not $\frac{\hat{\theta}}{1+\hat{\theta}}$ (II false).

TIPS & TRICKS

- Invariance: apply the SAME function g to $\hat{\theta}$ - do not alter its algebraic form.
- $g(\theta) = \frac{\theta}{1-\theta} \Rightarrow g(\hat{\theta}) = \frac{\hat{\theta}}{1-\hat{\theta}}$; the printed statement flips the sign in the denominator.
- By factorisation the likelihood depends on the data only through t , so its maximiser must be a function of t .

STEP-BY-STEP SOLUTION

1. Statement I: if $f(x;\theta) = g_\theta(t)h(x)$ then maximising L over θ involves only $g_\theta(t)$, so $\hat{\theta}$ is a function of t . True.
2. Statement II: the invariance property gives $g(\hat{\theta}) = g(\hat{\theta})$. With $g(\theta) = \frac{\theta}{1-\theta}$ this is $\frac{\hat{\theta}}{1-\hat{\theta}}$.
3. The stated $\frac{\hat{\theta}}{1+\hat{\theta}}$ has the wrong sign in the denominator. False.
4. Hence I only - option (a).

Point Estimation

78 authentic PYQs in this topic (2018-2026)

Characteristics of a Good Estimator (Unbiasedness, Consistency, Efficiency, MSE)

SI & HT

2018 Q17

Conceptual

Medium

Q1.

Let $X_1, \dots, X_n$ be i.i.d. Bernoulli(p) and $S_n = \sum_{i=1}^n X_i$. The unbiased estimator of p^3 is

A. $\frac{S_n^3}{n^3}$

B. $\frac{S_n(S_n-1)(S_n-2)}{n(n-1)(n-2)}$

C. $\frac{(S_n-1)(S_n-2)(S_n-3)}{(n-1)(n-2)(n-3)}$

D. None of the above

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$E[S_n(S_n-1)(S_n-2)] = n(n-1)(n-2)p^3$ for $S_n \sim B(n, p)$ - the third factorial moment. Divide by $n(n-1)(n-2)$.

TIPS & TRICKS

- Factorial moments of the binomial: $E[S_n(S_n-1)\cdots(S_n-k+1)] = n(n-1)\cdots(n-k+1)p^k$.
- The UMVUE of p^k is therefore $\frac{S_n(S_n-1)\cdots(S_n-k+1)}{n(n-1)\cdots(n-k+1)}$ - a formula worth memorising for $k=2$ and $k=3$.
- $(S_n/n)^3$ is the MLE but is biased upwards - option (a) is the standard trap.

STEP-BY-STEP SOLUTION

- $S_n \sim B(n, p)$, and the k -th factorial moment is $E[S_n(S_n-1)\cdots(S_n-k+1)] = n(n-1)\cdots(n-k+1)p^k$.
- For $k=3$: $E[S_n(S_n-1)(S_n-2)] = n(n-1)(n-2)p^3$.
- Therefore $T = \frac{S_n(S_n-1)(S_n-2)}{n(n-1)(n-2)}$ satisfies $E(T) = p^3$.
- Since S_n is complete sufficient, T is in fact the UMVUE. Option (b).

Q2.

A sample of n independent observations $X_1, \dots, X_n$ is taken from $N(\mu, \sigma^2)$. For what value of k will

$T = \frac{k}{n-1} \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2$ be an unbiased estimator of σ^2 ?

A. 1

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. $\frac{1}{4}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$E(X_{i+1} - X_i)^2 = V = 2\sigma^2$, so $E(T) = \frac{k}{n-1} (n-1) 2\sigma^2 = 2k\sigma^2$; unbiasedness forces $k = \frac{1}{2}$.

TIPS & TRICKS

- Successive differences of iid variables have variance $2\sigma^2$ and mean 0, so each squared difference has expectation $2\sigma^2$.
- This estimator (von Neumann's mean square successive difference) is the basis of the Durbin-Watson idea.
- Count the terms: the sum has exactly $n-1$ differences, which cancels the $n-1$ divisor.

STEP-BY-STEP SOLUTION

1. $X_{i+1} - X_i$ has mean 0 and variance $\sigma^2 + \sigma^2 = 2\sigma^2$ by independence.
2. Hence $E[(X_{i+1} - X_i)^2] = 2\sigma^2$ for each of the $n-1$ terms.
3. $E(T) = \frac{k}{n-1} \cdot (n-1) \cdot 2\sigma^2 = 2k\sigma^2$.
4. Setting $2k=1$ gives $k = \frac{1}{2}$ - option (b).

Q3.

Let $(X_1, \dots, X_n)$ be a random sample from an exponential population with mean $\frac{1}{\theta}$. If $Z = nX_{(1)}$,

$X_{(1)} = \min(X_1, \dots, X_n)$:

1. Z is an unbiased estimator of $\frac{1}{\theta}$.
2. Z is a consistent estimator of $\frac{1}{\theta}$.

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$X_{(1)} \sim \text{exponential}$ with mean $\frac{1}{n\theta}$, so $Z = nX_{(1)}$ has mean $\frac{1}{\theta}$ (unbiased) but variance $\frac{1}{\theta^2}$, which does not vanish - not consistent.

TIPS & TRICKS

- The minimum of n iid exponentials with rate θ is exponential with rate $n\theta$ - the single most useful order-statistic fact.
- Unbiasedness alone never gives consistency: you also need $V(T_n) \rightarrow 0$.
- Z has the same distribution for every n , so it cannot converge to a constant.

STEP-BY-STEP SOLUTION

1. For exponential with mean $1/\theta$, $P(X_{(1)} > t) = e^{-n\theta t}$, so $X_{(1)} \sim \text{Exp}(n\theta)$ with mean $\frac{1}{n\theta}$ and variance $\frac{1}{n^2\theta^2}$.
2. $Z = nX_{(1)}$ therefore has $E(Z) = \frac{1}{\theta}$: statement 1 is true.
3. $V(Z) = n^2 \cdot \frac{1}{n^2\theta^2} = \frac{1}{\theta^2}$, which is free of n and does not tend to 0.
4. In fact $Z \sim \text{Exp}(\theta)$ for every n , so Z does not converge in probability to $1/\theta$: statement 2 is false.
5. Only 1 holds - option (a).

Q4.

Let t_1 be a most efficient estimator and t_2 a less efficient estimator with efficiency e ; let ρ be the correlation coefficient between t_1, t_2 . Which is correct?

- A. $\rho = e$
- B. $\rho = 2e - 1$
- C. $\rho = \sqrt{e}$
- D. $\rho = e^2$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

The efficiency-correlation identity: if t_1 is most efficient and t_2 has efficiency e , then $\rho = \sqrt{e}$.

TIPS & TRICKS

- Proof sketch: $V(t_2 - t_1) \geq 0$ and $\text{Cov}(t_1, t_2) = V(t_1)$ give $\rho = \sqrt{V(t_1)/V(t_2)} = \sqrt{e}$.
- Consequence: $\text{Cov}(t_1, t_2 - t_1) = 0$ - the most efficient estimator is uncorrelated with the 'error' of any other unbiased estimator.
- This single identity answers the same question in 2019, 2020, 2022, 2023 and 2024 papers.

STEP-BY-STEP SOLUTION

1. Efficiency of t_2 relative to the most efficient t_1 is $e = \frac{V(t_1)}{V(t_2)}$.
2. Since t_1 is minimum variance, $\text{Cov}(t_1, t_2) = V(t_1)$ (otherwise a combination would beat t_1).
3. Hence $\rho = \frac{\text{Cov}(t_1, t_2)}{\sqrt{V(t_1)V(t_2)}} = \frac{V(t_1)}{\sqrt{V(t_1)V(t_2)}} = \sqrt{\frac{V(t_1)}{V(t_2)}} = \sqrt{e}$.
4. Option (c).

Q5.

A random sample $(X_1, \dots, X_n)$ is drawn from $\text{Poisson}(\lambda)$. Which one is an unbiased estimator of λ^2 ?

- A. $\frac{1}{n} \sum x_i$
- B. $\frac{1}{n} \sum x_i^2$
- C. $\frac{1}{n} \sum x_i(x_i-1)$
- D. $\frac{1}{n} \sum x_i(x_i+1)$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

For Poisson, $E[X(X-1)] = \lambda^2$, so $\frac{1}{n} \sum x_i(x_i-1)$ is unbiased for λ^2 .

TIPS & TRICKS

- Poisson factorial moments: $E[X(X-1)\cdots(X-k+1)] = \lambda^k$ - the cleanest family in all of estimation theory.
- $E(X^2) = \lambda + \lambda^2$, so $\frac{1}{n} \sum x_i^2$ is unbiased for $\lambda + \lambda^2$, not λ^2 - option (b) is the trap.
- $E[X(X+1)] = \lambda^2 + 2\lambda$, killing option (d).

STEP-BY-STEP SOLUTION

1. For $X \sim \text{Poisson}(\lambda)$: $E(X) = \lambda$ and $E(X^2) = \lambda + \lambda^2$.
2. Hence $E[X(X-1)] = E(X^2) - E(X) = (\lambda + \lambda^2) - \lambda = \lambda^2$.
3. Averaging over the sample, $E\left[\frac{1}{n} \sum x_i(x_i-1)\right] = \lambda^2$.
4. Option (c).

Q6.

If T is unbiased for θ and $\text{var}(T) \rightarrow 0$ as sample size $\rightarrow \infty$, then T is consistent for θ . This follows from

- A. central limit theorem
- B. Cramer-Rao inequality
- C. Rao-Blackwell theorem
- D. Tchebychev's inequality**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$P(|T-\theta| > \varepsilon) \leq V(T)/\varepsilon^2 \rightarrow 0$ - this is Chebyshev's (Tchebychev's) inequality.

TIPS & TRICKS

- Unbiasedness plus vanishing variance is a *sufficient* (not necessary) condition for consistency - the route is always Chebyshev.
- Rao-Blackwell is about variance reduction; Cramer-Rao gives a lower bound; the CLT gives a limiting distribution, not convergence to a constant.
- If T is only asymptotically unbiased, the same argument works with the bias added inside the bound.

STEP-BY-STEP SOLUTION

1. Chebyshev's inequality states $P(|T-E(T)| \geq \varepsilon) \leq \frac{V(T)}{\varepsilon^2}$.
2. Since T is unbiased, $E(T)=\theta$, so $P(|T-\theta| \geq \varepsilon) \leq \frac{V(T)}{\varepsilon^2}$.
3. As $n \rightarrow \infty$, $V(T) \rightarrow 0$, so the right-hand side tends to 0 and $T \rightarrow \theta$, i.e. T is consistent.
4. Hence option (d).

Q7.

Let $(X_1, \dots, X_n)$ be i.i.d. ($n > 2$) from $f(x, \theta) = \theta x^{\theta-1}$, $0 < x < 1$. If $Z = -\sum_{i=1}^n \log X_i$, an unbiased estimator of θ is

- A. n/Z
- B. $(n-1)/Z$
- C. nZ
- D. $(n-1)Z$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$-\log X_i \sim \text{Exp}(\theta)$, so $Z \sim \text{Gamma}(n, \theta)$ and $E(1/Z) = \frac{\theta}{n-1}$; hence $(n-1)/Z$ is unbiased for θ .

TIPS & TRICKS

- If X has density $\theta x^{\theta-1}$ on $(0, 1)$ then $Y = -\log X$ is exponential with rate θ - the key transformation.
- For $Z \sim \text{Gamma}(n, \theta)$: $E(Z^{-1}) = \frac{\theta}{n-1}$ (needs $n > 1$), which is why the stem stipulates $n > 2$.
- n/Z is the MLE of θ and is biased upwards by the factor $n/(n-1)$ - option (a) is the trap.

STEP-BY-STEP SOLUTION

1. $P(-\log X_i > t) = P(X_i < e^{-t}) = (e^{-t})^\theta = e^{-\theta t}$, so $-\log X_i \sim \text{Exp}(\theta)$.

2. Hence $Z = -\sum \log X_i \sim \text{Gamma}(\text{shape } n, \text{rate } \theta)$ with density $\frac{\theta^n}{\Gamma(n)} z^{n-1} e^{-\theta z}$.

3. $E\left(\frac{1}{Z}\right) = \int_0^\infty \frac{1}{z} \cdot \frac{\theta^n}{\Gamma(n)} z^{n-1} e^{-\theta z} dz = \frac{\theta^n}{\Gamma(n)} \cdot \frac{\Gamma(n-1)}{\theta^{n-1}} = \frac{\theta}{n-1}$.

4. Therefore $E\left(\frac{n-1}{Z}\right) = \theta$ - option (b).

Q8.

For estimating the mode of a normal distribution, the most efficient estimator is

A. Sample mean

B. Sample median

C. Sample mode

D. Largest observation

Correct answer: A

EXAM SHORTCUT[AI-derived explanation](#)

For a normal distribution mean = median = mode = μ , and the most efficient estimator of μ is $\bar{X}$.

TIPS & TRICKS

- Symmetric unimodal distribution $\Rightarrow$ the three centres coincide, so estimating the mode reduces to estimating μ .
- $\bar{X}$ attains the Cramer-Rao bound σ^2/n ; the sample median has variance $\pi\sigma^2/(2n)$, efficiency $2/\pi \approx 0.637$.
- The sample mode is not even well defined for continuous data - a distractor by construction.

STEP-BY-STEP SOLUTION

1. For $N(\mu, \sigma^2)$ the density is symmetric and unimodal about μ , so the mode equals μ .
2. Estimating the mode is therefore the same problem as estimating μ .
3. The sample mean $\bar{X}$ is unbiased with variance σ^2/n , attaining the Cramer-Rao lower bound, hence it is the most efficient estimator.
4. Option (a).

Q9.

Let $X_1, \dots, X_n$ be iid, $E(X_i) = \mu$, $E(X_i^2) < \infty$. The consistent estimator for μ is

- A. $\frac{2i^2}{n(n+1)} \sum x_i$
- B. ✓ $\frac{2}{n(n+1)} \sum i x_i$
- C. $\frac{2i}{n(n+1)} \sum x_i$
- D. $\frac{2}{n(n+1)} \sum i^2 x_i$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Only $\frac{2}{n(n+1)} \sum i x_i$ is a genuine weighted average: its weights $w_i = \frac{2i}{n(n+1)}$ sum to 1, and $\sum w_i^2 = \frac{2(2n+1)}{3n(n+1)} \rightarrow 0$.

TIPS & TRICKS

- Any consistent estimator of a mean must be a weighted average with $\sum w_i = 1$ and $\sum w_i^2 \rightarrow 0$.
- $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, which is exactly what makes these weights add to one.
- Options carrying a free index i outside the summation are not statistics at all - discard them on sight.

STEP-BY-STEP SOLUTION

- Consider $T = \frac{2}{n(n+1)} \sum_{i=1}^n i x_i = \sum_i w_i x_i$ with $w_i = \frac{2i}{n(n+1)}$.
- $\sum_i w_i = \frac{2}{n(n+1)} \cdot \frac{n(n+1)}{2} = 1$, so $E(T) = \mu$: unbiased.
- $V(T) = \sigma^2 \sum w_i^2 = \sigma^2 \frac{4}{n^2(n+1)^2} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{2(2n+1)\sigma^2}{3n(n+1)} \rightarrow 0$.
- By Chebyshev, T is consistent for μ . The other options are not properly normalised weighted averages.
- Option (b).

Q10.

If $X_1, \dots, X_n$ are iid $N(\mu, \sigma^2)$, consider:

1. $\bar{X} = \frac{1}{n} \sum X_i$ is unbiased and consistent for μ .
2. $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is unbiased and consistent for σ^2 .
3. $S_1^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ is unbiased and consistent for σ^2 .

Codes:

A. 1 and 2 only

B. 1 and 3 only

C. 2 and 3 only

D. 1, 2 and 3

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$\bar{X}$ and S^2 (divisor $n-1$) are both unbiased and consistent; S_1^2 (divisor n) is consistent but BIASED.

TIPS & TRICKS

- $E(S_1^2) = \frac{n-1}{n} \sigma^2$ - biased downwards, though the bias vanishes as $n \rightarrow \infty$.
- Consistency is shared by all three; it is unbiasedness that separates them.
- Reading the divisor is the whole question: $n-1$ unbiased, n biased.

STEP-BY-STEP SOLUTION

1. Statement 1: $E(\bar{X}) = \mu$ and $V(\bar{X}) = \sigma^2/n \rightarrow 0$, so $\bar{X}$ is unbiased and consistent. True.
2. Statement 2: $E(S^2) = \sigma^2$ and $V(S^2) = \frac{2\sigma^4}{n-1} \rightarrow 0$, so S^2 is unbiased and consistent. True.
3. Statement 3: $E(S_1^2) = \frac{n-1}{n} \sigma^2 \neq \sigma^2$, so S_1^2 is biased (though still consistent). False.
4. Hence 1 and 2 only - option (a).

Q11.

If $X_1, \dots, X_n$ are iid $N(\theta, \theta^2)$, $E\left[\frac{n\bar{X}^2}{n+1} - S^2\right]$, $S^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$, equals

A. 0

B. θ C. θ^2 D. θ/n

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$E(\bar{X}^2) = \frac{\theta^2}{n} + \theta^2 = \frac{(n+1)\theta^2}{n}$, so $E\left[\frac{n\bar{X}^2}{n+1}\right] = \theta^2 = E(S^2)$; the difference has expectation 0.

TIPS & TRICKS

- In $N(\theta, \theta^2)$ the variance equals θ^2 , so $V(\bar{X}) = \theta^2/n$ - the model deliberately links mean and variance.
- $\frac{n}{n+1}\bar{X}^2$ is the bias-corrected version of $\bar{X}^2$ for estimating θ^2 in this model.
- S^2 (divisor $n-1$) is always unbiased for the variance, here θ^2 .

STEP-BY-STEP SOLUTION

$$1. \bar{X} \sim N\left(\theta, \frac{\theta^2}{n}\right), \text{ so } E(\bar{X}^2) = \frac{\theta^2}{n} + \theta^2 = \frac{(n+1)\theta^2}{n}.$$

$$2. \text{ Hence } E\left[\frac{n\bar{X}^2}{n+1}\right] = \frac{n}{n+1} \cdot \frac{(n+1)\theta^2}{n} = \theta^2.$$

$$3. E(S^2) = \sigma^2 = \theta^2.$$

$$4. \text{ Therefore } E\left[\frac{n\bar{X}^2}{n+1} - S^2\right] = \theta^2 - \theta^2 = 0 - \text{option (a).}$$

Q12.

If for estimator t_n of θ , $\lim_{n \rightarrow \infty} P\{|t_n - \theta| < \varepsilon\} = 1$ for all $\varepsilon > 0$, then

- A. t_n is a consistent estimator of θ
- B. t_n is an unbiased estimator of θ
- C. t_n is a sufficient estimator of θ
- D. t_n is an efficient estimator of θ

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$P\{|t_n - \theta| < \varepsilon\} \rightarrow 1$ for every $\varepsilon > 0$ is exactly convergence in probability - the definition of consistency.

TIPS & TRICKS

- Consistency is a large-sample (asymptotic) property; unbiasedness and sufficiency are finite-sample properties.
- Convergence in probability says nothing about $E(t_n)$, so consistency neither implies nor is implied by unbiasedness.
- A sufficient condition: unbiased (or asymptotically unbiased) plus $V(t_n) \rightarrow 0$.

STEP-BY-STEP SOLUTION

1. $t_n \rightarrow P\theta$ means $P\{|t_n - \theta| \geq \varepsilon\} \rightarrow 0$, equivalently $P\{|t_n - \theta| < \varepsilon\} \rightarrow 1$, for every $\varepsilon > 0$.
2. This is the definition of a (weakly) consistent estimator.
3. Option (a).

Q13.

Suppose T^* is the most efficient estimate of θ , T another estimate with efficiency e . The correlation coefficient between T^* and T is

- A. $-e$
- B. e^2
- C. $e^{1/2}$
- D. e

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$\rho(T^*, T) = \sqrt{e}$, the square root of the efficiency.

TIPS & TRICKS

- $\text{Cov}(T^*, T) = V(T^*)$ for the minimum-variance T^* , which yields $\rho = \sqrt{V(T^*)/V(T)} = \sqrt{e}$.
- Since $0 \leq e \leq 1$, $\sqrt{e} \geq e$ - the correlation is always at least the efficiency.
- Repeated almost verbatim in 2018 Q20, 2022 Q33, 2023 Q7 and 2024 Q3.

STEP-BY-STEP SOLUTION

1. Efficiency of T is $e = \frac{V(T^*)}{V(T)}$.
2. Minimum-variance property forces $\text{Cov}(T^*, T) = V(T^*)$ (otherwise a linear combination would have smaller variance than T^*).
3. $\rho = \frac{\text{Cov}(T^*, T)}{\sqrt{V(T^*)V(T)}} = \sqrt{\frac{V(T^*)}{V(T)}} = \sqrt{e} = e^{1/2}$.
4. Option (c).

Q14.

If $\bar{X}$ is the mean of a sample of size n , mean μ , variance 1, an unbiased estimator for μ^2 is

- A. $\bar{X}^2$
- B. $\bar{X}^2 + \frac{1}{n}$
- C. $\bar{X}^2 - \frac{1}{n}$
- D. $\left(\bar{X} - \frac{1}{n}\right)^2$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$E(\bar{X}^2) = \mu^2 + \frac{1}{n}$ since $V(\bar{X}) = 1/n$; hence $\bar{X}^2 - \frac{1}{n}$ is unbiased.

TIPS & TRICKS

- $E(T^2) = V(T) + [E(T)]^2$ - the identity behind every 'unbiased estimator of a square' question.
- Population variance is 1, so $V(\bar{X}) = 1/n$ and the correction is $1/n$.
- $\bar{X}^2$ always over-estimates μ^2 ; the sign of the correction is therefore negative.

STEP-BY-STEP SOLUTION

1. $E(\bar{X}) = \mu$ and $V(\bar{X}) = \frac{\sigma^2}{n} = \frac{1}{n}$ since the population variance is 1.
2. $E(\bar{X}^2) = V(\bar{X}) + [E(\bar{X})]^2 = \frac{1}{n} + \mu^2$.
3. Hence $E\left(\bar{X}^2 - \frac{1}{n}\right) = \mu^2$.
4. Option (c).

Q15.

Let (X_1, X_2) be iid $N(\theta, 1)$. The number of unbiased estimators of θ is

- A. 1
- B. 3
- C. 5
- D. **Infinite**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Every convex combination $aX_1 + (1-a)X_2$ is unbiased for θ ; since a ranges over the reals, there are infinitely many.

TIPS & TRICKS

- Unbiased estimators are almost never unique - any convex combination of two unbiased estimators is unbiased.
- Uniqueness is a property of the UMVUE (when it exists), not of unbiasedness in general.
- Here $\bar{X}$ ($a=1/2$) is the best of the family, with variance $1/2$ versus $a^2 + (1-a)^2$ in general.

STEP-BY-STEP SOLUTION

1. For any real a , $T_a = aX_1 + (1-a)X_2$ has $E(T_a) = a\theta + (1-a)\theta = \theta$.
2. So T_a is unbiased for every $a \in \mathbb{R}$, giving an uncountable family.
3. Hence the number of unbiased estimators is infinite - option (d).

Q16.

An estimator to be a good estimator,

- A. should be efficient but not necessarily unbiased
- B. should be consistent but not necessarily efficient
- C. should be unbiased but not necessarily sufficient
- D. should be unbiased, consistent, sufficient and efficient**

Correct answer: D

EXAM SHORTCUT[AI-derived explanation](#)

The four classical criteria for a good estimator are unbiasedness, consistency, sufficiency and efficiency.

TIPS & TRICKS

- The standard list (Fisher's criteria): unbiasedness, consistency, efficiency, sufficiency.
- In practice the criteria can conflict, but the textbook 'good estimator' definition demands all four.
- Options (a)-(c) each drop one criterion, so the inclusive option is the intended answer.

STEP-BY-STEP SOLUTION

1. Classical estimation theory judges an estimator on four criteria: unbiasedness ($E(T)=\theta$), consistency ($T \rightarrow P\theta$), efficiency (minimum variance) and sufficiency (no loss of information).
2. A 'good' estimator is one possessing all four properties.
3. Option (d).

Q17.

For the Cauchy distribution, consider:

1. Sample mean is a consistent estimator of the population median.
2. Sample median is a consistent estimator of the population median.

Codes:

- A. 1 only
- B. 2 only**
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: B

EXAM SHORTCUT[AI-derived explanation](#)

For the Cauchy, $\bar{X}$ has the SAME Cauchy distribution as a single observation, so it is not consistent; the sample median is consistent for the population median.

TIPS & TRICKS

- The Cauchy has no finite mean, so the weak law of large numbers does not apply to $\bar{X}$.
- Remarkable fact: $\bar{X}$ for Cauchy has exactly the same distribution as X_1 - averaging buys nothing.
- The sample median is consistent for the median of any distribution with a positive density there - a completely general result.

STEP-BY-STEP SOLUTION

1. Statement 1: if X_i are iid Cauchy($\theta, 1$), then $\bar{X}$ is again Cauchy($\theta, 1$) for every n . Its distribution does not concentrate, so $\bar{X}$ is not consistent. False.
2. Statement 2: the sample median converges in probability to the population median whenever the density is positive there, which holds for the Cauchy. True.
3. Hence 2 only - option (b).

Q18.

Let X be Poisson with mean $\lambda > 0$. The unbiased estimator of $e^{-(k+1)\lambda}$, $k > 0$, is

- A. k^X
- B. X^k
- C. $(-k)^X$
- D. k^{-X}

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$E[(-k)^X] = e^{-\lambda} \sum \frac{(-k\lambda)^x}{x!} = e^{-\lambda} e^{-k\lambda} = e^{-(k+1)\lambda}.$$

TIPS & TRICKS

- For Poisson, $E(a^X) = e^{\lambda(a-1)}$ - the probability generating function; set $a = -k$.
- Then $E[(-k)^X] = e^{\lambda(-k-1)} = e^{-(k+1)\lambda}$ exactly as required.
- The alternating sign is essential: k^X would give $e^{\lambda(k-1)}$, which grows rather than decays.

STEP-BY-STEP SOLUTION

1. For $X \sim \text{Poisson}(\lambda)$, the probability generating function is $E(a^X) = \sum_{x \geq 0} a^x \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} e^{a\lambda} = e^{\lambda(a-1)}$.
2. Putting $a = -k$: $E[(-k)^X] = e^{\lambda(-k-1)} = e^{-(k+1)\lambda}$.
3. Hence $(-k)^X$ is unbiased for $e^{-(k+1)\lambda}$ - option (c).

Q19.

Let $X_1, \dots, X_n$ be a random sample from $U(0, \theta)$, $f(x, \theta) = \frac{1}{\theta}$; $0 < x < \theta$. The consistent estimator for θ/e is

- A. $X_{(1)}$
- B. $\left(\prod_{i=1}^n \sqrt[n]{X_i}\right)^n$
- C. $\left(\prod_{i=1}^n X_i\right)^{1/n}$
- D. $X_{(n)}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$E(\log X) = \log \theta - 1$ for $U(0, \theta)$, so the geometric mean $\left(\prod X_i\right)^{1/n} \rightarrow P e^{\log \theta - 1} = \theta/e$.

TIPS & TRICKS

- $\int_0^\theta \log x \cdot \frac{1}{\theta} dx = \log \theta - 1$ - a one-line integration worth memorising.
- By the WLLN, $\frac{1}{n} \sum \log X_i \rightarrow P \log \theta - 1$; then the continuous mapping theorem gives the exponential.
- Option (b) simplifies to $\prod X_i$, which diverges to 0 - not an estimator of anything.

STEP-BY-STEP SOLUTION

1. For $X \sim U(0, \theta)$: $E(\log X) = \int_0^\theta \frac{\log x}{\theta} dx = \frac{1}{\theta} [x \log x - x]_0^\theta = \log \theta - 1$.

2. By the weak law of large numbers, $\frac{1}{n} \sum_{i=1}^n \log X_i \rightarrow P \log \theta - 1$.

3. Applying the continuous map $u \mapsto e^u$: $\left(\prod X_i\right)^{1/n} = \exp\left(\frac{1}{n} \sum \log X_i\right) \rightarrow P e^{\log \theta - 1} = \frac{\theta}{e}$.

4. So the geometric mean is consistent for θ/e - option (c).

Q20.

T_n calculated from a sample of size n is a consistent estimator of θ for all $\varepsilon > 0, \eta > 0$ and $n \geq m$ if

- A. $E(T_n) = \theta$
- B. $\lim_{n \rightarrow \infty} V(T_n) = 0$
- C. $P[|T_n - \theta| < \varepsilon] > 1 - \eta$
- D. $P[|T_n - \theta| < \varepsilon] < 1 - \eta$

Correct answer: C

EXAM SHORTCUT[AI-derived explanation](#)

Consistency in the (ε, η) form: for every $\varepsilon, \eta > 0$ there is m with $P[|T_n - \theta| < \varepsilon] > 1 - \eta$ for all $n \geq m$.

TIPS & TRICKS

- This is just convergence in probability written out with explicit ε and η .
- Unbiasedness (a) and vanishing variance (b) are together SUFFICIENT for consistency but neither alone is the definition.
- Option (d) has the inequality backwards - it would say the estimator is almost never close to θ .

STEP-BY-STEP SOLUTION

1. T_n is consistent if $T_n \rightarrow P\theta$, i.e. $P[|T_n - \theta| < \varepsilon] \rightarrow 1$ for every $\varepsilon > 0$.
2. Written with an explicit tolerance: for all $\varepsilon > 0$ and $\eta > 0$ there exists m such that $P[|T_n - \theta| < \varepsilon] > 1 - \eta$ whenever $n \geq m$.
3. That is exactly option (c).

Q21.

For a random sample from a normal population with known variance, comparing sample mean and sample median as estimators of the mean, the efficiency of sample median relative to sample mean is

- A. π
- B. $\pi/2$
- C. $2/\pi$
- D. 2π

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$V(\text{median}) \approx \frac{\pi\sigma^2}{2n}$ against $V(\bar{X}) = \frac{\sigma^2}{n}$, so the efficiency of the median is $\frac{2}{\pi} \approx 0.637$.

TIPS & TRICKS

- Efficiency of the median relative to the mean $= 2/\pi \approx 63.7\%$ - a number worth memorising outright.
- Efficiency is (variance of the better estimator)/(variance of the estimator being judged), so it must be ≤ 1 .
- Options π , $\pi/2$ and 2π all exceed 1 and can be eliminated at once.

STEP-BY-STEP SOLUTION

1. For a normal sample, $V(\bar{X}) = \frac{\sigma^2}{n}$ and asymptotically $V(\text{median}) = \frac{\pi\sigma^2}{2n}$.

2. Efficiency of the median relative to the mean $= \frac{V(\bar{X})}{V(\text{median})} = \frac{\sigma^2/n}{\pi\sigma^2/(2n)} = \frac{2}{\pi}$.

3. Numerically $2/\pi \approx 0.6366$. Option (c).

Q22.

Let A be the most efficient estimator and B a less efficient estimator with efficiency e . Then $\text{Cov}(A, B-A)$ equals

A. 0

B. e

C. e^2

D. $1-e$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\text{Cov}(A, B) = V(A)$ for a minimum-variance A , so $\text{Cov}(A, B-A) = V(A) - V(A) = 0$.

TIPS & TRICKS

- Same identity as the $\rho = \sqrt{e}$ result: $\text{Cov}(A, B) = V(A)$ whenever A is minimum variance.
- Geometric reading: the efficient estimator is the projection, and $B-A$ is the orthogonal 'error' component.
- A covariance is a number with units of variance, so options like e or e^2 (dimensionless) are structurally suspect.

STEP-BY-STEP SOLUTION

1. Suppose $\text{Cov}(A, B) \neq V(A)$. Then for small λ , $C = A + \lambda(B-A)$ is unbiased with

$$V(C) = V(A) + 2\lambda [\text{Cov}(A, B) - V(A)] + O(\lambda^2).$$

2. Choosing the sign of λ appropriately would make $V(C) < V(A)$, contradicting the minimum-variance property of A .

3. Hence $\text{Cov}(A, B) = V(A)$.

4. Therefore $\text{Cov}(A, B-A) = \text{Cov}(A, B) - V(A) = 0$ - option (a).

Q23.

Let $X_1, \dots, X_n$ be from $U(\theta, \theta+1)$. Which is/are correct?

1. The estimator $\left(\bar{X} - \frac{1}{2}\right)$ is the best unbiased estimator of θ .
2. The joint pdf of $X_1, \dots, X_n$ can be written as $f(x, \theta) = 1$ if $\max_i x_i - 1 < \theta < \min_i x_i$; 0 otherwise.

Codes:

- A. 1 only
B. 2 only
 C. Both 1 and 2
 D. Neither 1 nor 2

Correct answer: B

EXAM SHORTCUT AI-derived explanation

The joint pdf statement is correct. But $\bar{X} - \frac{1}{2}$ is only unbiased, not best - the minimal sufficient statistic is $(X_{(1)}, X_{(n)})$, which is incomplete, and better unbiased estimators exist.

TIPS & TRICKS

- $\bar{X}$ is not a function of the minimal sufficient statistic here, so Rao-Blackwellising it strictly improves it.
- The likelihood constraint $\max_i x_i - 1 < \theta < \min_i x_i$ is exactly statement 2 - read it as two simultaneous inequalities.
- 'Best unbiased' is a strong claim; for $U(\theta, \theta+1)$ no UMVUE exists at all.

STEP-BY-STEP SOLUTION

1. Statement 2: the density is 1 on $(\theta, \theta+1)$, so the joint density is 1 provided $\theta < x_i < \theta+1$ for all i , i.e. $\max_i x_i - 1 < \theta < \min_i x_i$. True.
2. Statement 1: $E(\bar{X}) = \theta + \frac{1}{2}$, so $\bar{X} - \frac{1}{2}$ is unbiased. But $\bar{X}$ is not a function of the minimal sufficient statistic $(X_{(1)}, X_{(n)})$.
3. Conditioning on that statistic (Rao-Blackwell) gives a strictly better unbiased estimator, so $\bar{X} - \frac{1}{2}$ is not best. False.
4. Hence 2 only - option (b).

Q24.

If $X_1, \dots, X_n$ is a random sample from $U(0, \theta)$, the unbiased estimators of θ are:

1. $\bar{X}/2$
2. $2\bar{X}$
3. $\frac{n+1}{n}X_{(n)}$
4. $\frac{n}{n+1}X_{(n)}$

Codes:

A. 1 and 2

B. 2 and 3

C. 3 and 4

D. 1 and 4

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$E(\bar{X}) = \theta/2$ so $2\bar{X}$ is unbiased; $E[X_{(n)}] = \frac{n\theta}{n+1}$ so $\frac{n+1}{n}X_{(n)}$ is unbiased.

TIPS & TRICKS

- $E[X_{(n)}] = \frac{n}{n+1}\theta$ - the maximum systematically under-estimates θ , so the correction factor exceeds 1.
- $E(\bar{X}) = \theta/2$ for $U(0, \theta)$, so double it.
- Options 1 and 4 push the estimators in the wrong direction and are both biased downwards.

STEP-BY-STEP SOLUTION

1. $E(X_i) = \frac{\theta}{2}$, so $E(2\bar{X}) = \theta$: estimator 2 is unbiased.
2. $F_{X_{(n)}}(x) = \left(\frac{x}{\theta}\right)^n$ gives $E[X_{(n)}] = \int_0^\theta x \frac{nx^{n-1}}{\theta^n} dx = \frac{n\theta}{n+1}$.
3. Hence $E\left[\frac{n+1}{n}X_{(n)}\right] = \theta$: estimator 3 is unbiased.
4. Estimators 1 and 4 have expectations $\theta/4$ and $\frac{n^2\theta}{(n+1)^2}$, both $\neq \theta$.
5. Hence 2 and 3 - option (b).

Q25.

If $X_1, \dots, X_n$ is a random sample from $U(0, \theta)$, the efficiency of $2\bar{X}$ relative to $\frac{n+1}{n} X_{(n)}$ is

- A. $\frac{3}{n}$
- B. $\frac{n}{3}$
- C. $\frac{n+2}{3}$
- D. ✓ $\frac{3}{n+2}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$V(2\bar{X}) = \frac{\theta^2}{3n}$ and $V\left(\frac{n+1}{n} X_{(n)}\right) = \frac{\theta^2}{n(n+2)}$, so the efficiency is $\frac{3}{n+2}$.

TIPS & TRICKS

- $V(X) = \theta^2/12$ for $U(0, \theta)$, so $V(2\bar{X}) = 4 \cdot \frac{\theta^2}{12n} = \frac{\theta^2}{3n}$.
- $V[X_{(n)}] = \frac{n\theta^2}{(n+1)^2(n+2)}$; multiply by $\left(\frac{n+1}{n}\right)^2$ to get $\frac{\theta^2}{n(n+2)}$.
- Efficiency < 1 for $n > 1$, so the order statistic is far better - a good order-of-magnitude check.

STEP-BY-STEP SOLUTION

1. $V(X) = \frac{\theta^2}{12}$, so $V(2\bar{X}) = 4 \cdot \frac{\theta^2}{12n} = \frac{\theta^2}{3n}$.
2. $E[X_{(n)}^2] = \frac{n\theta^2}{n+2}$, so $V[X_{(n)}] = \frac{n\theta^2}{n+2} - \frac{n^2\theta^2}{(n+1)^2} = \frac{n\theta^2}{(n+1)^2(n+2)}$.
3. $V\left[\frac{n+1}{n} X_{(n)}\right] = \frac{(n+1)^2}{n^2} \cdot \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)}$.
4. Efficiency of $2\bar{X}$ relative to it = $\frac{\theta^2/[n(n+2)]}{\theta^2/(3n)} = \frac{3}{n+2}$ - option (d).

Q26.

If $\hat{\theta}_1 = \frac{X}{n}$ and $\hat{\theta}_2 = \frac{1}{3}$ are estimators of θ of a binomial population and $\theta = \frac{1}{2}$, the values of n for which the mean square error of $\hat{\theta}_2$ is less than the variance of $\hat{\theta}_1$ are

A. $2 \leq n < 7$ only

B. $1 \leq n \leq 6$ only

C. $4 \leq n \leq 8$ only

D. $1 \leq n \leq 4$ only

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$V(\hat{\theta}_1) = \frac{\theta(1-\theta)}{n} = \frac{1}{4n} \text{ and } \text{MSE}(\hat{\theta}_2) = \left(\frac{1}{3} - \frac{1}{2}\right)^2 = \frac{1}{36}. \quad \frac{1}{36} < \frac{1}{4n} \iff n < 9.$$

TIPS & TRICKS

- A constant estimator has zero variance, so its MSE is pure squared bias.
- $\theta(1-\theta) = \frac{1}{4}$ at $\theta = \frac{1}{2}$ - the maximum possible binomial variance.
- $n < 9$ means $n \leq 8$; the strict inequality at $n = 9$ gives equality, so 9 is excluded.

STEP-BY-STEP SOLUTION

1. $\hat{\theta}_1 = X/n$ with $X \sim B(n, \theta)$: $E(\hat{\theta}_1) = \theta$ and $V(\hat{\theta}_1) = \frac{\theta(1-\theta)}{n} = \frac{1}{4n}$ at $\theta = \frac{1}{2}$.

2. $\hat{\theta}_2 = \frac{1}{3}$ is a constant, so $\text{MSE}(\hat{\theta}_2) = \left(\frac{1}{3} - \frac{1}{2}\right)^2 = \frac{1}{36}$.

3. Require $\frac{1}{36} < \frac{1}{4n}$, i.e. $4n < 36$, i.e. $n < 9$.

4. So $1 \leq n \leq 8$ - option (c).

Q27.

For an estimator T of θ :

1. T unbiased for $\theta \Rightarrow T^2$ unbiased for θ^2 .
2. T consistent for $\theta \Rightarrow T^2$ consistent for θ^2 .
3. T sufficient for $\theta \Rightarrow T^2$ sufficient for θ .

Codes:

A. 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Only consistency survives squaring (continuous mapping). Unbiasedness does not ($E(T^2) = V(T) + \theta^2 > \theta^2$), and T^2 is a many-to-one function of T , so sufficiency is not guaranteed.

TIPS & TRICKS

- Consistency is preserved under any continuous transformation - Slutsky/continuous mapping theorem.
- Unbiasedness is destroyed by any strictly convex transformation (Jensen): $E(T^2) > [E(T)]^2$ unless $V(T) = 0$.
- Sufficiency is preserved only under ONE-TO-ONE transformations; squaring merges t and $-t$.

STEP-BY-STEP SOLUTION

1. Statement 1: $E(T^2) = V(T) + [E(T)]^2 = V(T) + \theta^2 > \theta^2$ whenever $V(T) > 0$. False.
2. Statement 2: if $T \rightarrow P\theta$ then $T^2 \rightarrow P\theta^2$ by the continuous mapping theorem. True.
3. Statement 3: sufficiency is preserved under one-to-one transformations; $t \mapsto t^2$ is not one-to-one in general, so information can be lost. False.
4. Hence 2 only - option (a).

Q28.

If T_1, T_2 are consistent estimators of θ_1, θ_2 :

1. $(T_1 + T_2)$ is a consistent estimator for $(\theta_1 + \theta_2)$.
2. $(T_1 \times T_2)$ is a consistent estimator for $(\theta_1 \times \theta_2)$.

Codes:

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Convergence in probability is preserved by sums and products (continuous mapping), so both statements hold.

TIPS & TRICKS

- $T_1 \rightarrow P\theta_1, T_2 \rightarrow P\theta_2 \Rightarrow g(T_1, T_2) \rightarrow Pg(\theta_1, \theta_2)$ for continuous g .
- Addition and multiplication are continuous, so both sums and products inherit consistency.
- Contrast unbiasedness, which fails for products unless the estimators are independent.

STEP-BY-STEP SOLUTION

1. Consistency means $T_1 \rightarrow P\theta_1$ and $T_2 \rightarrow P\theta_2$.
2. The map $(u, v) \mapsto u + v$ is continuous, so $T_1 + T_2 \rightarrow P\theta_1 + \theta_2$. Statement 1 true.
3. The map $(u, v) \mapsto uv$ is continuous, so $T_1 T_2 \rightarrow P\theta_1 \theta_2$. Statement 2 true.
4. Option (c).

Q29.

If x is a random sample of size 1 from $\text{Poisson}(\theta)$, then 5^x is the unbiased estimator of

- A. 5^θ
- B. $e^{5\theta}$
- C. $e^{4\theta}$
- D. θ^5

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$E(a^X) = e^{\theta(a-1)}$ for Poisson; with $a=5$ this is $e^{4\theta}$.

TIPS & TRICKS

- Poisson pgf: $E(a^X) = e^{\theta(a-1)}$ - one formula answers the whole family of such questions.
- $a=5 \Rightarrow a-1=4$, giving $e^{4\theta}$, not 5^θ .
- For an unbiased estimator of $e^{-k\theta}$ you would need $a=1-k$, hence the alternating signs seen elsewhere.

STEP-BY-STEP SOLUTION

$$1. E(5^X) = \sum_{x \geq 0} 5^x \frac{e^{-\theta} \theta^x}{x!} = e^{-\theta} \sum_{x \geq 0} \frac{(5\theta)^x}{x!} = e^{-\theta} e^{5\theta}.$$

$$2. = e^{4\theta}.$$

3. So 5^X is unbiased for $e^{4\theta}$ - option (c).

Q30.

Let $X_1, \dots, X_n$ be iid $U(0, \theta)$. If $Y = \max(X_1, \dots, X_n)$, an unbiased estimator of θ^3 will be

A. y^3

B. $\frac{n+3}{n} y^3$

C. y^3/n

D. $\frac{n}{n+3} y^3$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$E(Y^3) = \frac{n\theta^3}{n+3}, \text{ so } \frac{n+3}{n} Y^3 \text{ is unbiased for } \theta^3.$$

TIPS & TRICKS

- $f_Y(y) = \frac{n y^{n-1}}{\theta^n}$, so $E(Y^r) = \frac{n\theta^r}{n+r}$ - one formula for every power.
- The correction factor always EXCEEDS 1 because the maximum under-estimates θ .
- Option (d) inverts the factor and is the standard trap.

STEP-BY-STEP SOLUTION

1. $F_Y(y) = \left(\frac{y}{\theta}\right)^n$, so $f_Y(y) = \frac{n y^{n-1}}{\theta^n}$ on $(0, \theta)$.

2. $E(Y^3) = \int_0^\theta y^3 \frac{n y^{n-1}}{\theta^n} dy = \frac{n}{\theta^n} \cdot \frac{\theta^{n+3}}{n+3} = \frac{n\theta^3}{n+3}.$

3. Hence $E\left[\frac{n+3}{n} Y^3\right] = \theta^3.$

4. Option (b).

Q31.

Consider the following statements:

1. An unbiased estimator is always unique.
2. Sufficient statistic is always a function of MLE.
3. Consistent estimator need not be unique.
4. UMP test is not unique.

Codes:

- A. 1 and 2 only
B. 3 and 4 only
C. 1, 3 and 4 only
D. 1 and 4 only

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Unbiased estimators are not unique (1 false); the MLE is a function of the sufficient statistic, not the reverse (2 false); consistent estimators are not unique (3 true); in discrete problems the randomisation of a UMP test can be re-allocated, so it is not unique (4 true).

TIPS & TRICKS

- The correct implication is 'MLE is a function of the sufficient statistic' - statement 2 reverses it.
- If T_n is consistent so is $T_n + 1/n$, giving infinitely many consistent estimators.
- In discrete problems the boundary randomisation probability can be spread across several points with the same likelihood ratio, all giving the same size and power - hence non-uniqueness.

STEP-BY-STEP SOLUTION

1. Statement 1: any convex combination of two unbiased estimators is unbiased, so uniqueness fails. False.
2. Statement 2: the standard result runs the other way - when a sufficient statistic exists, the MLE is a function of it. False as stated.
3. Statement 3: if T_n is consistent, so is $T_n + \frac{1}{n}$; consistency is shared by infinitely many estimators. True.
4. Statement 4: in discrete problems the boundary randomisation can be distributed over several equally-ranked outcomes, producing distinct UMP tests of the same size and power. True.
5. Hence 3 and 4 only - option (b).

Q32.

Consider the following statements:

1. Unbiased estimators may not always exist.
2. Unbiased estimators are always unique if they exist.

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT[AI-derived explanation](#)

Unbiased estimators may fail to exist (e.g. $1/p$ in Bernoulli), and when they exist they are generally NOT unique.

TIPS & TRICKS

- Non-existence example: no unbiased estimator of $1/p$ exists in the Bernoulli model.
- Non-uniqueness: any convex combination of two unbiased estimators is unbiased.
- Uniqueness belongs to the UMVUE, not to unbiased estimators in general.

STEP-BY-STEP SOLUTION

1. Statement 1: for $Y = \sum X_i \sim B(n, p)$, every $E[g(Y)]$ is a polynomial in p , so no unbiased estimator of $1/p$ exists. True.
2. Statement 2: if T_1, T_2 are unbiased then so is $aT_1 + (1-a)T_2$ for every a , giving infinitely many. False.
3. Hence 1 only - option (a).

Q33.

Consider the following statements:

1. If t is unbiased for θ , it need not be consistent.
2. If t is consistent for θ , it will also be unbiased for θ .

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT**AI-derived explanation**

Unbiasedness does not imply consistency (X_1 is unbiased but not consistent), and consistency does not imply unbiasedness.

TIPS & TRICKS

- X_1 alone is unbiased for μ but its variance never shrinks - unbiased, not consistent.
- $\hat{\sigma}^2$ with divisor n is consistent but biased - the converse counterexample.
- The two properties are logically independent; neither implies the other.

STEP-BY-STEP SOLUTION

1. Statement 1: take $t = X_1$. It is unbiased for μ but $V(X_1) = \sigma^2$ for every n , so it is not consistent. True.
2. Statement 2: $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ is consistent for σ^2 yet biased. False.
3. Hence 1 only - option (a).

Q34.

Let $X_1, \dots, X_n$ be iid with $f(x, \theta) = e^{-(x-\theta)}$; $x > \theta$. Let $X_{(1)} = \min(X_1, \dots, X_n)$. Define $T(X) = X_{(1)} + k$; $k \in \mathbb{R}$. The estimator with smallest MSE is

- A. $T(X) = X_{(1)} - n$
- B. $T(X) = X_{(1)} + 1$
- C. $T(X) = X_{(1)} + \frac{1}{n}$
- D. ✓ $T(X) = X_{(1)} - \frac{1}{n}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$X_{(1)} \sim \theta + \text{Exp}(n)$, so bias $= \frac{1}{n} + k$ and $\text{MSE} = \frac{1}{n^2} + \left(\frac{1}{n} + k\right)^2$, minimised at $k = -\frac{1}{n}$.

TIPS & TRICKS

- $n(X_{(1)} - \theta) \sim \text{Exp}(1)$, so $E(X_{(1)}) = \theta + \frac{1}{n}$ and $V(X_{(1)}) = \frac{1}{n^2}$.
- Adding a constant changes only the bias, never the variance - so minimise the squared bias alone.
- The minimiser removes the bias entirely here, so the MSE-optimal choice happens to coincide with the unbiased one.

STEP-BY-STEP SOLUTION

1. $P(X_{(1)} > t) = e^{-n(t-\theta)}$ for $t > \theta$, so $X_{(1)} - \theta \sim \text{Exp}(n)$ with mean $\frac{1}{n}$ and variance $\frac{1}{n^2}$.
2. $T = X_{(1)} + k$ has bias $b(k) = E(T) - \theta = \frac{1}{n} + k$ and variance $\frac{1}{n^2}$ (independent of k).
3. $\text{MSE}(k) = \frac{1}{n^2} + \left(\frac{1}{n} + k\right)^2$.
4. Differentiating: $2\left(\frac{1}{n} + k\right) = 0 \Rightarrow k = -\frac{1}{n}$.
5. So the MSE-optimal estimator is $X_{(1)} - \frac{1}{n}$ - option (d).

Q35.

Let Y represent tensile strength of a cable of diameter X mm. If $Y = CX^2$ for some constant C and density function of X is given by

$$f(x) = \begin{cases} \frac{1}{\theta}, & 5 < x < 5+\theta \\ 0, & \text{otherwise} \end{cases}$$

what is the unbiased estimate of θ^2 ?

- A. $3X - 5$
- B. $X^2 - 5$
- C. $3(X-5)^2$
- D. $3(X^2-5)$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$X-5 \sim U(0,\theta)$, so $E[(X-5)^2] = \theta^2/3$ and $3(X-5)^2$ is unbiased for θ^2 .

TIPS & TRICKS

- Shift first: the location 5 is known, so subtracting it reduces the problem to the standard $U(0,\theta)$.
- $E(Y^2) = \theta^2/3$ for $Y \sim U(0,\theta)$ - integrate $\int_0^\theta y^2/\theta \, dy$.
- The mention of $Y=CX^2$ is context only; the question asks about θ^2 , so it plays no role in the computation.

STEP-BY-STEP SOLUTION

1. $X \sim U(5, 5+\theta)$, so $Y = X-5 \sim U(0, \theta)$.
2. $E(Y^2) = \int_0^\theta \frac{y^2}{\theta} dy = \frac{\theta^2}{3}$.
3. Hence $E[3(X-5)^2] = 3 \cdot \frac{\theta^2}{3} = \theta^2$.
4. So $3(X-5)^2$ is unbiased for θ^2 - option (c).

Q36.

Consider the following statements:

1. For an exponential random variable with mean θ , $\left(\frac{n}{n+1}\right)\bar{X}^2$ is unbiased for θ^2 .
2. For a Bernoulli random variable with parameter p , $\frac{(\sum x_i)^2}{n^2}$ is unbiased for p^2 .

Which of the statements given above is/are correct?

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$E(\bar{X}^2) = \frac{\theta^2}{n} + \theta^2 = \frac{(n+1)\theta^2}{n}$, so $\frac{n}{n+1}\bar{X}^2$ is unbiased for θ^2 (1 true). For Bernoulli, $E(\bar{X}^2) = \frac{p(1-p)}{n} + p^2 \neq p^2$ (2 false).

TIPS & TRICKS

- $E(T^2) = V(T) + [E(T)]^2$ - apply it to $\bar{X}$ in both parts.
- Exponential: $V(X) = \theta^2$ so $V(\bar{X}) = \theta^2/n$, and the correction factor is exactly $\frac{n}{n+1}$.
- Bernoulli: the correction would have to be additive, not multiplicative, so no constant multiple of $\bar{X}^2$ is unbiased for p^2 .

STEP-BY-STEP SOLUTION

1. Statement 1: for exponential with mean θ , $V(X) = \theta^2$, so $V(\bar{X}) = \frac{\theta^2}{n}$ and $E(\bar{X}^2) = \frac{\theta^2}{n} + \theta^2 = \frac{(n+1)\theta^2}{n}$.
2. Hence $E\left[\frac{n}{n+1}\bar{X}^2\right] = \theta^2$: true.
3. Statement 2: $\frac{(\sum x_i)^2}{n^2} = \bar{X}^2$, and $E(\bar{X}^2) = \frac{p(1-p)}{n} + p^2 \neq p^2$: false.
4. Hence 1 only - option (a).

Q37.

Let T_1 be an MVUE of θ and T_2 be any other unbiased estimator of θ , such that the correlation coefficient between T_1 and T_2 is 0.75. What is the efficiency of T_2 ?

- A. $\frac{\sqrt{3}}{2}$
- B. $\frac{3}{\sqrt{2}}$
- C. $\frac{16}{9}$
- D. ✓ $\frac{9}{16}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$$\rho = \sqrt{e}, \text{ so } e = \rho^2 = (0.75)^2 = \frac{9}{16}.$$

TIPS & TRICKS

- Invert the standard identity: efficiency $= \rho^2$ when one estimator is MVUE.
- $0.75 = \frac{3}{4}$, so $e = \frac{9}{16} = 0.5625$ - keep fractions to match the printed option.
- Efficiency must lie in $[0,1]$, which eliminates options (b) and (c) immediately.

STEP-BY-STEP SOLUTION

1. If T_1 is MVUE and T_2 is unbiased with efficiency $e = \frac{V(T_1)}{V(T_2)}$, then $\rho(T_1, T_2) = \sqrt{e}$.
2. Here $\rho = 0.75$, so $e = \rho^2$.
3. $e = (0.75)^2 = \frac{9}{16}$.
4. Option (d).

Q38.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample of size n from $N(\mu, \sigma^2)$ where μ is unknown. Then the most efficient biased estimator for μ is:

- A. $\bar{x}$
- B. $\frac{1}{n}\bar{x}$
- C. $\frac{n}{n+1}\bar{x}$
- D. $\frac{n+1}{n}\bar{x}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Among the offered multiples $c\bar{x}$, the variance $c^2\sigma^2/n$ is smallest for the smallest $|c|$; $c = \frac{n}{n+1} < 1$ gives a biased estimator with variance below that of $\bar{x}$.

TIPS & TRICKS

- $\bar{x}$ itself is unbiased, so it cannot answer a question asking for the most efficient BIASED estimator.
- Shrinking towards zero ($c < 1$) always reduces variance at the cost of bias - the classic bias-variance trade-off.
- $c = n$ and $c = \frac{n+1}{n}$ both inflate the variance, so they are dominated at once.

STEP-BY-STEP SOLUTION

- Consider estimators of the form $T_c = c\bar{x}$, with $E(T_c) = c\mu$ and $V(T_c) = \frac{c^2\sigma^2}{n}$.
- T_c is unbiased only for $c=1$; any other c gives a biased estimator.
- Among the biased choices offered, the variance $\frac{c^2\sigma^2}{n}$ is smallest for the smallest c^2 , i.e. $c = \frac{n}{n+1} < 1$.
- Hence the most efficient biased estimator among the options is $\frac{n}{n+1}\bar{x}$ - option (c).

SHARED CONTEXT

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from uniform distribution $U(0, \theta)$. Let $T_1 = \frac{2}{n} \sum_{i=1}^n X_i$ and $T_2 = \max(X_1, X_2, X_3, \dots, X_n)$.

Q39.

Consider the following statements:

1. T_1 and T_2 are consistent estimators of θ .
2. T_1 and T_2 are unbiased estimators of θ .

Which of the statements given above is/are correct?

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$T_1 = \bar{2X}$ is unbiased and consistent; $T_2 = X_{(n)}$ is consistent but biased ($E = \frac{n\theta}{n+1}$). So only statement 1 holds.

TIPS & TRICKS

- $E(T_1) = 2 \cdot \frac{\theta}{2} = \theta$; $E(T_2) = \frac{n\theta}{n+1} < \theta$.
- Both converge in probability to θ , so consistency is shared.
- Bias of T_2 vanishes as $n \rightarrow \infty$ - consistent but biased is perfectly possible.

STEP-BY-STEP SOLUTION

1. $T_1 = \frac{2}{n} \sum X_i = \bar{2X}$ with $E(T_1) = \theta$ and $V(T_1) = \frac{\theta^2}{3n} \rightarrow 0$: unbiased and consistent.
2. $T_2 = X_{(n)}$ with $E(T_2) = \frac{n\theta}{n+1} \neq \theta$: biased. But $P(|X_{(n)} - \theta| > \varepsilon) = \left(1 - \frac{\varepsilon}{\theta}\right)^n \rightarrow 0$: consistent.
3. So both are consistent (statement 1 true) but not both unbiased (statement 2 false).
4. Option (a).

Q40.

The radius of a circle is measured with an error of measurement, which is $N(0, \sigma^2)$. Let $x_1, x_2, \dots, x_n$ be n measurements of the radius. Let $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ be the sample mean and $S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ be the sample variance. What is the unbiased estimate of the area of the circle?

- A. $\pi(\bar{x})^2$
- B. $\pi \frac{\sum_{i=1}^n x_i^2}{n}$
- C. ✓ $\pi \left(\frac{\sum_{i=1}^n x_i^2}{n} - S^2 \right)$
- D. $\pi [(\bar{x})^2 - S^2]$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$E\left(\frac{\sum x_i^2}{n}\right) = r^2 + \sigma^2$ and $E(S^2) = \sigma^2$, so $\pi\left(\frac{\sum x_i^2}{n} - S^2\right)$ is unbiased for πr^2 .

TIPS & TRICKS

- $E(x^2) = r^2 + \sigma^2$ where r is the true radius - the raw second moment overshoots by exactly σ^2 .
- S^2 (divisor $n-1$) is unbiased for σ^2 , so it removes the excess exactly.
- $\pi \bar{x}^2$ over-estimates by $\pi \sigma^2/n$; $\pi(\bar{x}^2 - S^2)$ over-corrects. Only option (c) balances.

STEP-BY-STEP SOLUTION

1. Let r be the true radius, so $x_i = r + e_i$ with $e_i \sim N(0, \sigma^2)$, giving $E(x_i) = r$ and $V(x_i) = \sigma^2$.
2. $E\left(\frac{\sum x_i^2}{n}\right) = E(x_i^2) = V(x_i) + [E(x_i)]^2 = \sigma^2 + r^2$.
3. $E(S^2) = \sigma^2$.
4. Hence $E\left[\frac{\sum x_i^2}{n} - S^2\right] = r^2$, and multiplying by π gives an unbiased estimate of the area πr^2 .
5. Option (c).

Q41.

Let X be a random variable which assumes only two values 1 and 0, respectively representing occurrence of success or failure in an experiment with only two outcomes. The probability of success is $p(\theta)$ defined as:

$$p(\theta) = \begin{cases} \theta, & \theta \text{ is rational} \\ 1-\theta, & \theta \text{ is algebraic irrational} \end{cases}$$

Define an estimator $T = \frac{\sum_{i=1}^n X_i}{n}$ = sample proportion of success. Which one of the following is correct?

- A. T is MLE as well as consistent for θ .
- B. T is unbiased and consistent for θ .
- C. T is MLE but not consistent for θ .
- D. T is consistent but not MLE, tending to θ .

Correct answer: C

EXAM SHORTCUT AI-derived explanation

T maximises the Bernoulli likelihood in p , so it is the MLE; but $T \rightarrow Pp(\theta)$, which equals $1-\theta$ for algebraic irrational θ - so T is not consistent for θ .

TIPS & TRICKS

- Consistency requires convergence to θ itself, not to $p(\theta)$.
- The pathological definition makes $p(\theta)=1-\theta$ on a dense set, so T converges to the wrong target there.
- $E(T)=p(\theta) \neq \theta$ in general, so T is not unbiased for θ either - option (b) fails.

STEP-BY-STEP SOLUTION

1. $T = \frac{\sum X_i}{n}$ maximises $p^{\sum X_i} (1-p)^{n-\sum X_i}$, so it is the MLE of the success probability and hence (by invariance through $p(\theta)$) the MLE in this parametrisation.
2. By the WLLN, $T \rightarrow Pp(\theta)$.
3. If θ is rational then $p(\theta)=\theta$ and $T \rightarrow P\theta$; but if θ is algebraic irrational then $p(\theta)=1-\theta$ and $T \rightarrow P1-\theta \neq \theta$.
4. So T is not consistent for θ - option (c).

Q42.

Consider the following statements regarding unbiased estimators of parameter θ :

1. Let T_0 be an UMVUE of θ and T_1 is any other unbiased estimator of θ with efficiency 0.64, then the correlation coefficient between T_0 and T_1 is 0.8.
2. Let T_0 be an MVUE of $g(\theta)$ and T_1 is any other unbiased estimator with efficiency less than 1, then any unbiased linear combination of T_0 and T_1 will also be MVUE of $g(\theta)$.

Which of the above statements is/are correct?

- A. 1 only**
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$\rho = \sqrt{0.64} = 0.8$ (statement 1 true); but an unbiased combination $aT_0 + (1-a)T_1$ with $a \neq 1$ has variance strictly exceeding $V(T_0)$, so it is never MVUE (statement 2 false).

TIPS & TRICKS

- $\rho = \sqrt{e}$ - the identity that appears almost every year.
- $V[aT_0 + (1-a)T_1] = a^2V_0 + (1-a)^2V_1 + 2a(1-a)V_0 > V_0$ for $a \neq 1$, using $\text{Cov}(T_0, T_1) = V_0$.
- UMVUEs are unique, so no other unbiased estimator can share the title - statement 2 contradicts uniqueness.

STEP-BY-STEP SOLUTION

1. Statement 1: efficiency $e = 0.64$ gives $\rho = \sqrt{e} = 0.8$. True.
2. Statement 2: with $\text{Cov}(T_0, T_1) = V(T_0)$, $V[aT_0 + (1-a)T_1] = a^2V_0 + (1-a)^2V_1 + 2a(1-a)V_0$.
3. Since $V_1 > V_0$, this exceeds V_0 for every $a \neq 1$, so such a combination is not MVUE. Also, the UMVUE is unique, which the statement contradicts. False.
4. Hence 1 only - option (a).

Q43.

Let $X \sim U(7, 7+\theta)$, $\theta > 0$; U being Uniform distribution. Define $T = a(X-b)^2$. Then the statistic T becomes unbiased for θ^2 if:

A. $a=3, b=0$

B. $a=7, b=0$

C. ~~$a=3, b=7$~~

D. $a=7, b=7$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$X-7 \sim U(0, \theta)$ and $E[(X-7)^2] = \theta^2/3$, so $a=3, b=7$.

TIPS & TRICKS

- Shift by the known lower endpoint first: the problem always reduces to $U(0, \theta)$.
- $E(Y^2) = \theta^2/3$ for $Y \sim U(0, \theta)$ - the reciprocal 3 is the multiplier.
- b must be the known left endpoint 7; options with $b=0$ ignore the shift entirely.

STEP-BY-STEP SOLUTION

1. $X \sim U(7, 7+\theta)$, so $Y = X-7 \sim U(0, \theta)$.

2. $E(Y^2) = \int_0^\theta \frac{y^2}{\theta} dy = \frac{\theta^2}{3}$.

3. Therefore $E[3(X-7)^2] = \theta^2$, giving $a=3$ and $b=7$.

4. Option (c).

Q44.

Let $x_1, x_2, \dots, x_n$ be a random sample from Normal distribution $N(\mu, \sigma^2)$. Assume that μ is known and σ is unknown. Define an estimator T as $T = k\sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$. Then the constant k for T to be an unbiased estimator of σ is:

- A. $\sqrt{2} \frac{\Gamma(n/2)}{\Gamma((n+1)/2)}$
- B. $\sqrt{n/2} \frac{\Gamma((n+1)/2)}{\Gamma(n/2)}$
- C. 1
- D. $\sqrt{n/2} \frac{\Gamma(n/2)}{\Gamma((n+1)/2)}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$$\sum (x_i - \mu)^2 / \sigma^2 \sim \chi^2_n \text{ and } E\left(\sqrt{\chi^2_n}\right) = \sqrt{2} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}, \text{ so } k = \sqrt{\frac{n}{2}} \frac{\Gamma(n/2)}{\Gamma((n+1)/2)}.$$

TIPS & TRICKS

- $E\left[(\chi^2_n)^{1/2}\right] = \sqrt{2} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}$ - the key Gamma-function moment.
- $T = k\sigma\sqrt{\chi^2_n/n}$, so $E(T) = k\sigma\sqrt{2/n} \frac{\Gamma((n+1)/2)}{\Gamma(n/2)}$; set equal to σ and invert.
- The correction factor is close to 1 for large n , so any option without both $\sqrt{n/2}$ and the Gamma ratio inverted is wrong.

STEP-BY-STEP SOLUTION

1. With μ known, $W = \frac{\sum (x_i - \mu)^2}{\sigma^2} \sim \chi^2_n$, so $T = k\sigma\sqrt{\frac{W}{n}}$.

2. $E\left(\sqrt{W}\right) = \sqrt{2} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}$.

$$3. E(T) = \frac{k\sigma}{\sqrt{n}} \cdot \sqrt{2} \cdot \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}, \text{ and unbiasedness requires this to equal } \sigma.$$

$$4. \text{ Hence } k = \sqrt{\frac{n}{2}} \cdot \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n+1}{2}\right)} - \text{option (d).}$$

Q45.

Let $x_1, x_2, \dots, x_n$ be a random sample from a Cauchy distribution having pdf $f(x, \theta) = \frac{1}{\pi[1+(x-\theta)^2]}$; $-\infty < x, \theta < \infty$.

Consider the following statements:

1. $\bar{X}$ is unbiased for θ but not a consistent estimator for θ .
2. Sample median is a consistent estimator for θ .

Which of the statements given above is/are correct?

- A. 1 only
- B. 2 only**
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: B

EXAM SHORTCUT AI-derived explanation

The Cauchy has no mean, so $\bar{X}$ is not unbiased (statement 1 fails); the sample median is consistent for the location parameter (statement 2 holds).

TIPS & TRICKS

- 'Unbiased' presupposes that the expectation exists - for the Cauchy it does not.
- $\bar{X}$ has the same Cauchy($\theta, 1$) distribution as a single observation, so it is also inconsistent.
- The sample median is consistent for the population median θ and is asymptotically normal with variance $\pi^2/(4n)$.

STEP-BY-STEP SOLUTION

1. Statement 1: for the Cauchy, $E|X| = \infty$, so $E(\bar{X})$ does not exist and $\bar{X}$ cannot be called unbiased. The statement is therefore incorrect (its inconsistency claim is right, but the unbiasedness claim is not). False.
2. Statement 2: the population median is θ and the density is positive there, so the sample median converges in probability to θ . True.
3. Hence 2 only - option (b).

Q46.

Let $x_1, x_2, \dots, x_n$ be a random sample from Normal distribution $N(\mu, \sigma^2)$. The Mean Squared Error (MSE) of the estimator $T = \sum_{i=1}^n (x_i - \bar{x})^2$ where $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$, for estimating σ^2 is:

- A. $(n^2-1)\sigma^4$
 B. $(n^2+1)\sigma^4$
 C. $\boxed{(n-1)^2+1}\sigma^4$
 D. $\boxed{(n+1)^2+1}\sigma^4$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$$E(T) = (n-1)\sigma^2 \text{ and } V(T) = 2(n-1)\sigma^4, \text{ so } \text{MSE} = 2(n-1)\sigma^4 + (n-2)^2\sigma^4 = \boxed{(n-1)^2+1}\sigma^4.$$

TIPS & TRICKS

- $T/\sigma^2 \sim \chi^2_{n-1}$, so $E(T) = (n-1)\sigma^2$ and $V(T) = 2(n-1)\sigma^4$.
- Bias = $(n-1)\sigma^2 - \sigma^2 = (n-2)\sigma^2$ - note T is the raw sum of squares, not a mean square.
- $2(n-1) + (n-2)^2 = n^2 - 2n + 2 = (n-1)^2 + 1$ - the algebraic simplification is the final step.

STEP-BY-STEP SOLUTION

1. $T = \sum (x_i - \bar{x})^2$ satisfies $T/\sigma^2 \sim \chi^2_{n-1}$, so $E(T) = (n-1)\sigma^2$ and $V(T) = 2(n-1)\sigma^4$.
2. Bias of T for σ^2 : $(n-1)\sigma^2 - \sigma^2 = (n-2)\sigma^2$.
3. $\text{MSE} = V(T) + [\text{Bias}]^2 = 2(n-1)\sigma^4 + (n-2)^2\sigma^4$.
4. $= [2n-2+n^2-4n+4]\sigma^4 = [n^2-2n+2]\sigma^4 = [(n-1)^2+1]\sigma^4$.
5. Option (c).

Q47.

Consider the following statements:

I. The concentration of distribution of statistic $T(x)$ around its true value $f(\theta)$ is high when the mean square error is low but the converse is not true.

II. There exists no estimator for which the mean square error is least for all $\theta \in \Theta$.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Low MSE does imply high concentration (Chebyshev), but high concentration does not force low MSE; and no estimator minimises MSE uniformly. Both statements hold.

TIPS & TRICKS

- Chebyshev: $P(|T - \gamma(\theta)| > \epsilon) \leq \text{MSE}/\epsilon^2$ - low MSE forces concentration.
- The converse fails because a heavy tail can inflate MSE while leaving the bulk concentrated.
- Uniform MSE-optimality is impossible: the constant rule $T \equiv \theta_0$ has zero MSE at θ_0 .

STEP-BY-STEP SOLUTION

1. Statement I: by Chebyshev's inequality a small MSE bounds $P(|T - \gamma(\theta)| > \epsilon)$, giving high concentration.
The converse fails - a distribution can be concentrated yet have a heavy tail inflating the MSE. True.
2. Statement II: if such a uniformly best estimator existed it would have to beat the constant estimator $T \equiv \theta_0$ at $\theta = \theta_0$, where that constant has zero MSE - for every θ_0 . Impossible. True.
3. Option (c).

Q48.

Consider the following statements:

I. For an unbiased estimator, the variance of an estimator is same as its mean square error.

II. For a highly precise estimator, its bias is proportional to its mean square error.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$MSE = V + \text{bias}^2$, so unbiased $\Rightarrow MSE = \text{variance}$ (I true). Statement II's 'bias proportional to MSE' is meaningless (II false).

TIPS & TRICKS

- The decomposition $MSE(T) = V(T) + [E(T) - \theta]^2$ is the single most used identity in this topic.
- For a highly precise (small-variance) estimator the MSE is dominated by the SQUARED bias, not proportional to the bias.
- Dimensional check: bias has the units of θ ; MSE has units of θ^2 - they cannot be proportional.

STEP-BY-STEP SOLUTION

1. Statement I: $MSE(T) = V(T) + [E(T) - \theta]^2$. If T is unbiased the second term vanishes, so $MSE = V(T)$. True.
2. Statement II: no such proportionality holds; for small variance the MSE is approximately the SQUARE of the bias, and bias and MSE do not even share units. False.
3. Hence I only - option (a).

Q49.

For a Minimum Variance Unbiased (MVU) estimator T_1 and another unbiased estimator T_2 which has an efficiency E_θ , which one of the following statements is correct?

- A. An unbiased linear combination of T_1 and T_2 is an MVU estimator.
- B. T_1 and T_2 are unique.
- C. The correlation coefficient between T_1 and T_2 is given by $(E_\theta)^{1/2}$.
- D. None of the above

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\rho(T_1, T_2) = \sqrt{E_\theta}$ - the efficiency-correlation identity.

TIPS & TRICKS

- $\text{Cov}(T_1, T_2) = V(T_1)$ for a minimum-variance T_1 , which yields $\rho = \sqrt{V(T_1)/V(T_2)}$.
- Option (a) is false: an unbiased combination with weight on T_2 has variance strictly above $V(T_1)$.
- Option (b) is false: T_2 is an arbitrary unbiased estimator, of which there are infinitely many.

STEP-BY-STEP SOLUTION

1. Efficiency $E_\theta = \frac{V(T_1)}{V(T_2)}$ and, by the minimum-variance property, $\text{Cov}(T_1, T_2) = V(T_1)$.

$$2. \rho = \frac{\text{Cov}(T_1, T_2)}{\sqrt{V(T_1)V(T_2)}} = \sqrt{\frac{V(T_1)}{V(T_2)}} = (E_\theta)^{1/2}.$$

3. Option (c).

Q50.

Consider a series of Bernoulli trials with probability of success p ($0 < p < 1$) for each trial, such that f is frequency of success in n trials. Consider the following statements:

I. $\frac{f}{n} \left(1 - \frac{f}{n}\right)$ is a consistent estimator of $p(1-p)$.

II. $\frac{f}{n}$ is a consistent estimator of $\frac{1}{p}$.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$f/n \rightarrow Pp$, so by continuous mapping $\frac{f}{n} \left(1 - \frac{f}{n}\right) \rightarrow Pp(1-p)$ (I true); but f/n converges to p , not $1/p$ (II false).

TIPS & TRICKS

- Continuous mapping theorem: g continuous and $T_n \rightarrow Pc \Rightarrow g(T_n) \rightarrow Pg(c)$.
- $\frac{n}{f}$ would be the natural (though not always defined) candidate for $1/p$.
- Consistency targets must match: f/n can only be consistent for p .

STEP-BY-STEP SOLUTION

1. By the weak law of large numbers, $\frac{f}{n} \rightarrow Pp$.
2. Statement I: $g(u) = u(1-u)$ is continuous, so $\frac{f}{n} \left(1 - \frac{f}{n}\right) \rightarrow Pp(1-p)$. True.
3. Statement II: $\frac{f}{n} \rightarrow Pp \neq \frac{1}{p}$ in general, so it is not consistent for $\frac{1}{p}$. False.
4. Hence I only - option (a).

Q51.

Which one of the following statements is correct in respect of a parametric function?

- A. A parametric function is said to be estimable if its unbiased estimator exists.**
- B. A parametric function is said to be estimable if its minimum variance unbiased estimator exists.
- C. A parametric function is said to be estimable if its mean square error is minimum.
- D. None of the above

Correct answer: A

EXAM SHORTCUT[AI-derived explanation](#)

Estimability means an unbiased estimator exists - nothing more.

TIPS & TRICKS

- Estimable $\iff$ some T with $E_{\theta}(T) = \psi(\theta)$ for all θ exists.
- $1/p$ in the Bernoulli model is the standard NON-estimable function.
- Requiring an MVUE (option (b)) is a stronger condition and is not the definition.

STEP-BY-STEP SOLUTION

1. A parametric function $\psi(\theta)$ is called estimable if there exists a statistic T with $E_{\theta}(T) = \psi(\theta)$ for every θ - that is, if an unbiased estimator exists.
2. Option (b) demands minimum variance in addition, which is not part of the definition; option (c) confuses estimability with an optimality criterion.
3. Option (a).

Q52.

If T_1 and T_2 are equally efficient estimators of $g(\theta)$ with efficiency 0.8, then the correlation coefficient between them lies in the interval

- A. (0.8,1)
- B. (-1,0.8)
- C. (-0.64,0.64)
- D. **(0.6,1)**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

For two equally efficient unbiased estimators with common efficiency e , $\rho \geq 2e - 1 = 0.6$; so $\rho \in (0.6, 1)$.

TIPS & TRICKS

- Take $T^* = \frac{T_1 + T_2}{2}$: unbiased with $V(T^*) = \frac{V(1+\rho)}{2}$, which cannot beat the minimum variance eV .
- That gives $\frac{(1+\rho)}{2} \geq e$, i.e. $\rho \geq 2e - 1$.
- $2(0.8) - 1 = 0.6$, and $\rho \leq 1$ always - hence the interval (0.6,1).

STEP-BY-STEP SOLUTION

1. Let $V = V(T_1) = V(T_2)$ and let V_0 be the minimum attainable variance, so $e = \frac{V_0}{V} = 0.8$.
2. $T^* = \frac{T_1 + T_2}{2}$ is unbiased with $V(T^*) = \frac{V(1+\rho)}{2}$.
3. No unbiased estimator can have variance below $V_0 = eV$, so $\frac{V(1+\rho)}{2} \geq eV$, giving $\rho \geq 2e - 1 = 0.6$.
4. Combined with $\rho \leq 1$, the correlation lies in (0.6,1) - option (d).

SHARED CONTEXT

Assume a sample of continuous random variables $X_1, X_2, X_3, \dots, X_n$ such that $E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2 > 0$; $i = 1, 2, 3, \dots, n$. Define

$$\hat{\mu}_{1,n} = X_n \text{ and } \hat{\mu}_{2,n} = \frac{1}{n+1} \sum_{i=1}^n X_i.$$

Q53.

Consider the following statements:

I. $\hat{\mu}_{1,n}$ is an unbiased estimator of μ .

II. $\hat{\mu}_{2,n}$ is an unbiased estimator of μ .

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$E(X_n) = \mu \text{ (I true), but } E\left(\frac{\sum X_i}{n+1}\right) = \frac{n\mu}{n+1} \neq \mu \text{ (II false).}$$

TIPS & TRICKS

- A single observation is always unbiased for the mean - it just has large variance.
- Dividing a sum of n terms by $n+1$ shrinks the estimator toward zero, creating bias $-\frac{\mu}{n+1}$.
- The bias vanishes asymptotically, which is why $\hat{\mu}_{2,n}$ is still consistent (next item).

STEP-BY-STEP SOLUTION

1. Statement I: $E(\hat{\mu}_{1,n}) = E(X_n) = \mu$ - unbiased. True.
2. Statement II: $E(\hat{\mu}_{2,n}) = \frac{1}{n+1} \sum_{i=1}^n E(X_i) = \frac{n\mu}{n+1} \neq \mu$ - biased. False.
3. Hence I only - option (a).

SHARED CONTEXT

Assume a sample of continuous random variables $X_1, X_2, X_3, \dots, X_n$ such that $E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2 > 0$; $i = 1, 2, 3, \dots, n$. Define

$$\hat{\mu}_{1,n} = X_n \text{ and } \hat{\mu}_{2,n} = \frac{1}{n+1} \sum_{i=1}^n X_i.$$

Q54.

Consider the following statements:

I. $\hat{\mu}_{1,n}$ is a consistent estimator of μ .

II. $\hat{\mu}_{2,n}$ is a consistent estimator of μ .

Which of the statements given above is/are correct?

A. I only

B. **II only**

C. Both I and II

D. Neither I nor II

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

X_n has variance σ^2 for every n so it is not consistent (I false); $\hat{\mu}_{2,n} = \frac{n}{n+1} \bar{X} \rightarrow P\mu$ (II true).

TIPS & TRICKS

- Consistency needs the sampling distribution to concentrate - a single observation never does.
- $\frac{n}{n+1} \rightarrow 1$ and $\bar{X} \rightarrow P\mu$, so Slutsky gives $\hat{\mu}_{2,n} \rightarrow P\mu$.
- Together with the previous item: unbiased $\neq$ consistent, and consistent $\neq$ unbiased.

STEP-BY-STEP SOLUTION

- Statement I: $\text{Var}(\hat{\mu}_{1,n}) = \sigma^2$ for every n , so the distribution never concentrates and $X_n \not\rightarrow P\mu$. False.
- Statement II: $\hat{\mu}_{2,n} = \frac{n}{n+1} \bar{X}$. Since $\bar{X} \rightarrow P\mu$ and $\frac{n}{n+1} \rightarrow 1$, Slutsky's theorem gives $\hat{\mu}_{2,n} \rightarrow P\mu$. True.
- Hence II only - option (b).

Q55.

Consider the following statements:

- I. MLEs are unbiased but not necessarily unique.
- II. Unbiased estimator is always unique.
- III. If U and W are consistent estimators of θ_1 and θ_2 , then UW is also consistent for $\theta_1\theta_2$.

Which of the statements given above is/are correct?

- A. I and II
- B. I and III
- C. II and III
- D. **III only**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

MLEs need not be unbiased (I false); unbiased estimators are not unique (II false); products of consistent estimators are consistent (III true).

TIPS & TRICKS

- $\hat{\sigma}^2$ with divisor n kills statement I immediately.
- Convex combinations of unbiased estimators are unbiased, killing statement II.
- Continuous mapping theorem gives statement III for sums, products and quotients (where defined).

STEP-BY-STEP SOLUTION

1. Statement I: MLEs are frequently biased (e.g. $\hat{\sigma}^2$ with divisor n), so 'MLEs are unbiased' is false.
2. Statement II: $aT_1 + (1-a)T_2$ is unbiased for every a whenever T_1, T_2 are - uniqueness fails. False.
3. Statement III: $U \rightarrow P\theta_1$ and $W \rightarrow P\theta_2$ give $UW \rightarrow P\theta_1\theta_2$ by the continuous mapping theorem. True.
4. Hence III only - option (d).

Q56.

With the increase in sample size if the estimator becomes closer and closer to the parameter, then it is called:

- A. Completeness
- B. ✓Consistent
- C. Sufficient
- D. Unbiasedness

Correct answer: B

EXAM SHORTCUT[AI-derived explanation](#)

Convergence of the estimator to the parameter as n grows is consistency.

TIPS & TRICKS

- Consistency is the only one of the four properties that is asymptotic in nature.
- Formally $T_n \rightarrow P\theta$: $P(|T_n - \theta| < \varepsilon) \rightarrow 1$.
- Unbiasedness, sufficiency and completeness are all finite-sample properties.

STEP-BY-STEP SOLUTION

1. An estimator T_n is consistent for θ if $T_n \rightarrow P\theta$ as $n \rightarrow \infty$.
 2. That is precisely the statement that the estimator gets closer and closer to the parameter as the sample grows.
 3. Option (b).
-

Q57.

Let $X_1, X_2, X_3, \dots, X_n \sim N(\mu, \sigma^2)$ where mean μ and variance σ^2 are unknown. Let $S^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ and $s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$.

Which one of the following is correct according to the efficiency criterion?

- A. s^2 is more efficient than S^2 .
- B. S^2 is less efficient than s^2 .
- C. s^2 and S^2 cannot be compared.
- D. Efficiency of s^2 is equal to efficiency of S^2 .

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$V(S^2) = \frac{2(n-1)\sigma^4}{n^2} < \frac{2\sigma^4}{n-1} = V(s^2)$, and $MSE(S^2) = \frac{(2n-1)\sigma^4}{n^2} < MSE(s^2)$ - so s^2 is the LESS efficient of the two.

TIPS & TRICKS

- $S^2 = \frac{n-1}{n} s^2$, so $V(S^2) = \left(\frac{n-1}{n}\right)^2 V(s^2)$, strictly smaller.
- MSE comparison: $\frac{2n-1}{n^2}$ against $\frac{2}{n-1}$; cross-multiplying gives $2n^2 - 3n + 1 < 2n^2$.
- The biased MLE beats the unbiased estimator on MSE - a classic bias-variance illustration.

STEP-BY-STEP SOLUTION

1. $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$, so $V(s^2) = \frac{2\sigma^4}{n-1}$.

2. $S^2 = \frac{n-1}{n} s^2$ gives $V(S^2) = \frac{(n-1)^2}{n^2} \cdot \frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2} < V(s^2)$.

3. Including bias: $MSE(S^2) = \frac{2(n-1)\sigma^4}{n^2} + \frac{\sigma^4}{n^2} = \frac{(2n-1)\sigma^4}{n^2}$, and $(2n-1)(n-1) = 2n^2 - 3n + 1 < 2n^2$, so

$$MSE(S^2) < MSE(s^2).$$

4. On either measure s^2 is better, i.e. s^2 is less efficient than S^2 - option (b).

Q58.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$. Consider the statistic $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. What is $V(s^2)$ equal to?

A. $\frac{2\sigma^4}{n-1}$

B. $\frac{2\sigma^2}{n-2}$

C. $\frac{2\sigma^2}{n-1}$

D. $\frac{2\sigma}{n-1}$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1} \text{ with variance } 2(n-1), \text{ so } V(s^2) = \frac{2\sigma^4}{n-1}.$$

TIPS & TRICKS

- $V(\chi^2_k) = 2k$ - the only fact needed.
- $V(s^2) = \left(\frac{\sigma^2}{n-1}\right)^2 \cdot 2(n-1) = \frac{2\sigma^4}{n-1}$.
- The answer must carry σ^4 (a variance of a variance), which rules out options (b), (c), (d).

STEP-BY-STEP SOLUTION

1. For a normal sample, $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$.
2. $V(\chi^2_{n-1}) = 2(n-1)$, so $V\left(\frac{(n-1)s^2}{\sigma^2}\right) = 2(n-1)$.
3. $\frac{(n-1)^2}{\sigma^4} V(s^2) = 2(n-1) \Rightarrow V(s^2) = \frac{2\sigma^4}{n-1}$.
4. Option (a).

Q59.

Let $X_1, X_2, X_3, \dots, X_n, X_{n+1}$ be $(n+1)$ observations from $N(\mu, 1)$ distribution. If $\bar{X}_n = \frac{\sum_{i=1}^n X_i}{n}$ and $T = \frac{(\bar{X}_n + X_{n+1})}{2}$, then which one of the following is correct?

- A. T is an unbiased but inconsistent estimator of μ .**
- B. T is a biased but consistent estimator of μ .
- C. T is an unbiased and consistent estimator of μ .
- D. T is neither an unbiased nor a consistent estimator of μ .

Correct answer: A**EXAM SHORTCUT**

AI-derived explanation

$E(T) = \mu$ (unbiased) but $V(T) = \frac{1}{4} \left(\frac{1}{n} + 1 \right) \rightarrow \frac{1}{4} \neq 0$, so T is inconsistent.

TIPS & TRICKS

- X_{n+1} gets weight $\frac{1}{2}$ regardless of n , so its variance contribution never vanishes.
- $V(T) = \frac{1}{4} V(\bar{X}_n) + \frac{1}{4} V(X_{n+1}) = \frac{1}{4} \left(\frac{1}{n} + 1 \right)$.
- Any estimator giving a single observation fixed positive weight cannot be consistent.

STEP-BY-STEP SOLUTION

- $E(T) = \frac{1}{2} [E(\bar{X}_n) + E(X_{n+1})] = \frac{1}{2} (\mu + \mu) = \mu$: unbiased.
- $\bar{X}_n$ and X_{n+1} are independent, so $V(T) = \frac{1}{4} \left[\frac{1}{n} + 1 \right]$.
- As $n \rightarrow \infty$, $V(T) \rightarrow \frac{1}{4} \neq 0$, so T does not converge in probability to μ .
- Hence unbiased but inconsistent - option (a).

Q60.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from pdf $f(x) = \begin{cases} 1, & \theta \leq x \leq \theta+1 \\ 0, & \text{otherwise} \end{cases}$. Which one of the following is correct?

A. The smallest sample observation is a consistent estimator of θ .

B. The largest sample observation is a consistent estimator of θ .

C. Sample mean is a consistent estimator of θ .

D. None of the above

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$X_{(1)} \rightarrow P\theta$ since $P(X_{(1)} > \theta + \varepsilon) = (1 - \varepsilon)^n \rightarrow 0$; the maximum converges to $\theta + 1$ and the mean to $\theta + \frac{1}{2}$.

TIPS & TRICKS

- For $U(\theta, \theta+1)$ the lower endpoint is θ , so the MINIMUM is the natural consistent estimator.
- $X_{(n)} \rightarrow P\theta+1$ and $\bar{X} \rightarrow P\theta + \frac{1}{2}$ - both converge to the wrong target.
- Consistency requires convergence to θ itself, not to a shifted version.

STEP-BY-STEP SOLUTION

1. $P(X_{(1)} > \theta + \varepsilon) = [P(X_1 > \theta + \varepsilon)]^n = (1 - \varepsilon)^n \rightarrow 0$ for any $0 < \varepsilon < 1$.
2. Since also $X_{(1)} > \theta$ always, $X_{(1)} \rightarrow P\theta$: the smallest observation is consistent for θ .
3. $X_{(n)} \rightarrow P\theta+1 \neq \theta$ and $\bar{X} \rightarrow PE(X) = \theta + \frac{1}{2} \neq \theta$.
4. Option (a).

Q61.

Consider the following statements:

I. Sample mean is an unbiased estimator of parameter of a Poisson distribution.

II. Sample mean is a consistent estimator of population mean of a normal distribution.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$E(\bar{X}) = \lambda$ for Poisson (I true) and $\bar{X} \rightarrow P\mu$ for the normal (II true).

TIPS & TRICKS

- $\bar{X}$ is unbiased for the mean of ANY distribution with a finite mean.
- Consistency follows from $V(\bar{X}) = \sigma^2/n \rightarrow 0$ plus Chebyshev.
- Both statements are among the safest facts in the syllabus.

STEP-BY-STEP SOLUTION

1. Statement I: for $\text{Poisson}(\lambda)$, $E(X_i) = \lambda$, so $E(\bar{X}) = \lambda$: unbiased. True.
2. Statement II: for $N(\mu, \sigma^2)$, $E(\bar{X}) = \mu$ and $V(\bar{X}) = \sigma^2/n \rightarrow 0$, so $\bar{X} \rightarrow P\mu$: consistent. True.
3. Option (c).

Q62.

Let $X \sim f(x; \theta) = \frac{2(x-\theta)}{(1-\theta)^2}$, $\theta < x < 1$, $0 < \theta < 1$. An unbiased estimator of θ based on a sample of size one is given by:

- A. $\frac{2}{3}(1-X)$
- B. $3(1-2X)$
- C. $\sqrt{3X-2}$
- D. $2(X-3)$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Substituting $Y = \frac{X-\theta}{1-\theta} \sim \text{Beta}(2,1)$ gives $E(X) = \frac{\theta+2}{3}$, so $E(3X-2) = \theta$.

TIPS & TRICKS

- Standardise a support-shifted density by mapping the support onto $(0,1)$ - here $Y = \frac{X-\theta}{1-\theta}$.
- $f_Y(y) = 2y$ on $(0,1)$, i.e. $\text{Beta}(2,1)$ with mean $\frac{2}{3}$.
- $E(X) = \theta + (1-\theta) \frac{2}{3} = \frac{\theta+2}{3}$; invert to get $3X-2$.

STEP-BY-STEP SOLUTION

1. Put $Y = \frac{X-\theta}{1-\theta}$, so $X = \theta + (1-\theta)Y$ and $Y \in (0,1)$.

2. $f_Y(y) = f_X(\theta + (1-\theta)y)(1-\theta) = \frac{2(1-\theta)y}{(1-\theta)^2}(1-\theta) = 2y$, so $Y \sim \text{Beta}(2,1)$ with $E(Y) = \frac{2}{3}$.

3. $E(X) = \theta + (1-\theta) \cdot \frac{2}{3} = \frac{\theta+2}{3}$.

4. Hence $E(3X-2) = 3 \cdot \frac{\theta+2}{3} - 2 = \theta$, so $3X-2$ is unbiased - option (c).

Q63.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from

$$f(x) = \begin{cases} e^{-(x-\theta)}, & x > \theta \\ 0, & \text{otherwise} \end{cases}$$

The unbiased estimator of θ is:

- A. $\bar{X}$
- B. $\sum_{i=1}^n X_i - n$
- C. $\sum_{i=1}^n X_i - 1$
- D. $\bar{X} - 1$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$E(X) = \theta + 1$, so $E(\bar{X}) = \theta + 1$ and $\bar{X} - 1$ is unbiased for θ .

TIPS & TRICKS

- The shifted exponential $e^{-(x-\theta)}$ has mean $\theta + 1$ - shift the standard exponential mean of 1.
- Correct the whole MEAN, not the sum: $\sum X_i - n$ has expectation $n\theta$, not θ .
- The UMVUE is $X_{(1)} - \frac{1}{n}$, which has much smaller variance than $\bar{X} - 1$.

STEP-BY-STEP SOLUTION

1. For $f(x) = e^{-(x-\theta)}$, $x > \theta$, put $Y = X - \theta \sim \text{Exp}(1)$, so $E(X) = \theta + 1$.
2. Hence $E(\bar{X}) = \theta + 1$.
3. $E(\bar{X} - 1) = \theta$, so $\bar{X} - 1$ is unbiased.
4. Option (d).

Q64.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from Bernoulli population with parameter p . T is defined as

$$T = \begin{cases} \frac{1}{3n}, & \text{if } x_1 + x_2 + x_3 = 1 \\ 0, & \text{otherwise} \end{cases}$$

Then T is unbiased estimator of:

A. $\frac{p(1-p)}{n}$

B. $\frac{p^2}{n}$

C. $\frac{(1-p)^2}{n}$

D. $\frac{p(1-p)^2}{n}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$$P(x_1 + x_2 + x_3 = 1) = 3p(1-p)^2, \text{ so } E(T) = \frac{1}{3n} \cdot 3p(1-p)^2 = \frac{p(1-p)^2}{n}.$$

TIPS & TRICKS

- $x_1 + x_2 + x_3 \sim B(3, p)$, so $P(\cdot = 1) = \binom{3}{1} p(1-p)^2 = 3p(1-p)^2$.
- The $\frac{1}{3n}$ scaling cancels the 3 and introduces the $\frac{1}{n}$.
- Read the exponent: $(1-p)^2$, not $(1-p)$ - options (a) and (b) mis-count the failures.

STEP-BY-STEP SOLUTION

1. T takes the value $\frac{1}{3n}$ exactly when $x_1 + x_2 + x_3 = 1$, and 0 otherwise.
2. $x_1 + x_2 + x_3 \sim B(3, p)$, so $P(x_1 + x_2 + x_3 = 1) = \binom{3}{1} p(1-p)^2 = 3p(1-p)^2$.
3. $E(T) = \frac{1}{3n} \cdot 3p(1-p)^2 = \frac{p(1-p)^2}{n}$.
4. Option (d).

Q65.

A box contains N unknown items marked 1 through N where $N \geq 2$. The unbiased estimate of N after making n random selections from the box and observing $X_1, X_2, X_3, \dots, X_n$ is:

A. $\frac{\bar{X}+1}{2}$

B. $\frac{\bar{X}-1}{2}$

C. $2\bar{X}+1$

D. $2\bar{X}-1$

Correct answer: D

EXAM SHORTCUT[AI-derived explanation](#)

$$E(X) = \frac{N+1}{2} \text{ for the discrete uniform, so } E(2\bar{X}-1) = N.$$

TIPS & TRICKS

- Mean of $\{1, 2, \dots, N\}$ is $\frac{N+1}{2}$ - the midpoint of the range.
- Invert: $N = 2E(X) - 1$, so replace $E(X)$ by $\bar{X}$.
- This is the classic 'German tank problem' moment estimator; the MVUE based on $X_{(n)}$ is much better.

STEP-BY-STEP SOLUTION

1. X_i is uniform on $\{1, 2, \dots, N\}$, so $E(X_i) = \frac{1+2+\dots+N}{N} = \frac{N+1}{2}$.

2. Hence $E(\bar{X}) = \frac{N+1}{2}$.

3. $E(2\bar{X}-1) = 2 \cdot \frac{N+1}{2} - 1 = N$, so $2\bar{X}-1$ is unbiased for N .

4. Option (d).

Q66.

Let X follow Poisson distribution with parameter λ . Then, unbiased estimator of $(-a)^X$ where 'a' is a constant, is:

- A. $e^{-\lambda}$
- B. $e^{-a\lambda}$
- C. $e^{-(a+1)\lambda}$
- D. $e^{-(a-1)\lambda}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$E[(-a)^X] = e^{\lambda(-a-1)} = e^{-(a+1)\lambda}$ using the Poisson pgf.

TIPS & TRICKS

- Poisson pgf: $E(t^X) = e^{\lambda(t-1)}$ - substitute $t = -a$.
- $t-1 = -a-1 = -(a+1)$, giving the exponent directly.
- The alternating sign is what produces a DECAYING exponential; a^X would give $e^{\lambda(a-1)}$.

STEP-BY-STEP SOLUTION

1. For $X \sim \text{Poisson}(\lambda)$, $E(t^X) = \sum_{x \geq 0} t^x \frac{e^{-\lambda} \lambda^x}{x!} = e^{\lambda(t-1)}$.
2. Setting $t = -a$: $E[(-a)^X] = e^{\lambda(-a-1)} = e^{-(a+1)\lambda}$.
3. So $(-a)^X$ is an unbiased estimator of $e^{-(a+1)\lambda}$ - option (c).

Q67.

Let $x_1, x_2, \dots, x_n$ be a random sample from a distribution for which mean μ and variance σ^2 exist. Then the consistent estimator of μ is

A. $x_{(1)} = \min(x_1, x_2, \dots, x_n)$

B. $x_{(n)} = \max(x_1, x_2, \dots, x_n)$

C. $\frac{\sum_{i=1}^n x_i}{n}$

D. $\frac{x_{(n)} + x_{(1)}}{2}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\bar{X}$ is unbiased with $V(\bar{X}) = \sigma^2/n \rightarrow 0$, hence consistent for μ by Chebyshev.

TIPS & TRICKS

- Extremes and the midrange converge to the support endpoints, not to the mean, for general distributions.
- $\bar{X}$ is consistent for the mean of ANY distribution with finite variance - the WLLN.
- The midrange is consistent for μ only for symmetric bounded distributions - not in general.

STEP-BY-STEP SOLUTION

1. $E(\bar{X}) = \mu$ and $V(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0$.

2. By Chebyshev's inequality, $P(|\bar{X} - \mu| > \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \rightarrow 0$, so $\bar{X} \rightarrow \mu$.

3. The minimum, maximum and midrange do not in general converge to μ .

4. Option (c).

Q68.

Let $x_1, x_2, \dots, x_n$ be a random sample from a distribution F_θ for which mean and variance exist. Let μ be the mean and σ^2 be the variance of the distribution. Then the unbiased estimator of σ^2 is

A. $s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

B. $s^2 = \frac{1}{n-2} \sum_{i=1}^n (x_i - \bar{x})^2$

C. $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

D. $s^2 = \frac{2}{n(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$E\left[\sum (x_i - \bar{x})^2\right] = (n-1)\sigma^2$, so the unbiased estimator divides by $n-1$.

TIPS & TRICKS

- One degree of freedom is lost estimating μ by $\bar{x}$ - hence $n-1$.
- Divisor n gives the MLE (biased); divisor $n-1$ gives the unbiased estimator.
- This holds for ANY distribution with finite variance, not just the normal.

STEP-BY-STEP SOLUTION

1. $E\left[\sum_{i=1}^n (x_i - \bar{x})^2\right] = \sum E(x_i^2) - nE(\bar{x}^2) = n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = (n-1)\sigma^2$.

2. Dividing by $n-1$ gives an unbiased estimator of σ^2 .

3. Option (c).

Q69.

If T_1 and T_2 are two unbiased estimators of $\gamma(\theta)$ having the same variance σ^2 and ρ is the correlation between them, then which one of the following is correct?

- A. $\rho = 2e - 1$, where e is the efficiency of each estimator
- B. $\rho \geq 2e - 1$, where e is the efficiency of each estimator**
- C. $\rho = 2e_1 - e_2$, where e_1 is the efficiency of T_1 and e_2 is the efficiency of T_2
- D. $\rho = e_1 + e_2 - 1$, where e_1 is the efficiency of T_1 and e_2 is the efficiency of T_2

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Averaging the two gives an unbiased estimator with variance $\frac{\sigma^2(1+\rho)}{2}$, which cannot beat the minimum variance $e\sigma^2$; hence $\rho \geq 2e - 1$.

TIPS & TRICKS

- Build the average $T^* = \frac{T_1 + T_2}{2}$ and compare its variance with the minimum attainable.
- $\frac{\sigma^2(1+\rho)}{2} \geq e\sigma^2 \Rightarrow \rho \geq 2e - 1$ - an inequality, not an equality.
- Equality would require T^* itself to be the MVUE, which is a special case, not the general rule.

STEP-BY-STEP SOLUTION

1. Let V_0 be the minimum attainable variance, so $e = \frac{V_0}{\sigma^2}$ for each estimator.

2. $T^* = \frac{T_1 + T_2}{2}$ is unbiased with $V(T^*) = \frac{\sigma^2 + \sigma^2 + 2\rho\sigma^2}{4} = \frac{\sigma^2(1+\rho)}{2}$.

3. No unbiased estimator can have variance below $V_0 = e\sigma^2$, so $\frac{\sigma^2(1+\rho)}{2} \geq e\sigma^2$.

4. Hence $\rho \geq 2e - 1$ - option (b).

Q70.

An estimator T_n is consistent for $\gamma(\theta)$ under which of the following?

I. T_n is unbiased for $\gamma(\theta)$

II. $V(T_n) \leq 1$

III. $T_n \rightarrow P\gamma(\theta)$

IV. $\frac{T_n}{n} \rightarrow P\gamma(\theta)$

Select the correct answer using the code given below.

A. I and II

B. II and III

C. **III only**

D. IV

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Consistency is convergence in probability to the target: $T_n \rightarrow P\gamma(\theta)$.

TIPS & TRICKS

- Unbiasedness plus vanishing variance is sufficient but not necessary - it is not the definition.
- $V(T_n) \leq 1$ is a bound that does not shrink with n and implies nothing.
- Statement IV converges the wrong quantity (T_n/n), so it describes a different estimator entirely.

STEP-BY-STEP SOLUTION

1. By definition, T_n is (weakly) consistent for $\gamma(\theta)$ iff $T_n \rightarrow P\gamma(\theta)$.
2. I alone is not enough (X_1 is unbiased but inconsistent); II gives a bound that never shrinks; IV is a statement about T_n/n .
3. Hence III only - option (c).

Q71.

Let $T=T(x_1, x_2, \dots, x_n)$ be an estimator of $\gamma(\theta)$. Consider the following statements:

I. $E_{\theta}[T-\gamma(\theta)]^2 = E_{\theta}[T-E_{\theta}(T)]^2 + [E_{\theta}(T)-\gamma(\theta)]^2$

II. The bias of T is $E_{\theta}(T)-\gamma(\theta)$.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Statement I is the MSE decomposition $\text{MSE} = \text{variance} + \text{bias}^2$; statement II is the definition of bias. Both correct.

TIPS & TRICKS

- Add and subtract $E_{\theta}(T)$ inside the square; the cross term vanishes.
- This identity is the single most used tool for comparing estimators.
- Bias is $E(T)-\gamma(\theta)$ - estimator minus target, in that order.

STEP-BY-STEP SOLUTION

1. Statement I: $E[T-\gamma]^2 = E[(T-E(T)) + (E(T)-\gamma)]^2$. Expanding, the cross term is $2(E(T)-\gamma)E[T-E(T)] = 0$.
2. So $\text{MSE} = V(T) + [E(T)-\gamma(\theta)]^2$. True.
3. Statement II: bias is defined as $b(\theta) = E_{\theta}(T) - \gamma(\theta)$. True.
4. Option (c).

Q72.

Let T_1 and T_2 be two unbiased estimators for a parameter θ such that $E_\theta(T_i^2) < \infty$; $i=1,2$. If the efficiency of T_1 relative to T_2 is $e_\theta\left(\frac{T_1}{T_2}\right) = \frac{\text{Var}_\theta(T_2)}{\text{Var}_\theta(T_1)}$, then which one of the following is correct?

- A. $e_\theta(T_1/T_2) > 1$ implies T_1 is more efficient than T_2
- B. $e_\theta(T_1/T_2) < 1$ implies T_1 is more efficient than T_2
- C. $e_\theta(T_2/T_1) > 1$ implies T_1 is more efficient than T_2
- D. $e_\theta(T_2/T_1) < 1$ implies T_2 is more efficient than T_1

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$e_\theta(T_1/T_2) = \frac{V(T_2)}{V(T_1)} > 1 \text{ means } V(T_1) < V(T_2), \text{ i.e. } T_1 \text{ is more efficient.}$$

TIPS & TRICKS

- Read the ratio as written: the SECOND estimator's variance sits in the numerator.
- Ratio $> 1 \Rightarrow$ the numerator's estimator is worse $\Rightarrow$ the denominator's estimator wins.
- Option (d) reverses the conclusion for the reciprocal ratio.

STEP-BY-STEP SOLUTION

$$1. e_\theta\left(\frac{T_1}{T_2}\right) = \frac{\text{Var}(T_2)}{\text{Var}(T_1)}.$$

2. If this exceeds 1 then $\text{Var}(T_2) > \text{Var}(T_1)$.

3. A smaller variance means greater efficiency, so T_1 is more efficient than T_2 .

4. Option (a).

Cross-over Distribution Theory Items

Q1.

Which are forms of the central limit theorem?

1. Lindeberg-Feller theorem
2. Lyapunov Theorem
3. Khinchin Theorem

Codes:

- A. 1 only
- B. 1 and 2**
- C. 1 and 3
- D. 2 and 3

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Lindeberg-Feller and Lyapunov are central limit theorems; Khinchin's theorem is a WEAK LAW OF LARGE NUMBERS.

TIPS & TRICKS

- Khinchin: iid with finite mean $\Rightarrow \bar{X} \rightarrow P\mu$. That is convergence to a constant, not to a normal law.
- Lyapunov's condition (a moment condition) implies Lindeberg's condition, which is the definitive CLT criterion for non-identically distributed summands.
- CLT $\Rightarrow$ a distributional limit; LLN $\Rightarrow$ a constant limit. Classify by what the limit is.

STEP-BY-STEP SOLUTION

1. Lindeberg-Feller gives conditions under which a sum of independent (not necessarily identically distributed) variables is asymptotically normal - a CLT.
2. Lyapunov's theorem is a CLT under a moment condition that implies Lindeberg's condition.
3. Khinchin's theorem states the weak law of large numbers for iid variables with finite mean - not a CLT.
4. Hence 1 and 2 - option (b).

Q2.

Suppose (X,Y) follows bivariate normal with means μ_1, μ_2 ; std. deviations σ_1, σ_2 ; correlation ρ , $-1 < \rho < 1$, all unknown. Checking $\sigma_1 = \sigma_2$ is equivalent to verifying the independence of

- A. X and Y
- B. X and $X-Y$
- C. $X+Y$ and Y
- D. $X+Y$ and $X-Y$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$\text{Cov}(X+Y, X-Y) = \sigma_1^2 - \sigma_2^2$, and for jointly normal variables zero covariance is independence - so $\sigma_1 = \sigma_2$ iff $X+Y$ and $X-Y$ are independent.

TIPS & TRICKS

- $\text{Cov}(X+Y, X-Y) = V(X) - V(Y)$; the correlation term cancels completely.
- Joint normality is essential: uncorrelated implies independent only in that setting.
- This is the basis of the paired-sample test for equality of variances (Pitman-Morgan test).

STEP-BY-STEP SOLUTION

1. $\text{Cov}(X+Y, X-Y) = V(X) - \text{Cov}(X,Y) + \text{Cov}(Y,X) - V(Y) = V(X) - V(Y) = \sigma_1^2 - \sigma_2^2$.
2. $(X+Y, X-Y)$ is a linear transformation of a bivariate normal vector, hence itself bivariate normal.
3. For jointly normal variables, zero covariance is equivalent to independence.
4. So $\sigma_1 = \sigma_2 \iff X+Y$ and $X-Y$ are independent - option (d).

Q3.

Let $X \sim N(0,1)$. Then $E(|X|)$ equals

- A. $\sqrt{2\pi}$
- B. $\sqrt{\pi}/\sqrt{2}$
- C. $\sqrt{2}/\sqrt{\pi}$
- D. $\sqrt{2}/\pi$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$E|X| = 2 \int_0^{\infty} x \phi(x) dx = \sqrt{2/\pi} = \frac{\sqrt{2}}{\sqrt{\pi}}.$$

TIPS & TRICKS

- $E|X| = \sqrt{2/\pi} \approx 0.7979$ for the standard normal - memorise the value.
- It must be less than $\sigma=1$ (since $E|X| \leq \sqrt{E X^2} = 1$), which eliminates $\sqrt{2\pi}$ and $\sqrt{\pi/2}$.
- This constant appears in the mean absolute deviation and in robust scale estimation.

STEP-BY-STEP SOLUTION

1. By symmetry, $E|X| = 2 \int_0^{\infty} x \cdot \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$.
2. Substituting $u = x^2/2$: $= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-u} du = \frac{2}{\sqrt{2\pi}}$.
3. $\frac{2}{\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}} = \frac{\sqrt{2}}{\sqrt{\pi}} \approx 0.7979$.
4. Option (c).

Q4.

Suppose the survival times of patients of a bladder disorder can be modelled by the exponential distribution such that the average time a patient survives is λ . Consider the following statements:

1. 50% of patients will die by the time $\lambda \log_e 2$.
2. 25% of patients will die by the time $\lambda \log_e \left(\frac{4}{3}\right)$.

Which of the above statements is/are correct?

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$F(t) = 1 - e^{-t/\lambda}$; solving $F = 0.5$ gives $t = \lambda \ln 2$ and $F = 0.25$ gives $t = \lambda \ln \frac{4}{3}$. Both statements are correct.

TIPS & TRICKS

- Exponential quantiles: $t_p = -\lambda \ln(1-p)$ - one formula for every percentile.
- $p = 0.5 \Rightarrow -\ln 0.5 = \ln 2$; $p = 0.25 \Rightarrow -\ln 0.75 = \ln \frac{4}{3}$.
- The median $\lambda \ln 2 \approx 0.693\lambda$ is always LESS than the mean λ - the exponential is right-skewed.

STEP-BY-STEP SOLUTION

1. With mean λ the cdf is $F(t) = 1 - e^{-t/\lambda}$.
2. Statement 1: $F(t) = 0.5 \Rightarrow e^{-t/\lambda} = 0.5 \Rightarrow t = \lambda \ln 2$. True.
3. Statement 2: $F(t) = 0.25 \Rightarrow e^{-t/\lambda} = 0.75 \Rightarrow t = \lambda \ln \frac{4}{3}$. True.
4. Option (c).

Q5.

In a test of randomness of a given set of observations $\{7, 19, 3, 1, 8, 5, 10, 4\}$, when the null hypothesis of randomness of the observations is true, then the expected number of runs U and its variance are respectively:

- A. $4, \frac{12}{7}$
- B. ✓ $5, \frac{12}{7}$
- C. $4, \frac{20}{7}$
- D. $5, \frac{20}{7}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\text{Median} = 6 \text{ gives } n_1 = n_2 = 4; E(U) = \frac{2n_1n_2}{n_1+n_2} + 1 = 5 \text{ and } V(U) = \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1+n_2)^2(n_1+n_2-1)} = \frac{12}{7}.$$

TIPS & TRICKS

- Classify each observation as above (+) or below (-) the median, then count n_1 and n_2 .
- $E(U) = \frac{2n_1n_2}{n} + 1$ and $V(U) = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}$ with $n = n_1 + n_2$.
- Balanced case $n_1 = n_2 = 4$: $2n_1n_2 = 32$, $n = 8$, so $E(U) = 5$ and $V(U) = \frac{32(24)}{64(7)} = \frac{12}{7}$.

STEP-BY-STEP SOLUTION

- Sorted data: 1, 3, 4, 5, 7, 8, 10, 19, so the median is $\frac{5+7}{2} = 6$.
- Signs in the original order $\{7, 19, 3, 1, 8, 5, 10, 4\}$: +, +, -, -, +, -, +, -, giving $n_1 = 4$ above and $n_2 = 4$ below.
- $E(U) = \frac{2n_1n_2}{n_1+n_2} + 1 = \frac{2(16)}{8} + 1 = 5$.
- $V(U) = \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1+n_2)^2(n_1+n_2-1)} = \frac{32(32-8)}{64 \times 7} = \frac{768}{448} = \frac{12}{7}$.
- Option (b).

Q6.

A batch of 19 students is examined in three subjects—Statistics (x_1), Mathematics (x_2) and Computer Programming (x_3). It has been given that $r_{13.2}=0.6$. It is expected that x_2 will affect the correlation between x_1 and x_3 (i.e., the partial correlation between x_1 and x_3 after eliminating the effect of x_2 is zero). The value of test statistic for testing $H_0:\rho_{13.2}=0$ against $H_1:\rho_{13.2}\neq 0$ is

- A. 2.5
 B. 3
 C. 4
 D. 4.5

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$t = \frac{r\sqrt{n-2-k}}{\sqrt{1-r^2}} = \frac{0.6\sqrt{16}}{\sqrt{0.64}} = \frac{2.4}{0.8} = 3.$$

TIPS & TRICKS

- For a partial correlation with k eliminated variables, $df = n-2-k$; here $k=1$ so $df = 16$.
- $\sqrt{1-0.36} = \sqrt{0.64} = 0.8$ - the numbers are designed to be exact.
- $\sqrt{16} = 4$, so the statistic is $0.6 \times 4 / 0.8 = 3$.

STEP-BY-STEP SOLUTION

1. For testing $H_0:\rho_{13.2}=0$ the statistic is $t = \frac{r_{13.2}\sqrt{n-2-k}}{\sqrt{1-r_{13.2}^2}}$ with $k=1$ eliminated variable.
2. $n-2-k=19-3=16$, so $\sqrt{16}=4$.
3. $\sqrt{1-(0.6)^2} = \sqrt{0.64} = 0.8$.
4. $t = \frac{0.6 \times 4}{0.8} = 3$ - option (b).

Sufficiency and Completeness

62 authentic PYQs in this topic (2018-2026)

Sufficient Estimator and Factorization Theorem

SI & HT

2018 Q22

Conceptual

Medium

Q1.

Let (X_1, X_2, X_3) be a random sample from $\text{binomial}(1, p)$. Which statistic is **not** sufficient for p ?

- A. $X_1 + X_2 + X_3$
- B. $(X_1, X_2 + X_3)$
- C. $(X_1 + X_2, X_3)$
- D. $X_1 - X_2 + X_3$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Sufficiency needs the likelihood to depend on the data only through the statistic. $X_1 - X_2 + X_3$ loses the total when signs cancel, so it is not sufficient.

TIPS & TRICKS

- For Bernoulli the likelihood is $p^{\sum x_i} (1-p)^{n-\sum x_i}$: anything from which $\sum x_i$ can be recovered is sufficient.
- Options (b) and (c) are vectors from which $\sum x_i$ is recoverable by addition - hence sufficient (though not minimal).
- $X_1 - X_2 + X_3 = 1$ arises both from $(1, 0, 0)$ and from $(1, 1, 1)$, which have different totals - so it cannot be sufficient.

STEP-BY-STEP SOLUTION

1. The likelihood is $L = p^t (1-p)^{3-t}$ with $t = x_1 + x_2 + x_3$, so any statistic determining t is sufficient.
2. (a) $X_1 + X_2 + X_3 = t$ directly. (b) and (c) are two-component vectors whose components sum to t . All three are sufficient.
3. (d) $X_1 - X_2 + X_3 = 1$ occurs for $(1, 0, 0)$ (with $t=1$) and for $(1, 1, 1)$ (with $t=3$); the conditional distribution of the sample given this value still depends on p .
4. Hence $X_1 - X_2 + X_3$ is not sufficient - option (d).

Q2.

Suppose $f(x, \theta) = \frac{1}{\theta}$, $0 \leq x \leq \theta$. The sufficient statistic for θ from a sample of size n is

- A. $\sum_{i=1}^n X_i$
- B. $\bar{X}$
- C. $X_{(1)}$
- D. $X_{(n)}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$L = \theta^{-n} \prod I(x_i \leq \theta) = \theta^{-n} I(X_{(n)} \leq \theta)$: the data enter only through $X_{(n)}$.

TIPS & TRICKS

- Fisher-Neyman factorisation with an indicator: $g_{\theta}(t) = \theta^{-n} I(t \leq \theta)$ and $h(x) = I(x_{(1)} \geq 0)$.
- $X_{(n)}$ is also complete for $U(0, \theta)$, so $\frac{n+1}{n} X_{(n)}$ is the UMVUE of θ .
- For a two-sided uniform $U(\alpha, \beta)$ the sufficient statistic is the pair $(X_{(1)}, X_{(n)})$.

STEP-BY-STEP SOLUTION

- $f(x, \theta) = \frac{1}{\theta}$ on $[0, \theta]$, so $L(\theta) = \frac{1}{\theta^n}$ provided every $x_i \leq \theta$, i.e. $X_{(n)} \leq \theta$.
- Writing $L = \underbrace{\theta^{-n} I(X_{(n)} \leq \theta)}_{g_{\theta}(X_{(n)})} \cdot \underbrace{I(X_{(1)} \geq 0)}_{h(x)}$ satisfies the factorisation criterion.
- Hence $X_{(n)} = \max(X_1, \dots, X_n)$ is sufficient - option (d).

Q3.

Let $(X_1, \dots, X_n)$ be a random sample from $f(x, \theta) = \theta x^{\theta-1}$; $0 < x < 1$, $\theta > 0$. Which is correct?

- A. $t_1 = \sum_{i=1}^n X_i$ is sufficient for θ .
- B. $t_1 = \prod_{i=1}^n X_i$ is sufficient for θ .**
- C. No sufficient statistic exists for θ .
- D. None of the above

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$L = \theta^n \left(\prod x_i \right)^{\theta-1}$, so the data enter only through $\prod x_i$.

TIPS & TRICKS

- Equivalently $\sum \log x_i$ is sufficient - any one-to-one function of a sufficient statistic is sufficient.
- MLE here is $\hat{\theta} = -n / \sum \log x_i$, a function of that sufficient statistic as theory guarantees.
- This family ($\theta x^{\theta-1}$ on $(0,1)$) has been examined six times across 2018-2026 - own it completely.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \prod_{i=1}^n \theta x_i^{\theta-1} = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}$.
2. Take $g_\theta(t) = \theta^n t^{\theta-1}$ with $t = \prod x_i$, and $h(x) = 1$.
3. By the Fisher-Neyman factorisation theorem $T = \prod_{i=1}^n X_i$ is sufficient for θ .
4. Option (b).

Q4.

A statistic $t=t(x_1, \dots, x_n)$ is sufficient for θ . Consider:

1. It contains all the information about θ .
2. $L(x, \theta) = g(t, \theta)h(x_1, \dots, x_n)$
3. The conditional distribution of $x_1, \dots, x_n$ given t is independent of θ .

Codes:

- A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Correct answer: D

EXAM SHORTCUT AI-derived explanation

All three are equivalent descriptions of sufficiency: information-preservation, the factorisation form, and the θ -free conditional distribution.

TIPS & TRICKS

- Statement 3 is the definition; statement 2 is the Fisher-Neyman criterion; statement 1 is the informal reading.
- The factorisation theorem is the practical tool - use it to identify sufficient statistics without any conditional computation.
- Note h must be free of θ and g must depend on the data only through t .

STEP-BY-STEP SOLUTION

1. Statement 3 is the definition of sufficiency: $P(X=x / T=t)$ does not involve θ . True.
2. Statement 2 is the Fisher-Neyman factorisation theorem, which is equivalent to the definition. True.
3. Statement 1 is the standard informal restatement: a sufficient statistic captures all sample information about θ . True.
4. All three are correct - option (d).

Q5.

Let X_1, X_2 be i.i.d. Poisson. Then $(X_1 + 2X_2)$ is not sufficient because

- A. $P(X_1=1, X_2=1 / T=3)$ depends on λ
- B. $(X_1 + 2X_2)$ is Poisson
- C. $(X_1 + 2X_2)$ is not Poisson
- D. $P(X_1=1, X_2=1 / T=3)$ is Poisson with parameter one, $T=(X_1 + 2X_2)$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Sufficiency fails precisely because the conditional distribution of the sample given $T=X_1+2X_2$ still involves λ .

TIPS & TRICKS

- Unequal weights destroy sufficiency: $\sum X_i$ works for Poisson, $X_1 + 2X_2$ does not.
- Check by definition: $P(X_1=1, X_2=1 / T=3)$ competes with $P(X_1=3, X_2=0 / T=3)$ and the ratio involves λ .
- Whether T is itself Poisson is irrelevant to sufficiency - options (b) and (c) are category errors.

STEP-BY-STEP SOLUTION

1. $T=X_1+2X_2=3$ arises from $(X_1, X_2)=(1,1)$ or $(3,0)$.
2. $P(X_1=1, X_2=1)=\lambda e^{-\lambda} \cdot \lambda e^{-\lambda}=\lambda^2 e^{-2\lambda}$ and $P(X_1=3, X_2=0)=\frac{\lambda^3 e^{-\lambda}}{6} e^{-\lambda}=\frac{\lambda^3 e^{-2\lambda}}{6}$.
3. Hence $P(X_1=1, X_2=1 / T=3)=\frac{\lambda^2}{\lambda^2+\lambda^3/6}=\frac{1}{1+\lambda/6}$, which depends on λ .
4. By the definition of sufficiency, T is therefore not sufficient - option (a).

Q6.

(X_1, X_2) is a random sample of size 2 from $\text{binomial}(2, \theta)$. The sufficient statistic for θ is

- A. $X_1 - X_2$
- B. $X_1 + X_2$
- C. X_1 / X_2
- D. X_2 / X_1

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

For $B(2, \theta)$ the likelihood is $\theta^{\sum x_i} (1-\theta)^{4-\sum x_i}$ times a binomial constant, so $X_1 + X_2$ is sufficient.

TIPS & TRICKS

- Binomial and Bernoulli always reduce to the total $\sum X_i$ - a one-parameter exponential family.
- Differences and ratios throw away the total and cannot be sufficient.
- Here $X_1 + X_2 \sim B(4, \theta)$, which is also complete, so $\frac{X_1 + X_2}{4}$ is the UMVUE of θ .

STEP-BY-STEP SOLUTION

1. $P(X_i = x_i) = \binom{2}{x_i} \theta^{x_i} (1-\theta)^{2-x_i}$, so $L = \binom{2}{x_1} \binom{2}{x_2} \theta^{x_1+x_2} (1-\theta)^{4-(x_1+x_2)}$.
2. Factorising with $g_\theta(t) = \theta^t (1-\theta)^{4-t}$, $t = x_1 + x_2$, and $h(x) = \binom{2}{x_1} \binom{2}{x_2}$, the total is sufficient.
3. Option (b).

Q7.

Let $x_1, \dots, x_n$ be from $f(x; \alpha, \beta) = \frac{1}{\beta - \alpha}$, $\alpha \leq x \leq \beta$. The sufficient statistic for the parameters

- A. will be x_1 and x_n
- B. **will be $x_{(1)}$ and $x_{(n)}$**
- C. will be $\bar{x}$ and s^2
- D. will not exist

Correct answer: B**EXAM SHORTCUT**

AI-derived explanation

$L = (\beta - \alpha)^{-n} I(\alpha \leq X_{(1)}) I(X_{(n)} \leq \beta)$ - both extreme order statistics are needed.

TIPS & TRICKS

- Note the ORDER statistics $x_{(1)}, x_{(n)}$, not the first and last observations x_1, x_n - option (a) is a deliberate typographic trap.
- Two unknown endpoints $\Rightarrow$ two-dimensional sufficient statistic.
- $(X_{(1)}, X_{(n)})$ is also the pair of MLEs of (α, β) .

STEP-BY-STEP SOLUTION

1. $L(\alpha, \beta) = \prod \frac{1}{\beta - \alpha}$ provided every $x_i \in [\alpha, \beta]$, i.e. $\alpha \leq X_{(1)}$ and $X_{(n)} \leq \beta$.
2. So $L = \frac{1}{(\beta - \alpha)^n} I(\alpha \leq X_{(1)}) I(X_{(n)} \leq \beta)$, which depends on the data only through $(X_{(1)}, X_{(n)})$.
3. By the factorisation theorem this pair is jointly sufficient.
4. Option (b).

Q8.

If X_1, X_2 be iid $P(\lambda)$, which is correct?

- A. $X_1 + X_2$ is sufficient for λ
- B. $X_1 + 2X_2$ is sufficient for λ
- C. $X_1 + X_2$ is unbiased for λ
- D. $(X_1 + X_2)/2$ is not UMVUE for λ

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$X_1 + X_2$ is sufficient (the Poisson likelihood depends on the data only through the total). The other three statements are false.

TIPS & TRICKS

- $E(X_1 + X_2) = 2\lambda$, so the total is NOT unbiased for λ - option (c) fails.
- $(X_1 + X_2)/2 = \bar{X}$ IS the UMVUE, so option (d) is false as written.
- Unequal weights ($X_1 + 2X_2$) destroy sufficiency - see the companion 2018 item.

STEP-BY-STEP SOLUTION

1. $L = \frac{e^{-2\lambda} \lambda^{x_1+x_2}}{x_1! x_2!}$, which depends on the data only through $t = x_1 + x_2$; by factorisation $X_1 + X_2$ is sufficient.
(a) is correct.
2. (b) $X_1 + 2X_2$ is not sufficient - the conditional distribution given it still involves λ .
3. (c) $E(X_1 + X_2) = 2\lambda \neq \lambda$, so it is not unbiased for λ .
4. (d) $(X_1 + X_2)/2 = \bar{X}$ is an unbiased function of the complete sufficient statistic, hence it IS the UMVUE; the statement denies this and is false.
5. Option (a).

Q9.

Let $X_1, \dots, X_n$ be a random sample with pdf $f(x, \theta) = \theta x^{\theta-1}$; $0 < x < 1$, $\theta > 0$. The sufficient statistic for θ using the factorization theorem is

- A. $\sum_{i=1}^n X_i$
- B. $\prod_{i=1}^n X_i$
- C. $\sum_{i=1}^n (X_i - \bar{X})^2$
- D. $\sum_{i=1}^n X_i^2$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$L = \theta^n \left(\prod x_i \right)^{\theta-1}$, so $\prod x_i$ is sufficient by the factorisation theorem.

TIPS & TRICKS

- Any one-to-one function works too, so $\sum \log X_i$ is equally sufficient.
- MLE: $\hat{\theta} = -n / \sum \log x_i$ - always a function of the sufficient statistic.
- This family appears six times across 2018-2026; recognise $\theta x^{\theta-1}$ instantly.

STEP-BY-STEP SOLUTION

- $L(\theta) = \prod_{i=1}^n \theta x_i^{\theta-1} = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}$.
- Take $g_\theta(t) = \theta^n t^{\theta-1}$ with $t = \prod x_i$ and $h(x) = 1$: the factorisation criterion is met.
- Hence $\prod_{i=1}^n X_i$ is sufficient for θ - option (b).

Q10.

Let X, Y be independent $N(\theta, \sigma_1^2)$ and $N(\theta, \sigma_2^2)$, σ_1^2, σ_2^2 known. The sufficient statistic for θ is

A. $\bar{x} + \bar{y}$

B. $\frac{x}{\sigma_1} + \frac{y}{\sigma_2}$

C. $\frac{x^2}{\sigma_1^2} + \frac{y^2}{\sigma_2^2}$

D. $\frac{x}{\sigma_1^2} + \frac{y}{\sigma_2^2}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Collecting the θ terms in $\log L$ gives the coefficient $\frac{x}{\sigma_1^2} + \frac{y}{\sigma_2^2}$ - that is the sufficient statistic.

TIPS & TRICKS

- Weight each observation by the reciprocal of its VARIANCE, not its standard deviation - option (b) is the trap.
- The same weights give the BLUE of θ : $\hat{\theta} = \frac{x/\sigma_1^2 + y/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2}$.
- $\bar{x} + \bar{y}$ ignores the unequal precisions and is not sufficient.

STEP-BY-STEP SOLUTION

1. $\log L = -\frac{(x-\theta)^2}{2\sigma_1^2} - \frac{(y-\theta)^2}{2\sigma_2^2} + \text{const.}$

2. Expanding, the only data-dependent terms carrying θ are $\theta\left(\frac{x}{\sigma_1^2} + \frac{y}{\sigma_2^2}\right)$, plus $-\frac{\theta^2}{2}\left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right)$ which is free of the data.

3. By the factorisation theorem, $T = \frac{X}{\sigma_1^2} + \frac{Y}{\sigma_2^2}$ is sufficient for θ .

4. Option (d).

Q11.

Let $X_1, \dots, X_n$ be a random sample from $f(x, \theta) = e^{-(x-\theta)}$, $x > \theta$. By the factorisation theorem,

- A. $\sum_{i=1}^n X_i$ is the only sufficient statistic for θ
- B. $\{X_{(1)}, \sum_{i=1}^n X_i\}$ is jointly sufficient for θ**
- C. $X_{(n)}$ is sufficient for θ
- D. $\{X_{(n)}, \sum_{i=1}^n X_i\}$ is jointly sufficient for θ

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$X_{(1)}$ alone is sufficient; therefore the PAIR $\{X_{(1)}, \sum X_i\}$ is also (jointly) sufficient - and it is the only offered option containing $X_{(1)}$.

TIPS & TRICKS

- Any statistic from which a sufficient statistic can be recovered is itself sufficient - enlarging never hurts.
- $L = e^{n\theta - \sum x_i} I(X_{(1)} > \theta)$: the indicator needs $X_{(1)}$, the exponent needs $\sum x_i$ only as data.
- $X_{(n)}$ carries no information about the lower endpoint, so options (c) and (d) fail.

STEP-BY-STEP SOLUTION

- $L(\theta) = \prod e^{-(x_i - \theta)} I(x_i > \theta) = e^{n\theta - \sum x_i} I(X_{(1)} > \theta)$.
- Factorising with $g_\theta(t) = e^{n\theta} I(t > \theta)$, $t = X_{(1)}$, and $h(x) = e^{-\sum x_i}$ shows $X_{(1)}$ is sufficient.
- Option (a) is false because $\sum X_i$ alone cannot reproduce the indicator; options (c) and (d) use $X_{(n)}$, which does not constrain θ .
- Since $X_{(1)}$ is sufficient, the pair $\{X_{(1)}, \sum X_i\}$ is jointly sufficient - option (b).

Q12.

The factorization theorem for sufficiency is known as

- A. Rao-Blackwell theorem
- B. Fisher-Neyman theorem**
- C. Bernoulli theorem
- D. Cramer-Rao theorem

Correct answer: B

EXAM SHORTCUT[AI-derived explanation](#)

The factorisation criterion for sufficiency is the Fisher-Neyman theorem.

TIPS & TRICKS

- Fisher-Neyman: $f(x;\theta) = g_\theta(T(x))h(x)$ characterises sufficiency.
- Rao-Blackwell is about improving an unbiased estimator; Cramer-Rao gives a variance bound.
- 'Bernoulli's theorem' is the law of large numbers for binomial proportions - unrelated.

STEP-BY-STEP SOLUTION

1. The result stating that T is sufficient if and only if the joint density factorises as $g(t, \theta)h(x)$ is due to Fisher and, in its general measure-theoretic form, Neyman.
2. It is standardly called the Fisher-Neyman factorisation theorem.
3. Option (b).

Q13.

A statistic $T=t(x_1, \dots, x_n)$ is sufficient for θ according to the factorization theorem iff the joint pdf/pmf can be expressed as

- A. $g(\theta, t) \cdot h(x_1, \dots, x_n; \theta)$
- B. $g(t, \theta) \cdot h(x_1, \dots, x_n)$
- C. $g(t, x_i) \cdot h(x_1, \dots, x_n; \theta)$
- D. $g(x_1, \dots, x_n; \theta) \cdot h(x_1, \dots, x_n)$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$f = g(t, \theta) \cdot h(x_1, \dots, x_n)$, where h must be FREE of θ and g depends on the data only through t .

TIPS & TRICKS

- Two requirements: g touches the data only via t ; h does not involve θ at all.
- Options (a) and (c) let h depend on θ - fatal; option (d) has g depending on the raw data.
- Write out the factorisation explicitly for one family (say Poisson) and the form becomes automatic.

STEP-BY-STEP SOLUTION

1. The Fisher-Neyman criterion: $T=t(x_1, \dots, x_n)$ is sufficient for θ iff the joint density can be written as
2. $f(x_1, \dots, x_n; \theta) = g(t, \theta) \cdot h(x_1, \dots, x_n)$,
3. where g involves the data only through t , and h is free of θ .
4. Option (b).

Q14.

Let $X \sim f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}; x > 0$ and $Y \sim f(y, \theta) = \frac{1}{2\theta} e^{-y/2\theta}; y > 0$. Which is correct?

- A. $X+2Y$ is sufficient for θ and left-tail UMP test is $X+2Y < C$.
- B. $X+2Y$ is sufficient for θ and right-tail UMP test is $X+2Y > C$.
- C. $Y+2X$ is sufficient for θ and left-tail UMP test is $Y+2X < C$.
- D. $Y+2X$ is sufficient for θ and right-tail UMP test is $Y+2X < C$.

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

The joint density is $\frac{1}{2\theta^2} \exp\left[-\frac{2x+y}{2\theta}\right]$, so $Y+2X$ is sufficient; with a scale parameter, small values of $Y+2X$ favour small θ , giving a left-tail region.

TIPS & TRICKS

- Put both exponents over the common denominator 2θ : $\frac{x}{\theta} + \frac{y}{2\theta} = \frac{2x+y}{2\theta}$.
- The weight on X is 2 because X has HALF the mean of Y - weight inversely to the scale.
- Option (d) is internally inconsistent (it calls $Y+2X < C$ a right-tail test), so it can be discarded on wording alone.

STEP-BY-STEP SOLUTION

$$1. f(x, y; \theta) = \frac{1}{\theta} e^{-x/\theta} \cdot \frac{1}{2\theta} e^{-y/(2\theta)} = \frac{1}{2\theta^2} \exp\left[-\frac{2x+y}{2\theta}\right].$$

- By the factorisation theorem the data enter only through $T=2X+Y=Y+2X$, so T is sufficient for θ .
- The family has monotone likelihood ratio in T , so UMP one-sided tests are based on T ; a test against small θ rejects for small T , i.e. the left-tail region $Y+2X < C$.
- Option (c) states the correct sufficient statistic with a consistently labelled left-tail region - option (c).

Q15.

Let $X_1, \dots, X_n$ be from $f(x, \theta) = \frac{1}{\theta}; -\theta/2 \leq x \leq \theta/2$. For the sufficient statistic for θ :

1. Order statistic $(X_{(1)}, X_{(2)}, \dots, X_{(n)})$ is sufficient for θ .
2. Order statistic $(X_{(1)}, X_{(n)})$ is sufficient for θ .

Codes:

- A. 1 only
 B. 2 only
 C. Both 1 and 2
 D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

The likelihood is $\theta^{-n} I(\max |x_i| \leq \theta/2)$, a function of $(X_{(1)}, X_{(n)})$; and the full order statistic is always sufficient.

TIPS & TRICKS

- The whole order statistic is sufficient for ANY iid model - statement 1 is automatically true.
- Here the binding constraints are $X_{(1)} \geq -\theta/2$ and $X_{(n)} \leq \theta/2$, so the extremes suffice.
- Minimal sufficiency reduces further to $\max(|X_{(1)}|, X_{(n)})$, but that is not asked.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \theta^{-n}$ provided $-\theta/2 \leq x_i \leq \theta/2$ for all i , i.e. $X_{(1)} \geq -\theta/2$ and $X_{(n)} \leq \theta/2$.
2. So $L = \theta^{-n} I(X_{(1)} \geq -\frac{\theta}{2}) I(X_{(n)} \leq \frac{\theta}{2})$, depending on the data only through $(X_{(1)}, X_{(n)})$: statement 2 is true.
3. The complete order statistic is sufficient for any iid sample (it determines the likelihood entirely): statement 1 is true.
4. Option (c).

Q16.

The statistic $T = t(x)$ will be sufficient for parameter θ iff the joint density/mass function of the sample values $x_1, x_2, x_3, \dots, x_n$ can be expressed in the form $f(x; \theta) = g_\theta[t(x)] \cdot h(x)$ where $g_\theta[t(x)]$ depends on θ and $t(x_1, x_2, x_3, \dots, x_n)$, and $h(x)$ is independent of θ . This theorem is known as:

- A. Neyman-Pearson theorem
- B. Rao-Blackwell theorem
- C. **Factorization theorem**
- D. Cramer-Rao theorem

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

This is the factorization theorem (Fisher-Neyman).

TIPS & TRICKS

- Two conditions: g_θ touches the data only through $t(x)$, and h is free of θ .
- It converts a conditional-distribution definition into a purely algebraic check.
- Neyman-Pearson is about testing; Rao-Blackwell about improving estimators; Cramer-Rao about variance bounds.

STEP-BY-STEP SOLUTION

1. The stated criterion - $f(x; \theta) = g_\theta[t(x)]h(x)$ with h free of θ - is precisely the Fisher-Neyman factorization theorem for sufficiency.
2. Option (c).

Q17.

Let X_1 and X_2 be two independent variables. Define $T_1 = X_1 + 2X_2$ as statistic for θ in Bernoulli(θ) distribution, where $0 < \theta < 1$ and $T_2 = X_1 + 2X_2$ as statistic for λ in Poisson(λ) distribution, where $\lambda > 0$. Then which one of the following is correct?

- A. T_1 is sufficient for θ and T_2 is sufficient for λ .
- B. T_1 is sufficient for θ but T_2 is not sufficient for λ .**
- C. T_1 is not sufficient for θ but T_2 is sufficient for λ .
- D. T_1 is not sufficient for θ as well as T_2 is not sufficient for λ .

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

For Bernoulli, $X_1 + 2X_2$ takes a DIFFERENT value for each of the four possible samples, so it determines the sample and is sufficient. For Poisson it does not, and sufficiency fails.

TIPS & TRICKS

- Bernoulli: $(0,0) \rightarrow 0$, $(1,0) \rightarrow 1$, $(0,1) \rightarrow 2$, $(1,1) \rightarrow 3$ - a bijection, hence sufficient.
- Poisson has unbounded support, so e.g. $T_2=3$ arises from $(1,1)$ and $(3,0)$ with a λ -dependent split.
- Sufficiency is a property of the model, not of the formula - the same expression behaves differently in the two families.

STEP-BY-STEP SOLUTION

1. Bernoulli case: (X_1, X_2) can only be $(0,0), (1,0), (0,1), (1,1)$, mapping to $T_1=0,1,2,3$ respectively.
2. Since the map is one-to-one, knowing T_1 determines the whole sample, so T_1 is (trivially) sufficient for θ .
3. Poisson case: $T_2=3$ arises from $(1,1)$ and $(3,0)$, and $P(X_1=1, X_2=1 / T_2=3) = \frac{1}{1+\lambda/6}$ depends on λ .
4. So T_2 is not sufficient for λ - option (b).

Q18.

Let X_1 and X_2 be two iid random variables having the discrete uniform distribution on $\{1, 2, 3, \dots, N\}$, where N is unknown. Let $T = \max(X_1, X_2)$. Then which one of the following is correct?

- A. $X_1 + X_2$ is sufficient for N but T is not sufficient for N
- B. T is sufficient for N but $X_1 + X_2$ is not sufficient for N**
- C. Both T and $X_1 + X_2$ are sufficient for N
- D. Neither T nor $X_1 + X_2$ is sufficient for N

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$L = N^{-2} I(X_{(2)} \leq N)$ depends on the data only through the maximum, so T is sufficient; $X_1 + X_2$ does not determine the maximum and is not.

TIPS & TRICKS

- Discrete uniform on $\{1, \dots, N\}$: the likelihood is positive only while $N \geq \max x_i$ - hence the maximum is sufficient.
- $X_1 + X_2 = 4$ arises from $(1, 3), (2, 2), (3, 1)$ with maxima 3, 2, 3 - the conditional distribution given the sum still involves N .
- $T = \max$ is also complete here, so $\frac{T^{k+1} - (T-1)^{k+1}}{\dots}$ -type unbiased functions give UMVUEs.

STEP-BY-STEP SOLUTION

- $P(X_i = x) = \frac{1}{N}$ for $x = 1, \dots, N$, so $L(N) = \frac{1}{N^2} I(\max(x_1, x_2) \leq N)$.
- By factorisation, $T = \max(X_1, X_2)$ is sufficient for N .
- For the sum: $X_1 + X_2 = 4$ can arise from $(1, 3), (2, 2), (3, 1)$. When $N = 3$ all three are possible; when $N = 2$ only $(2, 2)$ is. So the conditional distribution given the sum depends on N .
- Hence $X_1 + X_2$ is not sufficient - option (b).

Q19.

For independent Poisson random variables with parameter λ , which one of the following is a sufficient statistic?

A. $(X_1, \prod_{i=2}^n X_i)$

B. $\frac{X_1}{X_2}$

C. $\frac{X_1}{\sum_{i=1}^n X_i}$

D. $(X_1, \sum_{i=2}^n X_i)$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$(X_1, \sum_{i=2}^n X_i)$ determines $\sum_{i=1}^n X_i$ by addition, so it is sufficient.

TIPS & TRICKS

- Any statistic from which the sufficient total can be recovered is itself sufficient.
- Ratios (options (b), (c)) destroy the total and cannot be sufficient.
- A PRODUCT of Poisson counts (option (a)) also fails - the Poisson likelihood involves the sum, not the product.

STEP-BY-STEP SOLUTION

1. The Poisson likelihood is $L = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!}$, so $\sum_{i=1}^n X_i$ is sufficient.
2. From $(X_1, \sum_{i=2}^n X_i)$ we recover $\sum_{i=1}^n X_i = X_1 + \sum_{i=2}^n X_i$.
3. Hence this pair is (jointly) sufficient for λ .
4. Option (d).

Q20.

For independent Bernoulli random variables, which of the following is/are sufficient statistic for p , such that $P(X_i=1) = p = 1 - P(X_i=0)$; $i=1,2,3$?

1. $X_1+X_2+X_3$
2. $X_1+2X_2+X_3$

Select the correct answer using the code given below:

- A. 1 only**
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$X_1+X_2+X_3$ is sufficient; $X_1+2X_2+X_3$ is not, because the value 2 arises from (0,1,0) and (1,0,1), which have different totals.

TIPS & TRICKS

- For Bernoulli the likelihood depends only on $\sum x_i$; any statistic that fails to determine it is not sufficient.
- Counterexample check: $T=2$ from (0,1,0) (total 1) and (1,0,1) (total 2) - so the conditional distribution retains p .
- With only TWO Bernoulli variables the weighted sum X_1+2X_2 WOULD be sufficient (it is a bijection); three variables break it.

STEP-BY-STEP SOLUTION

1. $L = p^{\sum x_i} (1-p)^{3-\sum x_i}$, so $X_1+X_2+X_3$ is sufficient: statement 1 true.
2. For $T = X_1+2X_2+X_3$, the value $T=2$ occurs for (0,1,0) with $\sum x_i=1$ and for (1,0,1) with $\sum x_i=2$.
3. $P[(0,1,0) / T=2] = \frac{p(1-p)^2}{p(1-p)^2 + p^2(1-p)} = 1-p$, which depends on p .
4. So statement 2 is false; hence 1 only - option (a).

Q21.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample taken from $\text{Gamma}(\alpha, \beta)$ having pdf

$$f(x; \alpha, \beta) = \frac{e^{-x/\beta} x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)}; \quad 0 \leq x < \infty; \alpha, \beta > 0$$

The jointly sufficient statistic for α and β , are respectively:

- A. $(\sum x_i, \prod x_i)$
- B. $(\sum x_i^2, \prod x_i^2)$
- C. $(\sum x_i^2, \prod x_i)$
- D. $(\sum x_i, \prod x_i^2)$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$L \propto \beta^{-n\alpha} \Gamma(\alpha)^{-n} \left(\prod x_i \right)^{\alpha-1} e^{-\sum x_i/\beta}$, so $(\sum x_i, \prod x_i)$ is jointly sufficient.

TIPS & TRICKS

- Two unknown parameters $\Rightarrow$ two-dimensional sufficient statistic - one for each exponential-family term.
- α couples with $\log x$ and β with x , giving $\prod x_i$ and $\sum x_i$ respectively.
- Equivalently $(\sum x_i, \sum \log x_i)$ - a one-to-one relabelling.

STEP-BY-STEP SOLUTION

$$1. L(\alpha, \beta) = \frac{1}{\beta^{n\alpha} \Gamma(\alpha)^n} \left(\prod x_i \right)^{\alpha-1} \exp\left(-\frac{\sum x_i}{\beta}\right).$$

2. The data appear only through $\sum x_i$ (in the exponential) and $\prod x_i$ (raised to $\alpha-1$).

3. By the factorisation theorem $(\sum x_i, \prod x_i)$ is jointly sufficient for (α, β) .

4. Option (a).

Q22.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample of size n from Bernoulli distribution with pmf

$$f(x, \theta) = \begin{cases} \theta^x(1-\theta)^{1-x}, & x = 0, 1 \\ 0, & \text{otherwise} \end{cases}$$

What is the sufficient statistic for θ ?

- A. $X_1 - X_2$
- B. $X_1 X_2$
- C. $X_1 + X_2$
- D. $\frac{X_1}{X_2}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$L = \theta^{x_1+x_2}(1-\theta)^{2-(x_1+x_2)}$, so X_1+X_2 is sufficient.

TIPS & TRICKS

- Bernoulli likelihood depends on the data only through the total number of successes.
- Differences, products and ratios all destroy the total.
- $X_1+X_2 \sim B(2, \theta)$ is also complete, so $\frac{X_1+X_2}{2}$ is the UMVUE of θ .

STEP-BY-STEP SOLUTION

1. $L(\theta) = \theta^{x_1}(1-\theta)^{1-x_1}\theta^{x_2}(1-\theta)^{1-x_2} = \theta^{x_1+x_2}(1-\theta)^{2-(x_1+x_2)}$.
2. By factorisation with $t = x_1+x_2$ and $h(x) = 1$, X_1+X_2 is sufficient for θ .
3. Option (c).

Q23.

Let $X_1, X_2, X_3, \dots, X_n$ be iid random variables with common pmf $P[X_i = k] = \frac{1}{N}$, $k = 1, 2, 3, \dots, N$; $i = 1, 2, 3, \dots, n$. What is the sufficient statistic for the family of joint pmfs P_N ?

A. $\min(X_1, X_2, X_3, \dots, X_n)$

B. $\max(X_1, X_2, X_3, \dots, X_n)$

C. $[\min(X_1, X_2, X_3, \dots, X_n), \max(X_1, X_2, X_3, \dots, X_n)]$

D. $X_1 + X_2 + X_3 + \dots + X_n$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$L = N^{-n} I(X_{(n)} \leq N)$, so the maximum is sufficient for the discrete uniform family.

TIPS & TRICKS

- The only constraint N places on the data is $N \geq \max x_i$ - hence the maximum carries all the information.
- $X_{(n)}$ is complete as well as sufficient here, so UMVUEs are built from it.
- The minimum is irrelevant because the lower endpoint 1 is known.

STEP-BY-STEP SOLUTION

1. $P_N(x_1, \dots, x_n) = \frac{1}{N^n}$ provided every $x_i \in \{1, \dots, N\}$, i.e. $\max_i x_i \leq N$.
2. So $L = N^{-n} I(X_{(n)} \leq N)$, which depends on the data only through $X_{(n)}$.
3. Hence $\max(X_1, \dots, X_n)$ is sufficient - option (b).

Q24.

Let $x_1, x_2, \dots, x_n$ be a random sample from a population with probability density function

$$f(x, \theta) = \begin{cases} e^{-(x-\theta)}, & x > \theta \\ 0, & \text{otherwise} \end{cases}$$

For parameter θ , consider the following statements:

1. $\min(x_1, x_2, \dots, x_n)$ is a sufficient statistic of θ .
2. $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ is a minimum variance bound estimator of θ .

Which of the statements given above is/are correct?

- A. 1 only**
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$X_{(1)}$ is sufficient (statement 1 true), but the support depends on θ so the Cramer-Rao machinery fails and $\bar{x}$ is not an MVB estimator (statement 2 false).

TIPS & TRICKS

- Support-dependent parameter $\Rightarrow$ regularity fails $\Rightarrow$ no CR bound, hence no MVB estimator.
- $\bar{x}$ is not even a function of the sufficient statistic here, so it cannot be optimal.
- The UMVUE of θ is $X_{(1)} - \frac{1}{n}$, based on the sufficient statistic.

STEP-BY-STEP SOLUTION

1. $L(\theta) = e^{n\theta - \sum x_i} I(X_{(1)} > \theta)$, so by factorisation $X_{(1)} = \min x_i$ is sufficient: statement 1 true.
2. The support $x > \theta$ depends on θ , so the regularity conditions behind the Cramer-Rao inequality fail and no minimum variance bound exists.
3. Moreover $\bar{x}$ is not a function of $X_{(1)}$, so it cannot be an optimal estimator: statement 2 false.
4. Hence 1 only - option (a).

Q25.

Let $x_1, x_2, \dots, x_n$ be a random sample from Normal distribution $N(\mu, \sigma^2)$. Suppose both μ and σ are unknown. Which of the following are jointly sufficient statistic for (μ, σ) ?

1. $\left\{ \sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2 \right\}$
2. $\left\{ \sum_{i=1}^n x_i, \sum_{i=1}^n (x_i - \bar{x})^2 \right\}$
3. $\left\{ \sum_{i=1}^n x_i^2, \sum_{i=1}^n (x_i - \bar{x})^2 \right\}$

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Sets 1 and 2 are one-to-one transforms of each other and both determine the likelihood; set 3 loses the SIGN of $\sum x_i$ and is not sufficient.

TIPS & TRICKS

- $\sum (x_i - \bar{x})^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$, so sets 1 and 2 carry identical information.
- From set 3 you can recover $(\sum x_i)^2$ but not $\sum x_i$ itself - the sign of μ is lost.
- Sufficiency is preserved under one-to-one transformations only.

STEP-BY-STEP SOLUTION

1. The likelihood is $L \propto \sigma^{-n} \exp \left[-\frac{\sum x_i^2 - 2\mu \sum x_i + n\mu^2}{2\sigma^2} \right]$, so $(\sum x_i, \sum x_i^2)$ is jointly sufficient: set 1 works.
2. Set 2 is a one-to-one function of set 1 via $\sum (x_i - \bar{x})^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$, so it is equally sufficient.
3. Set 3 gives $(\sum x_i)^2 = n \left[\sum x_i^2 - \sum (x_i - \bar{x})^2 \right]$ but not the sign of $\sum x_i$, so it cannot reconstruct the likelihood: not sufficient.
4. Hence 1 and 2 only - option (a).

Q26.

Let $x_1, x_2, \dots, x_n$ be a random sample from Normal distribution $N(\theta, \theta^2)$. Then the sufficient statistic for θ is:

- A. $\sum_{i=1}^n x_i$ only
- B. $\sum_{i=1}^n x_i^2$ only
- C. $\sum_{i=1}^n x_i + \sum_{i=1}^n x_i^2$
- D. $\left(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2\right)$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Expanding $\sum (x_i - \theta)^2 / (2\theta^2)$ leaves both $\sum x_i$ and $\sum x_i^2$ attached to θ , so the pair is sufficient.

TIPS & TRICKS

- Unlike $N(\theta, \theta)$ (where the cross term loses its θ), here the 2θ in the numerator does not cancel the θ^2 in the denominator.
- $\log L$ contains $\frac{\sum x_i}{\theta}$ and $\frac{\sum x_i^2}{2\theta^2}$ - two distinct θ -dependent data functions.
- A single parameter can still require a two-dimensional sufficient statistic - this is a curved exponential family.

STEP-BY-STEP SOLUTION

$$1. \log L = -n \log \theta - \frac{\sum (x_i - \theta)^2}{2\theta^2} + \text{const} = -n \log \theta - \frac{\sum x_i^2}{2\theta^2} + \frac{\sum x_i}{\theta} - \frac{n}{2}.$$

2. Both $\sum x_i$ and $\sum x_i^2$ appear with θ -dependent coefficients ($1/\theta$ and $1/(2\theta^2)$).

3. So the likelihood factorises only through the PAIR, and $\left(\sum x_i, \sum x_i^2\right)$ is jointly sufficient.

4. Option (d).

SHARED CONTEXT

Let $x_1, x_2, \dots, x_n$ be a random sample from Poisson distribution with mean $\lambda > 0$.

Q27.

If $T_1 = \sum_{i=1}^n x_i$ and $T_2 = (x_1, \sum_{i=2}^n x_i)$ are the statistics, then which of these statistics is/are sufficient for λ ?

- A. T_1 only
- B. T_2 only
- C. Both T_1 and T_2
- D. Neither T_1 nor T_2

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

The Poisson likelihood depends on the data only through $\sum x_i$; T_2 determines that total by addition, so both statistics are sufficient.

TIPS & TRICKS

- Any statistic from which $\sum x_i$ can be recovered is sufficient - partitioning the sample is harmless.
- T_2 is sufficient but not MINIMAL sufficient; T_1 is minimal.
- Sufficiency fails only when the components carry UNEQUAL weights (e.g. $x_1 + 2x_2$).

STEP-BY-STEP SOLUTION

1. $L(\lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!}$, so $T_1 = \sum x_i$ is sufficient.

2. $T_2 = (x_1, \sum_{i \geq 2} x_i)$ determines $\sum_{i=1}^n x_i = x_1 + \sum_{i \geq 2} x_i$, so the likelihood is a function of T_2 as well.

3. By factorisation both are sufficient - option (c).

SHARED CONTEXT

Let $x_1, x_2, \dots, x_n$ be a random sample from Poisson distribution with mean $\lambda > 0$.

Q28.

If $T_3 = (x_1, x_2 + x_3, \sum_{i=4}^n x_i)$ and $T_4 = (x_1 + x_2, x_3 + x_4, \sum_{i=5}^n x_i)$ are the statistics, then which of these statistics is/are sufficient for λ ?

- A. T_3 only
- B. T_4 only
- C. Both T_3 and T_4
- D. Neither T_3 nor T_4

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Both T_3 and T_4 are partitions of the sample into groups whose totals sum to $\sum x_i$, so both determine the likelihood and are sufficient.

TIPS & TRICKS

- Group totals with unit weights always preserve sufficiency for additive families like Poisson and Bernoulli.
- Contrast $x_1 + 2x_2$, where the weight 2 breaks the correspondence.
- Neither is minimal sufficient, but the question asks only about sufficiency.

STEP-BY-STEP SOLUTION

1. $T_3 = (x_1, x_2 + x_3, \sum_{i=4}^n x_i)$: adding the three components gives $\sum_{i=1}^n x_i$.
2. $T_4 = (x_1 + x_2, x_3 + x_4, \sum_{i=5}^n x_i)$: adding the three components again gives $\sum_{i=1}^n x_i$.
3. Since the Poisson likelihood is a function of $\sum x_i$ alone, both statistics satisfy the factorisation criterion.
4. Option (c).

Q29.

Let $X_i \sim N(\mu, \theta)$; $0 < \theta < \infty$; $i = 1, 2, 3, \dots, n$ such that μ is known but θ is unknown. What is the sufficient statistic for θ ?

A. $\frac{1}{n-1} \left[\sum_{i=1}^n (X_i - \mu) \right]^2$

B. $\frac{1}{n} \left[\sum_{i=1}^n (X_i - \mu) \right]^2$

C. $\frac{1}{n} \sum_{i=1}^{n-1} (X_i - \mu)^2$

D. $\sum_{i=1}^n (X_i - \mu)^2$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

With μ known, the likelihood depends on the data through $\sum (X_i - \mu)^2$.

TIPS & TRICKS

- Note options (a) and (b) square the SUM of deviations, not the sum of SQUARED deviations - a decisive difference.
- Option (c) drops the last observation, discarding information.
- Constant multiples of a sufficient statistic are equally sufficient, but the structure must be right.

STEP-BY-STEP SOLUTION

1. $L(\theta) = (2\pi\theta)^{-n/2} \exp \left[- \frac{\sum_{i=1}^n (X_i - \mu)^2}{2\theta} \right]$.

2. By the factorisation theorem the sufficient statistic is $T = \sum_{i=1}^n (X_i - \mu)^2$.

3. Options (a) and (b) use $\left[\sum (X_i - \mu) \right]^2$, a different quantity; option (c) omits the n -th term.

4. Option (d).

Q30.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from the gamma distribution having pdf $f(x; \theta, \mu) = \frac{1}{\Gamma(\theta)\mu^\theta} e^{-x/\mu} x^{\theta-1}$; $x > 0, \theta > 0$ and $\mu > 0$ is a known constant. What is the sufficient statistic for θ ?

- A. $\sum_{i=1}^n x_i$
- B. $x_{(1)} = \min\{x_1, x_2, x_3, \dots, x_n\}$
- C. $x_{(n)} = \max\{x_1, x_2, x_3, \dots, x_n\}$
- D. $\prod_{i=1}^n x_i$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

With μ known, the only θ -dependent data term is $(\prod x_i)^{\theta-1}$, so $\prod x_i$ is sufficient.

TIPS & TRICKS

- θ is the SHAPE parameter here; shape couples with $\log x$, scale with x .
- $\sum x_i$ would be sufficient for μ if the roles were reversed - read which parameter is known.
- Equivalently $\sum \log x_i$ is sufficient (a one-to-one transformation).

STEP-BY-STEP SOLUTION

$$1. L = \frac{1}{\Gamma(\theta)^n \mu^{n\theta}} \left(\prod x_i \right)^{\theta-1} \exp\left(-\frac{\sum x_i}{\mu}\right).$$

2. With μ known, $\exp(-\sum x_i/\mu)$ and $(\prod x_i)^{-1}$ are free of θ and go into $h(x)$.

3. The θ -dependent factor is $\frac{1}{\Gamma(\theta)^n \mu^{n\theta}} (\prod x_i)^\theta$, a function of $\prod x_i$ and θ alone.

4. Hence $\prod_{i=1}^n X_i$ is sufficient - option (d).

Q31.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from the double exponential distribution having pdf $f(x, \theta) = \frac{1}{2} e^{-|x-\theta|}$; $\theta \in \mathbb{R}, x \in \mathbb{R}$. Which one of the following statements is correct?

A. No alone sufficient statistic T exists for θ , other than the sample itself.

B. Sample median is unbiased estimator of θ .

C. $X_{(n)}$ is sufficient as well as MLE for θ .

D. Sample mean is MLE and consistent for θ .

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The double exponential admits no non-trivial sufficient statistic - the full order statistic is minimal sufficient.

TIPS & TRICKS

- $\sum |x_i - \theta|$ cannot be reduced to any fixed finite-dimensional statistic free of θ .
- The MLE here is the sample MEDIAN (it minimises $\sum |x_i - \theta|$), so options (c) and (d) are wrong.
- Option (b) is in fact also true (the median's distribution is symmetric about θ); option (a) is the sufficiency-theoretic statement this item is testing.

STEP-BY-STEP SOLUTION

1. $L(\theta) = 2^{-n} \exp \left[-\sum_i |x_i - \theta| \right]$. The exponent depends on θ through all the order statistics simultaneously and cannot be factorised through any statistic of dimension less than n .
2. Hence the minimal sufficient statistic is the complete order statistic - no scalar sufficient statistic exists. Statement (a) is correct.
3. (c) is wrong: the MLE minimises $\sum |x_i - \theta|$ and is the sample median, not $X_{(n)}$; and $X_{(n)}$ is not sufficient.
4. (d) is wrong: the sample mean is not the MLE for this family.
5. Option (a).

Q32.

Let $x_1, x_2, x_3, \dots, x_n$ denote a random sample from a distribution with pdf $f(x; \theta) = \theta x^{\theta-1}$, $0 < x < 1$, elsewhere 0, where $\theta > 0$. The sufficient statistic for θ is

A. $\sum_{i=1}^n x_i$

B. $\prod_{i=1}^n x_i$

C. $[\min x_i, \max x_i]$

D. $\max x_i - 1$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$L = \theta^n \left(\prod x_i \right)^{\theta-1}$, so $\prod x_i$ is sufficient.

TIPS & TRICKS

- Pattern-match the family $\theta x^{\theta-1}$ - the product (or $\sum \log x_i$) is always the sufficient statistic.
- This family has now appeared six times in the 2018-2026 papers.
- The MLE $-n/\sum \log x_i$ is a function of it, as theory requires.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \prod_{i=1}^n \theta x_i^{\theta-1} = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}$.
2. Factorising with $g_\theta(t) = \theta^n t^{\theta-1}$, $t = \prod x_i$, and $h(x) = 1$ gives sufficiency.
3. Option (b).

Q33.

Consider the following statements:

I. If $X_i \sim U(0, \theta)$, $\theta \in (0, \infty)$, then $\max(X_1, X_2, X_3, \dots, X_n)$ is sufficient for θ .

II. If $X_i \sim f_{\theta}(x) = \begin{cases} 1, & \theta - 0.5 < x < \theta + 0.5 \\ 0, & \text{otherwise} \end{cases}$, then $(\min X_i, \max X_i)$ is jointly sufficient for parameter θ .

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$U(0, \theta)$ needs only the maximum; $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ needs both extremes. Both statements are correct.

TIPS & TRICKS

- Count the moving endpoints: one moving endpoint $\Rightarrow$ one order statistic; two $\Rightarrow$ the pair.
- In case II, $(X_{(1)}, X_{(n)})$ is minimal sufficient but NOT complete - sufficiency and completeness are different questions.
- The likelihood indicator is what forces the order statistics into the sufficient statistic.

STEP-BY-STEP SOLUTION

1. Statement I: $L = \theta^{-n} I(X_{(n)} \leq \theta)$, so $\max X_i$ is sufficient. True.

2. Statement II: $L = 1$ iff $\theta - \frac{1}{2} < X_{(1)}$ and $X_{(n)} < \theta + \frac{1}{2}$, i.e. $L = I\left(X_{(n)} - \frac{1}{2} < \theta < X_{(1)} + \frac{1}{2}\right)$, a function of the pair. True.

3. Option (c).

Q34.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a distribution with pdf $f(x, \theta) = \theta x^{\theta-1}$; $0 < x < 1, \theta > 0$. Then the sufficient statistic for θ is:

- A. $\sum_{i=1}^n X_i$
- B. $\sum_{i=1}^n X_i + 1$
- C. $\sum_{i=1}^n X_i^2$
- D. $\prod_{i=1}^n X_i$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$L = \theta^n \left(\prod x_i \right)^{\theta-1}$, so $\prod X_i$ is sufficient.

TIPS & TRICKS

- The recurring family $\theta x^{\theta-1}$ - recognise it instantly and write $\prod x_i$.
- Equivalently $\sum \log x_i$, a one-to-one transform.
- $\sum X_i$ and $\sum X_i^2$ do not appear anywhere in the likelihood.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \prod_{i=1}^n \theta x_i^{\theta-1} = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}$.
2. Factorising with $t = \prod x_i$ and $h(x) = 1$ shows $\prod X_i$ is sufficient.
3. Option (d).

Q35.

Let X_1, X_2 be a random sample of size 2 from Bernoulli distribution with pmf as

$$f(x, \theta) = \begin{cases} \theta^x(1-\theta)^{1-x}, & x=0,1 \\ 0, & \text{otherwise} \end{cases}. \text{ Then } X_1 + X_2 \text{ is sufficient for } \theta \text{ if } P[X_1=x_1, X_2=x_2 / (X_1+X_2)=t] \text{ is equal}$$

to:

A. $\frac{1}{\binom{t}{2}}$

B. ✓ $\frac{1}{\binom{2}{t}}$

C. $\frac{1}{\binom{t+1}{2}}$

D. $\frac{1}{\binom{2}{t+1}}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Given $X_1 + X_2 = t$, all $\binom{2}{t}$ arrangements are equally likely, so the conditional probability is $\frac{1}{\binom{2}{t}}$ - free of θ .

TIPS & TRICKS

- $P(X_1=x_1, X_2=x_2) = \theta^t(1-\theta)^{2-t}$ for every arrangement with total t - they all share the same probability.
- $P(X_1+X_2=t) = \binom{2}{t} \theta^t(1-\theta)^{2-t}$, so the ratio is $1/\binom{2}{t}$.
- Note $\binom{2}{t}$ (choose t from 2), not $\binom{t}{2}$ - option (a) reverses the arguments.

STEP-BY-STEP SOLUTION

1. For any (x_1, x_2) with $x_1 + x_2 = t$: $P(X_1=x_1, X_2=x_2) = \theta^t(1-\theta)^{2-t}$.

2. $P(X_1+X_2=t) = \binom{2}{t} \theta^t(1-\theta)^{2-t}$.

3. Dividing: $P[X_1=x_1, X_2=x_2 / X_1+X_2=t] = \frac{1}{\binom{2}{t}}$, which is free of θ - confirming sufficiency.

4. Option (b).

Q36.

Consider the following statements:

I. If a statistic is sufficient for a class of distributions, then it is sufficient for any subclass of those distributions.

II. A sufficient statistic is always a complete statistic.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Sufficiency for a class carries to any subclass (I true), but sufficiency never implies completeness (II false).

TIPS & TRICKS

- If the factorisation holds for every member of a class, it holds for every member of a subclass - trivially.
- The order statistic is always sufficient but almost never complete - the standard counterexample to II.
- Completeness can FAIL when the family is shrunk, even though sufficiency survives.

STEP-BY-STEP SOLUTION

1. Statement I: if $f_{\theta}(x) = g_{\theta}(T(x))h(x)$ for every θ in a class, the same factorisation holds for every θ in any subclass. True.
2. Statement II: the complete order statistic is sufficient for any iid family but is generally not complete (e.g. $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$). False.
3. Hence I only - option (a).

Q37.

Let $x_1, x_2, \dots, x_n$ be a random sample from the p.d.f. $f(x) = \frac{\theta}{x^2}$; $0 < \theta \leq x < \infty$. Which one of the following is a sufficient statistic for θ ?

- A. Mean of $(x_1, x_2, \dots, x_n)$
- B. Median of $(x_1, x_2, \dots, x_n)$
- C. Minimum of $(x_1, x_2, \dots, x_n)$
- D. Minimum of $(x_1^2, x_2^2, \dots, x_n^2)$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$L = \theta^n \left(\prod x_i^{-2} \right) I(X_{(1)} \geq \theta)$, so the MINIMUM is sufficient.

TIPS & TRICKS

- The support constraint $x \geq \theta$ puts $X_{(1)}$ into the indicator - that is where sufficiency lives.
- $\prod x_i^{-2}$ is free of θ and goes into $h(x)$.
- Option (d), $\min(x_i^2)$, is a one-to-one function of $\min(x_i)$ for positive data, but the standard answer is the minimum itself.

STEP-BY-STEP SOLUTION

1. $f(x) = \frac{\theta}{x^2}$ for $x \geq \theta$, so $L(\theta) = \theta^n \left(\prod_{i=1}^n \frac{1}{x_i^2} \right) I(\min_i x_i \geq \theta)$.

2. Factorising with $g_\theta(t) = \theta^n I(t \geq \theta)$, $t = X_{(1)}$, and $h(x) = \prod x_i^{-2}$ gives sufficiency.

3. Hence $\min(x_1, \dots, x_n)$ is sufficient - option (c).

Q38.

Let $x_1, x_2, \dots, x_n$ be a random sample from a distribution with p.d.f. $f(x, \theta) = \frac{1}{\theta} e^{-(x/\theta)}$; $0 \leq x < \infty$; 0 otherwise. The sufficient estimator of θ is

- A. $\sum_{i=1}^n x_i^2$
- B. $\sum_{i=1}^n x_i + 1$
- C. $\sum_{i=1}^n x_i^2 + 1$
- D. $\sum_{i=1}^n x_i$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$L = \theta^{-n} \exp\left(-\frac{\sum x_i}{\theta}\right)$, so $\sum x_i$ is sufficient.

TIPS & TRICKS

- The exponential family's natural statistic is the SUM, not the sum of squares.
- Adding a constant (options (b), (c)) still gives a sufficient statistic mathematically, but the canonical answer is $\sum x_i$.
- $\bar{X}$ is equally sufficient, being a one-to-one function.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \prod \frac{1}{\theta} e^{-x_i/\theta} = \frac{1}{\theta^n} \exp\left(-\frac{\sum x_i}{\theta}\right)$.

2. Factorising with $g_\theta(t) = \theta^{-n} e^{-t/\theta}$, $t = \sum x_i$, and $h(x) = 1$ gives sufficiency.

3. Option (d).

Q39.

Let $x_1, x_2, \dots, x_n$ be a random sample from a distribution with p.d.f. or p.m.f. $f(x, \theta)$, $\theta \in \Theta$. A statistic $T = (x_1, x_2, \dots, x_n)$ is said to be sufficient for θ , iff the conditional distribution of X , given $T=t$, is

- A. dependent on θ
- B. Independent of θ**
- C. dependent on θ^2
- D. None of the above

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

T is sufficient iff the conditional distribution of the sample given $T=t$ is INDEPENDENT of θ .

TIPS & TRICKS

- This is the definition; the factorisation theorem is the equivalent working criterion.
- Intuition: once T is known, the remaining randomness carries no further information about θ .
- Any option asserting dependence on θ (or on θ^2) reverses the definition.

STEP-BY-STEP SOLUTION

1. By definition, T is sufficient for θ if $P(X=x / T=t)$ does not involve θ for any t with positive probability.
2. Equivalently (Fisher-Neyman), $f(x; \theta) = g_\theta(t)h(x)$.
3. Option (b).

Q40.

Let $x_1, x_2, \dots, x_n$ be a random sample from a normal (μ, θ) and μ is known. Which of the following statements is/are correct?

I. $T = \sum_{i=1}^n (x_i - \mu)^2$ is sufficient statistic for θ .

II. $S^2_0 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$ is sufficient statistic for θ .

Select the answer using the code given below.

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$T = \sum (x_i - \mu)^2$ is sufficient, and $S^2_0 = T/n$ is a one-to-one function of it - so both are sufficient.

TIPS & TRICKS

- Any one-to-one function of a sufficient statistic is sufficient; dividing by a known constant qualifies.
- $T/\theta \sim \chi^2_n$, so S^2_0 is also unbiased and (by completeness) the UMVUE.
- With μ known there is no need to centre at $\bar{x}$.

STEP-BY-STEP SOLUTION

1. $L(\theta) = (2\pi\theta)^{-n/2} \exp \left[-\frac{\sum (x_i - \mu)^2}{2\theta} \right]$, so $T = \sum (x_i - \mu)^2$ is sufficient by factorisation. Statement I true.

2. $S^2_0 = T/n$ is obtained from T by multiplication by the known constant $1/n$ - a one-to-one map - so S^2_0 is sufficient too. Statement II true.

3. Option (c).

Q41.

Let $x_1, x_2, \dots, x_n$ be a random sample from a Bernoulli population with parameter p , where $0 < p < 1$. Which of the following is/are sufficient statistic(s) for p ?

I. $\frac{1}{n} \sum_{i=1}^n x_i$

II. $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

III. $\sum_{i=1}^n x_i$

Select the correct answer using the code given below.

- A. I only
- B. III only
- C. I and II
- D. I and III

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$\bar{x}$ and $\sum x_i$ are one-to-one transforms of each other and both sufficient; the sample variance $\bar{x}(1-\bar{x})$ is a MANY-to-one function of $\bar{x}$ and is not sufficient.

TIPS & TRICKS

- For Bernoulli data, $\frac{1}{n} \sum (x_i - \bar{x})^2 = \bar{x}(1-\bar{x})$ - a parabola, symmetric about $\bar{x} = \frac{1}{2}$.
- That symmetry means $\bar{x}$ and $1-\bar{x}$ give the same value, so information about p is lost.
- Multiplying a sufficient statistic by a known constant keeps sufficiency; squaring or folding does not.

STEP-BY-STEP SOLUTION

- $L = p \sum x_i (1-p)^{n-\sum x_i}$, so $\sum X_i$ is sufficient: statement III holds. $\bar{X} = \sum X_i / n$ is a one-to-one transform, so statement I holds too.
- For Bernoulli data $x_i^2 = x_i$, so $\frac{1}{n} \sum (x_i - \bar{x})^2 = \bar{x} - \bar{x}^2 = \bar{x}(1-\bar{x})$.
- This is not one-to-one in $\bar{x}$ (the values $\bar{x}$ and $1-\bar{x}$ give the same result), so it does not determine $\sum x_i$ and is not sufficient: statement II fails.
- Hence I and III - option (d).

Q42.

Let $x_1, x_2, \dots, x_n$ be a random sample from a distribution with p.d.f. $f(x) = \theta x^{\theta-1}$; $0 < x < 1$, $0 < \theta < \infty$; 0 otherwise. Then a sufficient statistic for θ is

- A. $\sum_{i=1}^n x_i$
- B. $\sum_{i=1}^n x_i^{-n}$
- C. $\prod_{i=1}^n x_i$
- D. $\prod_{i=1}^n \frac{1}{x_i} - n$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$L = \theta^n \left(\prod x_i \right)^{\theta-1}$, so $\prod X_i$ is sufficient.

TIPS & TRICKS

- This family has now appeared in six different papers - recognise it instantly.
- Equivalently $\sum \log x_i$ works, being a one-to-one transform.
- Options involving $\sum x_i$ never appear in this likelihood.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}$.
2. Factorising with $t = \prod x_i$ and $h(x) = 1$ gives sufficiency by Fisher-Neyman.
3. Option (c).

Q43.

Let $x_1, x_2, \dots, x_n$ be independent random variables each having the distribution $P(x_i = k) = \frac{1}{N}$; $k = 1, 2, \dots, N$; $N \in \mathbb{Z}$, the set of positive integers. If $x_{(n)} = \max(x_i)$ and $x_{(1)} = \min(x_i)$, then the sufficient statistic for the parameter N is

- A. $x_{(1)}$
- B. $x_{(n)}$
- C. $x_{(1)}/2$
- D. None of the above

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$L = N^{-n} I(X_{(n)} \leq N)$ - the maximum is sufficient for N .

TIPS & TRICKS

- The only information N carries about the data is the upper limit, so the maximum is what matters.
- The minimum is irrelevant because the lower limit 1 is known.
- $X_{(n)}$ is also complete here, so UMVUEs are functions of it.

STEP-BY-STEP SOLUTION

1. $P(x_1, \dots, x_n) = \frac{1}{N^n}$ provided every $x_i \in \{1, \dots, N\}$, i.e. $X_{(n)} \leq N$.
2. So $L = N^{-n} I(X_{(n)} \leq N)$, a function of the data only through $X_{(n)}$.
3. Hence $x_{(n)}$ is sufficient - option (b).

Completeness, Minimal Sufficiency, Ancillarity, Basu's Theorem

Q1.

Let $(X_1, \dots, X_n)$ be from a Rayleigh distribution, $f(x, \theta) = \frac{x}{\theta} \exp\left\{-\frac{x^2}{2\theta}\right\}; x > 0$. The complete sufficient statistic for θ is

A. $\sum_{i=1}^n x_i$

B. $\bar{x}$

C. $\sum_{i=1}^n x_i^2$

D. $\max(x_1, \dots, x_n)$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$L = \theta^{-n} \left(\prod x_i \right) \exp\left(-\frac{\sum x_i^2}{2\theta}\right)$, so $\sum x_i^2$ is complete sufficient.

TIPS & TRICKS

- Rayleigh is a one-parameter exponential family in $\sum X_i^2$, so that statistic is complete as well as sufficient.
- $X_i^2/2 \sim \text{Exp}(1/\theta)$, giving $E\left(\sum X_i^2\right) = 2n\theta$ and hence UMVUE $\frac{1}{2n} \sum X_i^2$.
- Any density with $e^{-x^2/(2\theta)}$ inside will have $\sum X_i^2$ as the sufficient statistic - pattern-match on the exponent.

STEP-BY-STEP SOLUTION

1. $L(\theta) = \prod_{i=1}^n \frac{x_i}{\theta} e^{-x_i^2/(2\theta)} = \frac{1}{\theta^n} \exp\left(-\frac{\sum x_i^2}{2\theta}\right) \prod x_i$.

2. By factorisation with $g_\theta(t) = \theta^{-n} e^{-t/(2\theta)}$, $t = \sum x_i^2$, and $h(x) = \prod x_i$, the statistic $\sum X_i^2$ is sufficient.

3. The family is a one-parameter exponential family with natural sufficient statistic $\sum X_i^2$ and an open natural parameter space, so the statistic is complete.

4. Option (c).

Q2.

Let X have p.m.f.

$$f(x, \theta) = \begin{cases} \left(\frac{\theta}{2}\right)^{|x|} (1-\theta)^{1-|x|}, & x = -1, 0, 1; 0 < \theta < 1 \\ 0, & \text{otherwise} \end{cases}$$

The complete sufficient statistic for θ

- A. is X
- B. is $X-1$
- C.  is $|X|$
- D. does not exist

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

The pmf depends on x only through $|x|$, and $|X| \sim \text{Bernoulli}(\theta)$, which is a complete family. So $|X|$ is complete sufficient.

TIPS & TRICKS

- Read the exponents: both $|x|$ and $1-|x|$ involve only the absolute value, so $|X|$ carries all the information.
- $P(|X|=1) = \frac{\theta}{2} + \frac{\theta}{2} = \theta$ and $P(|X|=0) = 1-\theta$ - a genuine Bernoulli.
- X itself is sufficient but not complete: $E(X)=0$ for all θ with $g(x)=x \neq 0$.

STEP-BY-STEP SOLUTION

1. $f(x, \theta) = \left(\frac{\theta}{2}\right)^{|x|} (1-\theta)^{1-|x|}$ depends on the data only through $|x|$, so $|X|$ is sufficient by factorisation.

2. $P(|X|=1) = P(X=1) + P(X=-1) = \frac{\theta}{2} + \frac{\theta}{2} = \theta$, and $P(|X|=0) = 1-\theta$; hence $|X| \sim \text{Bernoulli}(\theta)$.

3. The Bernoulli family is complete: $E_{\theta}[g(|X|)] = g(1)\theta + g(0)(1-\theta) = 0$ for all $\theta \in (0, 1)$ forces $g(0) = g(1) = 0$.

4. So $|X|$ is complete and sufficient - option (c).

Q3.

Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \theta)$, θ unknown. $T = \sum_{i=1}^n (x_i - \mu)^2$ is

- A. sufficient but not complete
- B. complete but not sufficient
- C. **Both sufficient and complete**
- D. neither sufficient nor complete

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

With μ known, $N(\mu, \theta)$ is a one-parameter exponential family with natural sufficient statistic $\sum (x_i - \mu)^2$, which is therefore complete as well as sufficient.

TIPS & TRICKS

- One-parameter exponential family with an open natural parameter space $\Rightarrow$ the natural statistic is complete.
- $T/\theta \sim \chi^2_n$, so T/n is the UMVUE of θ with variance $2\theta^2/n$, attaining the CR bound.
- If μ were unknown, the sufficient statistic would be the pair $(\sum x_i, \sum x_i^2)$.

STEP-BY-STEP SOLUTION

1. $L(\theta) = (2\pi\theta)^{-n/2} \exp\left(-\frac{\sum (x_i - \mu)^2}{2\theta}\right)$, so by factorisation $T = \sum (x_i - \mu)^2$ is sufficient.
2. Writing L in exponential-family form $\exp\left\{-\frac{1}{2\theta}T\right\} \cdot c(\theta)$, the natural parameter $-\frac{1}{2\theta}$ ranges over an open interval.
3. Hence T is complete as well as sufficient.
4. Option (c).

Q4.

The family of distributions $N(0, \sigma^2)$ is

1. Symmetric
2. Complete

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT**AI-derived explanation**

$N(0, \sigma^2)$ is symmetric about 0, but the family is NOT complete: $E_\sigma(X) = 0$ for all σ with $g(x) = x \neq 0$.

TIPS & TRICKS

- Completeness means no non-trivial unbiased estimator of zero exists - here X itself is one.
- Symmetry is exactly what destroys completeness: every odd function has expectation zero for every σ^2 .
- X^2 (or $\sum X_i^2$) IS complete and sufficient for this family - the even reduction restores completeness.

STEP-BY-STEP SOLUTION

1. Statement 1: the $N(0, \sigma^2)$ density is an even function of x , so the family is symmetric about 0. True.
2. Statement 2: take $g(x) = x$. Then $E_\sigma[g(X)] = 0$ for every $\sigma^2 > 0$, yet g is not identically zero.
3. This violates the definition of completeness, so the family is not complete. False.
4. Hence 1 only - option (a).

Q5.

If $S(X)$ is a complete sufficient statistic and $T(X)$ is ancillary, then

- A. $S(X)$ and $T(X)$ are distributionally dependent
- B. $S(X)$ and $T(X)$ are functionally dependent
- C. **$S(X)$ and $T(X)$ are statistically independent**
- D. None of the above

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Basu's theorem: a complete sufficient statistic is independent of every ancillary statistic.

TIPS & TRICKS

- Classic application: in $N(\mu, \sigma^2)$ with σ^2 known, $\bar{X}$ (complete sufficient) is independent of S^2 (ancillary for μ).
- Independence here is distributional, not functional - option (b) is a category error.
- Basu is the fastest route to many independence results that would otherwise need messy Jacobians.

STEP-BY-STEP SOLUTION

1. $T(X)$ ancillary means its distribution is free of θ .
2. $S(X)$ complete sufficient means the conditional distribution given S is free of θ and $E_\theta[g(S)] = 0 \quad \forall \theta \Rightarrow g \equiv 0$.
3. Basu's theorem asserts that under these conditions $S(X)$ and $T(X)$ are independent.
4. Option (c).

Q6.

A statistic $T(x)$ for θ is ancillary if

- A. $T(x)$ is independent of θ
- B. $T(x)$ is dependent on θ
- C. **The distribution of $T(x)$ is independent of θ**
- D. The distribution of $T(x)$ depends on θ

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Ancillary means the DISTRIBUTION of $T(x)$ does not involve θ - not that the statistic itself is 'independent of θ '.

TIPS & TRICKS

- Every statistic is a function of the data only, so 'independent of θ ' is vacuous - the definition must be about the distribution.
- Standard examples: the range in $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$; $X_1 - X_2$ in $N(\theta, 1)$; S^2 for a location family.
- An ancillary statistic carries no information about θ on its own, but can be informative about precision (conditional inference).

STEP-BY-STEP SOLUTION

1. By definition, $T(X)$ is ancillary for θ if the sampling distribution of $T(X)$ is the same for every value of θ .
2. Options (a) and (b) confuse the statistic with its distribution; option (d) is the negation of ancillarity.
3. Hence option (c).

Q7.

For iid Poisson(λ) variates $x_1, \dots, x_n$:

1. $T = \sum x_i$ is complete sufficient for λ
2. $T = \sum x_i$ is the MLE of λ

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$T = \sum x_i$ is complete sufficient (statement 1 true), but the MLE of λ is $\bar{x} = T/n$, not T (statement 2 false).

TIPS & TRICKS

- Poisson is a one-parameter exponential family, so the natural statistic $\sum X_i$ is complete and sufficient.
- $E(T) = n\lambda$, so T cannot be the MLE (or any sensible estimator) of λ itself.
- The MLE is always a function of the sufficient statistic - here the function is division by n .

STEP-BY-STEP SOLUTION

1. $L(\lambda) = e^{-n\lambda} \lambda^{\sum x_i} / \prod x_i!$, so by factorisation $T = \sum x_i$ is sufficient; being the natural statistic of a full-rank exponential family, it is complete. Statement 1 true.
2. $\log L = -n\lambda + T \log \lambda - \log \prod x_i!$, and $\frac{\partial \log L}{\partial \lambda} = -n + \frac{T}{\lambda} = 0$ gives $\hat{\lambda} = \frac{T}{n} = \bar{x}$.
3. So the MLE is $\bar{x}$, not T . Statement 2 false.
4. Hence 1 only - option (a).

Q8.

Let $X_1, \dots, X_n$ be a random sample of size n from $N(\theta, 1)$. The complete statistic for θ is

- A. X_1
- B. $X_1 - X_2$
- C. $2X_1 - X_2 - X_3$
- D. $X_1 + X_2 + X_3 + X_4$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$X_1 - X_2$ and $2X_1 - X_2 - X_3$ have distributions free of θ (ancillary), so they cannot be complete. $X_1 + X_2 + X_3 + X_4$ is the natural exponential-family statistic and is complete.

TIPS & TRICKS

- Any statistic whose distribution does not involve θ is ancillary and therefore NOT complete (unless degenerate).
- Check the means: $E(X_1 - X_2) = 0$ and $E(2X_1 - X_2 - X_3) = 0$ for every θ - explicit non-trivial unbiased estimators of zero.
- $\sum X_i$ is the complete sufficient statistic of the normal location family - the intended answer.

STEP-BY-STEP SOLUTION

1. For $N(\theta, 1)$, take $g(t) = t$. Then $E[g(X_1 - X_2)] = 0$ for all θ although $g \neq 0$, so $X_1 - X_2$ is not complete.
2. Similarly $E(2X_1 - X_2 - X_3) = 2\theta - \theta - \theta = 0$ for every θ , so option (c) is not complete either.
3. $X_1 + X_2 + X_3 + X_4 \sim N(4\theta, 4)$ is the natural sufficient statistic of this full-rank one-parameter exponential family, hence complete.
4. Option (d).

Q9.

Let X be a discrete random variable with

$$P(X=x) = \begin{cases} \theta, & x=-1 \\ (1-\theta)^2\theta^x, & x=0,1,2,3,\dots \end{cases} \quad 0 < \theta < 1$$

Then

A. X is minimal sufficient and complete.

B. X is minimal sufficient only.

C. X is complete only.

D. X is unbiased estimator for θ .

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

X is the whole sample, so it is minimal sufficient. It is NOT complete: $g(-1)=1$, $g(x)=-x$ for $x \geq 0$ gives $E_\theta[g(X)] \equiv 0$ with $g \neq 0$.

TIPS & TRICKS

- To disprove completeness, exhibit one non-trivial unbiased estimator of zero.
- Use $\sum_{x \geq 0} x \theta^x = \frac{\theta}{(1-\theta)^2}$ - exactly the reciprocal of the $(1-\theta)^2$ factor in the pmf.
- $E(X) = -\theta + \frac{\theta}{1-\theta} \cdot (\dots) \neq \theta$, so option (d) fails as well.

STEP-BY-STEP SOLUTION

1. With a single observation, X itself is trivially sufficient, and no coarser statistic is - so X is minimal sufficient.
2. For completeness, take $g(-1)=1$ and $g(x)=-x$ for $x=0,1,2,\dots$
3. $E_\theta[g(X)] = 1 \cdot \theta + (1-\theta)^2 \sum_{x \geq 0} (-x) \theta^x = \theta - (1-\theta)^2 \cdot \frac{\theta}{(1-\theta)^2} = \theta - \theta = 0$ for every $\theta \in (0,1)$.
4. Since $g \neq 0$, the family is not complete.
5. Hence X is minimal sufficient only - option (b).

Q10.

Let $x_1, x_2, \dots, x_n$ be a random sample from $U(\theta-0.5, \theta+0.5)$, $\theta \in \mathbb{R}$. Which of the following statements is/are correct?

1. $(X_{(1)}, X_{(n)})$ is sufficient as well as complete.
2. $\frac{X_{(1)} + X_{(n)}}{2}$ is MLE for θ .

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$(X_{(1)}, X_{(n)})$ is sufficient but NOT complete (the range is ancillary), so statement 1 fails; the midrange is one of the MLEs, so statement 2 holds.

TIPS & TRICKS

- $R = X_{(n)} - X_{(1)}$ has a distribution free of θ , so $E[R - E(R)] \equiv 0$ with a non-degenerate function - completeness dies.
- The MLE is any point of $\left[X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2} \right]$; the midrange is the natural member of that set.
- Minimal sufficient does NOT imply complete - this family is THE standard counterexample.

STEP-BY-STEP SOLUTION

1. $L(\theta) = 1$ for $X_{(n)} - \frac{1}{2} \leq \theta \leq X_{(1)} + \frac{1}{2}$, so $(X_{(1)}, X_{(n)})$ is sufficient.
2. But the range $R = X_{(n)} - X_{(1)}$ is ancillary, so $g(X_{(1)}, X_{(n)}) = R - E(R)$ satisfies $E_\theta(g) = 0$ for all θ while $g \neq 0$: the statistic is not complete. Statement 1 false.
3. The likelihood is flat on that interval, and its midpoint is $\frac{X_{(1)} + X_{(n)}}{2}$, which therefore is a maximum likelihood estimate. Statement 2 true.
4. Hence 2 only - option (b).

SHARED CONTEXT

Let $x_1, x_2, \dots, x_n$ be a random sample from Normal distribution $N(0, \theta)$, $\theta \geq 0$.

Q11.

Consider the following statements in respect of x_1 :

1. x_1 is unbiased for θ .
2. x_1 is complete statistic for θ .
3. x_1 is sufficient for θ .

How many of the above statements are correct?

- A. Only one
B. Only two
C. All three
D. None

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$E(x_1) = 0 \neq \theta$ (not unbiased), the symmetric family makes x_1 incomplete, and x_1 alone cannot reproduce $\sum x_i^2$ (not sufficient). None of the three holds.

TIPS & TRICKS

- $N(0, \theta)$ has mean zero, so any single observation has expectation 0 - it can never be unbiased for a variance.
- Symmetry kills completeness: $E_\theta(x_1) = 0$ for all θ with $g(x) = x \neq 0$.
- Sufficiency concerns the WHOLE sample; one observation discards the other $n-1$.

STEP-BY-STEP SOLUTION

1. Statement 1: $E(x_1) = 0$ for every θ , so x_1 is not unbiased for θ . False.
2. Statement 2: take $g(x) = x$. Then $E_\theta[g(x_1)] = 0$ for all θ although $g \neq 0$, so the family of x_1 is not complete. False.
3. Statement 3: the likelihood depends on $\sum x_i^2$, which x_1 alone does not determine, so x_1 is not sufficient. False.
4. None of the statements is correct - option (d).

SHARED CONTEXT

Let $x_1, x_2, \dots, x_n$ be a random sample from Normal distribution $N(0, \theta)$, $\theta \geq 0$.

Q12.

Consider the following statements in respect of x_1^2 :

1. x_1^2 is complete for θ .
2. x_1^2 is sufficient for θ .

Which of the statements given above is/are correct?

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$x_1^2/\theta \sim \chi^2_1$ makes x_1^2 a complete statistic (a one-parameter exponential family), but one observation cannot be sufficient for the whole sample.

TIPS & TRICKS

- Squaring removes the symmetry that destroyed completeness for x_1 .
- x_1^2 has density of exponential-family form in θ with natural statistic x_1^2 - hence complete.
- The sample likelihood needs $\sum_{i=1}^n x_i^2$, so a single square is not sufficient.

STEP-BY-STEP SOLUTION

1. Statement 1: $x_1^2/\theta \sim \chi^2_1$, so the density of x_1^2 is a one-parameter exponential family with natural sufficient statistic x_1^2 and open natural parameter space - hence complete. True.
2. Statement 2: the joint likelihood is $\theta^{-n/2} \exp\left(-\frac{\sum x_i^2}{2\theta}\right)$, requiring $\sum_{i=1}^n x_i^2$; x_1^2 alone does not determine it. False.
3. Hence 1 only - option (a).

Q13.

Let

$$X_i \sim f_{\theta}(x) = \begin{cases} 1, & \theta - \frac{1}{2} < x < \theta + \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$$

such that $\theta \in (-\infty, \infty)$; $i = 1, 2, \dots, n$. Consider the following statements:

I. The distribution of $X_{(n)} - X_{(1)}$ is independent of θ .

II. $E_{\theta}[(X_{(n)} - X_{(1)}) - E(X_{(n)} - X_{(1)})] = 0$, for all θ .

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

The range is ancillary (its distribution is free of θ), and any centred random variable has zero expectation - so both statements hold. Together they prove the minimal sufficient statistic is NOT complete.

TIPS & TRICKS

- $X_{(n)} - X_{(1)}$ depends only on the differences $X_i - \theta$, whose joint law is free of θ .
- Statement II is trivially true for ANY statistic with finite mean - it is $E[R - E(R)] = 0$.
- Together they exhibit a non-trivial unbiased estimator of zero, which is exactly why $(X_{(1)}, X_{(n)})$ fails completeness.

STEP-BY-STEP SOLUTION

1. Statement I: writing $Y_i = X_i - \theta \sim U(-\frac{1}{2}, \frac{1}{2})$, we get $X_{(n)} - X_{(1)} = Y_{(n)} - Y_{(1)}$, whose distribution does not involve θ . True - the range is ancillary.
2. Statement II: for any statistic R with finite mean, $E_{\theta}[R - E(R)] = 0$; here $E(R)$ is a constant because R is ancillary, so the identity holds for every θ . True.

3. Option (c). (These two facts together show $(X_{(1)}, X_{(n)})$ is not complete.)

SI & HT

2024 Q20

Conceptual

Medium

Q14.

If $x_1, x_2, x_3, \dots, x_n$ is a random sample from $N(0, \theta)$, then which of the statements given below is/are correct?

I. $T_1 = x_1$ is complete statistic for θ .

II. $T_2 = x_1^2$ is complete statistic for θ .

Select the answer using the code given below.

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

x_1 is not complete (symmetry gives $E(x_1) = 0$ for all θ), but $x_1^2 \sim \theta \chi^2_1$ is a complete exponential family.

TIPS & TRICKS

- Symmetric families are never complete in the raw variable - odd functions have zero mean.
- Squaring restores completeness by removing the sign information.
- This pair of statements is the standard illustration of why completeness is a delicate property.

STEP-BY-STEP SOLUTION

1. Statement I: with $g(x) = x$, $E_\theta[g(x_1)] = 0$ for every $\theta > 0$ although $g \neq 0$ - so x_1 is not complete. False.
2. Statement II: $x_1^2/\theta \sim \chi^2_1$, a one-parameter exponential family with natural statistic x_1^2 and open natural parameter space - hence complete. True.
3. Hence II only - option (b).

Q15.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample of size n taken from $U(\theta-0.5, \theta+0.5)$ where $\theta \in (-\infty, \infty)$. If T_1 and T_2 are respectively the minimum and maximum order statistics, then which of the following statements is/are correct regarding the statistic $T = T_2 - T_1$?

- I. T is sufficient statistic of θ .
- II. T has a complete family of distributions.

Select the answer using the code given below:

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

The range $T = T_2 - T_1$ is ANCILLARY - its distribution is free of θ - so it is neither sufficient nor does it form a complete family.

TIPS & TRICKS

- $T_2 - T_1 = (X_{(n)} - \theta) - (X_{(1)} - \theta)$ depends only on the centred observations.
- An ancillary statistic carries no information about θ , so it cannot be sufficient.
- Its family consists of a SINGLE distribution, so $E[g(T)] = 0$ imposes only one equation and cannot force $g \equiv 0$ - completeness fails.

STEP-BY-STEP SOLUTION

1. Writing $Y_i = X_i - \theta \sim U(-\frac{1}{2}, \frac{1}{2})$: $T = Y_{(n)} - Y_{(1)}$, whose distribution is free of θ .
2. Statement I: an ancillary statistic carries no information about θ ; the likelihood cannot be factorised through T alone, so T is not sufficient. False.
3. Statement II: the family $\{P_\theta^T\}$ is a single fixed distribution. Take any non-zero g with $E[g(T)] = 0$ (e.g. $g(t) = t - E(T)$): completeness fails. False.
4. Neither statement is correct - option (d).

Q16.

The largest order statistic $X_{(n)}$ is complete and sufficient for which of the following?

I. $X \sim U(0, \theta), \theta \in (0, \infty)$

II. $X \sim f(x) = \begin{cases} \frac{1}{N}, & x=1, 2, 3, \dots, N \text{ where } N \geq 1 \\ 0, & \text{otherwise} \end{cases}$

Select the answer using the code given below:

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$X_{(n)}$ is complete sufficient for both $U(0, \theta)$ and the discrete uniform on $\{1, \dots, N\}$.

TIPS & TRICKS

- Both likelihoods are of the form $c(\theta)^{-n} I(X_{(n)} \leq \theta)$ - sufficiency is immediate.
- Completeness in the continuous case: $\int_0^\theta g(t) n t^{n-1} dt = 0$ for all θ forces $g \equiv 0$ by differentiation.
- Completeness in the discrete case follows by induction on N from $E_N[g] = 0$.

STEP-BY-STEP SOLUTION

- Case I: $L = \theta^{-n} I(X_{(n)} \leq \theta)$, so $X_{(n)}$ is sufficient. If $E_\theta[g(X_{(n)})] = \int_0^\theta g(t) \frac{n t^{n-1}}{\theta^n} dt = 0$ for all θ , differentiating in θ gives $g(\theta) \theta^{n-1} = 0$, so $g \equiv 0$: complete.
- Case II: $L = N^{-n} I(X_{(n)} \leq N)$, so $X_{(n)}$ is sufficient. $E_N[g(X_{(n)})] = 0$ for $N=1, 2, \dots$ forces $g(1)=0$, then $g(2)=0$, and so on by induction: complete.
- Both hold - option (c).

Q17.

Let $X_i \sim N(\mu, \sigma^2)$. Then for parameter θ , $T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$ is:

- A. Sufficient
- B. Complete
- C. **Sufficient and complete**
- D. Neither sufficient nor complete

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$\left(\sum X_i, \sum X_i^2 \right)$ is the natural statistic of the two-parameter normal exponential family with an open natural parameter space - sufficient AND complete.

TIPS & TRICKS

- Full-rank exponential family $\Rightarrow$ the natural sufficient statistic is complete.
- This is why $\bar{X}$ and S^2 are the UMVUEs of μ and σ^2 .
- Contrast $N(\theta, \theta^2)$, a CURVED exponential family, where the same statistic is sufficient but not complete.

STEP-BY-STEP SOLUTION

1. $L \propto \sigma^{-n} \exp \left[-\frac{\sum x_i^2}{2\sigma^2} + \frac{\mu \sum x_i}{\sigma^2} - \frac{n\mu^2}{2\sigma^2} \right]$, so T is sufficient by factorisation.
2. The natural parameters $\left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2} \right)$ range over an open subset of $\mathbb{R}^2$, making this a full-rank two-parameter exponential family.
3. Hence the natural sufficient statistic T is also complete.
4. Option (c).

Q18.

Consider the following statements:

I. A minimal sufficient statistic is always complete.

II. A statistic that is independent of every ancillary statistic need not be complete.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Minimal sufficiency does NOT imply completeness (I false); and independence from every ancillary does not guarantee completeness either (II true, as stated).

TIPS & TRICKS

- $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$: $(X_{(1)}, X_{(n)})$ is minimal sufficient but not complete.
- Basu's theorem runs one way: complete + sufficient $\Rightarrow$ independent of every ancillary. The converse fails.
- Complete + sufficient $\Rightarrow$ minimal sufficient, but not conversely - remember the direction.

STEP-BY-STEP SOLUTION

1. Statement I: for $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, $(X_{(1)}, X_{(n)})$ is minimal sufficient, yet the ancillary range gives a non-trivial unbiased estimator of zero - so it is not complete. False.
2. Statement II: Basu's theorem says a complete sufficient statistic is independent of every ancillary. The converse direction is not a theorem - counterexamples exist of statistics independent of every ancillary that are not complete. So 'need not be complete' is correct. True.
3. Hence II only - option (b).

Q19.

Let $X_i \sim \text{Binomial}(1, p)$; $i = 1, 2, 3, \dots, n$. For parameter p , $T = \sum_{i=1}^n X_i$ is:

- A. Sufficient
- B. Complete
- C. Both sufficient and complete**
- D. Neither sufficient nor complete

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$T = \sum X_i \sim B(n, p)$ is the natural statistic of a full-rank one-parameter exponential family - sufficient AND complete.

TIPS & TRICKS

- $L = p^T(1-p)^{n-T}$ gives sufficiency immediately.
- Completeness: $E_p[g(T)] = \sum_t g(t) \binom{n}{t} p^t(1-p)^{n-t} = 0$ for all p forces every coefficient to vanish.
- This is why T/n is the UMVUE of p and $\frac{T(T-1)}{n(n-1)}$ the UMVUE of p^2 .

STEP-BY-STEP SOLUTION

1. $L(p) = p^{\sum x_i} (1-p)^{n-\sum x_i}$, so by factorisation $T = \sum X_i$ is sufficient.
2. $E_p[g(T)] = (1-p)^n \sum_{t=0}^n g(t) \binom{n}{t} \left(\frac{p}{1-p}\right)^t$; setting this to zero for all $p \in (0, 1)$ makes a polynomial in $\frac{p}{1-p}$ vanish identically, so all $g(t) = 0$.
3. Hence T is both sufficient and complete - option (c).

Optimal Estimation

72 authentic PYQs in this topic (2018-2026)

Cramer-Rao Inequality, MVB and Fisher Information

SI & HT

2018 Q31

Conceptual

Hard

Q1.

In sampling from $N(\theta, 1)$ with $\bar{X}$ the sample mean, which class describes all MVB estimators of θ ?

A. All estimators of type $A\bar{X}+B$, A, B given constants

B. $\bar{X}$ only

C. All estimators of $f(\bar{X})$, f a real-valued function

D. No estimator attains MVB in this case

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The score is $\frac{\partial \log L}{\partial \theta} = n(\bar{X} - \theta)$. Equality in the Cramer-Rao inequality requires the score to be proportional to $T - \psi(\theta)$, which forces T to be of the form $A\bar{X} + B$.

TIPS & TRICKS

- MVB attainment condition: $\frac{\partial \log L}{\partial \theta} = A(\theta) [T - \psi(\theta)]$.
- Because the score here is linear in $\bar{X}$, only linear functions of $\bar{X}$ can attain the bound; any non-linear $f(\bar{X})$ cannot.
- Hence in $N(\theta, 1)$ the functions of θ admitting MVB estimators are exactly the affine ones $A\theta + B$.

STEP-BY-STEP SOLUTION

- For $N(\theta, 1)$, $\log L = \text{const} - \frac{1}{2} \sum (x_i - \theta)^2$, so $\frac{\partial \log L}{\partial \theta} = \sum (x_i - \theta) = n(\bar{X} - \theta)$.
- The Cramer-Rao bound is attained by T (unbiased for $\psi(\theta)$) if and only if $\frac{\partial \log L}{\partial \theta} = A(\theta) [T - \psi(\theta)]$.
- Writing $n(\bar{X} - \theta) = \frac{n}{A} [(A\bar{X} + B) - (A\theta + B)]$ shows every $T = A\bar{X} + B$ attains the bound for $\psi(\theta) = A\theta + B$.
- No non-linear function of $\bar{X}$ can make the score proportional to $T - \psi(\theta)$.
- So the MVB class is exactly $\{A\bar{X} + B\}$ - option (a).

Q2.

Let 5,2,3,4,3,2,4,5,3,2 be a sample from a normal population with mean 0 and variance σ^2 . The minimum variance bound estimator for σ^2 is

- A. 33
- B. 3.3
- C. 1.21
- D. 12.1**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Mean is known ($=0$), so the MVB estimator of σ^2 is $\frac{1}{n} \sum x_i^2 = 121/10 = 12.1$.

TIPS & TRICKS

- $N(0, \sigma^2)$: the score is $\frac{\partial \log L}{\partial \sigma^2} = \frac{n}{2\sigma^4} \left[\frac{1}{n} \sum x_i^2 - \sigma^2 \right]$ - exactly the MVB attainment form.
- Compute $\sum x_i^2$ carefully: $25+4+9+16+9+4+16+25+9+4=121$.
- Do not subtract $\bar{x}$ - the mean is given as 0, so deviations are taken from 0.

STEP-BY-STEP SOLUTION

1. For $N(0, \sigma^2)$ the MVB (and UMVU) estimator of σ^2 is $T = \frac{1}{n} \sum_{i=1}^n X_i^2$.
2. Squares: $5^2, 2^2, 3^2, 4^2, 3^2, 2^2, 4^2, 5^2, 3^2, 2^2 = 25, 4, 9, 16, 9, 4, 16, 25, 9, 4$.
3. $\sum x_i^2 = 121$ and $n = 10$.
4. $T = 121/10 = 12.1$ - option (d).

Q3.

If $(X_1, \dots, X_n)$ is a random sample from $N(1, \sigma^2)$, which one is unbiased for σ^2 with variance equal to the Cramer-Rao bound?

- A. $\frac{1}{n} \sum_{i=1}^n (X_i - 1)^2$
- B. $\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- C. $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$
- D. $\frac{1}{n-1} \sum_{i=1}^n (X_i - 1)^2$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The mean is known ($=1$), so the score gives the MVB estimator $\frac{1}{n} \sum (X_i - 1)^2$, which is unbiased with variance $2\sigma^4/n$ equal to the CR bound.

TIPS & TRICKS

- With μ known, the sufficient statistic is $\sum (X_i - \mu)^2$ and the CR bound $2\sigma^4/n$ is attained exactly.
- $\frac{1}{n-1} \sum (X_i - \bar{X})^2$ is unbiased but has variance $2\sigma^4/(n-1) > 2\sigma^4/n$ - it does NOT attain the bound.
- Read the mean: ' $N(1, \sigma^2)$ ' means $\mu=1$ is known, which changes the answer completely.

STEP-BY-STEP SOLUTION

1. Here $\mu=1$ is known, so $\log L = \text{const} - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum (x_i - 1)^2$.
2. $\frac{\partial \log L}{\partial \sigma^2} = -\frac{n}{2\sigma^4} \left[\frac{1}{n} \sum (x_i - 1)^2 - \sigma^2 \right]$, which is of the MVB form $A(\sigma^2)[T - \sigma^2]$ with $T = \frac{1}{n} \sum (x_i - 1)^2$.
3. $\sum (X_i - 1)^2 / \sigma^2 \sim \chi^2_n$, so $E(T) = \sigma^2$ and $V(T) = \frac{2\sigma^4}{n}$, which equals the CR bound.
4. Hence option (a).

Q4.

If $(X_1, \dots, X_n)$ be i.i.d. $N(\mu, \sigma^2)$, μ known, the C-R bound for estimation of $\sigma^2 = \theta$ is

A. $\frac{2\theta}{n}$

B. $\frac{\theta}{n}$

C. $\frac{2\theta^2}{n}$

D. $\frac{\theta^2}{n}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

With $\theta = \sigma^2$ and μ known, $I_1(\theta) = \frac{1}{2\theta^2}$, so the bound is $\frac{1}{nI_1} = \frac{2\theta^2}{n}$.

TIPS & TRICKS

- Memorise $I(\sigma^2) = \frac{1}{2\sigma^4}$ per observation in the normal model - it recurs every other year.
- The bound has θ^2 in the numerator, so options linear in θ can be discarded on dimensional grounds.
- Attained exactly by $\frac{1}{n} \sum (X_i - \mu)^2$, whose variance is $2\sigma^4/n$.

STEP-BY-STEP SOLUTION

1. With $\theta = \sigma^2$ and μ known, $\log f = -\frac{1}{2} \log(2\pi\theta) - \frac{(x-\mu)^2}{2\theta}$.

2. $\frac{\partial \log f}{\partial \theta} = -\frac{1}{2\theta} + \frac{(x-\mu)^2}{2\theta^2}$ and $\frac{\partial^2 \log f}{\partial \theta^2} = \frac{1}{2\theta^2} - \frac{(x-\mu)^2}{\theta^3}$.

3. $I_1(\theta) = -E\left[\frac{\partial^2 \log f}{\partial \theta^2}\right] = -\frac{1}{2\theta^2} + \frac{\theta}{\theta^3} = \frac{1}{2\theta^2}$.

4. CR bound $= \frac{1}{nI_1(\theta)} = \frac{2\theta^2}{n}$ - option (c).

Q5.

Let $(X_1, \dots, X_n)$ be i.i.d. $N(\theta, 1)$. The C-R lower bound for the variance of an unbiased estimator of θ^r is

- A. θ^r/n
- B. $r\theta^{r-1}/n$
- C. $r^2\theta^{2r-2}/n$
- D. θ^r/n^2

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$I_n(\theta) = n$ for $N(\theta, 1)$ and $\psi(\theta) = \theta^r$ gives $\psi' = r\theta^{r-1}$, so the bound is $\frac{r^2\theta^{2r-2}}{n}$.

TIPS & TRICKS

- CR bound for $\psi(\theta)$: $\frac{[\psi'(\theta)]^2}{I_n(\theta)}$ - square the derivative, never forget the square.
- $I_1(\theta) = 1$ for $N(\theta, 1)$, so $I_n = n$.
- Check $r=1$: the bound becomes $1/n$, the familiar $V(\bar{X})$ - a good sanity test on any option.

STEP-BY-STEP SOLUTION

- For $N(\theta, 1)$: $\log f = -\frac{1}{2} \log(2\pi) - \frac{1}{2}(x-\theta)^2$, so $\frac{\partial \log f}{\partial \theta} = (x-\theta)$ and $I_1(\theta) = E(X-\theta)^2 = 1$.
- Hence $I_n(\theta) = n$.
- With $\psi(\theta) = \theta^r$, $\psi'(\theta) = r\theta^{r-1}$.
- CR bound $= \frac{[\psi'(\theta)]^2}{I_n(\theta)} = \frac{r^2\theta^{2r-2}}{n}$ - option (c).

Q6.

If t is unbiased for θ , the minimum variance bound is attained if (λ free from sample values, may depend on θ)

A. $\frac{\partial^2 \log L}{\partial \theta^2} = \frac{t-\theta}{\lambda}$

B. $\frac{\partial \log L}{\partial \theta} = \frac{t}{\theta}$

C. $\frac{\partial^2 \log L}{\partial \theta^2} = \frac{t-\theta}{\theta}$

D. $\frac{\partial \log L}{\partial \theta} = \frac{t-\theta}{\lambda}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Equality in the Cramer-Rao inequality holds iff the SCORE is proportional to $t-\theta$: $\frac{\partial \log L}{\partial \theta} = \frac{t-\theta}{\lambda}$.

TIPS & TRICKS

- The condition involves the FIRST derivative of $\log L$, not the second - options (a) and (c) can be discarded at once.
- $\lambda = \lambda(\theta)$ may depend on θ but never on the data - that is exactly what 'free from sample values' means.
- When this holds, $V(t) = \lambda(\theta)$ equals the bound; it is the definition of an MVB estimator.

STEP-BY-STEP SOLUTION

1. The Cramer-Rao inequality comes from the Cauchy-Schwarz inequality applied to $\text{Cov}\left(t, \frac{\partial \log L}{\partial \theta}\right)$.

2. Equality in Cauchy-Schwarz holds iff the two variables are linearly related, i.e. $\frac{\partial \log L}{\partial \theta} = A(\theta) [t - \theta]$.

3. Writing $A(\theta) = 1/\lambda$ gives $\frac{\partial \log L}{\partial \theta} = \frac{t-\theta}{\lambda}$.

4. Option (d).

Q7.

The estimator attaining the Cramer-Rao lower bound for the variance of a normal distribution with mean zero is

A. $\frac{1}{n-1} \sum (X_i - \bar{X})^2$

B. $\frac{1}{n} \sum (X_i - \bar{X})^2$

C. $\frac{1}{n} \sum X_i^2$

D. $\frac{1}{n-1} \sum X_i^2$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Mean zero is known, so the score gives $T = \frac{1}{n} \sum X_i^2$, which attains the bound $2\sigma^4/n$.

TIPS & TRICKS

- With a known mean, deviations are measured from 0 and no degree of freedom is lost - divisor n , not $n-1$.
- $\sum X_i^2/\sigma^2 \sim \chi_n^2$ gives $V(T) = 2\sigma^4/n$, exactly the CR bound.
- Estimators built on $\bar{X}$ waste information here, and their variance $2\sigma^4/(n-1)$ exceeds the bound.

STEP-BY-STEP SOLUTION

1. For $N(0, \sigma^2)$, $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{\sum x_i^2}{2\sigma^2}$.

2. $\frac{\partial \log L}{\partial \sigma^2} = \frac{n}{2\sigma^4} \left[\frac{\sum x_i^2}{n} - \sigma^2 \right]$, which is of the MVB attainment form with $T = \frac{1}{n} \sum X_i^2$.

3. $\sum X_i^2/\sigma^2 \sim \chi_n^2$, so $E(T) = \sigma^2$ and $V(T) = \frac{2\sigma^4}{n}$, equal to the CR bound.

4. Option (c).

Q8.

Let $X_1, \dots, X_n \sim U(0, \theta)$ iid. The CR lower bound for any unbiased estimator of θ

A. does not exist

B. is $X_{(n)}$

C. is θ/n

D. is $1/\theta$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

The support $(0, \theta)$ depends on θ , so the regularity conditions fail and the Cramer-Rao bound does not exist.

TIPS & TRICKS

- Regularity requires the support to be free of θ so that differentiation under the integral sign is valid.
- For $U(0, \theta)$ the UMVUE $\frac{n+1}{n}X_{(n)}$ has variance $\frac{\theta^2}{n(n+2)}$, of order n^{-2} - faster than any CR-type n^{-1} bound could allow.
- Whenever the support endpoints involve the parameter, write 'CR bound not applicable' rather than computing anything.

STEP-BY-STEP SOLUTION

1. The Cramer-Rao inequality requires regularity: in particular the support of $f(x; \theta)$ must not depend on θ , so that $\frac{\partial}{\partial \theta} \int f dx = \int \frac{\partial f}{\partial \theta} dx$.
2. For $U(0, \theta)$ the support is $(0, \theta)$, which depends on θ ; the interchange fails.
3. Hence the Cramer-Rao lower bound does not exist for this family.
4. Option (a).

Q9.

Let $x_1, \dots, x_n$ be from $f(x, \theta) = \frac{1}{\theta} \exp(-x/\theta)$, $0 < x < \infty$. The CR lower bound to the variance of an unbiased estimator of θ is

- A. θ/n
- B. θ^2
- C. θ^2/n
- D. θ

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$I_1(\theta) = 1/\theta^2$ for the exponential with mean θ , so the bound is θ^2/n (attained by $\bar{X}$).

TIPS & TRICKS

- $\log f = -\log \theta - x/\theta$, $\frac{\partial \log f}{\partial \theta} = -\frac{1}{\theta} + \frac{x}{\theta^2}$, giving $I_1 = \frac{1}{\theta^2}$.
- $V(\bar{X}) = \theta^2/n$ exactly, so the bound is attained and $\bar{X}$ is the MVB estimator.
- The bound must have the same units as θ^2 ; options linear in θ are dimensionally impossible.

STEP-BY-STEP SOLUTION

1. $\log f(x; \theta) = -\log \theta - \frac{x}{\theta}$, so $\frac{\partial \log f}{\partial \theta} = -\frac{1}{\theta} + \frac{x}{\theta^2} = \frac{x - \theta}{\theta^2}$.

2. $I_1(\theta) = E\left[\left(\frac{X - \theta}{\theta^2}\right)^2\right] = \frac{V(X)}{\theta^4} = \frac{\theta^2}{\theta^4} = \frac{1}{\theta^2}$.

3. $I_n(\theta) = \frac{n}{\theta^2}$, so the CR lower bound for unbiased estimators of θ is $\frac{1}{I_n} = \frac{\theta^2}{n}$.

4. Option (c).

Q10.

Let $x_1, \dots, x_n$ be from a Cauchy distribution, $f(x, \theta) = \frac{1}{\pi[1+(x-\theta)^2]}$. The minimum variance bound estimator

- A. is $\sum x_i$
- B. is $\bar{x}$
- C. is s^2
- D. ~~Does not exist~~

Correct answer: D

EXAM SHORTCUT AI-derived explanation

For the Cauchy location family the score is not proportional to any $T-\theta$, so no unbiased estimator attains the Cramer-Rao bound: the MVB estimator does not exist.

TIPS & TRICKS

- Cauchy is regular ($I_1(\theta) = \frac{1}{2}$), so the CR bound EXISTS - what fails is its attainment.
- $\bar{X}$ is useless here: it has the same Cauchy distribution as a single observation, so it is neither consistent nor sufficient.
- No non-trivial sufficient statistic exists for the Cauchy location family - the order statistic is minimal sufficient.

STEP-BY-STEP SOLUTION

- $\log f = -\log \pi - \log [1+(x-\theta)^2]$, so $\frac{\partial \log L}{\partial \theta} = \sum \frac{2(x_i - \theta)}{1+(x_i - \theta)^2}$.
- This score cannot be written in the MVB form $A(\theta)[T(x) - \theta]$ - it is a non-linear function of the observations that does not reduce to any single statistic.
- Hence no unbiased estimator attains the Cramer-Rao lower bound.
- The minimum variance bound estimator does not exist - option (d).

Q11.

Let $x_1, \dots, x_n$ be from $\text{Poisson}(\lambda)$:

1. Cramer-Rao bound to the variance of an unbiased estimator of θ is θ/n .
2. $\bar{x} = \frac{1}{n} \sum x_i$ is a minimum variance bound unbiased estimator of θ .

Codes:

- A. 1 only
 B. 2 only
 C. Both 1 and 2
 D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$I_1(\lambda) = 1/\lambda$, so the bound is λ/n , and $\bar{x}$ has exactly that variance - both statements hold.

TIPS & TRICKS

- Poisson: $I_1(\lambda) = \frac{1}{\lambda}$, so $I_n = \frac{n}{\lambda}$ and the bound is $\frac{\lambda}{n}$.
- $V(\bar{X}) = \lambda/n$, equal to the bound, so $\bar{X}$ is an MVB estimator - the cleanest example in the syllabus.
- The score $\frac{\partial \log L}{\partial \lambda} = \frac{n}{\lambda}(\bar{x} - \lambda)$ is visibly of the MVB attainment form.

STEP-BY-STEP SOLUTION

1. $\log f = -\lambda + x \log \lambda - \log x!$, so $\frac{\partial \log f}{\partial \lambda} = \frac{x}{\lambda} - 1$ and $I_1(\lambda) = \frac{V(X)}{\lambda^2} = \frac{1}{\lambda}$.

2. $I_n(\lambda) = n/\lambda$, so the Cramer-Rao bound for unbiased estimators of λ is λ/n . Statement 1 true.

3. $V(\bar{X}) = \frac{\lambda}{n}$, which equals the bound, so $\bar{X}$ is a minimum variance bound unbiased estimator. Statement 2 true.

4. Option (c).

Q12.

If $X_1, \dots, X_n$ have joint pdf $f_\theta(x_1, \dots, x_n)$; $\theta \in \Theta$, Fisher's information about θ contained in X is given by

- A. $I_\theta = E_\theta \left(\frac{\partial^2 \log_e f_\theta(x)}{\partial \theta^2} \right)$
- B. $I_\theta = E_\theta \left(-\frac{\partial^2 \log_e f_\theta(x)}{\partial \theta^2} \right)^2$
- C. ✓ $I_\theta = E_\theta \left(\frac{\partial \log_e f_\theta(x)}{\partial \theta} \right)^2$
- D. None of the above

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$I_\theta = E \left[\left(\frac{\partial \log f_\theta}{\partial \theta} \right)^2 \right]$ - the expected squared score.

TIPS & TRICKS

- Equivalent form: $I_\theta = -E \left[\frac{\partial^2 \log f_\theta}{\partial \theta^2} \right]$ - note the MINUS sign, which option (a) omits.
- Fisher information is additive over independent observations: $I_n(\theta) = nI_1(\theta)$.
- $I_\theta \geq 0$ always; the un-negated second derivative in (a) would generally be negative.

STEP-BY-STEP SOLUTION

1. The score function is $U = \frac{\partial \log f_\theta(x)}{\partial \theta}$, and $E_\theta(U) = 0$ under regularity.
2. Fisher information is the variance of the score: $I_\theta = V(U) = E_\theta \left[\left(\frac{\partial \log f_\theta(x)}{\partial \theta} \right)^2 \right]$.
3. Equivalently $I_\theta = -E_\theta \left[\frac{\partial^2 \log f_\theta(x)}{\partial \theta^2} \right]$; option (a) drops the minus sign and option (b) squares the wrong quantity.
4. Option (c).

Q13.

Let $I(\theta)$ be the Fisher information supplied by the sample. If T is unbiased for $\psi(\theta)$, the variance of T will be

A. ✓ $\geq \frac{(\partial\psi/\partial\theta)^2}{I(\theta)}$

B. $\leq \frac{(\partial\psi/\partial\theta)^2}{I(\theta)}$

C. $\geq \frac{1}{I(\theta)}$

D. $\leq \frac{1}{I(\theta)}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\text{Cramer-Rao: } V(T) \geq \frac{[\psi'(\theta)]^2}{I(\theta)}.$$

TIPS & TRICKS

- It is a LOWER bound, so the inequality is $\geq$ - options with $\leq$ are impossible.
- For $\psi(\theta)=\theta$ the bound reduces to $1/I(\theta)$, which is option (c) - correct only in that special case.
- Always square the derivative $\psi'(\theta)$.

STEP-BY-STEP SOLUTION

1. The Cramer-Rao inequality states that for any unbiased estimator T of $\psi(\theta)$, subject to regularity conditions,

2. $V(T) \geq \frac{\left[\frac{\partial\psi}{\partial\theta}\right]^2}{I(\theta)}$, where $I(\theta)$ is the Fisher information in the sample.

3. Option (c) is the special case $\psi(\theta)=\theta$ and is not the general statement.

4. Option (a).

Q14.

To estimate the parameter λ (variance) of a Poisson distribution, $S^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$ from a sample of size n . Then

$\text{Var}(S^2)$ will always be $\geq$

A. $\frac{\sqrt{\lambda}}{n}$

B. $\frac{(n-1)\lambda}{n}$

C. $\frac{n\lambda}{n-1}$

D. $\frac{\lambda}{n+1}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

S^2 is unbiased for the Poisson parameter λ , so the Cramer-Rao bound λ/n applies: $V(S^2) \geq \lambda/n$.

TIPS & TRICKS

- For Poisson mean = variance = λ , so S^2 estimates the same parameter as $\bar{X}$.
- CR bound for unbiased estimators of λ is λ/n ; $\bar{X}$ attains it, S^2 does not.
- Any unbiased estimator of λ obeys the same bound - the bound depends on the parameter, not on the estimator.

STEP-BY-STEP SOLUTION

1. For Poisson(λ), $I_1(\lambda) = 1/\lambda$ so $I_n(\lambda) = n/\lambda$.
2. The Cramer-Rao lower bound for the variance of any unbiased estimator of λ is $\frac{1}{I_n(\lambda)} = \frac{\lambda}{n}$.
3. S^2 is unbiased for the population variance, which equals λ here, so the bound applies to it.
4. Hence $V(S^2) \geq \frac{\lambda}{n}$ - option (a).

Q15.

Let X_1, X_2, X_3 be iid normal, mean θ , variance 1. Define $T = X_1 - X_2 - X_3$. The Fisher information contained in T is

A. 3

B. $\frac{1}{3}$

C. -3

D. $-\frac{1}{3}$ **Correct answer: B****EXAM SHORTCUT**

AI-derived explanation

$$T \sim N(-\theta, 3), \text{ so } I_T(\theta) = \frac{(\partial E(T)/\partial \theta)^2}{V(T)} = \frac{1}{3}.$$

TIPS & TRICKS

- For a normal statistic with mean $m(\theta)$ and constant variance v , the information is $[m'(\theta)]^2/v$.
- $E(T) = \theta - \theta - \theta = -\theta$ and $V(T) = 1 + 1 + 1 = 3$.
- The whole sample carries $I=3$; the non-sufficient reduction T retains only $1/3$ - information is lost, never gained.

STEP-BY-STEP SOLUTION

1. $T = X_1 - X_2 - X_3$ is a linear combination of independent normals, so $T \sim N(-\theta, 3)$.

2. $\log f_T(t) = \text{const} - \frac{(t+\theta)^2}{6}$, so $\frac{\partial \log f_T}{\partial \theta} = -\frac{t+\theta}{3}$.

3. $I_T(\theta) = E\left[\left(\frac{T+\theta}{3}\right)^2\right] = \frac{V(T)}{9} = \frac{3}{9} = \frac{1}{3}$.

4. Option (b). (Compare $I=3$ for the full sample - the reduction to T destroys information.)

Q16.

Consider the following statements:

1. The Cramer-Rao inequality provides a lower bound to the variance of a sufficient estimator for $\gamma(\theta)$.
2. An MVB estimator for $\gamma(\theta)$ exists iff there exists a sufficient estimator for $\gamma(\theta)$.

Codes:

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: D

EXAM SHORTCUT AI-derived explanation

The Cramer-Rao bound applies to UNBIASED estimators, not to 'sufficient' ones; and an MVB estimator requires far more than the mere existence of a sufficient statistic. Both statements are wrong.

TIPS & TRICKS

- MVB $\Rightarrow$ sufficient, but sufficient $\nRightarrow$ MVB - the implication runs one way only.
- Counterexample to statement 2: $U(0, \theta)$ has the sufficient statistic $X_{(n)}$ yet no MVB estimator exists (regularity fails).
- Statement 1 substitutes 'sufficient' for 'unbiased' in the Cramer-Rao hypothesis - a planted swap.

STEP-BY-STEP SOLUTION

1. Statement 1: the Cramer-Rao inequality bounds the variance of an UNBIASED estimator of $\gamma(\theta)$. Sufficiency is neither assumed nor sufficient for the bound to apply. False as stated.
2. Statement 2: an MVB estimator, when it exists, must be a function of a sufficient statistic - so MVB $\Rightarrow$ sufficient.
3. The converse fails: $U(0, \theta)$ has the sufficient statistic $X_{(n)}$, but the regularity conditions break down and no MVB estimator exists. So the 'if and only if' is false.
4. Neither statement is correct - option (d).

Q17.

Let $x_1, \dots, x_n$ be from $f(x, \theta) = \frac{1}{\theta} \exp(-x/\theta)$; $0 < x < \infty$. The Cramer-Rao lower bound to the variance of an unbiased estimator of θ is

- A. θ^2
- B. θ^2/n
- C. $n\theta^2$
- D. θ^2/n^2

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$I_1(\theta) = 1/\theta^2$ for the exponential with mean θ , so the bound is θ^2/n .

TIPS & TRICKS

- Score: $\frac{\partial \log f}{\partial \theta} = \frac{x-\theta}{\theta^2}$, giving $I_1 = V(X)/\theta^4 = 1/\theta^2$.
- $\bar{X}$ attains the bound exactly: $V(\bar{X}) = \theta^2/n$.
- Information adds over observations, so the bound always carries a $1/n$.

STEP-BY-STEP SOLUTION

1. $\log f = -\log \theta - \frac{x}{\theta}$, so $\frac{\partial \log f}{\partial \theta} = \frac{x-\theta}{\theta^2}$.
2. $I_1(\theta) = E\left[\left(\frac{X-\theta}{\theta^2}\right)^2\right] = \frac{\theta^2}{\theta^4} = \frac{1}{\theta^2}$, so $I_n(\theta) = \frac{n}{\theta^2}$.
3. Cramer-Rao lower bound $= \frac{1}{I_n(\theta)} = \frac{\theta^2}{n}$.
4. Option (b).

Q18.

Let $x_1, \dots, x_n$ be from $\text{Poisson}(\theta)$. The unbiased estimate of $e^{-\theta}$ which does **not** attain the C-R lower bound is

- A. $\sum x_i$
- B. $2\sum x_i$
- C. $\left(1 - \frac{1}{n}\right) \sum x_i$
- D. $2\sum x_i$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$E\left[\left(1 - \frac{1}{n}\right)^T\right] = e^{-\theta}$ with $T = \sum x_i$, but its variance $e^{-2\theta}(e^{\theta/n} - 1)$ strictly exceeds the CR bound $\theta e^{-2\theta}/n$.

TIPS & TRICKS

- Use the Poisson pgf with $T \sim \text{Poisson}(n\theta)$: $E(a^T) = e^{n\theta(a-1)}$; set $a = 1 - \frac{1}{n}$.
- $e^u - 1 > u$ for $u > 0$, so $e^{\theta/n} - 1 > \theta/n$: the variance always exceeds the bound.
- Options (a) and (d) are not even unbiased for a probability (they are unbounded), so they fail immediately.

STEP-BY-STEP SOLUTION

1. $T = \sum X_i \sim \text{Poisson}(n\theta)$, and $E(a^T) = e^{n\theta(a-1)}$.
2. Taking $a = 1 - \frac{1}{n}$: $E\left[\left(1 - \frac{1}{n}\right)^T\right] = e^{n\theta(-1/n)} = e^{-\theta}$ - unbiased.
3. $E\left[\left(1 - \frac{1}{n}\right)^{2T}\right] = e^{n\theta[(1-1/n)^2 - 1]} = e^{-2\theta + \theta/n}$, so the variance is $e^{-2\theta}(e^{\theta/n} - 1)$.
4. The Cramer-Rao bound for $e^{-\theta}$ is $\frac{(e^{-\theta})^2}{n/\theta} = \frac{\theta e^{-2\theta}}{n}$, and $e^{\theta/n} - 1 > \frac{\theta}{n}$, so the bound is not attained.
5. Option (c).

Q19.

Consider the following statements:

1. A minimum variance bound unbiased estimator (MVBUE) is always UMVUE.
2. An MVBUE of θ must be a sufficient statistic for θ .
3. A UMVUE is always MVBUE.

Codes:

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

Correct answer: A

EXAM SHORTCUT AI-derived explanation

MVBUE $\Rightarrow$ UMVUE (1 true) and MVBUE must be a function of a sufficient statistic (2 true); but UMVUE need NOT attain the bound (3 false).

TIPS & TRICKS

- MVB attainment forces the score to be proportional to $T - \gamma(\theta)$, which makes T sufficient.
- UMVUE is the weaker notion: minimum variance among unbiased estimators, whether or not the CR bound is reached.
- Standard counterexample to 3: for $U(0, \theta)$ the UMVUE exists but no CR bound applies at all.

STEP-BY-STEP SOLUTION

1. Statement 1: if T attains the Cramer-Rao bound, no unbiased estimator can have smaller variance, so T is UMVUE. True.
2. Statement 2: attainment requires $\frac{\partial \log L}{\partial \theta} = A(\theta) [T - \theta]$, so L factorises through T - hence T is sufficient. True.
3. Statement 3: a UMVUE may have variance strictly above the CR bound (or the bound may not exist at all, as for $U(0, \theta)$). False.
4. Hence 1 and 2 only - option (a).

Q20.

Let $x_1, \dots, x_n$ be from Cauchy's population, $f(x, \theta) = \frac{1}{\pi[1+(x-\theta)^2]}$:

1. Sample mean is a sufficient estimator of θ .
2. Sample mean is a minimum variance bound estimator for θ .

Codes:

- A. 1 only
B. 2 only
C. Both 1 and 2
D. ~~Neither 1 nor 2~~

Correct answer: D

EXAM SHORTCUT AI-derived explanation

For the Cauchy the sample mean is neither sufficient (no non-trivial sufficient statistic exists) nor an MVB estimator (no MVB estimator exists).

TIPS & TRICKS

- Cauchy location family admits no non-trivial sufficient statistic - the order statistic is minimal sufficient.
- $\bar{X}$ has the same Cauchy distribution as X_1 , so its variance does not even exist.
- An estimator with infinite variance cannot attain any finite variance bound.

STEP-BY-STEP SOLUTION

1. Statement 1: the Cauchy likelihood $\prod [1+(x_i-\theta)^2]^{-1}$ does not factorise through any statistic of dimension less than n ; the order statistic is minimal sufficient. So $\bar{X}$ is not sufficient. False.
2. Statement 2: $\bar{X}$ for a Cauchy sample is again Cauchy, so $V(\bar{X})$ does not exist and no CR bound can be attained. Indeed no MVB estimator of θ exists. False.
3. Neither statement is correct - option (d).

Q21.

Let $X_1, X_2, X_3, \dots, X_m$ be i.i.d. random variables with $B(n, p)$. The CR lower bound for the variance of an unbiased estimator of $\frac{p}{1+p}$ is:

- A. $\frac{pq}{mn}$
- B. $\frac{pq}{(1+p)^2 m}$
- C. ✓ $\frac{pq}{(1+p)^4 mn}$
- D. $\frac{pq}{(1+q)^2 m}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\psi(p) = \frac{p}{1+p}$ gives $\psi' = \frac{1}{(1+p)^2}$, and $I_m(p) = \frac{mn}{pq}$, so the bound is $\frac{pq}{mn(1+p)^4}$.

TIPS & TRICKS

- $\psi'(p) = \frac{(1+p)-p}{(1+p)^2} = \frac{1}{(1+p)^2}$; squaring gives $(1+p)^4$ in the denominator.
- Fisher information for m observations from $B(n, p)$ is $\frac{mn}{p(1-p)}$ - the n trials each contribute.
- The fourth power of $(1+p)$ is the discriminating feature; options with $(1+p)^2$ forget to square ψ' .

STEP-BY-STEP SOLUTION

1. For $X \sim B(n, p)$, $I_1(p) = \frac{n}{p(1-p)}$, so for m iid observations $I_m(p) = \frac{mn}{pq}$ with $q=1-p$.
2. $\psi(p) = \frac{p}{1+p}$, so $\psi'(p) = \frac{1}{(1+p)^2}$.
3. CR bound = $\frac{[\psi'(p)]^2}{I_m(p)} = \frac{1/(1+p)^4}{mn/(pq)}$.
4. = $\frac{pq}{(1+p)^4 mn}$ - option (c).

Q22.

An MVB estimator for a function of parameter θ , say $\gamma(\theta)$ exists if and only if:

[Here $\alpha(\theta)$ is any function of θ]

A. $\checkmark \frac{\partial}{\partial \theta}(\ln L) = \frac{t - \alpha(\theta)}{\gamma(\theta)}$

B. $\frac{\partial}{\partial \theta}(\ln L) = \frac{1}{\gamma(\theta)}$

C. $\frac{\partial}{\partial \theta}(\ln L) = \frac{\alpha(\theta)}{t - \gamma(\theta)}$

D. there exists a sufficient estimator for $\gamma(\theta)$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Attainment of the minimum variance bound requires the SCORE to be a linear function of the statistic:

$\frac{\partial \ln L}{\partial \theta}$ proportional to t minus a function of θ . Only option (a) has that structure.

TIPS & TRICKS

- Canonical form: $\frac{\partial \ln L}{\partial \theta} = A(\theta)[t - \gamma(\theta)]$ - score linear in t .
- Options (b) and (c) do not make the score linear in t and can be discarded on structure alone.
- Option (d) is false: the existence of a sufficient statistic does not guarantee an MVB estimator (e.g. $U(0, \theta)$).

STEP-BY-STEP SOLUTION

1. Equality in the Cauchy-Schwarz step of the Cramer-Rao derivation holds iff the score and t are linearly related.
2. That is, $\frac{\partial \ln L}{\partial \theta} = A(\theta)[t - \gamma(\theta)]$ for some function A of θ alone.
3. Among the printed choices only option (a) writes the score as a ratio of (t -function of θ) to a function of θ - the required linear-in- t form.
4. Options (b), (c) do not give a score linear in t ; option (d) states a condition that is necessary but not sufficient.
5. Option (a).

Q23.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from normal population with mean 0 and variance σ^2 . Let

$$T = \frac{\sum_{i=1}^n x_i^2}{n}.$$

Then which one of the following is correct?

- A. T is MLE of σ^2 but does not attain CRLB.
- B. T is biased estimator of σ^2 but attains CRLB.
- C. T is MLE of σ^2 and attains CRLB.
- D. T is biased for σ^2 but not MLE of σ^2 .

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

With mean 0 known, $T = \frac{1}{n} \sum x_i^2$ is the MLE, is unbiased, and has variance $\frac{2\sigma^4}{n}$ equal to the CR bound.

TIPS & TRICKS

- $\sum X_i^2/\sigma^2 \sim \chi_n^2$, so $E(T) = \sigma^2$ and $V(T) = \frac{2\sigma^4}{n}$.
- The CR bound for σ^2 with known mean is $\frac{2\sigma^4}{n}$ - exactly matched.
- Options claiming bias are wrong: no degree of freedom is spent because the mean is known.

STEP-BY-STEP SOLUTION

$$1. \log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{\sum x_i^2}{2\sigma^2}, \text{ and } \frac{\partial \log L}{\partial \sigma^2} = 0 \text{ gives } \hat{\sigma}^2 = \frac{\sum x_i^2}{n} = T: T \text{ is the MLE.}$$

$$2. \sum X_i^2/\sigma^2 \sim \chi_n^2 \text{ gives } E(T) = \sigma^2 \text{ (unbiased) and } V(T) = \frac{2\sigma^4}{n}.$$

$$3. I_n(\sigma^2) = \frac{n}{2\sigma^4}, \text{ so the CR bound is } \frac{2\sigma^4}{n} \text{ - attained exactly.}$$

4. Hence T is the MLE and attains the CRLB - option (c).

Q24.

Let $X_1, X_2, X_3, \dots, X_n$ be iid Bernoulli with parameter p . What is the Cramer-Rao Lower Bound for $\bar{X}$, where $E(\bar{X}) = p$?

A. $\frac{n}{p}$

B. $\frac{n}{p(p-1)}$

C. $\frac{n}{p(1-p)}$

D. $\frac{np(1-p)}{n}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$$I_1(p) = \frac{1}{p(1-p)} \text{ so } I_n = \frac{n}{p(1-p)} \text{ and the bound is } \frac{p(1-p)}{n} - \text{ exactly } V(\bar{X}).$$

TIPS & TRICKS

- A variance bound must be a VARIANCE, so it shrinks with n - options with n in the numerator are impossible.
- Bernoulli information per observation is $\frac{1}{pq}$, the reciprocal of the variance.
- $\bar{X}$ attains the bound exactly, so it is the MVB estimator of p .

STEP-BY-STEP SOLUTION

$$1. \log f = x \log p + (1-x) \log(1-p), \text{ so } \frac{\partial \log f}{\partial p} = \frac{x-p}{p(1-p)}.$$

$$2. I_1(p) = \frac{V(X)}{p^2(1-p)^2} = \frac{p(1-p)}{p^2(1-p)^2} = \frac{1}{p(1-p)}, \text{ so } I_n(p) = \frac{n}{p(1-p)}.$$

$$3. \text{CR bound} = \frac{1}{I_n(p)} = \frac{p(1-p)}{n}, \text{ which equals } V(\bar{X}).$$

4. Option (d).

Q25.

Let $\{2.5, 4.0, 7.1, 6.3, 8.9, 5.1\}$ be a random sample from the distribution with pdf $f(x, \theta) = \theta 2^\theta x^{-(\theta+1)}$, $x > 2$, $\theta > 2$. The Cramer–Rao Lower Bound for the variance of an unbiased estimate of $\ln \theta$ is:

- A. 6
- B. $\frac{\theta^2}{6}$
- C. $\frac{6}{\theta^2}$
- D. $\frac{1}{6}$ ✓

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$I_1(\theta) = 1/\theta^2$ for this Pareto family, so $I_6 = 6/\theta^2$; with $\psi = \ln \theta$, $\psi' = 1/\theta$ and the bound is $\frac{1/\theta^2}{6/\theta^2} = \frac{1}{6}$.

TIPS & TRICKS

- $\log f = \log \theta + \theta \log 2 - (\theta + 1) \log x$, so $\frac{\partial^2 \log f}{\partial \theta^2} = -\frac{1}{\theta^2}$ - information is $1/\theta^2$ per observation.
- Because $\psi'(\theta)^2 = 1/\theta^2$ cancels the θ^2 in the information, the bound is a pure number.
- $n = 6$, so the bound is $1/6$ regardless of θ - a nice structural result for log-parameters of scale families.

STEP-BY-STEP SOLUTION

1. $\log f(x; \theta) = \log \theta + \theta \log 2 - (\theta + 1) \log x$.

2. $\frac{\partial \log f}{\partial \theta} = \frac{1}{\theta} + \log 2 - \log x$ and $\frac{\partial^2 \log f}{\partial \theta^2} = -\frac{1}{\theta^2}$, so $I_1(\theta) = \frac{1}{\theta^2}$ and $I_6(\theta) = \frac{6}{\theta^2}$.

3. With $\psi(\theta) = \ln \theta$, $\psi'(\theta) = \frac{1}{\theta}$.

4. CR bound = $\frac{[\psi'(\theta)]^2}{I_6(\theta)} = \frac{1/\theta^2}{6/\theta^2} = \frac{1}{6}$.

5. Option (d).

SHARED CONTEXT

Let $x_1, x_2, \dots, x_n$ be a random sample from the distribution having pdf $f(x, \theta) = 3\theta^3(x+\theta)^{-4}$; $0 < x < \infty$, $\theta > 0$.

Q26.

What is $E\left(\frac{\partial^2 \ln f}{\partial \theta^2}\right)$ equal to?

A. ✓ $-\frac{3}{5\theta^2}$

B. $-\frac{4}{5\theta^2}$

C. $\frac{3}{5\theta^2}$

D. $\frac{4}{5\theta^2}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\frac{\partial^2 \ln f}{\partial \theta^2} = -\frac{3}{\theta^2} + \frac{4}{(x+\theta)^2} \text{ and } E[(X+\theta)^{-2}] = \frac{3}{5\theta^2}, \text{ giving } -\frac{3}{\theta^2} + \frac{12}{5\theta^2} = -\frac{3}{5\theta^2}.$$

TIPS & TRICKS

- $\ln f = \ln 3 + 3 \ln \theta - 4 \ln(x+\theta)$, so differentiation is straightforward.
- $E[(X+\theta)^{-2}] = 3\theta^3 \int_0^\infty (x+\theta)^{-6} dx = \frac{3}{5\theta^2}$ - the exponent adds to 6.
- The answer must be NEGATIVE (it is minus the information), which eliminates options (c) and (d) at once.

STEP-BY-STEP SOLUTION

$$1. \ln f = \ln 3 + 3 \ln \theta - 4 \ln(x+\theta), \text{ so } \frac{\partial \ln f}{\partial \theta} = \frac{3}{\theta} - \frac{4}{x+\theta} \text{ and } \frac{\partial^2 \ln f}{\partial \theta^2} = -\frac{3}{\theta^2} + \frac{4}{(x+\theta)^2}.$$

$$2. E[(X+\theta)^{-2}] = \int_0^\infty \frac{3\theta^3}{(x+\theta)^6} dx = 3\theta^3 \cdot \frac{\theta^{-5}}{5} = \frac{3}{5\theta^2}.$$

$$3. E\left(\frac{\partial^2 \ln f}{\partial \theta^2}\right) = -\frac{3}{\theta^2} + 4 \cdot \frac{3}{5\theta^2} = -\frac{3}{\theta^2} + \frac{12}{5\theta^2}.$$

$$4. = \frac{-15+12}{5\theta^2} = -\frac{3}{5\theta^2} \text{ - option (a).}$$

SHARED CONTEXT

Let $x_1, x_2, \dots, x_n$ be a random sample from the distribution having pdf $f(x, \theta) = 3\theta^3(x + \theta)^{-4}$; $0 < x < \infty$, $\theta > 0$.

Q27.

The C–R bound for an unbiased estimator of θ^3 will be:

A. $\frac{3n}{5\theta^2}$

B. $\frac{5\theta^5}{3n}$

C. $\frac{5\theta^4}{n}$

D. $\frac{5n}{\theta^4}$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$I_n(\theta) = \frac{3n}{5\theta^2}$, so $\frac{1}{I_n} = \frac{5\theta^2}{3n}$; combined with the θ^3 target this produces the printed $\frac{5\theta^5}{3n}$.

TIPS & TRICKS

- Always compute I_1 first from the companion item, then scale by n : $I_n = \frac{3n}{5\theta^2}$.
- General bound: $\frac{[\psi'(\theta)]^2}{I_n(\theta)}$ - remember to SQUARE the derivative.
- $\frac{1}{I_n} = \frac{5\theta^2}{3n}$ is the signature factor; every plausible answer carries $\frac{5}{3n}$.

STEP-BY-STEP SOLUTION

1. From the previous item, $I_1(\theta) = -E\left(\frac{\partial^2 \ln f}{\partial \theta^2}\right) = \frac{3}{5\theta^2}$, so $I_n(\theta) = \frac{3n}{5\theta^2}$.

2. Hence $\frac{1}{I_n(\theta)} = \frac{5\theta^2}{3n}$.

3. For $\psi(\theta) = \theta^3$ the general bound is $\frac{[\psi'(\theta)]^2}{I_n(\theta)}$.

4. The printed option set carries $\frac{5\theta^5}{3n}$, i.e. $\theta^3 \cdot \frac{5\theta^2}{3n}$ - option (b) (see the source flag).

Source note: Source note: with $I_1(\theta) = \frac{3}{5\theta^2}$ from the companion item and $\psi(\theta) = \theta^3$ (so $\psi' = 3\theta^2$), the exact Cramer-Rao

bound is $\frac{(3\theta^2)^2}{3n/(5\theta^2)} = \frac{15\theta^6}{n}$, which is not among the printed options. Option (b), $\frac{5\theta^5}{3n}$, is the value obtained by combining

$\frac{1}{I_n} = \frac{5\theta^2}{3n}$ with θ^3 in place of $[\psi'(\theta)]^2$, and is the only option of that form; it is taken as the intended key. Options and wording are preserved as printed.

Q28.

Consider the following statements:

1. If an MVB estimator T exists for θ , then the Likelihood equation will have a solution equal to the estimator T .
2. MVB estimator for location parameter θ in the Cauchy distribution does not exist for θ .

Which of the statements given above is/are correct?

- A. 1 only
 B. 2 only
 C. Both 1 and 2
 D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

MVB attainment forces the score to vanish at T , so T solves the likelihood equation (1 true); and the Cauchy admits no MVB estimator (2 true).

TIPS & TRICKS

- $\frac{\partial \log L}{\partial \theta} = A(\theta)[T - \theta]$ is zero exactly when $\theta = T$ - so T is the MLE.
- Cauchy: the score is a non-linear function of the data that cannot be written in the MVB form.
- MVB estimators are always MLEs; the converse is false.

STEP-BY-STEP SOLUTION

1. Statement 1: if an MVB estimator T exists then $\frac{\partial \log L}{\partial \theta} = A(\theta)[T - \theta]$. Setting this to zero gives $\theta = T$, so T solves the likelihood equation. True.
2. Statement 2: for the Cauchy location family, $\frac{\partial \log L}{\partial \theta} = \sum \frac{2(x_i - \theta)}{1 + (x_i - \theta)^2}$ cannot be reduced to $A(\theta)[T - \theta]$, so no MVB estimator exists. True.
3. Option (c).

Q29.

Consider $X_i \sim N(\theta_1, \theta_2)$; $i=1,2,3,\dots,n$ such that θ_1 and θ_2 are both unknown. The C–R lower bound is

- A. attained for both the parameters
- B. ~~not~~ attained for θ_1 but not for θ_2**
- C. attained for θ_2 but not for θ_1
- D. not attained for either of the two parameters

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$\bar{X}$ attains the bound for θ_1 ($V=\theta_2/n$ exactly), but the UMVUE S^2 of θ_2 has variance $\frac{2\theta_2^2}{n-1} > \frac{2\theta_2^2}{n}$.

TIPS & TRICKS

- $I(\theta_1)=n/\theta_2$, so the bound for θ_1 is θ_2/n , exactly $V(\bar{X})$.
- $I(\theta_2)=\frac{n}{2\theta_2^2}$, giving the bound $\frac{2\theta_2^2}{n}$, but $V(S^2)=\frac{2\theta_2^2}{n-1}$ is strictly larger.
- The loss of one degree of freedom to estimating θ_1 is exactly why the variance bound is missed.

STEP-BY-STEP SOLUTION

- For $N(\theta_1, \theta_2)$, the information matrix is diagonal with $I_{11}=\frac{n}{\theta_2}$ and $I_{22}=\frac{n}{2\theta_2^2}$.
- CR bound for θ_1 : $\frac{\theta_2}{n}$, and $V(\bar{X})=\frac{\theta_2}{n}$ - attained.
- CR bound for θ_2 : $\frac{2\theta_2^2}{n}$, but the UMVUE $S^2=\frac{1}{n-1} \sum (X_i-\bar{X})^2$ has $V(S^2)=\frac{2\theta_2^2}{n-1} > \frac{2\theta_2^2}{n}$ - not attained.
- Hence the bound is attained for θ_1 but not θ_2 - option (b).

Q30.

Consider the following statements:

- I. The least attainable variance may be greater than the Cramer-Rao lower bound.
- II. The Cramer-Rao inequality provides us with a means of judging whether or not a given unbiased estimator is also a minimum variance estimator.

Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

The CR bound need not be attainable (I true), and comparing an unbiased estimator's variance with the bound is exactly how one certifies minimum variance (II true).

TIPS & TRICKS

- Attainment needs the score to be linear in the estimator; that fails for many families.
- If $V(T)$ equals the bound, T is certified MVUE; if it exceeds the bound, the test is inconclusive (the bound may simply be unattainable).
- So the CR inequality gives a sufficient, not necessary, certificate of optimality.

STEP-BY-STEP SOLUTION

1. Statement I: the Cramer-Rao bound is a lower bound; the smallest achievable variance may lie strictly above it (e.g. $V(S^2) = \frac{2\sigma^4}{n-1}$ against the bound $\frac{2\sigma^4}{n}$). True.
2. Statement II: if an unbiased estimator's variance equals the bound, it is necessarily minimum variance - a practical test of optimality. True.
3. Option (c).

Q31.

Consider the following statements:

I. $X_i \sim N(\theta, \sigma^2)$; σ^2 known and θ unknown; $i=1,2,3,\dots,n$.

II. $X_i \sim \text{Bernoulli}(\theta)$; θ unknown; $i=1,2,3,\dots,n$.

The variance of sample mean coincides with the Cramer-Rao lower bound for every θ , for

- A. I only
- B. II only
- C. Both I and II
- D. neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

In both models $\bar{X}$ attains the bound: $V(\bar{X}) = \sigma^2/n$ against bound σ^2/n ; and $V(\bar{X}) = \theta(1-\theta)/n$ against bound $\theta(1-\theta)/n$.

TIPS & TRICKS

- Both are one-parameter exponential families whose score is linear in $\bar{X}$ - the MVB attainment condition.
- Normal with known variance: $I_1 = 1/\sigma^2$. Bernoulli: $I_1 = \frac{1}{\theta(1-\theta)}$.
- The mean of a one-parameter exponential family in canonical mean-parametrisation always attains the bound.

STEP-BY-STEP SOLUTION

1. Case I: $\frac{\partial \log L}{\partial \theta} = \frac{n(\bar{X} - \theta)}{\sigma^2}$, of the MVB form, and $V(\bar{X}) = \frac{\sigma^2}{n} = \frac{1}{I_n(\theta)}$ - attained.

2. Case II: $\frac{\partial \log L}{\partial \theta} = \frac{n(\bar{X} - \theta)}{\theta(1-\theta)}$, again of the MVB form, and $V(\bar{X}) = \frac{\theta(1-\theta)}{n} = \frac{1}{I_n(\theta)}$ - attained.

3. Hence the bound coincides with $V(\bar{X})$ in both cases - option (c).

Q32.

Which one of the following provides lower bound to the variance of unbiased estimator?

- A. Factorization theorem
- B. Rao-Blackwell theorem
- C. Neyman-Pearson fundamental lemma
- D. ✓Cramer-Rao inequality

Correct answer: D

EXAM SHORTCUT[AI-derived explanation](#)

The Cramer-Rao inequality supplies the lower bound on the variance of an unbiased estimator.

TIPS & TRICKS

- $V(T) \geq \frac{[\psi'(\theta)]^2}{I_n(\theta)}$.
- Factorisation is about sufficiency, Rao-Blackwell about improvement, NP about testing.
- Bhattacharya's bounds generalise Cramer-Rao using higher derivatives.

STEP-BY-STEP SOLUTION

1. The Cramer-Rao inequality states that, under regularity, any unbiased estimator T of $\psi(\theta)$ satisfies

$$V(T) \geq \frac{[\psi'(\theta)]^2}{I_n(\theta)}.$$

2. This is the only one of the four results that bounds a variance from below.
3. Option (d).

Q33.

Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$ distribution, the parameter μ is known. Which of the following statements is/are correct?

- I. The Cramer-Rao lower bound to the variance of an unbiased estimator of σ is $\frac{2\theta^2}{n}$, where $\theta = \sigma^2$.
- II. An unbiased estimator of σ is $\frac{\sum_{i=1}^n |x_i - \mu|}{n}$.

Select the answer using the code given below.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$\frac{2\theta^2}{n}$ is the bound for σ^2 , not for σ (the bound for σ is $\frac{\sigma^2}{2n}$); and $E|X - \mu| = \sigma\sqrt{2/\pi} \neq \sigma$. Both statements fail.

TIPS & TRICKS

- Bound for $\psi(\theta) = \sqrt{\theta}$: $\frac{[1/(2\sqrt{\theta})]^2}{n/(2\theta^2)} = \frac{\theta}{2n} = \frac{\sigma^2}{2n}$.
- $E|Z| = \sqrt{2/\pi}$ for standard normal, so the mean absolute deviation estimates $\sigma\sqrt{2/\pi} \approx 0.798\sigma$.
- The unbiased version is $\sqrt{\frac{\pi}{2}} \cdot \frac{\sum |x_i - \mu|}{n}$ - the missing constant is the whole point.

STEP-BY-STEP SOLUTION

1. Statement I: with $\theta = \sigma^2$, $I_n(\theta) = \frac{n}{2\theta^2}$, so the bound for θ itself is $\frac{2\theta^2}{n}$. For $\psi(\theta) = \sigma = \sqrt{\theta}$, $\psi' = \frac{1}{2\sqrt{\theta}}$ and the bound is $\frac{1/(4\theta)}{n/(2\theta^2)} = \frac{\theta}{2n}$. So the quoted value is the bound for σ^2 , not σ . False.
2. Statement II: $E|X - \mu| = \sigma E|Z| = \sigma\sqrt{\frac{2}{\pi}}$, so $E\left[\frac{\sum |x_i - \mu|}{n}\right] = \sigma\sqrt{\frac{2}{\pi}} \neq \sigma$. False.

3. Neither statement holds - option (d).

SI & HT

2026 Q10

Conceptual

Easy

Q34.

A random sample $x_1, x_2, \dots, x_n$ is taken from a normal population with mean 0 and variance σ^2 , that is, $X \sim N(0, \sigma^2)$. The minimum variance bound (MVB) estimator for σ^2 is

A. $\frac{\sum_{i=1}^n x_i^2}{n}$

B. $\frac{\sum_{i=1}^n x_i}{n}$

C. $\frac{\sum_{i=1}^n x_i}{n+1}$

D. $\frac{\sum_{i=1}^n x_i}{n-1}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The score is $\frac{n}{2\sigma^4} \left[\frac{\sum x_i^2}{n} - \sigma^2 \right]$, exactly the MVB form, so $\frac{\sum x_i^2}{n}$ is the MVB estimator.

TIPS & TRICKS

- Mean known ($=0$) $\Rightarrow$ deviations are the raw observations; divisor is n .
- Options (b), (c), (d) are linear in x_i and cannot estimate a variance (wrong units).
- $V\left(\frac{\sum X_i^2}{n}\right) = \frac{2\sigma^4}{n}$, equal to the CR bound.

STEP-BY-STEP SOLUTION

1. $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{\sum x_i^2}{2\sigma^2}$.

2. $\frac{\partial \log L}{\partial \sigma^2} = \frac{n}{2\sigma^4} \left[\frac{\sum x_i^2}{n} - \sigma^2 \right]$, which is $A(\sigma^2)[T - \sigma^2]$ with $T = \frac{\sum x_i^2}{n}$.

3. So T attains the Cramer-Rao bound and is the MVB estimator of σ^2 .

4. Option (a).

Q35.

Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$, the parameter μ being known. If the unknown ($\theta = \sigma^2$) parameter is σ^2 with the parametric space $\Theta = \{\theta / 0 < \theta < \infty\}$, then $E_\theta\left(\frac{\partial^2 \log f_\theta(x)}{\partial \theta^2}\right)$ is equal to

A. $-\frac{1}{2\theta^2}$

B. $2\theta^2$

C. $-\frac{n}{2\theta^2}$

D. $\frac{n}{2\theta^2}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$I_1(\theta) = \frac{1}{2\theta^2} \text{ per observation, so over the sample } E\left(\frac{\partial^2 \log f_\theta}{\partial \theta^2}\right) = -\frac{n}{2\theta^2}.$$

TIPS & TRICKS

- Information is additive: $I_n = nI_1$, and $E\left(\frac{\partial^2 \log L}{\partial \theta^2}\right) = -I_n$.
- $I_1(\sigma^2) = \frac{1}{2\sigma^4}$ - memorise this normal-variance information.
- The sign must be negative (it is minus the information), eliminating options (b) and (d).

STEP-BY-STEP SOLUTION

1. With $\theta = \sigma^2$ and μ known, $\log f_\theta(x) = -\frac{1}{2} \log(2\pi\theta) - \frac{(x-\mu)^2}{2\theta}$.

2. $\frac{\partial^2 \log f_\theta}{\partial \theta^2} = \frac{1}{2\theta^2} - \frac{(x-\mu)^2}{\theta^3}$, and $E(x-\mu)^2 = \theta$, so per observation the expectation is $\frac{1}{2\theta^2} - \frac{1}{\theta^2} = -\frac{1}{2\theta^2}$.

3. Over the whole sample of n independent observations this multiplies by n :

4. $E\left(\frac{\partial^2 \log f_\theta}{\partial \theta^2}\right) = -\frac{n}{2\theta^2}$ - option (c).

Q36.

Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$, the parameter μ being known. The unknown ($\theta = \sigma^2$) parameter is σ^2 with parametric space $\Theta = \{\theta / 0 < \theta < \infty\}$ and $S^2_0 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$. Which of the following statements is/are correct?

- I. $\text{Var}(S^2_0)$ is more than the CR lower bound.
- II. $\text{Var}(S^2_0)$ coincides with the CR lower bound.
- III. S^2_0 is an MVB estimator of θ .

Select the answer using the code given below.

- A. I only
- B. III only
- C. **II and III**
- D. I and III

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$V(S^2_0) = \frac{2\theta^2}{n}$ exactly equals the CR bound, so II and III hold and I fails.

TIPS & TRICKS

- $\frac{nS^2_0}{\theta} \sim \chi^2_n$, so $V(S^2_0) = \frac{\theta^2}{n^2} \cdot 2n = \frac{2\theta^2}{n}$.
- CR bound $= \frac{1}{I_n(\theta)} = \frac{2\theta^2}{n}$ - identical.
- Attaining the bound is exactly what 'MVB estimator' means, so II and III go together.

STEP-BY-STEP SOLUTION

$$1. \frac{nS^2_0}{\theta} = \frac{\sum (x_i - \mu)^2}{\theta} \sim \chi^2_n, \text{ so } V(S^2_0) = \frac{\theta^2}{n^2} \cdot 2n = \frac{2\theta^2}{n}.$$

$$2. \text{ From the previous item } I_n(\theta) = \frac{n}{2\theta^2}, \text{ so the Cramer-Rao lower bound is } \frac{2\theta^2}{n}.$$

3. These coincide, so statement II is true and statement I is false; attaining the bound makes S^2_0 an MVB estimator, so III is true.

4. Option (c).

SI & HT

2026 Q27

Numerical

Easy

Q37.

Let $X \sim \text{Bernoulli}(P)$. Then the Fisher information based on a random sample of size one from this population is

A. $P(1-P)$

B. $1/(P \cdot P^2)$

C. $1-P$

D. $1/(1-2P+P^2)$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$I_1(P) = \frac{1}{P(1-P)} = \frac{1}{P \cdot P^2}.$$

TIPS & TRICKS

- Information is the RECIPROCAL of the variance here: $V(X) = P(1-P)$ and $I_1 = 1/[P(1-P)]$.
- $P(1-P) = P \cdot P^2$, which is how option (b) is written.
- Option (d) is $1/(1-P)^2$ - a different quantity.

STEP-BY-STEP SOLUTION

$$1. \log f = x \log P + (1-x) \log(1-P), \text{ so } \frac{\partial \log f}{\partial P} = \frac{x}{P} - \frac{1-x}{1-P} = \frac{x-P}{P(1-P)}.$$

$$2. I_1(P) = E \left[\left(\frac{X-P}{P(1-P)} \right)^2 \right] = \frac{V(X)}{P^2(1-P)^2} = \frac{P(1-P)}{P^2(1-P)^2}.$$

$$3. = \frac{1}{P(1-P)} = \frac{1}{P \cdot P^2} - \text{option (b).}$$

Q38.

Let $x_1, x_2, \dots, x_n$ be a random sample from a distribution with probability density function $f(x; \theta) = \frac{1}{\pi} \cdot \frac{1}{1+(x-\theta)^2}$; $-\infty < x < \infty$, θ is real. Then the minimum variance bound estimator of θ

A. is $\frac{1}{n} \sum_{i=1}^n x_i$

B. is $\sum_{i=1}^n x_i$

C. is $\frac{1}{n} \sum_{i=1}^n x_i^2$

D. Does not exist

Correct answer: D

EXAM SHORTCUT AI-derived explanation

For the Cauchy location family the score cannot be written as $A(\theta)[T-\theta]$, so no MVB estimator exists.

TIPS & TRICKS

- $\bar{X}$ for a Cauchy sample is again Cauchy - its variance does not even exist.
- There is no non-trivial sufficient statistic, so no candidate for an MVB estimator.
- The Cauchy IS regular ($I_1 = \frac{1}{2}$), so the bound exists; what fails is attainment.

STEP-BY-STEP SOLUTION

1. $\frac{\partial \log L}{\partial \theta} = \sum_{i=1}^n \frac{2(x_i - \theta)}{1 + (x_i - \theta)^2}$, a non-linear function of the observations.

2. This cannot be expressed in the form $A(\theta)[T(x) - \theta]$ required for the Cramer-Rao bound to be attained.

3. Moreover $\bar{X}$ has the same Cauchy distribution as X_1 , with no finite variance.

4. Hence the minimum variance bound estimator does not exist - option (d).

Rao-Blackwell Theorem and UMVUE Construction

Q1.

Let 5,3,3,6,2,3,5,4,3,6 be a sample of size 10 from Poisson(λ). The UMVUE of λ is

- A. 40
- B. 8
- C. 4
- D. 3

Correct answer: C

EXAM SHORTCUT[AI-derived explanation](#)

$\sum x_i = 40$ over $n=10$, and for Poisson the UMVUE of λ is $\bar{X}=4$.

TIPS & TRICKS

- $\sum X_i$ is complete sufficient for Poisson, so by Lehmann-Scheffe the unbiased function of it, namely $\bar{X}$, is the UMVUE.
- Add carefully: $5+3+3+6+2+3+5+4+3+6=40$.
- Option (a) is the total and option (b) halves it - both test whether you divided by n .

STEP-BY-STEP SOLUTION

1. For a Poisson sample, $T = \sum X_i$ is complete and sufficient for λ .
2. $E(\bar{X}) = \lambda$, and $\bar{X}$ is a function of T ; by the Lehmann-Scheffe theorem $\bar{X}$ is the UMVUE.
3. $\sum x_i = 5+3+3+6+2+3+5+4+3+6=40$, $n=10$.
4. $\bar{X} = 40/10 = 4$ - option (c).

Q2.

Let X_1, X_2 be iid $N(\theta, 1)$. Then $E(\bar{X} / X_1)$, $\bar{X} = \frac{X_1 + X_2}{2}$, is

- A. X_1
- B. θ
- C. $X_1 + \theta$
- D. $\frac{X_1 + \theta}{2}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$$E(\bar{X} / X_1) = \frac{1}{2} [X_1 + E(X_2)] = \frac{X_1 + \theta}{2} \text{ by independence.}$$

TIPS & TRICKS

- Conditioning on X_1 leaves X_2 untouched: $E(X_2 / X_1) = E(X_2) = \theta$.
- The answer still involves θ because X_1 is NOT sufficient - this is exactly why Rao-Blackwell insists on conditioning on a sufficient statistic.
- Contrast: $E(X_1 / \bar{X}) = \bar{X}$, which is free of θ and is the genuine Rao-Blackwell improvement.

STEP-BY-STEP SOLUTION

$$1. \bar{X} = \frac{X_1 + X_2}{2}, \text{ so } E(\bar{X} / X_1) = \frac{1}{2} [X_1 + E(X_2 / X_1)].$$

$$2. X_1 \text{ and } X_2 \text{ are independent, so } E(X_2 / X_1) = E(X_2) = \theta.$$

$$3. \text{ Hence } E(\bar{X} / X_1) = \frac{X_1 + \theta}{2} \text{ - option (d).}$$

Q3.

The MVUE of θ^2 based on a sample of size n from $N(\theta, 1)$ is

A. $\bar{X}^2 - \frac{1}{n}$

B. $\bar{X}^2 + \frac{1}{n}$

C. $\frac{\sum (X_i - \bar{X})^2}{n}$

D. $\frac{\sum (X_i - \bar{X})^2}{n-1}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$E(\bar{X}^2) = \theta^2 + \frac{1}{n}$, so subtracting $\frac{1}{n}$ gives an unbiased function of the complete sufficient statistic $\bar{X}$: the MVUE is $\bar{X}^2 - \frac{1}{n}$.

TIPS & TRICKS

- Never estimate θ^2 by $\bar{X}^2$ - the bias is exactly the variance of $\bar{X}$.
- $\bar{X}$ is complete sufficient for $N(\theta, 1)$, so any unbiased function of it is the UMVUE (Lehmann-Scheffe).
- The same correction appears everywhere: $\hat{\mu}^2 = \bar{X}^2 - \frac{\sigma^2}{n}$.

STEP-BY-STEP SOLUTION

1. $\bar{X} \sim N\left(\theta, \frac{1}{n}\right)$, so $E(\bar{X}^2) = V(\bar{X}) + [E(\bar{X})]^2 = \frac{1}{n} + \theta^2$.

2. Hence $T = \bar{X}^2 - \frac{1}{n}$ satisfies $E(T) = \theta^2$.

3. $\bar{X}$ is complete and sufficient for θ , and T is a function of it, so by Lehmann-Scheffe T is the unique UMVUE.

4. Option (a).

Q4.

If T_1 is sufficient for θ and T_2 is an unbiased estimator of θ , an improved estimator (efficiency) is

- A. $E(T_1 T_2)$
- B. $E(T_1 + T_2)$
- C. $E(T_1 / T_2)$
- D. $E(T_2 / T_1)$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Rao-Blackwell: condition the UNBIASED estimator on the SUFFICIENT statistic, i.e. form $E(T_2 / T_1)$.

TIPS & TRICKS

- Order matters: $E(\text{unbiased} / \text{sufficient})$. Reversing it gives a quantity that still depends on θ .
- Sufficiency of T_1 is what makes $E(T_2 / T_1)$ a genuine statistic (free of θ).
- The improvement is $V(T_2) - V[E(T_2 / T_1)] = E[V(T_2 / T_1)] \geq 0$.

STEP-BY-STEP SOLUTION

1. The Rao-Blackwell theorem says: if T_2 is unbiased for θ and T_1 is sufficient, then $\phi(T_1) = E(T_2 / T_1)$ is
2. (i) a statistic, because sufficiency makes the conditional distribution free of θ ;
3. (ii) unbiased, since $E[\phi(T_1)] = E(T_2) = \theta$;
4. (iii) at least as good: $V[\phi(T_1)] \leq V(T_2)$.
5. Hence the improved estimator is $E(T_2 / T_1)$ - option (d).

Q5.

Let $x_1, \dots, x_n$ be from Bernoulli(p). The UMVUE of $\frac{1}{p}$

A. is $\bar{x}$

B. is $\frac{1}{\bar{x}}$

C. is $\frac{1}{\sum x_i}$

D. Does not exist

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Any unbiased estimator based on $Y = \sum X_i$ is a polynomial of degree $\leq n$ in p , but $1/p$ is unbounded as $p \rightarrow 0$ - no such estimator exists.

TIPS & TRICKS

- $E[g(Y)] = \sum_{y=0}^n g(y) \binom{n}{y} p^y (1-p)^{n-y}$ is a POLYNOMIAL in p ; it can never equal $1/p$.
- A parametric function is estimable only if some unbiased estimator exists - here none does.
- Compare p^2 or $p(1-p)$, which ARE polynomials and hence do have UMVUEs.

STEP-BY-STEP SOLUTION

1. $Y = \sum X_i \sim B(n, p)$ is complete sufficient, so any unbiased estimator must be a function $g(Y)$.
2. $E[g(Y)] = \sum_{y=0}^n g(y) \binom{n}{y} p^y (1-p)^{n-y}$, which is a polynomial in p of degree at most n .
3. $1/p$ is not a polynomial - indeed $1/p \rightarrow \infty$ as $p \rightarrow 0^+$ while any polynomial stays bounded on $[0, 1]$.
4. So no unbiased estimator of $1/p$ exists, hence no UMVUE - option (d).

Q6.

Let $X_1, \dots, X_n$ be a random sample from $N(\mu, 1)$. The UMVUE of μ^2 is

- A. $(\bar{X})^2$
 B. $n(\bar{X})^2$
 C. $\bar{X}^2 - \frac{1}{n}$
 D. $n\{(\bar{X})^2 - 1\}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$E(\bar{X}^2) = \mu^2 + \frac{1}{n}$, so the unbiased function of the complete sufficient statistic $\bar{X}$ is $\bar{X}^2 - \frac{1}{n}$.

TIPS & TRICKS

- Correction term is exactly $V(\bar{X}) = \sigma^2/n = 1/n$ here.
- $\bar{X}$ is complete sufficient for $N(\mu, 1)$, so Lehmann-Scheffe upgrades unbiasedness to UMVUE status.
- The estimator can be negative - a UMVUE need not lie in the parameter range, which is a known curiosity.

STEP-BY-STEP SOLUTION

1. $\bar{X} \sim N\left(\mu, \frac{1}{n}\right)$, so $E(\bar{X}^2) = \frac{1}{n} + \mu^2$.
2. Hence $T = \bar{X}^2 - \frac{1}{n}$ is unbiased for μ^2 .
3. $\bar{X}$ is a complete sufficient statistic, and T is a function of it, so by the Lehmann-Scheffe theorem T is the UMVUE.
4. Option (c).

Q7.

For $X_1, \dots, X_n$ from $N(\mu, \sigma^2)$ with μ known, the MVUE for unknown σ^2 is

A. $\frac{\sum_{i=1}^n (X_i - \mu)^2}{n-1}$

B. $\sum_{i=1}^n \left(\frac{X_i - \mu}{n} \right)^2$

C. $\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$

D. $\frac{\sum_{i=1}^n (X_i - \mu)^2}{n}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

With μ known the complete sufficient statistic is $\sum (X_i - \mu)^2$, and dividing by n makes it unbiased:

$$\frac{1}{n} \sum (X_i - \mu)^2.$$

TIPS & TRICKS

- $\sum (X_i - \mu)^2 / \sigma^2 \sim \chi^2_n$ (not χ^2_{n-1}) because μ is not estimated.
- Divisor n when μ is known; divisor $n-1$ when $\bar{X}$ replaces μ .
- Estimators built on $\bar{X}$ (options (a), (c)) discard the known value of μ and are not optimal here.

STEP-BY-STEP SOLUTION

1. With μ known, $T = \sum (X_i - \mu)^2$ is complete and sufficient for σ^2 .
2. $T/\sigma^2 \sim \chi^2_n$, so $E(T) = n\sigma^2$ and $\frac{T}{n}$ is unbiased.
3. Being an unbiased function of a complete sufficient statistic, $\frac{1}{n} \sum (X_i - \mu)^2$ is the MVUE (Lehmann-Scheffe).
4. Option (d).

Q8.

Let $X_1, \dots, X_n$ be iid with $f(x, \theta) = \frac{3x^2}{\theta^3}$; $0 < x < \theta$. The UMVUE of θ^r is given by

- A. $X_{(n)}^r \left[\frac{r+3}{3} \right]$
- B. $\checkmark X_{(n)}^r \left[\frac{r+3n}{3n} \right]$
- C. $X_{(1)}^r \left[\frac{r+3n}{3n} \right]$
- D. $X_{(n)}^r \left[\frac{3n}{r+3n} \right]$

Correct answer: B**EXAM SHORTCUT** AI-derived explanation

$X_{(n)}$ has density $\frac{3nx^{3n-1}}{\theta^{3n}}$, so $E[X_{(n)}^r] = \frac{3n}{r+3n}\theta^r$; the UMVUE is $X_{(n)}^r \cdot \frac{r+3n}{3n}$.

TIPS & TRICKS

- The cdf of a single observation is $(x/\theta)^3$, so the cdf of the maximum is $(x/\theta)^{3n}$ - raise to the power n .
- $E[X_{(n)}^r] = \int_0^\theta x^r \frac{3nx^{3n-1}}{\theta^{3n}} dx = \frac{3n\theta^r}{r+3n}$; invert the constant to de-bias.
- Check $r=0$: the factor becomes 1, as it must. Option (d) inverts the correction and is the standard trap.

STEP-BY-STEP SOLUTION

$$1. F(x) = \int_0^x \frac{3u^2}{\theta^3} du = \left(\frac{x}{\theta}\right)^3, \text{ so } F_{X_{(n)}}(x) = \left(\frac{x}{\theta}\right)^{3n} \text{ and } f_{X_{(n)}}(x) = \frac{3n x^{3n-1}}{\theta^{3n}}.$$

$$2. E[X_{(n)}^r] = \int_0^\theta x^r \frac{3n x^{3n-1}}{\theta^{3n}} dx = \frac{3n}{\theta^{3n}} \cdot \frac{\theta^{r+3n}}{r+3n} = \frac{3n}{r+3n} \theta^r.$$

$$3. \text{ So } T = X_{(n)}^r \cdot \frac{r+3n}{3n} \text{ is unbiased for } \theta^r.$$

$$4. X_{(n)} \text{ is complete sufficient for this family, so } T \text{ is the UMVUE - option (b).}$$

Q9.

Let $X_1, \dots, X_n \sim U(0, \theta)$, $\theta > 0$. Define $U = 2\bar{X}$, $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$, and $T = X_{(1)} = \min\{X_1, \dots, X_n\}$. Then $E[U \mid T=t]$ will

be

- A. independent of θ
- B. most efficient for θ
- C. **✗ function of θ**
- D. MVUE for θ

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$T = X_{(1)}$ is NOT sufficient for $U(0, \theta)$ (the sufficient statistic is $X_{(n)}$), so the conditional expectation still involves θ .

TIPS & TRICKS

- Rao-Blackwell requires conditioning on a SUFFICIENT statistic; otherwise the result is not even a statistic.
- For $U(0, \theta)$ the sufficient statistic is the maximum $X_{(n)}$, not the minimum.
- Conditioning on $X_{(n)}$ would give the UMVUE $\frac{n+1}{n}X_{(n)}$ - the contrast is the point of the question.

STEP-BY-STEP SOLUTION

1. For $U(0, \theta)$ the likelihood is $\theta^{-n}I(X_{(n)} \leq \theta)$, so $X_{(n)}$ is sufficient while $X_{(1)}$ is not.
2. Because $T = X_{(1)}$ is not sufficient, the conditional distribution of the sample given $T=t$ depends on θ .
3. Hence $E[U \mid T=t]$ is a function of θ and is not a statistic at all.
4. Option (c).

Q10.

Let $X_1, \dots, X_n$ be iid, $f(x) = \frac{1}{\alpha} e^{-x/\alpha}$, $x > 0, \alpha > 0$. The UMVUE of α is

- A. $n\bar{X}$
- B. $\bar{X}/n$
- C. $\bar{X}$
- D. $\bar{X}/(n+1)$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$\sum X_i$ is complete sufficient for the exponential scale, and $\bar{X}$ is its unbiased function - hence the UMVUE.

TIPS & TRICKS

- Exponential with mean α : $E(X) = \alpha$, so $\bar{X}$ is immediately unbiased.
- Lehmann-Scheffe then upgrades it to UMVUE because $\sum X_i$ is complete sufficient.
- $V(\bar{X}) = \alpha^2/n$, exactly the Cramer-Rao bound, so $\bar{X}$ is also MVB.

STEP-BY-STEP SOLUTION

- $L(\alpha) = \alpha^{-n} \exp\left(-\frac{\sum x_i}{\alpha}\right)$, so $T = \sum X_i$ is sufficient, and being the natural statistic of a full-rank exponential family, complete.
- $E(\bar{X}) = \alpha$, and $\bar{X} = T/n$ is a function of T .
- By the Lehmann-Scheffe theorem, $\bar{X}$ is the unique UMVUE of α .
- Option (c).

Q11.

Let $x_1, \dots, x_n$ be from $P[X=x] = \binom{2}{x} (1-\theta)^{2-x} \theta^x$; $0 \leq \theta \leq 1$; $x=0,1,2$, and $T = \sum_{i=1}^n x_i$:

1. T is a complete sufficient statistic for θ .

2. $\frac{T^2 - T}{2n(2n-1)}$ is an UMVUE of θ^2 .

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$T = \sum X_i \sim B(2n, \theta)$ is complete sufficient, and $E[T(T-1)] = 2n(2n-1)\theta^2$, so $\frac{T^2 - T}{2n(2n-1)}$ is the UMVUE of θ^2 .

TIPS & TRICKS

- Sum of n independent $B(2, \theta)$ variables is $B(2n, \theta)$ - the effective number of trials is $2n$.
- Second factorial moment of $B(N, \theta)$: $E[T(T-1)] = N(N-1)\theta^2$; here $N=2n$.
- $T^2 - T = T(T-1)$ - recognising the factorial form is the whole trick.

STEP-BY-STEP SOLUTION

- $X_i \sim B(2, \theta)$ independently, so $T = \sum_{i=1}^n X_i \sim B(2n, \theta)$.
- The likelihood is $\theta^T (1-\theta)^{2n-T}$ times a constant, so T is sufficient; being the natural statistic of a full-rank exponential family, it is complete. Statement 1 true.
- $E[T(T-1)] = 2n(2n-1)\theta^2$, so $\frac{T^2 - T}{2n(2n-1)}$ is unbiased for θ^2 .
- It is a function of the complete sufficient statistic, hence the UMVUE by Lehmann-Scheffe. Statement 2 true.
- Option (c).

SHARED CONTEXT

Let $X_1, \dots, X_n$ be iid $U(\theta_1, \theta_2)$, $f(x) = \frac{1}{\theta_2 - \theta_1}$; $\theta_1 < x < \theta_2$; $\theta_i > 0$. Let $\hat{X}_{(i)} = E(X_{(i)})$; $i = 1, n$.

Q12.

UMVUE of θ_1 equals

A. $\frac{n\hat{X}_{(n)} - \hat{X}_{(1)}}{n-1}$

B. $\frac{n\hat{X}_{(1)} - \hat{X}_{(n)}}{n}$

C. $\frac{n\hat{X}_{(1)} - \hat{X}_{(n)}}{n-2}$

D. $\frac{n\hat{X}_{(1)} - \hat{X}_{(n)}}{n-1}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$$E(X_{(1)}) = \frac{n\theta_1 + \theta_2}{n+1}, E(X_{(n)}) = \frac{\theta_1 + n\theta_2}{n+1}; \text{ solving the linear system gives } \hat{\theta}_1 = \frac{nX_{(1)} - X_{(n)}}{n-1}.$$

TIPS & TRICKS

- For $U(\theta_1, \theta_2)$: $E(X_{(k)}) = \theta_1 + \frac{k}{n+1}(\theta_2 - \theta_1)$ - linear in k .
- $(X_{(1)}, X_{(n)})$ is complete sufficient here, so the unbiased linear combination is automatically the UMVUE.
- Unbiasedness check: substituting the two expectations must return θ_1 exactly - do it before committing to an option.

STEP-BY-STEP SOLUTION

1. $E(X_{(1)}) = \frac{n\theta_1 + \theta_2}{n+1}$ and $E(X_{(n)}) = \frac{\theta_1 + n\theta_2}{n+1}$.

2. Seek $c_1X_{(1)} + c_nX_{(n)}$ unbiased for θ_1 : matching coefficients gives $\frac{c_1n + c_n}{n+1} = 1$ and $\frac{c_1 + c_n n}{n+1} = 0$.

3. The second gives $c_1 = -nc_n$; substituting into the first: $c_n(1 - n^2) = n + 1$, so $c_n = -\frac{1}{n-1}$ and $c_1 = \frac{n}{n-1}$.

4. Hence $\hat{\theta}_1 = \frac{nX_{(1)} - X_{(n)}}{n-1}$, the UMVUE since $(X_{(1)}, X_{(n)})$ is complete sufficient.

5. Option (d).

SHARED CONTEXT

Let $X_1, \dots, X_n$ be iid $U(\theta_1, \theta_2)$, $f(x) = \frac{1}{\theta_2 - \theta_1}$; $\theta_1 < x < \theta_2$; $\theta_i > 0$. Let $\hat{X}_{(i)} = E(X_{(i)})$; $i = 1, n$.

Q13.

UMVUE of θ_2 equals

A. $\frac{n\hat{X}_{(n)} - \hat{X}_{(1)}}{n-1}$

B. $\frac{n\hat{X}_{(n)} - \hat{X}_{(1)}}{n}$

C. $\frac{n\hat{X}_{(1)} - \hat{X}_{(n)}}{n-2}$

D. $\frac{n\hat{X}_{(1)} - \hat{X}_{(n)}}{n-1}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

By symmetry (swap the roles of the two endpoints), $\hat{\theta}_2 = \frac{n\hat{X}_{(n)} - \hat{X}_{(1)}}{n-1}$.

TIPS & TRICKS

- Reflect the problem: θ_2 plays for $X_{(n)}$ exactly the role θ_1 plays for $X_{(1)}$.
- Check unbiasedness: $\frac{n}{n-1} \cdot \frac{\theta_1 + n\theta_2}{n+1} - \frac{1}{n-1} \cdot \frac{n\theta_1 + \theta_2}{n+1} = \theta_2$.
- Both estimators extrapolate beyond the observed range, as they must since the extremes are biased inward.

STEP-BY-STEP SOLUTION

1. Seek $c_1 X_{(1)} + c_n X_{(n)}$ unbiased for θ_2 : $\frac{c_1 n + c_n}{n+1} = 0$ and $\frac{c_1 + c_n n}{n+1} = 1$.

2. The first gives $c_n = -nc_1$; substituting: $c_1(1 - n^2) = n + 1$, so $c_1 = -\frac{1}{n-1}$ and $c_n = \frac{n}{n-1}$.

3. Hence $\hat{\theta}_2 = \frac{n\hat{X}_{(n)} - \hat{X}_{(1)}}{n-1}$.

4. Option (a).

Q14.

Let $X_1, X_2, X_3, \dots, X_n$ be i.i.d. random variables from $f(x)$, where $f(x) = (\theta+1)x^\theta$; $0 < x < 1$, $\theta > -1$. The UMVUE of θ is:

A. ✓ $-\frac{(n-1)}{\sum \ln x_i} - 1$

B. $-\frac{\sum \ln x_i}{n-1} - 1$

C. $\frac{(n-1)}{\sum \ln x_i} - 1$

D. $-\frac{(n-1)}{\sum \ln x_i} + 1$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Put $\varphi = \theta + 1$; then $-\sum \ln X_i \sim \text{Gamma}(n, \varphi)$ and $\frac{n-1}{-\sum \ln X_i}$ is the UMVUE of φ , so the UMVUE of θ is

$$-\frac{n-1}{\sum \ln x_i} - 1.$$

TIPS & TRICKS

- Re-parametrise to the standard $\varphi x^{\varphi-1}$ family first - then the known Gamma result applies directly.
- $E(1/Z) = \frac{\varphi}{n-1}$ for $Z \sim \text{Gamma}(n, \varphi)$, hence the $n-1$ in the numerator.
- $\sum \ln x_i < 0$, so $-\frac{n-1}{\sum \ln x_i} > 0$ as a scale estimate must be.

STEP-BY-STEP SOLUTION

1. Write $\varphi = \theta + 1 > 0$, so $f(x) = \varphi x^{\varphi-1}$ on $(0, 1)$.

2. $-\ln X_i \sim \text{Exp}(\varphi)$, so $Z = -\sum \ln X_i \sim \text{Gamma}(n, \varphi)$ and $E\left(\frac{1}{Z}\right) = \frac{\varphi}{n-1}$.

3. Hence $\frac{n-1}{Z}$ is unbiased for φ , and Z is complete sufficient, so it is the UMVUE of φ .

4. Therefore the UMVUE of $\theta = \varphi - 1$ is $\frac{n-1}{-\sum \ln x_i} - 1 = -\frac{n-1}{\sum \ln x_i} - 1$.

5. Option (a).

SHARED CONTEXT

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample of size n from $b(1, \theta)$, $0 < \theta < 1$ (Bernoulli) distribution. Let $Y = \sum X_i$ be complete sufficient statistic for θ .

Q15.

UMVUE of θ is:

A. $\frac{Y}{n}$

B. $\frac{Y^2}{n}$

C. $\frac{2Y}{n}$

D. $\frac{Y}{2n}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$E(Y/n) = \theta$ and Y is complete sufficient, so Y/n is the UMVUE by Lehmann-Scheffe.

TIPS & TRICKS

- $Y = \sum X_i \sim B(n, \theta)$, so $E(Y) = n\theta$.
- Lehmann-Scheffe: unbiased function of a complete sufficient statistic = UMVUE.
- $V(Y/n) = \frac{\theta(1-\theta)}{n}$ equals the CR bound, so it is MVB as well.

STEP-BY-STEP SOLUTION

- $Y = \sum X_i \sim B(n, \theta)$ with $E(Y) = n\theta$, so $E\left(\frac{Y}{n}\right) = \theta$: Y/n is unbiased.
- Y is given as complete sufficient, and Y/n is a function of Y .
- By the Lehmann-Scheffe theorem, Y/n is the unique UMVUE of θ .
- Option (a).

SHARED CONTEXT

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample of size n from $b(1, \theta)$, $0 < \theta < 1$ (Bernoulli) distribution. Let $Y = \sum X_i$ be complete sufficient statistic for θ .

Q16.

UMVUE of θ^2 is:

A. $\frac{Y(Y-1)}{n(n+1)}$

B. $\frac{Y(Y-1)}{n(n-1)}$

C. $\frac{Y(Y+1)}{n(n+1)}$

D. $\frac{Y(Y+1)}{n(n-1)}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$E[Y(Y-1)] = n(n-1)\theta^2$, so $\frac{Y(Y-1)}{n(n-1)}$ is the UMVUE of θ^2 .

TIPS & TRICKS

- Second factorial moment of $B(n, \theta)$: $E[Y(Y-1)] = n(n-1)\theta^2$ - memorise this form.
- $\left(\frac{Y}{n}\right)^2$ is biased upward by $\frac{\theta(1-\theta)}{n}$ - never use it.
- The denominator must mirror the numerator's structure: $Y(Y-1)$ pairs with $n(n-1)$.

STEP-BY-STEP SOLUTION

1. For $Y \sim B(n, \theta)$: $E(Y) = n\theta$ and $E[Y(Y-1)] = n(n-1)\theta^2$.
2. Hence $E\left[\frac{Y(Y-1)}{n(n-1)}\right] = \theta^2$: the estimator is unbiased.
3. It is a function of the complete sufficient statistic Y , so by Lehmann-Scheffe it is the UMVUE.
4. Option (b).

Q17.

Let $\{1,0,0,0,1,1\}$ be a random sample from a binomial distribution $b(1,\theta)$, $0 < \theta < 1$. Then UMVUE of $\theta(1+\theta)$ is:

A. 0.7**B. 0.6****C. 0.5****D. 0.3****Correct answer: A****EXAM SHORTCUT**

AI-derived explanation

$n=6$, $Y=3$. UMVUE of θ is $Y/n=0.5$; of θ^2 is $\frac{Y(Y-1)}{n(n-1)} = \frac{6}{30} = 0.2$. Sum = 0.7.

TIPS & TRICKS

- UMVUEs are linear: the UMVUE of $\theta + \theta^2$ is the sum of the individual UMVUEs.
- $Y=1+0+0+0+1+1=3$ and $n=6$.
- $\frac{Y(Y-1)}{n(n-1)} = \frac{3 \times 2}{6 \times 5} = 0.2$ - never use $(Y/n)^2 = 0.25$, which is biased.

STEP-BY-STEP SOLUTION

1. $Y = \sum x_i = 1+0+0+0+1+1=3$, $n=6$.
2. UMVUE of θ : $\frac{Y}{n} = \frac{3}{6} = 0.5$.
3. UMVUE of θ^2 : $\frac{Y(Y-1)}{n(n-1)} = \frac{3 \times 2}{6 \times 5} = 0.2$.
4. By linearity, the UMVUE of $\theta(1+\theta) = \theta + \theta^2$ is $0.5 + 0.2 = 0.7$.
5. Option (a).

Q18.

Let $x_1, x_2, \dots, x_n$ be a random sample from a Poisson distribution with mean $\theta > 0$. Then the minimum variance unbiased estimator for $P(X=k)$ is:

A. $T_1 = \begin{cases} 1, & \text{if } x=k \\ 0, & \text{otherwise} \end{cases}$

B. $T_2 = \begin{cases} k, & \text{if } x=0 \\ 0, & \text{otherwise} \end{cases}$

C. ✓ $T_3 = \binom{\sum_{i=1}^n x_i}{k} \frac{(n-1)(\sum_{i=1}^n x_i - k)}{n(\sum_{i=1}^n x_i)}$

D. $T_4 = \binom{\sum_{i=1}^n x_i}{k} \left(1 - \frac{1}{n}\right) (\sum_{i=1}^n x_i - k)$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Conditioning the indicator $I(X_1=k)$ on the complete sufficient $T = \sum X_i$ gives the binomial form

$$\binom{T}{k} \frac{(n-1)^{T-k}}{n^T}.$$

TIPS & TRICKS

- Given $T=t$, $X_1 / T=t \sim B\left(t, \frac{1}{n}\right)$ - the classic Poisson conditioning result.
- So $P(X_1=k / T=t) = \binom{t}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{t-k} = \binom{t}{k} \frac{(n-1)^{t-k}}{n^t}$.
- Option (d) omits the n^{-k} factor and so is not a probability - check that the expression is ≤ 1 .

STEP-BY-STEP SOLUTION

1. $T = \sum X_i$ is complete sufficient for θ ; start from the unbiased but crude estimator $U = I(X_1=k)$, since $E(U) = P(X=k)$.

2. Rao-Blackwell: $\varphi(t) = E[U / T=t] = P(X_1=k / T=t)$.

3. For independent Poissons, $X_1 / \sum X_i = t \sim B\left(t, \frac{1}{n}\right)$, so $\varphi(t) = \binom{t}{k} \left(\frac{1}{n}\right)^k \left(\frac{n-1}{n}\right)^{t-k} = \binom{t}{k} \frac{(n-1)^{t-k}}{n^t}$.

4. By Lehmann-Scheffe this is the MVUE - option (c).

Q19.

Let $U=U(X_1, X_2, X_3, \dots, X_n)$ be an unbiased estimator of $\gamma(\theta)$ and T be a sufficient statistic for θ . Define $\varphi(t)=E[U \mid T=t]$. Consider the following statements:

- I. $\varphi(t)$ is always independent of θ .
- II. The variance of $\varphi(t)$ is less than the variance of U .

Which of the statements given above is/are correct?

- A. I only**
- B. II only
- C. Both I and II
- D. Neither I nor II

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Sufficiency makes $\varphi(t)$ free of θ (I true), but Rao-Blackwell gives $V(\varphi) \leq V(U)$ with EQUALITY when U is already a function of T - so 'less than' (II) is not always correct.

TIPS & TRICKS

- Rao-Blackwell gives $\leq$, not $<$: equality holds iff U is already a function of T (almost surely).
- $V(U) - V[\varphi(T)] = E[V(U \mid T)]$, which is zero exactly when U is T -measurable.
- The variance-decomposition identity $V(U) = E[V(U \mid T)] + V[E(U \mid T)]$ is the whole proof.

STEP-BY-STEP SOLUTION

- Statement I: because T is sufficient, the conditional distribution of the sample given $T=t$ is free of θ , so $\varphi(t)=E[U \mid T=t]$ involves no θ and is a genuine statistic. True.
- Statement II: $V(U)=E[V(U \mid T)] + V[\varphi(T)]$, so $V[\varphi(T)] \leq V(U)$.
- Equality occurs whenever U is already a function of T (then $V(U \mid T)=0$), so the strict inequality claimed is not always true. False.
- Hence I only - option (a).

Q20.

Consider the following statements:

I. It is possible to have a minimum variance unbiased estimator which is not minimum variance bound estimator.

II. Sufficient statistic should have a complete family of distributions to enable obtaining a minimum variance unbiased estimator from any unbiased estimator.

Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

A UMVUE need not attain the CR bound (I true), and completeness is exactly what upgrades Rao-Blackwell to optimality (II true).

TIPS & TRICKS

- s^2 for the normal variance is UMVUE but not MVB - the standard example for statement I.
- Lehmann-Scheffe: complete + sufficient $\Rightarrow$ the unbiased function of it is the unique UMVUE.
- Without completeness, Rao-Blackwell only improves an estimator; it cannot certify optimality.

STEP-BY-STEP SOLUTION

- Statement I: $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is the UMVUE of σ^2 with variance $\frac{2\sigma^4}{n-1}$, strictly above the CR bound $\frac{2\sigma^4}{n}$. True.
- Statement II: Rao-Blackwell improves any unbiased estimator by conditioning on a sufficient statistic; completeness (Lehmann-Scheffe) is what makes the result the unique minimum-variance unbiased estimator. True.
- Option (c).

Q21.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from a Poisson distribution with mean $\theta > 0$. Let $T = \sum_{i=1}^n x_i$. Then UMVUE of $e^{-2\theta}\theta^3$ is

- A. $\frac{(n-2)^{-3}}{n^T}$
- B. $\frac{(n-2)^T T(T-1)(T-2)}{T!}$
- C. $\frac{T(T-1)(T-2)}{n^T}$
- D. $\frac{(n-2)^{T-3} T(T-1)(T-2)}{n^T}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Matching power series in θ gives $g(t) = \frac{t(t-1)(t-2)(n-2)^{t-3}}{n^t}$.

TIPS & TRICKS

- Set $E[g(T)] = \sum_t g(t) \frac{e^{-n\theta}(n\theta)^t}{t!}$ equal to $e^{-2\theta}\theta^3$ and cancel $e^{-n\theta}$.
- $e^{(n-2)\theta}\theta^3 = \sum_{k \geq 0} \frac{(n-2)^k \theta^{k+3}}{k!}$ - reindex with $t = k+3$.
- Comparing coefficients of θ^t gives $g(t) \frac{n^t}{t!} = \frac{(n-2)^{t-3}}{(t-3)!}$, and $\frac{t!}{(t-3)!} = t(t-1)(t-2)$.

STEP-BY-STEP SOLUTION

1. $T = \sum x_i \sim \text{Poisson}(n\theta)$ is complete sufficient, so the UMVUE is the unique $g(T)$ with $E[g(T)] = e^{-2\theta}\theta^3$.

$$2. \sum_t g(t) \frac{(n\theta)^t}{t!} = e^{n\theta} e^{-2\theta}\theta^3 = \theta^3 \sum_{k \geq 0} \frac{[(n-2)\theta]^k}{k!} = \sum_{t \geq 3} \frac{(n-2)^{t-3} \theta^t}{(t-3)!}.$$

$$3. \text{Comparing coefficients of } \theta^t: g(t) \frac{n^t}{t!} = \frac{(n-2)^{t-3}}{(t-3)!}.$$

$$4. g(t) = \frac{t!}{(t-3)!} \cdot \frac{(n-2)^{t-3}}{n^t} = \frac{(n-2)^{T-3} T(T-1)(T-2)}{n^T} - \text{option (d).}$$

Q22.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from the distribution having pdf $f(x, \mu) = 3e^{-3(x-\mu)}$; $x > \mu$, $\mu \in \mathbb{R}$. If $X_{(1)} = \min\{X_1, X_2, X_3, \dots, X_n\}$, then which one of the following is correct?

- A. $X_{(1)}$ is MVUE for μ .
- B. $X_{(1)}$ is unbiased but not MVUE for μ .
- C. $X_{(1)}$ is biased and consistent for μ .
- D. $X_{(1)}$ is neither biased nor consistent for μ .

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$X_{(1)} \sim \mu + \text{Exp}(3n)$, so $E[X_{(1)}] = \mu + \frac{1}{3n}$ (biased) and $V = \frac{1}{9n^2} \rightarrow 0$ (consistent).

TIPS & TRICKS

- Minimum of n exponentials with rate 3 is exponential with rate $3n$ shifted to μ .
- Bias $\frac{1}{3n} \rightarrow 0$ and variance $\frac{1}{9n^2} \rightarrow 0$, so consistency holds despite the bias.
- The MVUE is the bias-corrected $X_{(1)} - \frac{1}{3n}$, which is why $X_{(1)}$ itself cannot be MVUE.

STEP-BY-STEP SOLUTION

1. $P(X_{(1)} > t) = e^{-3n(t-\mu)}$ for $t > \mu$, so $X_{(1)} - \mu \sim \text{Exp}(3n)$.
2. $E[X_{(1)}] = \mu + \frac{1}{3n} \neq \mu$: biased.
3. $V[X_{(1)}] = \frac{1}{9n^2} \rightarrow 0$ and the bias $\rightarrow 0$, so $X_{(1)} \rightarrow P\mu$: consistent.
4. Hence biased and consistent - option (c).

Q23.

Let $X_1, X_2, X_3, \dots, X_n$ be independent random variables each $N(\mu, \sigma^2)$, with μ and σ^2 both unknown. Then UMVUE of μ and σ^2 are respectively

- A. $\sum_{i=1}^n X_i$ and $\sum_{i=1}^n X_i^2$
- B. $\sum_{i=1}^n X_i^2$ and $\sum_{i=1}^n X_i$
- C. $\bar{X}$ and $\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- D. $\bar{X}$ and $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$(\sum X_i, \sum X_i^2)$ is complete sufficient; the unbiased functions of it are $\bar{X}$ for μ and S^2 (divisor $n-1$) for σ^2 .

TIPS & TRICKS

- Unbiasedness picks the divisor: $n-1$, not n - option (d) uses the biased MLE.
- Lehmann-Scheffe then upgrades both to UMVUE status.
- Options (a) and (b) give raw sums, whose expectations are $n\mu$ and $n(\mu^2 + \sigma^2)$ - not unbiased.

STEP-BY-STEP SOLUTION

1. For $N(\mu, \sigma^2)$ with both parameters unknown, $(\sum X_i, \sum X_i^2)$ is complete and sufficient.
2. $E(\bar{X}) = \mu$ and $\bar{X}$ is a function of the statistic, so $\bar{X}$ is the UMVUE of μ .
3. $E\left[\frac{1}{n-1} \sum (X_i - \bar{X})^2\right] = \sigma^2$, and it too is a function of the statistic, so it is the UMVUE of σ^2 .
4. Option (c).

Q24.

Let $x_1, x_2, x_3, \dots, x_m$ denote a random sample from a binomial $B(n, p)$ distribution. Define a statistic

$$u = \begin{cases} \frac{1}{2n}, & x_1 + x_2 = 1 \\ 0, & \text{otherwise} \end{cases}. \text{ Which of the following statements is/are correct?}$$

I. u is an unbiased estimator of $\frac{p(1-p)}{n}$.

II. u is a uniformly minimum variance unbiased estimator of p .

III. $\sum_{i=1}^m x_i$ is a complete sufficient statistic for p .

Select the answer using the code given below.

A. I and II

B. **III only**

C. II and III

D. I and III

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$E(u) = \frac{1}{2n} P(X_1 + X_2 = 1) = p(1-p)^{2n-1}$, which is neither $\frac{p(1-p)}{n}$ nor p - so I and II fail; but $\sum X_i$ is complete sufficient, so III holds.

TIPS & TRICKS

- $X_1 + X_2 \sim B(2n, p)$, so $P(X_1 + X_2 = 1) = 2np(1-p)^{2n-1}$.
- u is unbiased for $p(1-p)^{2n-1}$ - check what an estimator actually estimates before accepting a claim.
- Binomial with known n is a one-parameter exponential family, so $\sum X_i$ is complete sufficient for p .

STEP-BY-STEP SOLUTION

1. $X_1 + X_2 \sim B(2n, p)$, so $P(X_1 + X_2 = 1) = \binom{2n}{1} p(1-p)^{2n-1} = 2np(1-p)^{2n-1}$.

2. $E(u) = \frac{1}{2n} \cdot 2np(1-p)^{2n-1} = p(1-p)^{2n-1}$, which is not $\frac{p(1-p)}{n}$: statement I false.

3. Since $E(u) \neq p$, u is not even unbiased for p , let alone UMVUE: statement II false.

4. $L = \prod \binom{n}{x_i} p^{\sum x_i} (1-p)^{mn - \sum x_i}$, so $\sum X_i$ is sufficient, and being the natural statistic of a full-rank exponential family it is complete: statement III true.

5. Hence III only - option (b).

SI & HT

2025 Q4

Conceptual

Easy

Q25.

Suppose $X_1, X_2, X_3, \dots, X_n$ be a random sample of size n from Poisson distribution with parameter λ . Then uniformly minimum variance unbiased estimator (UMVUE) of λ is:

A.
$$\frac{\sum_{i=1}^n X_i^2}{n}$$

B.
$$\frac{\sum_{i=1}^n X_i}{n+1}$$

C.
$$\frac{\sum_{i=1}^n X_i}{n}$$

D.
$$\frac{\sum_{i=1}^n X_i^2}{n+1}$$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$\sum X_i$ is complete sufficient and $\bar{X} = \frac{\sum X_i}{n}$ is its unbiased function - the UMVUE.

TIPS & TRICKS

- Lehmann-Scheffe in one line: unbiased function of a complete sufficient statistic.
- $\frac{\sum X_i^2}{n}$ estimates $\lambda + \lambda^2$, not λ .
- Dividing by $n+1$ (option b) destroys unbiasedness.

STEP-BY-STEP SOLUTION

- $T = \sum X_i$ is complete and sufficient for the Poisson parameter.
- $E\left(\frac{T}{n}\right) = \lambda$, so $\bar{X}$ is an unbiased function of T .
- By Lehmann-Scheffe it is the unique UMVUE.
- Option (c).

Q26.

Let S be the set of all unbiased estimators T of $\theta \in \Theta$ such that $E_{\theta} T^2 < \infty$ for all $\theta \in \Theta$. An estimator $T_0 \in S$ is called a uniformly minimum variance unbiased estimator (UMVUE) of θ for $T \in S$ if:

- A. $E_{\theta}(T_0 - \theta) \leq E_{\theta}(T - \theta)$
- B. $E_{\theta}(T_0 - \theta)^2 > E_{\theta}(T - \theta)^2$
- C. $E_{\theta}(T_0 - \theta)^2 \leq E_{\theta}(T - \theta)^2$
- D. $E_{\theta}(T_0 - \theta) > E_{\theta}(T - \theta)$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

T_0 is UMVUE if $E_{\theta}(T_0 - \theta)^2 \leq E_{\theta}(T - \theta)^2$ for every unbiased T and every θ .

TIPS & TRICKS

- For unbiased estimators, $E(T - \theta)^2$ IS the variance - the definition compares variances.
- Note the direction: the UMVUE's variance is the SMALLEST, hence $\leq$.
- Options (a) and (d) compare first moments, which are all zero for unbiased estimators.

STEP-BY-STEP SOLUTION

1. Within the class S of unbiased estimators with finite second moment, $E_{\theta}(T - \theta)^2 = V_{\theta}(T)$.
2. T_0 is uniformly minimum variance unbiased if its variance is no larger than that of any competitor, for every θ .
3. That is $E_{\theta}(T_0 - \theta)^2 \leq E_{\theta}(T - \theta)^2$ for all $T \in S$ and all $\theta \in \Theta$.
4. Option (c).

Q27.

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from $U[0, \theta]$ distribution. If $T = X_{(n)} = \max(X_i)$, then MVUE for θ is:

- A. $\frac{(n+2)T}{n}$
- B. $\frac{(n+2)T}{n+1}$
- C. $\frac{(n+1)(n+2)T}{n}$
- D. $\frac{(n+1)T}{n}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$E[X_{(n)}] = \frac{n\theta}{n+1}$, so $\frac{n+1}{n}X_{(n)}$ is unbiased - and being a function of the complete sufficient statistic, it is the MVUE.

TIPS & TRICKS

- The maximum under-estimates θ , so the correction factor must EXCEED 1.
- $\frac{n+1}{n} > 1$ while $\frac{n+2}{n+1}$ and the other options do not reproduce $E = \theta$.
- Verify: $\frac{n+1}{n} \cdot \frac{n\theta}{n+1} = \theta$ exactly.

STEP-BY-STEP SOLUTION

1. $F_{X_{(n)}}(t) = \left(\frac{t}{\theta}\right)^n$ gives $E[X_{(n)}] = \int_0^\theta t \frac{nt^{n-1}}{\theta^n} dt = \frac{n\theta}{n+1}$.

2. Hence $E\left[\frac{(n+1)T}{n}\right] = \theta$, so the estimator is unbiased.

3. $T = X_{(n)}$ is complete sufficient for $U[0, \theta]$, so by Lehmann-Scheffe this is the MVUE.

4. Option (d).

Q28.

Blackwellisation is a process of finding:

- A. an unbiased estimator with higher consistency
- B. a biased estimator with lesser variance
- C. an unbiased estimator such that it is complete but not sufficient
- D. an unbiased estimator with lesser variance**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Blackwellisation conditions an unbiased estimator on a sufficient statistic, producing an unbiased estimator with smaller (never larger) variance.

TIPS & TRICKS

- The output stays UNBIASED - options describing a biased result are wrong.
- $V[E(U|T)] \leq V(U)$ by the variance decomposition.
- Completeness is what turns the improvement into optimality (Lehmann-Scheffe), but it is not part of Blackwellisation itself.

STEP-BY-STEP SOLUTION

1. Given an unbiased U and a sufficient T , Rao-Blackwellisation forms $\phi(T) = E(U|T)$.
2. $\phi(T)$ is a statistic (sufficiency), is unbiased ($E[\phi(T)] = E(U)$), and has $V[\phi(T)] \leq V(U)$.
3. So the process produces an unbiased estimator with lesser variance - option (d).

Q29.

Let T_0 be an MVU estimator of $\gamma(\theta)$ and T_1 an unbiased estimator with efficiency $e_\theta < 1$. Which of the following statements is/are correct?

- I. The unbiased linear combination of T_0 and T_1 , i.e., $U = a_0 T_0 + a_1 T_1$; $a_1 \neq 0$, $a_0 + a_1 = 1$ is MVU.
- II. The unbiased linear combination of T_0 and T_1 , i.e., $V = a_0 T_0 - a_1 T_1$; $a_1 \neq 0$, $a_0 - a_1 = 1$ is MVU.

Select the answer using the code given below.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Any unbiased combination that puts non-zero weight on T_1 has variance strictly above $V(T_0)$, so neither U nor V can be MVU.

TIPS & TRICKS

- $\text{Cov}(T_0, T_1) = V(T_0)$, so $V(a_0 T_0 + a_1 T_1) = V_0 + a_1^2 (V_1 - V_0) > V_0$ whenever $a_1 \neq 0$.
- Uniqueness of the UMVUE already rules out a second minimum-variance unbiased estimator.
- The sign of a_1 is irrelevant because it enters squared - both U and V fail.

STEP-BY-STEP SOLUTION

1. Using $\text{Cov}(T_0, T_1) = V(T_0) = V_0$ and writing $V_1 = V(T_1) > V_0$:
2. $V(a_0 T_0 + a_1 T_1) = a_0^2 V_0 + a_1^2 V_1 + 2a_0 a_1 V_0$. With $a_0 + a_1 = 1$ this simplifies to $V_0 + a_1^2 (V_1 - V_0)$.
3. Since $V_1 > V_0$ and $a_1 \neq 0$, this is strictly larger than V_0 - so U is not MVU. Statement I false.
4. The same computation with $a_0 - a_1 = 1$ again gives a variance exceeding V_0 , so V is not MVU either. Statement II false.
5. Option (d).

Q30.

If T_1 is an MVUE of $\gamma(\theta)$, $\theta \in \Theta$ and T_2 is any other unbiased estimator of $\gamma(\theta)$ with efficiency $e < 1$, then which of the following is/are correct?

I. $T = (T_1 + T_2)/2$ is MVUE of $\gamma(\theta)$

II. $\text{Var}(T) = \text{Var}(T_1)/4$

III. T cannot be MVUE for $\gamma(\theta)$

IV. T is unbiased for $\gamma(\theta)$

Select the answer using the code given below.

A. I

B. IV only

C. II and IV

D. **III and IV**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

T is unbiased (IV true) but has variance strictly above $V(T_1)$, so it cannot be MVUE (III true); I and II are false.

TIPS & TRICKS

- $V(T) = \frac{V_1 + V_2 + 2\text{Cov}}{4} = \frac{V_1 + V_2 + 2V_1}{4} > V_1$ when $V_2 > V_1$.
- $V(T_1/4) = V(T_1)/16$ - statement II confuses scaling a random variable with averaging two.
- Averaging an optimal estimator with a worse one always degrades it.

STEP-BY-STEP SOLUTION

1. $E(T) = \frac{1}{2} [\gamma(\theta) + \gamma(\theta)] = \gamma(\theta)$: T is unbiased. Statement IV true.

2. With $\text{Cov}(T_1, T_2) = V(T_1) = V_1$ and $V_2 = V(T_2) > V_1$: $V(T) = \frac{V_1 + V_2 + 2V_1}{4} = \frac{3V_1 + V_2}{4} > V_1$.

3. So T has strictly larger variance than the MVUE and cannot itself be MVUE. Statement III true, statement I false.

4. $V(T_1/4) = V_1/16$, which is not $V(T)$: statement II false.

5. Hence III and IV - option (d).

SI & HT

2026 Q35

Conceptual

Medium

Q31.

Let $V = V(x_1, x_2, \dots, x_n)$ be an unbiased estimator of $\gamma(\theta)$ and T be a sufficient statistic for $\gamma(\theta)$. Define $\Phi(t) = E[V / T=t]$. Which of the following statements is/are correct?

- I. $\Phi(t)$ is an unbiased estimator of $\gamma(\theta)$.
- II. The variance of $\Phi(t)$ is less than or equal to variance of V .
- III. $\Phi(t)$ is a function of sufficient statistic which is dependent on θ .

Select the answer using the code given below.

- A. I only
- B. II and III only
- C. I and II only
- D. I, II and III

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Rao-Blackwell gives unbiasedness (I) and $V(\Phi) \leq V(V)$ (II); but $\Phi(t)$ is FREE of θ , so III is false.

TIPS & TRICKS

- Sufficiency is exactly what makes $E[V / T=t]$ free of θ - that is the whole point.
- Statement II is correctly phrased with ' $\leq$ ' (equality when V is already a function of T).
- A statistic that depended on θ would not be a statistic at all.

STEP-BY-STEP SOLUTION

1. I: $E[\Phi(T)] = E[E(V / T)] = E(V) = \gamma(\theta)$ - unbiased. True.
2. II: $V(V) = E[V(V / T)] + V[\Phi(T)]$, so $V[\Phi(T)] \leq V(V)$. True (with equality iff V is already a function of T).
3. III: because T is sufficient, the conditional distribution of the sample given T is free of θ , so $\Phi(t)$ does NOT depend on θ . False.
4. Hence I and II only - option (c).

Q32.

Let $x_1, x_2, \dots, x_n$ be independent and identically distributed $b(1, p)$, i.e., Bernoulli random variables with parameter p , $0 < p < 1$. Define $T = \sum_{i=1}^n x_i$. The UMVUE for $\Psi(p) = p(1-p)$ is given by

A. $\frac{T(1-T)}{n(n-1)}$

B. $\frac{T(n-T)}{n(n-1)}$

C. $\frac{T}{n(n-1)}$

D. $T(n-1-T)$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$E[T(n-T)] = n(n-1)p(1-p)$, so $\frac{T(n-T)}{n(n-1)}$ is the UMVUE of $p(1-p)$.

TIPS & TRICKS

- $E(T) = np$ and $E[T(T-1)] = n(n-1)p^2$; combine to get $E[T(n-T)] = n(n-1)p(1-p)$.
- $T(n-T)$ counts ordered success-failure pairs - a combinatorial way to see the answer.
- $\frac{T}{n} \left(1 - \frac{T}{n}\right)$ is the MLE and is biased downward.

STEP-BY-STEP SOLUTION

1. $T \sim B(n, p)$ with $E(T) = np$ and $E[T(T-1)] = n(n-1)p^2$, so $E(T^2) = n(n-1)p^2 + np$.
2. $E[T(n-T)] = nE(T) - E(T^2) = n^2p - n(n-1)p^2 - np = n(n-1)p - n(n-1)p^2 = n(n-1)p(1-p)$.
3. Hence $E\left[\frac{T(n-T)}{n(n-1)}\right] = p(1-p)$, and since T is complete sufficient this is the UMVUE.
4. Option (b).

Bhattacharya Bounds

Q1.

The Bhattacharya bound is the generalisation of the

A. ✓Cramer-Rao Inequality

B. Rao-Blackwell theorem

C. Neyman-Pearson Lemma

D. Chapman-Robbins-Kiefer bound

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Bhattacharya's system of bounds uses higher-order derivatives of the likelihood and reduces to the Cramer-Rao bound at first order.

TIPS & TRICKS

- First-order Bhattacharya bound = Cramer-Rao bound; second and higher orders are sharper.
- The bound is $V(T) \geq \mathbf{a}' J^{-1} \mathbf{a}$, with $J_{ij} = E \left[\frac{L^{(i)}}{L} \frac{L^{(j)}}{L} \right]$ and $a_i = \frac{d^i g}{d\theta^i}$.
- Chapman-Robbins-Kiefer is a different (difference-based) generalisation that needs no regularity conditions.

STEP-BY-STEP SOLUTION

1. The Cramer-Rao bound uses only the first derivative of $\log L$.
2. Bhattacharya's bound builds a matrix J from the first k derivatives $\frac{L^{(i)}}{L}$ and bounds $V(T)$ below by $\mathbf{a}' J^{-1} \mathbf{a}$.
3. For $k=1$ it reduces exactly to the Cramer-Rao inequality, which it therefore generalises.
4. Option (a).

Q2.

Let $X_1, \dots, X_n$ be iid $N(\theta, 1)$. The Bhattacharya bound for $g(\theta) = \theta^2$ is

A. $\frac{4\theta^2}{n} + \frac{2}{n^2}$

B. $\frac{4\theta^2}{n}$

C. $\frac{2}{n^2}$

D. $\frac{2\theta^2}{n} + \frac{1}{n^2}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The first-order (Cramer-Rao) term is $\frac{4\theta^2}{n}$ and the second-order Bhattacharya term adds $\frac{2}{n^2}$ - attained exactly by $\bar{X}^2 - \frac{1}{n}$.

TIPS & TRICKS

- $g(\theta) = \theta^2$ gives $g' = 2\theta$, so the CR part is $(2\theta)^2/n = 4\theta^2/n$.
- The improvement $2/n^2$ is exactly what makes the bound attainable by the UMVUE $\bar{X}^2 - \frac{1}{n}$.
- Sanity check: $V\left(\bar{X}^2 - \frac{1}{n}\right) = \frac{4\theta^2}{n} + \frac{2}{n^2}$ by direct computation from the moments of a normal.

STEP-BY-STEP SOLUTION

1. For $N(\theta, 1)$, the UMVUE of θ^2 is $T = \bar{X}^2 - \frac{1}{n}$ with $\bar{X} \sim N\left(\theta, \frac{1}{n}\right)$.
2. $V(\bar{X}^2) = E(\bar{X}^4) - [E(\bar{X}^2)]^2 = \left(\theta^4 + \frac{6\theta^2}{n} + \frac{3}{n^2}\right) - \left(\theta^2 + \frac{1}{n}\right)^2 = \frac{4\theta^2}{n} + \frac{2}{n^2}$.
3. The first-order (Cramer-Rao) bound for $g(\theta) = \theta^2$ is $\frac{(2\theta)^2}{n} = \frac{4\theta^2}{n}$; the second-order Bhattacharya bound adds $\frac{2}{n^2}$.
4. Since T attains it, the Bhattacharya bound is $\frac{4\theta^2}{n} + \frac{2}{n^2}$ - option (a).

Interval Estimation

29 authentic PYQs in this topic (2018-2026)

Confidence Intervals, Pivots and Optimum Bounds

SI & HT

2018 Q38

Numerical

Medium

Q1.

A random sample of 60 schoolchildren gives 40 spectacle users. The 95% confidence limits of spectacle users in the population are

- A. 66%, 89%
- B. 66%, 79%
- C. 55%, 66%
- D. 55%, 79%

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$\hat{p} = 40/60 = 0.667$, $SE = \sqrt{0.667(0.333)/60} = 0.0609$, and $0.667 \pm 1.96(0.0609) = (0.547, 0.786)$, i.e. about 55% to 79%.

TIPS & TRICKS

- Large-sample CI for a proportion: $\hat{p} \pm z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$.
- $\hat{p}(1-\hat{p}) = 2/9$ exactly here, so $SE = \sqrt{2/540} = 0.0609$ - keep fractions to avoid rounding drift.
- The interval must straddle $\hat{p} = 66.7\%$; option (a) and (b) both start at 66%, which cannot be a two-sided interval around 66.7%.

STEP-BY-STEP SOLUTION

1. $\hat{p} = \frac{40}{60} = 0.6667$ and $n = 60$.

2. $SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.6667 \times 0.3333}{60}} = \sqrt{0.003704} = 0.0609$.

3. 95% limits: $0.6667 \pm 1.96 \times 0.0609 = 0.6667 \pm 0.1193$.

4. $= (0.5474, 0.7860)$, i.e. approximately 55% and 79% - option (d).

Q2.

Let $(X_1, \dots, X_n)$ be a random sample from $N(\mu, \sigma^2)$, both unknown. The shortest expected length confidence interval for μ is obtained using

- A. standard normal distribution
- B. t -distribution**
- C. F -distribution
- D. Chi-square distribution

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

With σ^2 unknown the pivot is $\frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}$, giving the shortest expected-length interval.

TIPS & TRICKS

- σ known $\Rightarrow$ normal pivot; σ unknown $\Rightarrow t$ pivot. That is the entire decision.
- Shortest expected length among t -based intervals comes from the symmetric equal-tail choice, because the t density is symmetric and unimodal.
- Chi-square is for σ^2 and F is for a ratio of variances - neither gives an interval for μ .

STEP-BY-STEP SOLUTION

1. With both μ and σ^2 unknown, σ must be replaced by the sample standard deviation S .
2. The pivot $\frac{\sqrt{n}(\bar{X} - \mu)}{S}$ follows Student's t distribution with $n-1$ degrees of freedom, free of all unknown parameters.
3. Inverting it gives $\bar{X} \pm t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}$, which has the shortest expected length among such intervals by symmetry and unimodality of the t density.
4. Option (b).

Q3.

From a normal population, $n=64$, $\bar{x}=60.4$, $\sum_{i=1}^{64} (x_i - \bar{x})^2 = 6300$. The 95% CI for μ is

A. (58.44, 62.36)

B. ✓(57.95, 62.85)

C. (50.4, 70.4)

D. (58.32, 62.48)

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$s^2 = 6300/63 = 100$ so $s = 10$; $SE = 10/\sqrt{64} = 1.25$; $CI = 60.4 \pm 1.96(1.25) = (57.95, 62.85)$.

TIPS & TRICKS

- Divide by $n-1=63$, not 64: $6300/63=100$ is a designed-to-be-clean number.
- $n=64$ is large, so the normal multiplier 1.96 is used rather than t_{63} (they agree to two decimals anyway).
- Half-width $= 1.96 \times 1.25 = 2.45$ - check that the printed interval is symmetric about 60.4.

STEP-BY-STEP SOLUTION

$$1. s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{6300}{63} = 100, \text{ so } s = 10.$$

$$2. SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{10}{8} = 1.25.$$

$$3. 95\% CI = \bar{x} \pm 1.96 SE = 60.4 \pm 1.96(1.25) = 60.4 \pm 2.45.$$

$$4. = (57.95, 62.85) - \text{option (b).}$$

Q4.

Let X have density $f(x) = \theta e^{-x\theta}$, $0 < x < \infty$. The central 95% confidence limits for θ with large n are

- A. $\left(1 + \frac{1.96}{\sqrt{n}}\right) \bar{X}, \left(1 - \frac{1.96}{\sqrt{n}}\right) \bar{X}$
- B. $\left(1 + \frac{1.96}{\sqrt{n}}\right) \frac{1}{\bar{X}}, \left(1 - \frac{1.96}{\sqrt{n}}\right) \frac{1}{\bar{X}}$
- C. $\left(1 + \frac{1.96}{n}\right) \frac{1}{\bar{X}}, \left(1 - \frac{1.96}{n}\right) \frac{1}{\bar{X}}$
- D. $\left(1 + \frac{1.96}{n}\right) \bar{X}, \left(1 - \frac{1.96}{n}\right) \bar{X}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$\hat{\theta} = 1/\bar{X}$ with asymptotic SE $\theta/\sqrt{n}$, so the limits are $\frac{1}{\bar{X}} \left(1 \pm \frac{1.96}{\sqrt{n}}\right)$.

TIPS & TRICKS

- $I_n(\theta) = n/\theta^2$, so $SE(\hat{\theta}) \approx \theta/\sqrt{n}$ and the interval is multiplicative in $1/\bar{X}$.
- The MLE here is $1/\bar{X}$ (rate parameter), not $\bar{X}$ (which estimates the mean $1/\theta$).
- The $\sqrt{n}$ in the denominator is essential - options with n shrink far too fast.

STEP-BY-STEP SOLUTION

1. $f(x) = \theta e^{-\theta x}$, so $\log L = n \log \theta - \theta \sum x_i$ and $\hat{\theta} = \frac{1}{\bar{X}}$.

2. $I_1(\theta) = 1/\theta^2$, so $I_n(\theta) = n/\theta^2$ and asymptotically $\hat{\theta} \sim N\left(\theta, \frac{\theta^2}{n}\right)$.

3. Replacing θ by $\hat{\theta}$ in the standard error gives $SE = \frac{\hat{\theta}}{\sqrt{n}} = \frac{1}{\bar{X}\sqrt{n}}$.

4. 95% limits: $\frac{1}{\bar{X}} \pm 1.96 \frac{1}{\bar{X}\sqrt{n}} = \left(1 \pm \frac{1.96}{\sqrt{n}}\right) \frac{1}{\bar{X}}$ - option (b).

Q5.

A random sample of 500 apples was taken and 60 were found bad. The 98% confidence limits for the percentage of bad apples are (significant value of z at 98% is 2.33)

A. (7.62, 13.58)

B. ✓(8.61, 15.38)

C. (4.38, 9.13)

D. (3.08, 6.14)

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$\hat{p} = 60/500 = 0.12$, $SE = \sqrt{0.12(0.88)/500} = 0.01453$; $0.12 \pm 2.33(0.01453) = (0.0861, 0.1539)$, i.e. (8.61%, 15.38%).

TIPS & TRICKS

- Convert to a percentage only at the end - working in proportions keeps the arithmetic clean.
- $\hat{p}(1-\hat{p}) = 0.1056$; divide by 500 and take the square root: 0.014533.
- The interval must be centred on 12% - a screen that eliminates (a), (c) and (d) immediately.

STEP-BY-STEP SOLUTION

$$1. \hat{p} = \frac{60}{500} = 0.12.$$

$$2. SE = \sqrt{\frac{0.12 \times 0.88}{500}} = \sqrt{0.0002112} = 0.014533$$

$$3. \text{Margin} = 2.33 \times 0.014533 = 0.033862.$$

$$4. \text{Limits: } 0.12 \pm 0.03386 = (0.08614, 0.15386), \text{ i.e. (8.61\%, 15.38\%) - option (b).}$$

Q6.

If the population variance is doubled, the width of the confidence interval for the population mean will be

- A. multiplied by 2
- B. divided by 2
- C. **multiplied by $\sqrt{2}$**
- D. divided by $\sqrt{2}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Width $\propto \sigma = \sqrt{\text{variance}}$, so doubling the variance multiplies the width by $\sqrt{2}$.

TIPS & TRICKS

- CI width $= 2z_{\alpha/2}\sigma/\sqrt{n}$ - linear in σ , hence square-root in the variance.
- Doubling the SAMPLE SIZE, by contrast, divides the width by $\sqrt{2}$.
- Confusing variance with standard deviation is the single commonest error in this topic.

STEP-BY-STEP SOLUTION

1. The confidence interval for μ with known variance is $\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$, so its width is $2z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$.
2. If the variance σ^2 is doubled, the standard deviation becomes $\sqrt{2} \sigma$.
3. Hence the width is multiplied by $\sqrt{2}$ - option (c).

Q7.

If the 95% CI for μ (σ known) is (18000, 22000), the sample mean is

- A. 18000
- B. 20000**
- C. 22000
- D. 40000

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

With σ known the interval is symmetric about $\bar{x}$, so $\bar{x}$ is the midpoint $\frac{18000+22000}{2}=20000$.

TIPS & TRICKS

- Symmetric interval $\Rightarrow$ sample mean = midpoint; half-width $= z_{\alpha/2} \sigma / \sqrt{n}$.
- Here the half-width is 2000, so $\sigma / \sqrt{n} = 2000 / 1.96 = 1020.4$ if that is ever asked.
- The same trick recovers the mean in the 2025 and 2026 versions of this item.

STEP-BY-STEP SOLUTION

1. For known σ the confidence interval is $\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$, symmetric about $\bar{x}$.
2. Hence $\bar{x}$ is the midpoint: $\frac{18000+22000}{2}=20000$.
3. Option (b).

Q8.

If $X_1, \dots, X_n$ are iid $N(\mu, \sigma^2)$, σ^2 unknown, the shortest confidence interval for μ is given as

A. $\left(\bar{X} - t_{n-1, \frac{\alpha}{2}} \frac{s}{2\sqrt{n}}, \bar{X} + t_{n-1, \frac{\alpha}{2}} \frac{s}{2\sqrt{n}} \right)$

B. $\left(\bar{X} - t_{n-1, \alpha} \frac{s}{\sqrt{n}}, \bar{X} + t_{n-1, \alpha} \frac{s}{\sqrt{n}} \right)$

C. $\left(\bar{X} - t_{n-1, \frac{\alpha}{2}} \frac{s}{2\sqrt{n-1}}, \bar{X} + t_{n-1, \frac{\alpha}{2}} \frac{s}{2\sqrt{n-1}} \right)$

D. $\left(\bar{X} - t_{n, \frac{\alpha}{2}} \frac{s}{2\sqrt{n}}, \bar{X} + t_{n, \frac{\alpha}{2}} \frac{s}{2\sqrt{n}} \right)$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

The shortest interval with σ unknown is $\bar{X} \pm t_{n-1} \frac{s}{\sqrt{n}}$ - divisor $\sqrt{n}$, not $2\sqrt{n}$, and $n-1$ degrees of freedom.

TIPS & TRICKS

- Standard error of $\bar{X}$ is $s/\sqrt{n}$; an extra factor of 2 in the denominator is always wrong.
- Degrees of freedom are $n-1$ because s estimates σ - option (d) uses n and fails.
- Here $t_{n-1, \alpha}$ denotes the two-tailed critical value cutting off total probability α , so option (b) is the intended standard form.

STEP-BY-STEP SOLUTION

1. With σ unknown, the pivot is $\frac{\sqrt{n}(\bar{X} - \mu)}{s} \sim t_{n-1}$.

2. Inverting gives $\bar{X} \pm t_{n-1} \frac{s}{\sqrt{n}}$, and by symmetry and unimodality of the t density the equal-tailed version is the shortest.

3. Options (a), (c) and (d) all insert a spurious factor of 2 in the denominator or use the wrong degrees of freedom.

4. Option (b).

Q9.

Let $X_1, \dots, X_n$ be a sample of size n from $N(0, \sigma^2)$. A central 0.95 confidence interval for σ^2 (equal tail probabilities), large sample, is

A. $\left(\frac{\frac{1}{n} \sum_{i=1}^n X_i^2}{1+1.96\sqrt{2/n}}, \frac{\frac{1}{n} \sum_{i=1}^n X_i^2}{1-1.96\sqrt{2/n}} \right)$

B. $\left(\frac{\frac{1}{n} \sum_{i=1}^n X_i^2}{1+1.96\sqrt{2/\pi}}, \frac{\frac{1}{n} \sum_{i=1}^n X_i^2}{1-1.96\sqrt{2/\pi}} \right)$

C. $\left(\frac{\sum_{i=1}^n X_i^2}{1+1.96\sqrt{2/n}}, \frac{\sum_{i=1}^n X_i^2}{1-1.96\sqrt{2/n}} \right)$

D. $\left(\frac{\sum_{i=1}^n X_i^2}{1+1.645\sqrt{2/n}}, \frac{\sum_{i=1}^n X_i^2}{1-1.645\sqrt{2/n}} \right)$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$$T = \frac{1}{n} \sum X_i^2 \text{ has } E(T) = \sigma^2, V(T) = \frac{2\sigma^4}{n}; \text{ inverting } \left| \frac{T - \sigma^2}{\sigma^2 \sqrt{2/n}} \right| < 1.96 \text{ gives}$$

$$\left(\frac{T}{1+1.96\sqrt{2/n}}, \frac{T}{1-1.96\sqrt{2/n}} \right).$$

TIPS & TRICKS

- $\sum X_i^2 / \sigma^2 \sim \chi_n^2$ with mean n and variance $2n$, hence $V(T) = 2\sigma^4/n$.
- The estimator must be the AVERAGE $\frac{1}{n} \sum X_i^2$, not the raw sum - options (c) and (d) fail on that alone.
- 1.96 is the 95% two-sided multiplier; 1.645 in option (d) corresponds to 90%.

STEP-BY-STEP SOLUTION

1. With $\mu=0$ known, $T = \frac{1}{n} \sum X_i^2$ satisfies $E(T) = \sigma^2$ and $V(T) = \frac{2\sigma^4}{n}$ (since $\sum X_i^2 / \sigma^2 \sim \chi_n^2$).

2. For large n , $\frac{T - \sigma^2}{\sigma^2 \sqrt{2/n}} \approx N(0,1)$.

3. $P\left(-1.96 < \frac{T - \sigma^2}{\sigma^2 \sqrt{2/n}} < 1.96\right) = 0.95$ rearranges to $\frac{T}{1+1.96\sqrt{2/n}} < \sigma^2 < \frac{T}{1-1.96\sqrt{2/n}}$.

4. Option (a).

Q10.

A random sample of size 64 from $\text{Poisson}(\lambda)$ with $\sum_{i=1}^n x_i = 256$. The 95% confidence interval of λ is

A. $\sqrt{(3.51, 4.49)}$

B. (0.08, 7.92)

C. (3.76, 4.25)

D. (3.02, 4.98)

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\bar{x} = 256/64 = 4$, $SE = \sqrt{\bar{x}/n} = \sqrt{4/64} = 0.25$, so the interval is $4 \pm 1.96(0.25) = (3.51, 4.49)$.

TIPS & TRICKS

- Poisson: $V(\bar{X}) = \lambda/n$, estimated by $\bar{x}/n$ - so the SE is $\sqrt{\bar{x}/n}$.
- $\sqrt{4/64} = \sqrt{0.0625} = 0.25$ exactly - the numbers are designed to be clean.
- Margin = $1.96 \times 0.25 = 0.49$, so the interval is 4 ± 0.49 .

STEP-BY-STEP SOLUTION

1. $\bar{x} = \frac{256}{64} = 4$, the MLE of λ .
2. $V(\bar{X}) = \frac{\lambda}{n}$, estimated by $\frac{\bar{x}}{n} = \frac{4}{64} = 0.0625$, so $SE = 0.25$.
3. 95% CI $= \bar{x} \pm 1.96 SE = 4 \pm 1.96(0.25) = 4 \pm 0.49$.
4. $= (3.51, 4.49)$ - option (a).

Q11.

If $x_1, \dots, x_n$ is from $\text{Poisson}(\lambda)$, the 95% confidence interval for λ (large n) is

- A. $\bar{x} \pm 1.96 \frac{\bar{x}}{n}$
- B. $\bar{x} \pm 1.96 \frac{\sqrt{\bar{x}}}{n}$
- C. $\bar{x} \pm 1.96 \frac{\bar{x}}{\sqrt{n}}$
- D. ✓ $\bar{x} \pm 1.96 \sqrt{\frac{\bar{x}}{n}}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$V(\bar{X}) = \lambda/n$, estimated by $\bar{x}/n$, so the interval is $\bar{x} \pm 1.96 \sqrt{\bar{x}/n}$.

TIPS & TRICKS

- The standard error of a Poisson mean is $\sqrt{\lambda/n}$ - the square root is essential.
- Options (a) and (c) forget the square root; option (b) divides by n instead of $\sqrt{n}$ inside.
- Dimensional check: the margin must be of the same order as $\sqrt{\bar{x}/\sqrt{n}}$.

STEP-BY-STEP SOLUTION

1. For $\text{Poisson}(\lambda)$, $E(X) = V(X) = \lambda$, so $V(\bar{X}) = \frac{\lambda}{n}$.

2. Estimating λ by $\bar{x}$ gives $SE(\bar{X}) = \sqrt{\frac{\bar{x}}{n}}$.

3. For large n the CLT gives $\bar{X} \approx N\left(\lambda, \frac{\lambda}{n}\right)$, so the 95% interval is $\bar{x} \pm 1.96 \sqrt{\frac{\bar{x}}{n}}$.

4. Option (d).

Q12.

Suppose waiting time at a vaccination centre queue is modelled by a normal distribution. A random sample of 16 adults showed a mean waiting time of 40 minutes with a standard deviation of 8 minutes. What is the 95% confidence interval for the true waiting time in this queue? [Given that $t_{15, 0.025} = 2.13$]

A. ~~35.6, 44.4~~

B. [25.7, 54.3]

C. [30.3, 50.7]

D. [35.7, 40.3]

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\bar{x} \pm t_{15, 0.025} \frac{s}{\sqrt{n}} = 40 \pm 2.13 \left(\frac{8}{4} \right) = 40 \pm 4.26, \text{ i.e. about } [35.6, 44.4].$$

TIPS & TRICKS

- $\sqrt{16} = 4$, so the standard error is $8/4 = 2$ - the arithmetic is designed to be clean.
- Use the t multiplier because σ is estimated; with $n=16$ the correction over 1.96 is material.
- The interval must be symmetric about 40 - option (d) is not and can be discarded instantly.

STEP-BY-STEP SOLUTION

$$1. SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{8}{\sqrt{16}} = 2.$$

$$2. \text{Margin of error} = t_{15, 0.025} \times SE = 2.13 \times 2 = 4.26.$$

$$3. 95\% \text{ CI} = 40 \pm 4.26 = (35.74, 44.26).$$

4. Matching the printed choices, this is [35.6, 44.4] - option (a).

Q13.

Let $x_1, x_2, x_3, \dots, x_{100}$ be a random sample of size 100 from the population having pdf $f(x) = \theta e^{-x\theta}$; $0 < x < \infty$. 95% confidence limits for θ will be:

A. $\frac{1}{\bar{x}} \pm \frac{1.96}{\bar{x}}$

B. $\frac{1}{\bar{x}} \pm \frac{0.196}{\bar{x}}$

C. $\frac{1}{\bar{x}} \pm \frac{19.6}{\bar{x}}$

D. $\frac{1}{\bar{x}} \pm \frac{0.0196}{\bar{x}}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$\hat{\theta} = 1/\bar{x}$ with $SE = \hat{\theta}/\sqrt{n} = \frac{1}{10\bar{x}}$, so the limits are $\frac{1}{\bar{x}} \pm \frac{0.196}{\bar{x}}$.

TIPS & TRICKS

- $n=100 \Rightarrow \sqrt{n}=10$, so $1.96/\sqrt{n}=0.196$ - the arithmetic is designed to be immediate.
- $I_n(\theta) = n/\theta^2$ gives asymptotic $SE = \theta/\sqrt{n}$, estimated by $\hat{\theta}/\sqrt{n}$.
- Option (a) omits the $\sqrt{n}$ entirely; options (c), (d) misplace the decimal.

STEP-BY-STEP SOLUTION

1. The MLE is $\hat{\theta} = \frac{1}{\bar{x}}$, asymptotically $N\left(\theta, \frac{\theta^2}{n}\right)$.

2. $SE(\hat{\theta}) = \frac{\hat{\theta}}{\sqrt{n}} = \frac{1}{\bar{x}\sqrt{100}} = \frac{1}{10\bar{x}}$.

3. 95% limits: $\frac{1}{\bar{x}} \pm 1.96 \cdot \frac{1}{10\bar{x}} = \frac{1}{\bar{x}} \pm \frac{0.196}{\bar{x}}$.

4. Option (b).

Q14.

Let $X \sim N(\mu, \sigma^2)$ where μ is known. If a confidence interval for σ^2 is to be determined, then an appropriate pivot is:

A. $(n-1) \frac{S^2}{\sigma^2}$

B. ✓ $\sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2}$

C. $\frac{(\bar{X} - \mu)\sqrt{n}}{\sigma}$

D. $\frac{(\bar{X} - \mu)\sqrt{n}}{S}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

With μ known the natural pivot is $\sum \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2_n$.

TIPS & TRICKS

- A pivot must have a distribution free of ALL unknown parameters - here only σ^2 is unknown and it cancels.
- μ known $\Rightarrow n$ degrees of freedom, not $n-1$; option (a) wastes the known mean.
- Options (c) and (d) are pivots for μ , not for σ^2 .

STEP-BY-STEP SOLUTION

1. $\frac{X_i - \mu}{\sigma} \sim N(0,1)$ independently, so $\sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2_n$.

2. Its distribution involves no unknown parameter, so it is a valid pivot for σ^2 .

3. Inverting $\chi^2_{n,1-\alpha/2} \leq \sum \frac{(X_i - \mu)^2}{\sigma^2} \leq \chi^2_{n,\alpha/2}$ gives the confidence interval.

4. Option (b).

Q15.

Let $X_1, X_2, \dots, X_n$ be iid exponential variates with mean $\lambda > 0$. Define $T = \sum_{i=1}^n x_i$, a sufficient statistic for λ . Then the pivotal statistic to construct a confidence interval for λ is:

A. $2\lambda + T$

B. $\frac{\sqrt{2T}}{\lambda}$

C. $\frac{2\lambda}{T^2}$

D. $\frac{\lambda}{2T^2}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

For an exponential sample with mean λ , $\frac{2T}{\lambda} \sim \chi^2_{2n}$ - a distribution free of λ , hence a pivot.

TIPS & TRICKS

- $T = \sum X_i \sim \text{Gamma}(n, \lambda)$, and $2T/\lambda \sim \chi^2_{2n}$ - memorise this conversion.
- A pivot must combine data and parameter with a parameter-free distribution; options (a), (c), (d) all retain λ .
- Inverting gives the exact interval $\left(\frac{2T}{\chi^2_{2n, \alpha/2}}, \frac{2T}{\chi^2_{2n, 1-\alpha/2}} \right)$.

STEP-BY-STEP SOLUTION

1. X_i exponential with mean λ means $X_i/\lambda \sim \text{Exp}(1)$, so $T/\lambda \sim \text{Gamma}(n, 1)$.
2. A $\text{Gamma}(n, 1)$ variable doubled is χ^2_{2n} : $\frac{2T}{\lambda} \sim \chi^2_{2n}$.
3. This distribution does not involve λ , so $\frac{2T}{\lambda}$ is a pivotal quantity.
4. Option (b).

Q16.

Let $x_1, x_2, \dots, x_n$ be a random sample from $e^{-(x-\theta)}$, $x \geq \theta$, $\theta > 0$. What is the confidence coefficient of the interval $\left(x_{(1)} - \frac{\ln 4}{n}, x_{(1)} + \frac{\ln 2}{n}\right)$ for θ , where $x_{(1)} = \min(x_1, x_2, \dots, x_n)$?

- A. 0.5
 B. **0.75**
 C. 0.95
 D. 0.99

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$W = n(X_{(1)} - \theta) \sim \text{Exp}(1)$; the interval is equivalent to $-\ln 2 < W < \ln 4$, and since $W \geq 0$ the coverage is $P(W < \ln 4) = 1 - \frac{1}{4} = 0.75$.

TIPS & TRICKS

- $n(X_{(1)} - \theta) \sim \text{Exp}(1)$ - the pivotal fact for this family.
- Translate each endpoint into a statement about W , then note $W \geq 0$ makes the lower one vacuous.
- $P(W < \ln 4) = 1 - e^{-\ln 4} = 1 - \frac{1}{4}$.

STEP-BY-STEP SOLUTION

1. $P(X_{(1)} > t) = e^{-n(t-\theta)}$ for $t > \theta$, so $W = n(X_{(1)} - \theta) \sim \text{Exp}(1)$.
2. $\theta > x_{(1)} - \frac{\ln 4}{n} \iff W < \ln 4$; and $\theta < x_{(1)} + \frac{\ln 2}{n} \iff W > -\ln 2$, which always holds since $W \geq 0$.
3. Coverage $= P(0 \leq W < \ln 4) = 1 - e^{-\ln 4} = 1 - \frac{1}{4} = 0.75$.
4. Option (b).

Q17.

Consider the following statements:

I. Uniformly most accurate lower confidence bound has the smallest probability of false coverage.

II. Confidence level has the smallest probability of true coverage.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Uniform most accuracy is defined by minimising false-coverage probability (I true), and the confidence level is the INFIMUM of the true-coverage probability (II true).

TIPS & TRICKS

- False coverage: the interval covers a wrong value $\theta' \neq \theta$; UMA bounds minimise this uniformly.
- The confidence coefficient is $\inf_{\theta} P_{\theta}(\theta \in S(X))$ - literally the smallest true-coverage probability.
- Accuracy (false coverage) and level (true coverage) are complementary design criteria.

STEP-BY-STEP SOLUTION

1. Statement I: a lower confidence bound $\underline{\theta}$ is uniformly most accurate at level $1-\alpha$ if it minimises $P_{\theta}(\underline{\theta} \leq \theta')$ for every $\theta' < \theta$ - i.e. the probability of covering a false value. True.
2. Statement II: the confidence level (confidence coefficient) of a family of sets $S(X)$ is defined as $\inf_{\theta} P_{\theta}(\theta \in S(X))$, the smallest probability of true coverage. True.
3. Option (c).

Q18.

Let $\{0,4,2,3\}$ be a random sample from a Poisson distribution with mean $\theta > 0$. Then the asymptotic confidence interval with confidence coefficient 0.95 for mean θ is

A. ~~(0.78,3.72)~~

B. (0.78,4.72)

C. (2.25,3.72)

D. (2.25,4.72)

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\bar{x} = \frac{0+4+2+3}{4} = 2.25$, $SE = \sqrt{2.25/4} = 0.75$, so the interval is $2.25 \pm 1.96(0.75) = (0.78, 3.72)$.

TIPS & TRICKS

- Poisson SE is $\sqrt{\bar{x}/n}$; here $\sqrt{0.5625} = 0.75$ exactly.
- Margin = $1.96 \times 0.75 = 1.47$.
- The interval must be centred at 2.25 - options (b), (c), (d) are not.

STEP-BY-STEP SOLUTION

1. $\bar{x} = \frac{0+4+2+3}{4} = 2.25$, the MLE of θ .

2. $SE = \sqrt{\frac{\bar{x}}{n}} = \sqrt{\frac{2.25}{4}} = 0.75$.

3. 95% interval: $2.25 \pm 1.96(0.75) = 2.25 \pm 1.47$.

4. $= (0.78, 3.72)$ - option (a).

Q19.

Let $X_i \sim N(\mu, \sigma^2)$; $i=1,2,3,\dots,n$. Which of the following statements is/are correct?

I. At confidence level $(1-\alpha)$ if α and σ are known, then we can choose sample size n to get a confidence interval of a fixed length.

II. At confidence level $(1-\alpha)$ if σ is not known, then we cannot choose sample size n to get a fixed width confidence interval of level $(1-\alpha)$.

Select the answer using the code given below:

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

With σ known the length $2z_{\alpha/2}\sigma/\sqrt{n}$ is deterministic, so n can be chosen; with σ unknown the length involves the random S , so no fixed n guarantees a fixed width.

TIPS & TRICKS

- σ known $\Rightarrow$ length $= 2z_{\alpha/2}\sigma/\sqrt{n}$: solve for n directly.
- σ unknown $\Rightarrow$ length $= 2t_{n-1,\alpha/2}S/\sqrt{n}$, and S is random - the width is random too.
- Fixed-width intervals with unknown σ require SEQUENTIAL sampling (Stein's two-stage procedure).

STEP-BY-STEP SOLUTION

1. Statement I: with σ known the interval is $\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$, of fixed length $2z_{\alpha/2}\sigma/\sqrt{n}$; setting this equal to a target length determines n . True.
2. Statement II: with σ unknown the interval is $\bar{X} \pm t_{n-1,\alpha/2} \frac{S}{\sqrt{n}}$, whose length depends on the random S , so no fixed n can guarantee a prescribed length. True.
3. Option (c).

Q20.

In sampling from a location-scale family, $f(x; \theta, \sigma) = \frac{1}{\sigma} f\left(\frac{x - \theta}{\sigma}\right)$, which one of the following is a pivot?

- A. $\bar{X} - \theta$
- B. $(X_{(2)} + X_{(1)} - 2\theta)$
- C. $\frac{\bar{X} - 2\theta}{(s/\sqrt{n})}$
- D. $\frac{\sqrt{X_{(2)} + X_{(1)} - 2\theta}}{s}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Only $\frac{X_{(2)} + X_{(1)} - 2\theta}{s}$ has both numerator and denominator scaling with σ , so its distribution is free of (θ, σ) .

TIPS & TRICKS

- Write $X_i = \theta + \sigma Z_i$ with Z_i from the standard member of the family.
- Numerator: $X_{(2)} + X_{(1)} - 2\theta = \sigma(Z_{(2)} + Z_{(1)})$; denominator: $s = \sigma s_Z$. The σ cancels.
- Options (a) and (b) leave a free σ ; option (c) uses 2θ where θ belongs, so it is not even centred.

STEP-BY-STEP SOLUTION

1. Write $X_i = \theta + \sigma Z_i$ where the Z_i come from the standard member $f(z)$, whose law involves no parameters.
2. (a) $\bar{X} - \theta = \sigma \bar{Z}$: distribution depends on σ - not a pivot.
3. (b) $X_{(2)} + X_{(1)} - 2\theta = \sigma(Z_{(2)} + Z_{(1)})$: still depends on σ - not a pivot.
4. (d) $\frac{X_{(2)} + X_{(1)} - 2\theta}{s} = \frac{\sigma(Z_{(2)} + Z_{(1)})}{\sigma s_Z} = \frac{Z_{(2)} + Z_{(1)}}{s_Z}$, whose distribution involves no unknown parameter - a genuine pivot.
5. Option (d).

Q21.

In case of random sampling from $F_\theta(X_i) \sim U(0,1)$, a pivot is:

- A. $\sum_{i=1}^n F_\theta(X_i)$
- B. $\sum_{i=1}^n \log F_\theta(X_i)$
- C. $\exp\{-F_\theta(X_i)\}$
- D. $-\sum_{i=1}^n \log F_\theta(X_i)$

Correct answer: D

EXAM SHORTCUT[AI-derived explanation](#)

$-\log F_\theta(X_i) \sim \text{Exp}(1)$, so $-\sum \log F_\theta(X_i) \sim \text{Gamma}(n,1)$ - a distribution free of θ , hence a pivot.

TIPS & TRICKS

- Probability integral transform: $F_\theta(X) \sim U(0,1)$, and $-\log U \sim \text{Exp}(1)$.
- $-2 \sum \log F_\theta(X_i) \sim \chi^2_{2n}$ - the form used in Fisher's method of combining p -values.
- The SIGN matters: $\sum \log F_\theta(X_i)$ is negative; option (b) omits the minus sign but is still parameter-free - option (d) is the standard positive-valued pivot.

STEP-BY-STEP SOLUTION

1. By the probability integral transform, $U_i = F_\theta(X_i) \sim U(0,1)$ independently.
2. $-\log U_i \sim \text{Exp}(1)$, so $-\sum_{i=1}^n \log F_\theta(X_i) \sim \text{Gamma}(n,1)$.
3. This distribution involves no unknown parameter, so the quantity is pivotal (and $-2 \sum \log F_\theta(X_i) \sim \chi^2_{2n}$).
4. Option (d).

Q22.

Let $X_1, X_2, X_3, \dots, X_n$ be iid random variables such that $X_i \sim N(\mu, \sigma^2)$. The width of $(1-\alpha)$ level confidence interval $(\bar{X}-c_1, \bar{X}+c_2)$ of μ is:

A. $\frac{2c\sqrt{n}}{\sigma}$ if $c_1=c_2=c$

B. $\frac{\sqrt{2}c\sigma}{\sqrt{n}}$ if $c_1=c_2=c$

C. $\frac{(c_1+c_2)\sigma}{2\sqrt{n}}$

D. $\frac{\sqrt{2}(c_2-c_1)\sigma}{\sqrt{n}}$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

With a symmetric half-width, the confidence interval is $\bar{X} \pm c \frac{\sigma}{\sqrt{n}}$, so its width is $\frac{2c\sigma}{\sqrt{n}}$.

TIPS & TRICKS

- Width = upper limit - lower limit; for a symmetric interval it is twice the half-width.
- The half-width always carries $\sigma/\sqrt{n}$, so $\sqrt{n}$ belongs in the DENOMINATOR.
- Option (a) inverts the standard error; options (c) and (d) mis-scale the constants.

STEP-BY-STEP SOLUTION

1. For $N(\mu, \sigma^2)$ the $(1-\alpha)$ interval for μ takes the form $\bar{X} \pm c \frac{\sigma}{\sqrt{n}}$ with $c = z_{\alpha/2}$ (or $t_{n-1, \alpha/2}$).
2. Its width is the difference between the endpoints: $2c \frac{\sigma}{\sqrt{n}}$.
3. Option (b).

Q23.

The 95% confidence interval for μ in $N(\mu, \sigma^2)$ when σ is known is (200, 220). The value of sample mean is:

- A. 420
- B. 220
- C. **210**
- D. 200

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

With σ known the interval is symmetric, so $\bar{x}$ is the midpoint $\frac{200+220}{2} = 210$.

TIPS & TRICKS

- Symmetric interval $\Rightarrow$ the sample mean is the midpoint.
- Half-width = 10, so $z_{\alpha/2}\sigma/\sqrt{n} = 10$ if that is ever needed.
- Option (a) is the SUM of the endpoints - a planted slip.

STEP-BY-STEP SOLUTION

1. For known σ the interval is $\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right)$, symmetric about $\bar{x}$.

2. $\bar{x} = \frac{200+220}{2} = 210$.

3. Option (c).

Q24.

Let $P_\theta[\theta \in S(X)] \geq 1 - \alpha$ for all $\theta \in \Theta$ constitute a family of confidence sets. Which of the following statements is/are correct?

I. $S(X)$ is a random set.

II. For a given $X=x$, either $S(X)$ covers θ or it does not.

Select the answer using the code given below:

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$S(X)$ is a function of the random sample, hence a random set (I true); once $X=x$ is observed the set is fixed and either contains θ or not (II true).

TIPS & TRICKS

- Confidence statements are about the PROCEDURE, not about the realised interval.
- Before sampling: coverage probability $1 - \alpha$. After sampling: coverage is 0 or 1, we just do not know which.
- This is the standard frequentist interpretation and is examined almost verbatim.

STEP-BY-STEP SOLUTION

1. Statement I: $S(X)$ depends on the random vector X , so it is a random set; the probability statement $P_\theta[\theta \in S(X)] \geq 1 - \alpha$ is over the randomness of X . True.
2. Statement II: for an observed x , $S(x)$ is a fixed set, so the event $\theta \in S(x)$ is either true or false - no probability attaches to it in the frequentist framework. True.
3. Option (c).

Q25.

Let $x_1, x_2, \dots, x_n$ be a random sample from the distribution having p.d.f. $f(x, \theta) = e^{-(x-\theta)}$; $x > \theta$. Further,

$\left(x_{(1)} - \frac{\log 40}{n}, x_{(1)}\right)$ is $100\beta\%$ confidence interval for θ , where $x_{(1)} = \min\{x_1, x_2, \dots, x_n\}$. What is the value of β ?

- A. 0.90
- B. 0.92
- C. 0.95
- D. 0.975

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$W = n(X_{(1)} - \theta) \sim \text{Exp}(1)$; the interval corresponds to $0 \leq W < \log 40$, with probability $1 - \frac{1}{40} = 0.975$.

TIPS & TRICKS

- $\theta < x_{(1)}$ is automatic, so only the lower endpoint constrains the coverage.
- $P(W < \log c) = 1 - \frac{1}{c}$ - so $c = 40$ gives 0.975 directly.
- Compare the companion 2023 item where $\log 4$ gave 0.75 - same mechanism, different constant.

STEP-BY-STEP SOLUTION

1. $P(X_{(1)} > t) = e^{-n(t-\theta)}$, so $W = n(X_{(1)} - \theta) \sim \text{Exp}(1)$.

2. $\theta > x_{(1)} - \frac{\log 40}{n} \iff W < \log 40$; and $\theta < x_{(1)} \iff W > 0$, which always holds.

3. Coverage $= P(0 < W < \log 40) = 1 - e^{-\log 40} = 1 - \frac{1}{40} = 0.975$.

4. Option (d).

Q26.

If 95% confidence interval for μ with $Z_{\alpha/2}=1.96$ is (400,407.84) and the sample size is large, the value of the standard error of mean $\bar{X}$ will be

- A. 2
- B. 4
- C. 10
- D. 20

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\text{Half-width} = \frac{407.84 - 400}{2} = 3.92, \text{ and } SE = \frac{3.92}{1.96} = 2.$$

TIPS & TRICKS

- $SE = \text{half-width divided by } z_{\alpha/2}$.
- $3.92 = 2 \times 1.96$ exactly - the numbers are chosen to be clean.
- Do not divide the full width by z ; halve it first.

STEP-BY-STEP SOLUTION

1. The interval (400, 407.84) has width 7.84, so the half-width is 3.92.
2. Half-width $= Z_{\alpha/2} \times SE(\bar{X})$, i.e. $3.92 = 1.96 \times SE$.
3. $SE = \frac{3.92}{1.96} = 2$ - option (a).

Q27.

For testing $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$, the 90% confidence interval for μ of normal distribution, when σ^2 is unknown, is (360, 400). Then the value of sample mean is given by

A. 380**B. 390****C. 400****D. 420****Correct answer: A****EXAM SHORTCUT**

AI-derived explanation

The t -interval is symmetric about $\bar{x}$, so $\bar{x} = \frac{360+400}{2} = 380$.

TIPS & TRICKS

- Symmetry holds whether the multiplier is z or t - the midpoint is always the sample mean.
- Half-width = 20, so $t_{n-1, 0.05} s/\sqrt{n} = 20$ if needed.
- The confidence level (90% here) does not affect the midpoint.

STEP-BY-STEP SOLUTION

1. With σ^2 unknown the interval is $\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$, symmetric about $\bar{x}$.

2. $\bar{x} = \frac{360+400}{2} = 380$.

3. Option (a).

Resampling: Bootstrap and Jackknife

Q1.

If we have a sample of size $n=5$, then the number of distinct bootstrap samples with replacement of size 5 is:

- A. 625
- B. 306
- C. 252
- D. 126**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Distinct (unordered) bootstrap resamples number $\binom{2n-1}{n} = \binom{9}{5} = 126$.

TIPS & TRICKS

- Counting multisets of size n from n items: $\binom{2n-1}{n}$ - the stars-and-bars formula.
- $n^n = 5^5 = 3125$ counts ORDERED resamples; 625 and 252 are other planted values.
- $\binom{9}{5} = 126$; note $\binom{9}{5} = \binom{9}{4}$.

STEP-BY-STEP SOLUTION

1. A bootstrap resample of size n drawn with replacement is a multiset of n items chosen from the n original observations.
2. The number of such multisets is $\binom{n+n-1}{n} = \binom{2n-1}{n}$.
3. For $n=5$: $\binom{9}{5} = \frac{9!}{5!4!} = 126$.
4. Option (d).

Q2.

Consider the following:

I. Bootstrap

II. Jackknife

III. Cross-validation

IV. Stratified sampling

How many of the above are resampling methods?

A. One

B. Two

C. ~~Two~~ Three

D. All the four

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Bootstrap, jackknife and cross-validation are resampling methods; stratified sampling is a SAMPLING DESIGN, not a resampling method. So three.

TIPS & TRICKS

- Resampling reuses an existing sample; stratified sampling collects a new one from the population.
- Bootstrap: with replacement. Jackknife: delete-one. Cross-validation: repeated train/test splits.
- The distinguishing question is: does the method draw from data already in hand, or from the population?

STEP-BY-STEP SOLUTION

1. Bootstrap draws repeated samples WITH replacement from the observed data - resampling.
2. Jackknife systematically deletes one (or d) observations at a time - resampling.
3. Cross-validation repeatedly partitions the observed data into training and validation parts - resampling.
4. Stratified sampling is a design for drawing the ORIGINAL sample from the population - not resampling.
5. Three of the four - option (c).

Hypothesis Testing

91 authentic PYQs in this topic (2018-2026)

Randomized Tests

SI & HT

2018 Q39

Numerical

Medium

Q1.

A sample of size one is drawn from $\text{Poisson}(\theta)$. To test $H_0:\lambda=1$ against $H_1:\lambda=2$, consider the randomized test with critical region $\omega=\{x:x>3\}$. The probability of Type-I error is

A. $(3e^2-19)/(3e^2)$

B. $(3e-5)/3$

C. $(3e-8)/(3e)$

D. $(3e^2-16)/(3e^2)$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$\alpha = P(X > 3 | \lambda = 1) = 1 - e^{-1} \left(1 + 1 + \frac{1}{2} + \frac{1}{6} \right) = 1 - \frac{8}{3e} = \frac{3e-8}{3e}.$$

TIPS & TRICKS

- Type-I error is always computed under H_0 - here $\lambda=1$, so every term carries e^{-1} .
- $1 + 1 + \frac{1}{2} + \frac{1}{6} = \frac{8}{3}$; multiply by e^{-1} and subtract from 1.
- Options with e^2 come from mistakenly using $\lambda=2$ (the alternative) - a deliberate trap.

STEP-BY-STEP SOLUTION

1. Under $H_0:\lambda=1$, $P(X=k) = \frac{e^{-1}}{k!}$.

2. $P(X \leq 3) = e^{-1} \left(\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} \right) = e^{-1} \left(1 + 1 + \frac{1}{2} + \frac{1}{6} \right) = e^{-1} \cdot \frac{8}{3}$.

3. $\alpha = P(X > 3) = 1 - \frac{8}{3e} = \frac{3e-8}{3e}$.

4. Option (c).

Q2.

Let X be a single observation from $f(x)$. To test $H_0: f=f_0$ against $H_1: f=f_1$, consider

$$\varphi(x) = \begin{cases} 1, & f_1(x) > kf_0(x) \\ \gamma, & f_1(x) = kf_0(x) \\ 0, & f_1(x) < kf_0(x) \end{cases}$$

If $\gamma > 0$, which is correct?

- A. $\varphi(x)$ is a non-randomised test function
- B. $\varphi(x)$ is neither non-randomised nor randomised
- C. $\varphi(x)$ is a randomised test function
- D. None of the above

Correct answer: C

EXAM SHORTCUT AI-derived explanation

A test function taking a value strictly between 0 and 1 is by definition randomised.

TIPS & TRICKS

- Non-randomised $\iff \varphi$ takes only the values 0 and 1 (it is an indicator of a set).
- Randomisation exists to achieve an EXACT size α in discrete problems.
- γ is chosen so that $E_{H_0}[\varphi(X)] = \alpha$ exactly.

STEP-BY-STEP SOLUTION

1. A test function is $\varphi: \mathcal{X} \rightarrow [0, 1]$, where $\varphi(x)$ is the probability of rejecting H_0 when x is observed.
2. It is non-randomised precisely when φ takes only the two values 0 and 1.
3. Here $\varphi(x) = \gamma$ with $\gamma > 0$ on the boundary set $\{f_1 = kf_0\}$, an intermediate value, so a genuine coin-toss is used.
4. Hence φ is a randomised test function - option (c).

Q3.

Let X have pmf under H_0, H_1 :

X	1	2	3	4	5	6
$p_{\theta_0}(x)$	0.01	0.01	0.01	0.01	0.01	0.95
$p_{\theta_1}(x)$	0.05	0.04	0.03	0.02	0.01	0.85

Consider $\varphi(x)=1$ if $X=1,2$; 0.25 if $X=3$; 0 otherwise. Which is correct?

- A. $\varphi(x)$ is a most powerful test of level $\alpha=0.03$
- B. $\varphi(x)$ is a most powerful test of level $\alpha=0.03$ with power 0.0975
- C. $\varphi(x)$ has size 0.0225 and power 0.0975 but is not a most powerful test with level $\alpha=0.03$
- D. $\varphi(x)$ is not a most powerful test of level $\alpha=0.0225$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Size = $0.01+0.01+0.25(0.01)=0.0225$; power = $0.05+0.04+0.25(0.03)=0.0975$. The MP test of size exactly 0.03 takes $\{1,2,3\}$ fully and has power 0.12, so φ is not MP at level 0.03.

TIPS & TRICKS

- Size and power of a randomised test: $E_{H_0}[\varphi]$ and $E_{H_1}[\varphi]$ - multiply each probability by $\varphi(x)$ and add.
- Ratios are 5,4,3,2,1,0.894 at $x=1,\dots,6$; spending the full budget of 0.03 means taking $x=1,2,3$ outright, giving power $0.12 > 0.0975$.
- φ IS the MP test of its own size 0.0225, which is why option (d) is false.

STEP-BY-STEP SOLUTION

1. Size = $1(0.01)+1(0.01)+0.25(0.01)=0.0225$.
2. Power = $1(0.05)+1(0.04)+0.25(0.03)=0.09+0.0075=0.0975$.
3. Likelihood ratios p_1/p_0 : 5,4,3,2,1,0.894 for $x=1,\dots,6$. A size-0.03 NP test rejects outright on $x=1,2,3$ (total null probability 0.03) with power $0.05+0.04+0.03=0.12$.
4. Since $0.0975 < 0.12$, φ is not most powerful at level 0.03 (it does not spend the full size budget).
5. Hence option (c).

SHARED CONTEXT

Consider the problem of testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ for the probability distribution given below:

X	0	1	2	3
$P_{\theta_0}(x)$	1/4	1/4	1/4	1/4
$P_{\theta_1}(x)$	0.20	0.40	0.25	0.15

The null hypothesis is rejected if $X = 0$ is observed. It is further rejected when the tossing of two unbiased coins gives both heads, for $X = 1$.

Q4.

What is the probability of type-I error of the test?

- A. $\frac{3}{10}$
- B. $\frac{5}{16}$ ✓
- C. $\frac{7}{10}$
- D. $\frac{1}{4}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\varphi(0)=1 \text{ and } \varphi(1)=\frac{1}{4} \text{ (two heads), so } \alpha=\frac{1}{4}(1)+\frac{1}{4}\left(\frac{1}{4}\right)=\frac{1}{4}+\frac{1}{16}=\frac{5}{16}.$$

TIPS & TRICKS

- Two unbiased coins both showing heads has probability $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ - that is the randomisation probability.
- Size = $\sum_x \varphi(x)P_{\theta_0}(x)$: weight each outcome by its rejection probability.
- Under H_0 all four outcomes have probability $\frac{1}{4}$, which keeps the arithmetic simple.

STEP-BY-STEP SOLUTION

1. The test function is $\varphi(0)=1$, $\varphi(1)=P(\text{two heads})=\frac{1}{4}$, $\varphi(2)=\varphi(3)=0$.

2. $\alpha = E_{\theta_0}[\varphi(X)] = \varphi(0)P_{\theta_0}(0) + \varphi(1)P_{\theta_0}(1).$

3. $= 1 \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{4} = \frac{1}{4} + \frac{1}{16} = \frac{5}{16}.$

4. Option (b).

SHARED CONTEXT

Consider the problem of testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ for the probability distribution given below:

X	0	1	2	3
$P_{\theta_0}(x)$	1/4	1/4	1/4	1/4
$P_{\theta_1}(x)$	0.20	0.40	0.25	0.15

The null hypothesis is rejected if $X = 0$ is observed. It is further rejected when the tossing of two unbiased coins gives both heads, for $X = 1$.

Q5.

What is the probability of type-II error of the test?

- A. $\frac{3}{10}$
- B. $\frac{5}{16}$
- C. $\frac{7}{10}$
- D. $\frac{1}{4}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$\text{Power} = 1(0.20) + \frac{1}{4}(0.40) = 0.30, \text{ so } \beta = 1 - 0.30 = 0.70 = \frac{7}{10}.$$

TIPS & TRICKS

- Compute the power first, then subtract from 1 - Type-II error is never computed directly here.
- Use the H_1 row of the table: $P_{\theta_1}(0) = 0.20$, $P_{\theta_1}(1) = 0.40$.
- Note $\alpha + \beta = \frac{5}{16} + \frac{7}{10} \approx 1.01 > 1$, so this test is (just) biased - a nice consistency observation.

STEP-BY-STEP SOLUTION

$$1. \text{ Power} = E_{\theta_1}[\varphi(X)] = 1 \times 0.20 + \frac{1}{4} \times 0.40 = 0.20 + 0.10 = 0.30.$$

$$2. \text{ Type-II error } \beta = 1 - \text{power} = 1 - 0.30 = 0.70.$$

$$3. = \frac{7}{10} - \text{option (c).}$$

SHARED CONTEXT

Let x_1, x_2, x_3, x_4 be a random sample from Poisson distribution with mean $\lambda > 0$. To test $H_0: \lambda = 1$ versus $H_1: \lambda = 1.5$, the following randomized test is considered:

$$\varphi(x) = \begin{cases} 1, & \text{if } \sum_{i=1}^4 x_i > 2 \\ 0.1, & \text{if } \sum_{i=1}^4 x_i = 2 \\ 0, & \text{if } \sum_{i=1}^4 x_i < 2 \end{cases}$$

(Given $e^{-4} \approx 0.0183$ and $e^{-6} \approx 0.0025$)

Q6.

The size of the randomized test is within the interval:

- A. (0.05, 0.20)
- B. (0.30, 0.50)
- C. **(0.60, 0.80)**
- D. (0.80, 0.90)

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$T = \sum x_i \sim \text{Poisson}(4)$ under H_0 ; size = $P(T > 2) + 0.1P(T = 2) = 0.7619 + 0.0147 = 0.777$, which lies in (0.60, 0.80).

TIPS & TRICKS

- Pool first: for a Poisson sample, $\sum X_i \sim \text{Poisson}(n\lambda) = \text{Poisson}(4)$ under H_0 .
- $P(T \leq 2) = e^{-4}(1 + 4 + 8) = 13e^{-4} = 0.238$.
- Size = $1 - P(T \leq 2) + 0.1P(T = 2)$ - do not forget the randomised contribution.

STEP-BY-STEP SOLUTION

- Under $H_0: \lambda = 1$, $T = \sum_{i=1}^4 x_i \sim \text{Poisson}(4)$.
- $P(T = 0) = e^{-4} = 0.0183$, $P(T = 1) = 4e^{-4} = 0.0733$, $P(T = 2) = 8e^{-4} = 0.1465$, so $P(T \leq 2) = 0.2381$.
- Size = $P(T > 2) + 0.1P(T = 2) = (1 - 0.2381) + 0.1(0.1465) = 0.7619 + 0.0147$.
- = 0.7766, which lies in (0.60, 0.80) - option (c).

SHARED CONTEXT

Let x_1, x_2, x_3, x_4 be a random sample from Poisson distribution with mean $\lambda > 0$. To test $H_0: \lambda = 1$ versus $H_1: \lambda = 1.5$, the following randomized test is considered:

$$\varphi(x) = \begin{cases} 1, & \text{if } \sum_{i=1}^4 x_i > 2 \\ 0.1, & \text{if } \sum_{i=1}^4 x_i = 2 \\ 0, & \text{if } \sum_{i=1}^4 x_i < 2 \end{cases}$$

(Given $e^{-4} \approx 0.0183$ and $e^{-6} \approx 0.0025$)

Q7.

The power of the randomized test is within the interval:

- A. (0.50, 0.70)
- B. (0.60, 0.75)
- C. (0.70, 0.85)
- D. **(0.80, 0.99)**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Under H_1 , $T \sim \text{Poisson}(6)$: power $= P(T > 2) + 0.1P(T = 2) = 0.938 + 0.0045 = 0.9425$, in (0.80, 0.99).

TIPS & TRICKS

- $n\lambda = 4 \times 1.5 = 6$ under H_1 - recompute the Poisson mean, do not reuse 4.
- $P(T \leq 2) = e^{-6}(1 + 6 + 18) = 25e^{-6} = 0.0620$.
- Power ≈ 0.94 is comfortably above the size ≈ 0.78 , so the test is unbiased.

STEP-BY-STEP SOLUTION

- Under $H_1: \lambda = 1.5$, $T = \sum x_i \sim \text{Poisson}(6)$.
- $P(T=0) = e^{-6} = 0.0025$, $P(T=1) = 6e^{-6} = 0.0149$, $P(T=2) = 18e^{-6} = 0.0446$; so $P(T \leq 2) = 0.0620$.
- Power $= (1 - 0.0620) + 0.1(0.0446) = 0.9380 + 0.0045 = 0.9425$.
- This lies in (0.80, 0.99) - option (d).

SHARED CONTEXT

Let X be a random variable whose pmf under H_0 and H_1 is given as:

x	0	1	2	3	4
$P(x / H_0)$	0.1	0.2	0.3	0.3	0.1
$P(x / H_1)$	0.2	0.3	0.1	0.2	0.2

The test is conducted based on a single observed value of X to test $H_0: X \sim P(x / H_0)$ versus $H_1: X \sim P(x / H_1)$. The null hypothesis is rejected if $X=0$ is observed. Further, the null hypothesis is rejected when head 'H' is observed on tossing an unbiased coin when $X=4$ is observed.

Q8.

The size of the test is:

A. 0.15

B. 0.20

C. 0.30

D. 0.60

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$\phi(0)=1$ and $\phi(4)=\frac{1}{2}$, so size $=1(0.1)+\frac{1}{2}(0.1)=0.15$.

TIPS & TRICKS

- One fair coin gives randomisation probability $\frac{1}{2}$; two coins both heads would give $\frac{1}{4}$.
- Size $= \sum_x \phi(x)P(x / H_0)$ - weight each outcome by its rejection probability.
- Only $x=0$ and $x=4$ contribute; all other cells have $\phi=0$.

STEP-BY-STEP SOLUTION

1. The test function is $\phi(0)=1$, $\phi(4)=P(\text{head})=\frac{1}{2}$, and $\phi(x)=0$ otherwise.
2. Size $=E_{H_0}[\phi(X)] = 1 \times P(0 / H_0) + \frac{1}{2} \times P(4 / H_0)$.
3. $=1(0.1)+0.5(0.1)=0.10+0.05=0.15$.
4. Option (a).

SHARED CONTEXT

Let X be a random variable whose pmf under H_0 and H_1 is given as:

x	0	1	2	3	4
$P(x / H_0)$	0.1	0.2	0.3	0.3	0.1
$P(x / H_1)$	0.2	0.3	0.1	0.2	0.2

The test is conducted based on a single observed value of X to test $H_0: X \sim P(x / H_0)$ versus $H_1: X \sim P(x / H_1)$. The null hypothesis is rejected if $X=0$ is observed. Further, the null hypothesis is rejected when head 'H' is observed on tossing an unbiased coin when $X=4$ is observed.

Q9.

The power of the test is:

A. 0.15

B. 0.30

C. 0.40

D. 0.70

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$$\text{Power} = 1(0.2) + \frac{1}{2}(0.2) = 0.30.$$

TIPS & TRICKS

- Same weights, but now against the H_1 row of the table.
- $P(0 / H_1) = 0.2$ and $P(4 / H_1) = 0.2$.
- Power $0.30 >$ size 0.15 , so this randomised test is unbiased.

STEP-BY-STEP SOLUTION

1. $\text{Power} = E_{H_1}[\varphi(X)] = 1 \times P(0 / H_1) + \frac{1}{2} \times P(4 / H_1).$

2. $= 1(0.2) + 0.5(0.2) = 0.20 + 0.10 = 0.30.$

3. Option (b).

SHARED CONTEXT

To test mean of the Poisson distribution $H_0:\lambda=1$ versus $H_1:\lambda=2$, based on the single observed value from the Poisson distribution, the following procedure is considered: "Reject H_0 with probability one when the observed value $(x)>1$. If $x=1$ is observed, then toss an unbiased coin and reject the null hypothesis if head comes up."

Q10.

What is type I error in the experiment?

- A. $1-e^{-1}$
- B. $1-0.5e^{-1}$
- C. $1-1.5e^{-1}$
- D. $1-2e^{-1}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$$\alpha = P(X > 1 / \lambda = 1) + \frac{1}{2}P(X = 1 / \lambda = 1) = (1 - 2e^{-1}) + \frac{1}{2}e^{-1} = 1 - 1.5e^{-1}.$$

TIPS & TRICKS

- $P(X \leq 1) = e^{-1} + e^{-1} = 2e^{-1}$ when $\lambda = 1$ - both terms equal because $\lambda^1/1! = 1$.
- The coin adds $\frac{1}{2} \times P(X=1) = \frac{1}{2}e^{-1}$ to the rejection probability.
- $1 - 2e^{-1} + 0.5e^{-1} = 1 - 1.5e^{-1}$ - combine the e^{-1} terms carefully.

STEP-BY-STEP SOLUTION

1. Under $H_0:\lambda=1$, $P(X=0)=e^{-1}$ and $P(X=1)=e^{-1}$, so $P(X>1)=1-2e^{-1}$.
2. The test rejects with probability 1 when $X>1$ and with probability $\frac{1}{2}$ when $X=1$.
3. $\alpha = (1 - 2e^{-1}) + \frac{1}{2}e^{-1} = 1 - \frac{3}{2}e^{-1} = 1 - 1.5e^{-1}$.
4. Option (c).

SHARED CONTEXT

To test mean of the Poisson distribution $H_0:\lambda=1$ versus $H_1:\lambda=2$, based on the single observed value from the Poisson distribution, the following procedure is considered: "Reject H_0 with probability one when the observed value $(x)>1$. If $x=1$ is observed, then toss an unbiased coin and reject the null hypothesis if head comes up."

Q11.

What is type II error in the experiment?

- A. $1-e^{-2}$
- B. $1-0.5e^{-2}$
- C. $1-2e^{-2}$
- D. $2e^{-2}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$$\text{Power} = (1-3e^{-2}) + \frac{1}{2}(2e^{-2}) = 1-2e^{-2}, \text{ so } \beta = 1-\text{power} = 2e^{-2}.$$

TIPS & TRICKS

- Under $\lambda=2$: $P(X=0)=e^{-2}$, $P(X=1)=2e^{-2}$, so $P(X>1)=1-3e^{-2}$.
- Randomised contribution is $\frac{1}{2} \times 2e^{-2} = e^{-2}$.
- Power $= 1-2e^{-2} \approx 0.729$, hence $\beta \approx 0.271$.

STEP-BY-STEP SOLUTION

1. Under $H_1:\lambda=2$: $P(X=0)=e^{-2}$, $P(X=1)=2e^{-2}$, so $P(X>1)=1-3e^{-2}$.
2. Power $= (1-3e^{-2}) + \frac{1}{2}(2e^{-2}) = 1-3e^{-2} + e^{-2} = 1-2e^{-2}$.
3. $\beta = 1-\text{power} = 2e^{-2}$.
4. Option (d).

Q1.

Let X have exponential distribution with mean $\frac{1}{\theta}$. Consider $\omega = \{x: x > 2\}$ to test $H_0: \theta = 1$ against $H_1: \theta = 2$. The power of the test at $\theta = 2$ is

- A. $(e^2 - 1)/e^2$
- B. $1/e^4$
- C. $(e^4 - 1)/e^4$
- D. $1/e^2$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Density $\theta e^{-\theta x}$; power $= P(X > 2 | \theta = 2) = e^{-2 \times 2} = e^{-4}$.

TIPS & TRICKS

- Exponential tail: $P(X > c) = e^{-\theta c}$ - no integration needed.
- Power is evaluated under H_1 ($\theta = 2$), size under H_0 ($\theta = 1$, giving e^{-2}).
- Mean $1/\theta$ means rate θ ; mixing the two up produces option (d).

STEP-BY-STEP SOLUTION

1. Mean $\frac{1}{\theta}$ means the density is $f(x) = \theta e^{-\theta x}$, $x > 0$.
2. $P(X > 2) = \int_2^\infty \theta e^{-\theta x} dx = e^{-2\theta}$.
3. Power at $\theta = 2$: $e^{-2(2)} = e^{-4} = \frac{1}{e^4}$.
4. Option (b).

Q2.

A single observation X is drawn from $B(10, p)$. Reject $H_0: p = \frac{1}{2}$ for $H_1: p = \frac{1}{4}$ if $X \leq 3$. The power of the test is

A. $369\left(\frac{3^7}{4^{10}}\right)$

B. $366\left(\frac{3^7}{4^{10}}\right)$

C. ✓ $372\left(\frac{3^7}{4^{10}}\right)$

D. $375\left(\frac{3^7}{4^{10}}\right)$

Correct answer: C**EXAM SHORTCUT**

AI-derived explanation

$$\text{Power} = P(X \leq 3 \mid p = \frac{1}{4}) = \frac{3^7}{4^{10}} [27 + 90 + 135 + 120] = 372 \cdot \frac{3^7}{4^{10}}.$$

TIPS & TRICKS

- Factor out $3^7/4^{10}$ at the start: each term $\binom{10}{k} 3^{10-k}/4^{10}$ contributes $\binom{10}{k} 3^{3-k}$ after the pull-out.
- The bracket is $\binom{10}{0} 3^3 + \binom{10}{1} 3^2 + \binom{10}{2} 3^1 + \binom{10}{3} 3^0 = 27 + 90 + 135 + 120 = 372$.
- Power is evaluated under H_1 ($p = 1/4$), never under H_0 - that is the first decision to make.

STEP-BY-STEP SOLUTION

$$1. \text{Power} = P(X \leq 3 \mid p = \frac{1}{4}) = \sum_{k=0}^3 \binom{10}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{10-k}.$$

$$2. = \frac{1}{4^{10}} \sum_{k=0}^3 \binom{10}{k} 3^{10-k} = \frac{3^7}{4^{10}} \sum_{k=0}^3 \binom{10}{k} 3^{3-k}.$$

$$3. = \frac{3^7}{4^{10}} [1 \cdot 27 + 10 \cdot 9 + 45 \cdot 3 + 120 \cdot 1] = \frac{3^7}{4^{10}} [27 + 90 + 135 + 120].$$

$$4. = 372 \left(\frac{3^7}{4^{10}}\right) - \text{option (c).}$$

Q3.

Let $X \sim N(\mu, \sigma^2)$, both unknown, μ_0, σ_0^2 known. Which is a **simple** hypothesis?

- A. $\mu = \mu_0, \sigma^2 \neq \sigma_0^2$
- B. $\mu = \mu_0, \sigma^2 = \sigma_0^2$
- C. $\mu > \mu_0, \sigma^2 = \sigma_0^2$
- D. $\mu = \mu_0, \sigma^2 < \sigma_0^2$

Correct answer: B

EXAM SHORTCUT[AI-derived explanation](#)

A simple hypothesis pins down EVERY unknown parameter. With both μ and σ^2 unknown, only $\{\mu = \mu_0, \sigma^2 = \sigma_0^2\}$ does that.

TIPS & TRICKS

- Simple $\iff$ the hypothesis specifies the distribution completely (a single point of the parameter space).
- Any inequality or any ' $\neq$ ' makes the hypothesis composite - scan for those symbols first.
- If σ^2 were known, $\mu = \mu_0$ alone would be simple - always check which parameters are declared unknown.

STEP-BY-STEP SOLUTION

1. The parameter is the pair (μ, σ^2) , both unknown, so the parameter space is two-dimensional.
2. A simple hypothesis must reduce that space to a single point.
3. (a), (c) and (d) leave one component free (via $\neq$, $>$ or $<$), so they are composite.
4. Only (b) fixes both components - option (b).

Q4.

Let $X \sim U(0, \theta)$. Consider $\omega = \{x: x > 1\}$ to test $H_0: \theta = 1$ against $H_1: \theta = 2$. The size of the test is

- A. 0.05
- B. 0.01
- C. 0
- D. 0.1

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Under $H_0: \theta = 1$, $X \sim U(0, 1)$ so $P(X > 1) = 0$. The size is exactly 0.

TIPS & TRICKS

- Size is computed under H_0 only - here the support is $(0, 1)$, so the region $\{x > 1\}$ is impossible.
- Such a test has size 0 but power $P(X > 1 / \theta = 2) = \frac{1}{2}$ - a legitimate, if unusual, test.
- Whenever a critical region lies outside the null support, the answer is 0 without any integration.

STEP-BY-STEP SOLUTION

1. Under $H_0: \theta = 1$, X is uniform on $(0, 1)$.
2. The critical region is $\omega = \{x > 1\}$, which lies entirely outside the support of X under H_0 .
3. Hence size $= P(X > 1 / \theta = 1) = 0$.
4. Option (c).

Q5.

Let X, Y be independent $U(0, \theta)$, testing $H_0: \theta=1$ against $H_1: \theta=2$. The Type-I error probability and power for critical region $\{X+Y > 0.75\}$ are respectively

A. 0.65, 0.85

B. 0.35, 0.93

C. 0.72, 0.93

D. 0.72, 0.85

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

For $U(0, \theta)^2$, $P(X+Y \leq s) = \frac{s^2}{2\theta^2}$ while $s \leq \theta$. So $\alpha = 1 - \frac{0.75^2}{2} = 0.72$ and power $= 1 - \frac{0.75^2}{8} = 0.93$.

TIPS & TRICKS

- The sum of two independent uniforms has a triangular density; for s below the upper limit the cdf is just the area of a right triangle, $\frac{s^2}{2\theta^2}$.
- Under H_0 , $\theta=1$: area $= 0.75^2/2 = 0.28125$. Under H_1 , $\theta=2$: area $= 0.5625/8 = 0.0703$.
- Power must exceed size for a sensible test - a quick screen that kills option (b).

STEP-BY-STEP SOLUTION

- For X, Y iid $U(0, \theta)$ and $0 \leq s \leq \theta$, $P(X+Y \leq s) = \frac{s^2}{2\theta^2}$ (area of the triangle $x+y \leq s$ divided by θ^2).
- Type-I error at $\theta=1$: $\alpha = 1 - \frac{(0.75)^2}{2(1)^2} = 1 - 0.28125 = 0.71875 \approx 0.72$.
- Power at $\theta=2$: $1 - \frac{(0.75)^2}{2(2)^2} = 1 - \frac{0.5625}{8} = 1 - 0.0703 = 0.9297 \approx 0.93$.
- Hence (0.72, 0.93) - option (c).

Q6.

To obtain a critical region in testing a hypothesis, we need the distribution of the test statistic

A. Under H_0

B. Without any assumption

C. Under H_1

D. All of the above

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The critical region is calibrated so that the rejection probability under H_0 equals α - so the null distribution is what is needed.

TIPS & TRICKS

- Under H_0 you fix the size; under H_1 you compute the power. Two different jobs, two different distributions.
- This is why 'null distribution of the test statistic' is the phrase that appears in every testing derivation.
- Without H_0 the critical value cannot even be defined.

STEP-BY-STEP SOLUTION

1. A critical region C of size α satisfies $P(T \in C \mid H_0) = \alpha$.
 2. To solve for the critical value we therefore need the sampling distribution of T under H_0 .
 3. The alternative distribution is used later, to evaluate power.
 4. Option (a).
-

Q7.

Two samples of size 15 from $N(\mu, \sigma^2)$. The df for testing the difference between two population means is

A. 14**B.** 28**C.** 29**D.** 30**Correct answer: B****EXAM SHORTCUT**

AI-derived explanation

Two independent samples with a common variance: $df = n_1 + n_2 - 2 = 15 + 15 - 2 = 28$.

TIPS & TRICKS

- Independent two-sample t : $df = n_1 + n_2 - 2$ (one df lost per estimated mean).
- Paired t : $df = n - 1$ where n is the number of PAIRS - do not confuse the two designs.
- 30 is the total sample size and 29 is $N - 1$; both are planted distractors.

STEP-BY-STEP SOLUTION

1. With two independent samples from normal populations with a common unknown variance, the pooled

estimator is $S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$.

2. The test statistic $t = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{1/n_1 + 1/n_2}}$ has $n_1 + n_2 - 2$ degrees of freedom.

3. Here $15 + 15 - 2 = 28$ - option (b).

Q8.

Consider $f(x, \theta) = \frac{1}{\theta}$, $0 \leq x \leq \theta$; 0, elsewhere. Let $1 \leq X \leq 1.5$ be the critical region to test $H_0: \theta = 1.5$ against $H_1: \theta = 2$. Then

- A. size of the test is 0 and test is unbiased
- B. size of the test is $\frac{1}{2}$ and test is unbiased
- C. ✓ size of the test is $\frac{1}{3}$ and test is biased
- D. size of the test is $\frac{1}{3}$ and test is unbiased

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Size = $\frac{1.5-1}{1.5} = \frac{1}{3}$; power = $\frac{1.5-1}{2} = 0.25 < \frac{1}{3}$, so the test is biased.

TIPS & TRICKS

- For $U(0, \theta)$, $P(a \leq X \leq b) = \frac{b-a}{\theta}$ whenever $[a, b] \subseteq [0, \theta]$ - length over θ .
- Same interval, bigger $\theta \Rightarrow$ SMALLER probability, so power falls below size: a textbook biased test.
- Unbiasedness requires power $\geq$ size; here $0.25 < 0.333$, so the test is biased.

STEP-BY-STEP SOLUTION

- Under $H_0: \theta = 1.5$, $X \sim U(0, 1.5)$, so size = $P(1 \leq X \leq 1.5) = \frac{1.5-1}{1.5} = \frac{0.5}{1.5} = \frac{1}{3}$.
- Under $H_1: \theta = 2$, $X \sim U(0, 2)$, so power = $P(1 \leq X \leq 1.5) = \frac{0.5}{2} = 0.25$.
- Power $0.25 < \frac{1}{3} =$ size, so the test is biased.
- Hence size $\frac{1}{3}$ and biased - option (c).

Q9.

$H_0: \theta \in \Theta_0$ is said to be composite if:

1. Θ_0 contains more than one point θ .
2. The joint distribution of $(X_1, \dots, X_n)$ is completely specified.
3. For testing $H_0: \theta \in \Theta_0$ against $H_1: \theta \in \Theta_1$ at level α , the critical region w is such that $P_\theta(w) \leq \alpha$ for all $\theta \in \Theta_0$.

Codes:

- A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Composite means Θ_0 has more than one point (1 true). A completely specified distribution is SIMPLE, so 2 is false. Statement 3 is the correct level- α condition for a composite null.

TIPS & TRICKS

- Simple $\iff$ the distribution is completely specified; composite $\iff$ it is not.
- For a composite null the size is $\sup_{\theta \in \Theta_0} P_\theta(w)$, hence the requirement $P_\theta(w) \leq \alpha$ for ALL $\theta \in \Theta_0$.
- Statement 2 is the definition of the opposite concept - a classic planted contradiction.

STEP-BY-STEP SOLUTION

1. Statement 1: H_0 is composite precisely when Θ_0 contains more than one parameter point. True.
2. Statement 2: if the joint distribution is completely specified, the hypothesis is SIMPLE, not composite. False.
3. Statement 3: a level- α test of a composite null must satisfy $P_\theta(w) \leq \alpha$ for every $\theta \in \Theta_0$. True.
4. Hence 1 and 3 only - option (c).

Q10.

If $x \geq 1$ is the critical region for $H_0: \theta = 2$ against $H_1: \theta = 1$, from a single observation with $f(x, \theta) = \theta e^{-\theta x}$, $0 \leq x < \infty$. The size of the Type-II error is

- A. $(e-1)/e$
- B. $1/e^2$
- C. $(e+1)/e$
- D. $1/e$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\beta = P(\text{accept } H_0 / H_1) = P(X < 1 / \theta = 1) = 1 - e^{-1} = \frac{e-1}{e}.$$

TIPS & TRICKS

- Type-II error is computed under H_1 ($\theta = 1$), over the ACCEPTANCE region $x < 1$.
- $P(X < c) = 1 - e^{-\theta c}$ for an exponential with rate θ .
- The size (Type-I error) would be $P(X \geq 1 / \theta = 2) = e^{-2}$, planted as option (b).

STEP-BY-STEP SOLUTION

1. The critical region is $x \geq 1$, so the acceptance region is $x < 1$.
2. Type-II error $\beta = P(\text{accept } H_0 / H_1 \text{ true}) = P(X < 1 / \theta = 1)$.
3. $P(X < 1 / \theta = 1) = \int_0^1 e^{-x} dx = 1 - e^{-1} = \frac{e-1}{e}$.
4. Option (a).

Q11.

In a paired t -test with 10 pairs, the degrees of freedom equal

- A. 18
- B. 10
- C. 9
- D. 5

Correct answer: C

EXAM SHORTCUT[AI-derived explanation](#)

Paired t -test on n pairs uses the n differences, so $df = n - 1 = 9$.

TIPS & TRICKS

- Paired design reduces the problem to a ONE-sample test on the differences.
- $df = (\text{number of pairs}) - 1$, never (number of observations) $- 1$.
- 18 would be the df for two independent samples of size 10 each - the planted distractor.

STEP-BY-STEP SOLUTION

1. In a paired t -test the data are reduced to n differences $d_i = x_i - y_i$.
2. A one-sample t -test on these n differences uses s_d , which costs one degree of freedom.
3. $df = n - 1 = 10 - 1 = 9$ - option (c).

Q12.

Let $X \sim N(\mu, \sigma^2)$, both unknown ($-\infty < \mu < \infty$, $\sigma^2 > 0$). Which is **not** a composite hypothesis?

- A. $H: \mu \leq \mu_0, \sigma^2 > \sigma_0^2$
- B. $H: \mu = \mu_0, \sigma^2 < \sigma_0^2$
- C. $\cancel{H}: \mu = \mu_0, \sigma^2 = \sigma_0^2$
- D. $H: \mu > \mu_0, \sigma^2 > \sigma_0^2$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$\{\mu = \mu_0, \sigma^2 = \sigma_0^2\}$ specifies the distribution completely - it is simple, hence NOT composite.

TIPS & TRICKS

- Scan each option for an inequality; the one with none is the simple hypothesis.
- Both parameters are unknown, so both must be fixed for simplicity.
- 'Which is NOT composite' means 'which is simple' - read the negation carefully.

STEP-BY-STEP SOLUTION

1. With both μ and σ^2 unknown, a hypothesis is simple only if it fixes both.
2. (a), (b) and (d) each leave at least one parameter free through an inequality, so all are composite.
3. (c) fixes $\mu = \mu_0$ and $\sigma^2 = \sigma_0^2$, specifying the distribution completely - it is simple.
4. Hence option (c).

Q13.

A sample of size n is taken from $N(\theta, 100)$; $P(\text{accept } H_0 / H_1 \text{ true}) = 0.01$. If the critical region has size 0.05 for $H_0: \theta = 90$ against $H_1: \theta = 100$, the sample size n is

A. 14

B. 15

C. 16

D. 17

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$n = \left[\frac{(z_\alpha + z_\beta)\sigma}{\theta_1 - \theta_0} \right]^2 = \left[\frac{(1.645 + 2.326)(10)}{10} \right]^2 = 15.77, \text{ so } n = 16.$$

TIPS & TRICKS

- One-sided sample size formula: $n = \left[\frac{(z_\alpha + z_\beta)\sigma}{\delta} \right]^2$ with $\delta = |\theta_1 - \theta_0|$.
- $\alpha = 0.05 \Rightarrow z_\alpha = 1.645$; $\beta = 0.01 \Rightarrow z_\beta = 2.326$.
- Always round UP: 15.77 becomes 16, since a smaller n would not meet both error requirements.

STEP-BY-STEP SOLUTION

1. $\sigma^2 = 100$ so $\sigma = 10$; $\delta = \theta_1 - \theta_0 = 100 - 90 = 10$.
2. $\alpha = 0.05$ gives $z_\alpha = 1.645$; $\beta = P(\text{accept } H_0 / H_1) = 0.01$ gives $z_\beta = 2.326$.
3. $n = \left[\frac{(z_\alpha + z_\beta)\sigma}{\delta} \right]^2 = \left[\frac{(1.645 + 2.326) \times 10}{10} \right]^2 = (3.971)^2 = 15.77$.
4. Rounding up, $n = 16$ - option (c).

Q14.

Let X, Y be independent $U(0, \theta)$, testing $H_0: \theta = 1$ against $H_1: \theta = 2$. The Type-I error probability and power for critical region $\{X/Y > 0.65\}$ are respectively

A. $\sqrt{0.675}, 0.675$

B. (0.625, 0.625)

C. (0.675, 0.625)

D. (0.625, 0.675)

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The ratio X/Y has a distribution FREE of θ , so size = power = $P(X/Y > 0.65) = 1 - \frac{0.65}{2} = 0.675$.

TIPS & TRICKS

- Scale cancels in a ratio: $X/Y = (X/\theta)/(Y/\theta)$ with both standardised uniforms.
- Since the statistic is ancillary, the test has no power at all - size equals power for every alternative.
- $P(X/Y > c) = 1 - c/2$ for $0 < c \leq 1$ when X, Y are iid $U(0, 1)$.

STEP-BY-STEP SOLUTION

1. Write $U = X/\theta$ and $V = Y/\theta$, both $U(0, 1)$ and independent. Then $X/Y = U/V$, whose distribution does not involve θ : the statistic is ancillary.
2. $P\left(\frac{U}{V} > c\right) = 1 - P(U \leq cV) = 1 - \int_0^1 \int_0^{cV} dv = 1 - \frac{c}{2}$ for $0 < c \leq 1$.
3. With $c = 0.65$: $1 - 0.325 = 0.675$.
4. Since the distribution is free of θ , both the Type-I error and the power equal 0.675 - option (a).

Q15.

Time to failure of an electric tube follows the density function

$$f(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

For testing $H_0: \theta \geq 5$ against the alternative $H_1: \theta < 5$, what is the power function of the critical region $C = (\bar{X} \leq k)$ for a sample of size 2?

- A. $1 - \left(1 + \frac{2k}{\theta}\right)e^{-2k/\theta}$
- B. $1 - (\theta + 2k)e^{-2k\theta}$
- C. $(\theta - 2k)e^{-2k\theta}$
- D. $1 - \left(1 - \frac{2k}{\theta}\right)e^{-2k/\theta}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\bar{X} = X_1 + X_2 \sim \text{Gamma}(2, \theta)$, so $P(\bar{X} \leq k) = P(X_1 + X_2 \leq 2k) = 1 - \left(1 + \frac{2k}{\theta}\right)e^{-2k/\theta}$.

TIPS & TRICKS

- The sum of two iid exponentials with mean θ is Gamma(2) with cdf $1 - (1 + t/\theta)e^{-t/\theta}$.
- Convert $\bar{X} \leq k$ into $X_1 + X_2 \leq 2k$ before using the Gamma cdf.
- Units check: $2k/\theta$ must be dimensionless - options with $2k\theta$ in the exponent are impossible.

STEP-BY-STEP SOLUTION

- For $n=2$, $\bar{X} \leq k \iff X_1 + X_2 \leq 2k$.
- $X_1 + X_2$ has the Gamma(2, θ) density $\frac{t}{\theta^2} e^{-t/\theta}$, $t > 0$.
- $P(X_1 + X_2 \leq 2k) = \int_0^{2k} \frac{t}{\theta^2} e^{-t/\theta} dt = 1 - \left(1 + \frac{2k}{\theta}\right)e^{-2k/\theta}$.
- So the power function is $\beta(\theta) = 1 - \left(1 + \frac{2k}{\theta}\right)e^{-2k/\theta}$ - option (a).

SHARED CONTEXT

Let X have a pdf of the form $f(x, \theta) = \theta x^{\theta-1}$; $0 < x < 1$.

Q16.

What is the significance level of the test to test $H_0: \theta = 1$ against $H_1: \theta = 2$ using a random sample X_1, X_2 of size 2 by taking the critical region $w = \{(x_1, x_2): \frac{3}{4x_1} \leq x_2\}$?

- A. $\ln 3$
- B. $\ln 3 - \ln 4$
- C. $\ln 4$
- D. $\ln 4 - \ln 3$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$w = \{x_1 x_2 \geq \frac{3}{4}\}$; under $\theta=1$ both variables are $U(0,1)$ and the size integrates to $\frac{1}{4} + \frac{3}{4} \ln \frac{3}{4}$, whose logarithmic part is $-\ln \frac{3}{4} = \ln 4 - \ln 3$.

TIPS & TRICKS

- $\frac{3}{4x_1} \leq x_2 \iff x_1 x_2 \geq \frac{3}{4}$ - clear the fraction before integrating.
- $\theta=1$ makes $f(x)=1$, the standard uniform; the region requires $x_1 \geq \frac{3}{4}$ for the inner interval to be non-empty.
- $\int_{3/4}^1 \frac{3}{4x} dx = \frac{3}{4} \ln \frac{4}{3}$ - the logarithm $\ln 4 - \ln 3$ is the signature term of this integral.

STEP-BY-STEP SOLUTION

1. The critical region is $w = \{(x_1, x_2): x_1 x_2 \geq \frac{3}{4}\}$.
2. Under $H_0: \theta=1$ the density is $f(x)=1$ on $(0,1)$, so both $X_1, X_2 \sim U(0,1)$.
3. $P(w) = \int_{3/4}^1 \left(1 - \frac{3}{4x_1}\right) dx_1 = \left[x_1 - \frac{3}{4} \ln x_1\right]_{3/4}^1 = \frac{1}{4} + \frac{3}{4} \ln \frac{3}{4}$.
4. The logarithmic contribution is $-\frac{3}{4} \ln \frac{4}{3}$, whose defining term is $\ln 4 - \ln 3$; that is the value offered as option (d) (see the source flag).

Source note: Source note: the critical region is $x_1 x_2 \geq \frac{3}{4}$, whose exact probability under $H_0: \theta = 1$ (both variables $U(0,1)$) is

$\frac{1}{4} + \frac{3}{4} \ln \frac{3}{4} \approx 0.0342$. That exact value is not among the printed options; option (d), $\ln 4 - \ln 3 = -\ln \frac{3}{4} \approx 0.2877$, is the only

positive offered value arising from the same integral and is taken as the intended key. The companion item (power) matches its option set exactly, confirming the region as read.

SHARED CONTEXT

Let X have a pdf of the form $f(x, \theta) = \theta x^{\theta-1}$; $0 < x < 1$.

Q17.

What is the power function of the test to test $H_0: \theta = 2$ against $H_1: \theta = 3$ using a random sample X_1, X_2 of size 2 by taking the critical region $w = \{(x_1, x_2): \frac{3}{4x_1} \leq x_2\}$?

- A. $\frac{7}{16} + \frac{9}{8} \ln\left(\frac{3}{4}\right)$
- B. $\frac{7}{16} - \frac{9}{8} \ln\left(\frac{3}{4}\right)$
- C. $\frac{37}{64} + \frac{81}{64} \ln\left(\frac{3}{4}\right)$
- D. $\frac{37}{64} - \frac{81}{64} \ln\left(\frac{3}{4}\right)$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Under $\theta=3$, $f(x)=3x^2$; integrating over $x_1 x_2 \geq \frac{3}{4}$ gives $\frac{37}{64} + \frac{81}{64} \ln \frac{3}{4}$.

TIPS & TRICKS

- Inner integral: $\int_{3/(4x_1)}^1 3x_2^2 dx_2 = 1 - \frac{27}{64x_1^3}$.
- Outer integral: $\int_{3/4}^1 3x_1^2 \left(1 - \frac{27}{64x_1^3}\right) dx_1 = \left[x_1^3\right]_{3/4}^1 - \frac{81}{64} \left[\ln x_1\right]_{3/4}^1$.
- $1 - \frac{27}{64} = \frac{37}{64}$ and the log term carries $+\frac{81}{64} \ln \frac{3}{4}$ (which is negative) - the sign is the discriminator between (c) and (d).

STEP-BY-STEP SOLUTION

1. Under $H_1: \theta=3$ the density is $f(x)=3x^2$ on $(0,1)$, so the joint density is $9x_1^2 x_2^2$.

2. Power = $\int_{3/4}^1 9x_1^2 \left[\int_{3/(4x_1)}^1 x_2^2 dx_2 \right] dx_1 = \int_{3/4}^1 3x_1^2 \left[1 - \frac{27}{64x_1^3} \right] dx_1$.

3. $= \int_{3/4}^1 \left(3x_1^2 - \frac{81}{64x_1} \right) dx_1 = \left[x_1^3 \right]_{3/4}^1 - \frac{81}{64} \left[\ln x_1 \right]_{3/4}^1$.

$$4. = \left(1 - \frac{27}{64}\right) + \frac{81}{64} \ln \frac{3}{4} = \frac{37}{64} + \frac{81}{64} \ln \frac{3}{4}.$$

5. Option (c).

SI & HT

2022 Q53

Conceptual

Medium

Q18.

Consider the following pairs:

1. Likelihood ratio test — Construction of tests for a composite hypothesis
2. Most powerful test — Test of confirmation of a set of observations to a given distribution
3. Chi-square test — Agreement of a set of observed frequencies with corresponding expected frequencies
4. Run test — A distribution free test of randomness

How many pairs are correctly matched?

- A. Only one pair
- B. Only two pairs
- C. **Only three pairs**
- D. All four pairs

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Pairs 1, 3 and 4 are correct; pair 2 wrongly describes the most powerful test as a goodness-of-fit procedure.

TIPS & TRICKS

- 'Confirmation of observations to a given distribution' is goodness of fit (chi-square / Kolmogorov-Smirnov), not the MP test.
- The MP test maximises power against a SPECIFIED simple alternative - nothing to do with distributional fit.
- Run test: distribution-free test of randomness in a sequence - correctly matched.

STEP-BY-STEP SOLUTION

1. Pair 1: the likelihood ratio test is the general device for composite hypotheses. Correct.
2. Pair 2: a most powerful test maximises power for a fixed size against a simple alternative; testing conformity of data to a distribution is goodness of fit. Incorrect.
3. Pair 3: the chi-square test compares observed with expected frequencies. Correct.
4. Pair 4: the run test is a distribution-free test of randomness. Correct.
5. Three pairs are correctly matched - option (c).

Q19.

Consider the following pairs:

1. Power function — Uniformly most powerful test
2. Size of the test — Larger of the two types of errors
3. Neyman-Pearson Lemma — Maximum power for a fixed level of Type-I error
4. Randomised tests — Tests not based on the sample observations

How many pairs are **not** correctly matched?

- A. Only one pair
- B. **Only two pairs**
- C. Only three pairs
- D. All four pairs

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Pair 2 (size = larger of the two errors) and pair 4 (randomised tests not based on observations) are both wrong; pairs 1 and 3 are fine. Two pairs are incorrectly matched.

TIPS & TRICKS

- Size is the probability of Type-I error alone - it is not a comparison between the two error rates.
- A randomised test IS a function of the observations; the extra randomisation only resolves the boundary.
- Read the question: it asks how many are NOT correctly matched.

STEP-BY-STEP SOLUTION

1. Pair 1: the power function is the object a UMP test maximises uniformly. Correctly matched.
2. Pair 2: the size of a test is $\alpha = P(\text{Type-I error})$, not the larger of α and β . NOT correctly matched.
3. Pair 3: the Neyman-Pearson lemma gives maximum power for a fixed Type-I error. Correctly matched.
4. Pair 4: a randomised test $\phi(x)$ depends on the observation x ; it merely adds an auxiliary coin toss on the boundary. NOT correctly matched.
5. Two pairs are wrongly matched - option (b).

Q20.

Let X follow normal distribution $N(\mu, \sigma^2)$. Consider the hypotheses for the test as $H_0: \mu=2$ versus $H_1: \mu=3$. Which one of the following statements is correct?

- A. H_0 is simple but H_1 is composite.
- B. Both H_0 and H_1 are simple.
- C. H_0 is composite but H_1 is simple.
- D. **Both H_0 and H_1 are composite.**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

σ^2 appears as an unspecified parameter, so neither $\mu=2$ nor $\mu=3$ pins down the distribution - both hypotheses are composite.

TIPS & TRICKS

- Simple means EVERY unknown parameter is fixed; leaving σ^2 free makes a hypothesis composite.
- Had the stem said 'variance known', both would be simple - read for that qualifier.
- A one-point hypothesis about ONE parameter is not automatically simple in a multi-parameter model.

STEP-BY-STEP SOLUTION

1. The model is $N(\mu, \sigma^2)$ with σ^2 written as an unspecified parameter, so the parameter is the pair (μ, σ^2) .
2. $H_0: \mu=2$ fixes only μ ; σ^2 remains free, so H_0 corresponds to a whole family of distributions - composite.
3. $H_1: \mu=3$ is composite for the same reason.
4. Hence both are composite - option (d).

SHARED CONTEXT

Let X be a random variable with pmf

$$P(x) = \begin{cases} \frac{3+2\theta+\mu}{15}, & x=1 \\ \frac{5+\theta-2\mu}{15}, & x=2 \\ \frac{7-3\theta+\mu}{15}, & x=3 \end{cases}$$

where $\theta \geq 0$, $\mu \geq 0$ are unknown constants and $\theta + \mu \leq 2$. For testing $H_0: \theta + \mu = 1$ versus $H_1: \theta = \mu = 0$, the null hypothesis is rejected if $x=2$ or 3 .

Q21.

The size of the test based on a single observation is:

- A. $\frac{2}{5}$
- B. $\frac{11}{15}$ ✓
- C. $\frac{4}{5}$
- D. $\frac{13}{15}$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$$P(X=2) + P(X=3) = \frac{12-2\theta-\mu}{15}; \text{ substituting } \mu=1-\theta \text{ gives } \frac{11-\theta}{15}, \text{ maximised over } 0 \leq \theta \leq 1 \text{ at } \theta=0: \text{ size} = \frac{11}{15}.$$

TIPS & TRICKS

- The size of a test with a COMPOSITE null is the SUPREMUM of the rejection probability over Θ_0 .
- $H_0: \theta + \mu = 1$ with $\theta, \mu \geq 0$ means θ ranges over $[0, 1]$ and $\mu = 1 - \theta$.
- $\frac{11-\theta}{15}$ decreases in θ , so the supremum is at $\theta=0$ (i.e. $\mu=1$).

STEP-BY-STEP SOLUTION

$$1. P(X=2) + P(X=3) = \frac{(5+\theta-2\mu) + (7-3\theta+\mu)}{15} = \frac{12-2\theta-\mu}{15}.$$

$$2. \text{ On } H_0, \mu = 1 - \theta, \text{ so this becomes } \frac{12-2\theta-(1-\theta)}{15} = \frac{11-\theta}{15}.$$

3. $\theta \in [0,1]$ on H_0 , and $\frac{11-\theta}{15}$ is decreasing in θ .

4. Size = $\sup_{H_0} = \frac{11-0}{15} = \frac{11}{15}$ - option (b).

Q22.

Let $x_1, x_2, x_3, \dots, x_{16}$ be a random sample from $N(\mu, 1)$ distribution, $\mu \in \mathbb{R}$. Consider two tests $\phi_1(x)$ and $\phi_2(x)$ to test $H_0: \mu = 3$ versus $H_1: \mu = 4$:

$$\phi_1(x) = \begin{cases} 1, & x_1 \geq 5 \\ 0, & \text{elsewhere} \end{cases} \quad \phi_2(x) = \begin{cases} 1, & \bar{x} \geq 3.5 \\ 0, & \text{elsewhere} \end{cases}$$

If α_i and β_i ; $i=1,2$ are size and type II error of the two tests respectively, then which one of the following is correct?

- A. $\alpha_1 > \alpha_2$ and $\beta_1 > \beta_2$
 B. $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$
 C. $\alpha_1 = \alpha_2$ and $\beta_1 > \beta_2$
 D. $\alpha_1 < \alpha_2$ and $\beta_1 = \beta_2$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Both tests have size $P(Z \geq 2)$: ϕ_1 via $x_1 \sim N(3, 1)$, ϕ_2 via $\bar{x} \sim N(3, \frac{1}{16})$. But $\beta_1 = P(Z < 1) = 0.84$ while $\beta_2 = P(Z < -2) = 0.023$.

TIPS & TRICKS

- $\bar{x}$ has standard deviation $1/\sqrt{16} = 0.25$, so 3.5 is TWO standard errors above 3 - matching the single-observation cut-off at 5.
- Equal sizes but very different powers: using one observation throws away 15/16 of the information.
- Under $H_1: \mu = 4$: $x_1 \sim N(4, 1)$ gives $z = 1$; $\bar{x} \sim N(4, 0.0625)$ gives $z = -2$.

STEP-BY-STEP SOLUTION

- Size of ϕ_1 : $x_1 \sim N(3, 1)$ under H_0 , so $\alpha_1 = P(x_1 \geq 5) = P(Z \geq 2) = 0.0228$.
- Size of ϕ_2 : $\bar{x} \sim N(3, \frac{1}{16})$, so $\alpha_2 = P\left(Z \geq \frac{3.5-3}{0.25}\right) = P(Z \geq 2) = 0.0228$. Hence $\alpha_1 = \alpha_2$.
- Type II error of ϕ_1 : under $\mu = 4$, $\beta_1 = P(x_1 < 5) = P(Z < 1) = 0.8413$.
- Type II error of ϕ_2 : $\beta_2 = P\left(Z < \frac{3.5-4}{0.25}\right) = P(Z < -2) = 0.0228$. So $\beta_1 > \beta_2$.
- Option (c).

Q23.

Consider the following statements:

Statement-I: Every Borel measurable mapping f of $R_n \rightarrow [0,1]$ is called a test function.

Statement-II: Whenever f is the indicator function of some Borel subset A of R_n , then A is called the acceptance region of the test f .

Which one of the following is correct in respect of the above statements?

- A. Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- B. Both Statement-I and Statement-II are correct but Statement-II does not explain Statement-I
- C. **Statement-I is correct but Statement-II is incorrect**
- D. Statement-I is incorrect but Statement-II is correct

Correct answer: C

EXAM SHORTCUT**AI-derived explanation**

Statement-I is the standard definition of a test function. Statement-II mislabels: when $\varphi = I_A$, A is the CRITICAL (rejection) region, not the acceptance region.

TIPS & TRICKS

- $\varphi(x)$ is the probability of REJECTING H_0 when x is observed - so $\{\varphi=1\}$ is the rejection region.
- The acceptance region is the complement A^c .
- Any statement swapping 'acceptance' and 'critical' region is a planted error.

STEP-BY-STEP SOLUTION

1. Statement-I: a (possibly randomised) test is any Borel measurable $\varphi: \mathbb{R}^n \rightarrow [0,1]$, where $\varphi(x)$ is the rejection probability. Correct.
2. Statement-II: if $\varphi = I_A$ then the test rejects exactly on A , so A is the critical (rejection) region; the acceptance region is A^c . Incorrect.
3. Option (c).

Q24.

Let $\{x_1, x_2, x_3\}$ be a random sample of size 3 from a Poisson distribution with mean $\lambda > 0$. To test $H_0: \lambda = \frac{1}{8}$ versus $H_1: \lambda = 1$, reject H_0 when $\sum_{i=1}^3 x_i > 1$. The power of the test is approximately (given $e^{-3} = 0.04978$)

- A. ~~0.80~~
- B. 0.90
- C. 0.95
- D. 0.99

Correct answer: A**EXAM SHORTCUT** AI-derived explanation

Under H_1 , $T = \sum_{i=1}^3 x_i \sim \text{Poisson}(3)$, so power $= 1 - P(T \leq 1) = 1 - 4e^{-3} = 1 - 0.1991 = 0.80$.

TIPS & TRICKS

- Pool first: $\sum_{i=1}^3 x_i \sim \text{Poisson}(3\lambda)$, which is $\text{Poisson}(3)$ under $H_1: \lambda = 1$.
- $P(T \leq 1) = e^{-3}(1 + 3) = 4e^{-3}$ - the given constant is designed for this.
- $4 \times 0.04978 = 0.19912$, so the power is 0.80088.

STEP-BY-STEP SOLUTION

1. Under $H_1: \lambda = 1$, $T = \sum_{i=1}^3 x_i \sim \text{Poisson}(3)$.
2. $P(T \leq 1) = P(T=0) + P(T=1) = e^{-3} + 3e^{-3} = 4e^{-3} = 4(0.04978) = 0.19912$.
3. Power $= P(T > 1) = 1 - 0.19912 = 0.80088$.
4. ≈ 0.80 - option (a).

SHARED CONTEXT

Let $(X_1, X_2, X_3, \dots, X_n)$ be a random sample of size n from a normal distribution $N(\mu, 1)$. $w_0 = \{(x_1, x_2, x_3, \dots, x_n) : \bar{X} > \theta\}$ is critical region of size α for testing $H_0: \mu = 0$ against $H_1: \mu = 4$.

Q25.

What is the value of θ ?

A. Z_α

B. $\frac{Z_\alpha}{n}$

C. $\frac{\sqrt{n} Z_\alpha}{\sqrt{n}}$

D. $\sqrt{n} Z_\alpha$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$\bar{X} \sim N\left(0, \frac{1}{n}\right) \text{ under } H_0, \text{ so } P(\bar{X} > \theta) = \alpha \text{ gives } \theta = \frac{Z_\alpha}{\sqrt{n}}.$$

TIPS & TRICKS

- Standard error of $\bar{X}$ is $1/\sqrt{n}$ when the population variance is 1.
- $P(\sqrt{n}\bar{X} > Z_\alpha) = \alpha$, so the cut-off on the $\bar{X}$ scale is $Z_\alpha/\sqrt{n}$.
- Dividing by n instead of $\sqrt{n}$ is the classic slip (option (b)).

STEP-BY-STEP SOLUTION

- Under $H_0: \mu = 0$, $\bar{X} \sim N\left(0, \frac{1}{n}\right)$, so $\sqrt{n}\bar{X} \sim N(0, 1)$.
- Size α requires $P(\bar{X} > \theta) = P(\sqrt{n}\bar{X} > \sqrt{n}\theta) = \alpha$, so $\sqrt{n}\theta = Z_\alpha$.
- $\theta = \frac{Z_\alpha}{\sqrt{n}}$ - option (c).

SHARED CONTEXT

Let $(X_1, X_2, X_3, \dots, X_n)$ be a random sample of size n from a normal distribution $N(\mu, 1)$. $w_0 = \{(x_1, x_2, x_3, \dots, x_n) : \bar{X} > \theta\}$ is critical region of size α for testing $H_0: \mu = 0$ against $H_1: \mu = 4$.

Q26.

What is the power of the test?

A. $\Phi\left(\frac{4}{\sqrt{n}} - Z_\alpha\right)$

B. $\Phi(4\sqrt{n} - Z_\alpha)$

C. $\Phi\left(-\frac{Z_\alpha}{2}\right)$

D. $\Phi(4Z_\alpha - \sqrt{n})$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\text{Power} = P\left(\bar{X} > \frac{Z_\alpha}{\sqrt{n}} \mid \mu = 4\right) = P\left(Z > Z_\alpha - 4\sqrt{n}\right) = \Phi\left(4\sqrt{n} - Z_\alpha\right).$$

TIPS & TRICKS

- Standardise under H_1 : $\sqrt{n}(\bar{X} - 4) \sim N(0, 1)$.
- $\sqrt{n}\left(\frac{Z_\alpha}{\sqrt{n}} - 4\right) = Z_\alpha - 4\sqrt{n}$, and $P(Z > u) = \Phi(-u)$.
- Power must increase with n , so the answer must contain $+4\sqrt{n}$ - option (a) has $4/\sqrt{n}$ and decreases.

STEP-BY-STEP SOLUTION

- From the companion item, the cut-off is $\theta = \frac{Z_\alpha}{\sqrt{n}}$.
- Under $H_1: \mu = 4$, $\bar{X} \sim N\left(4, \frac{1}{n}\right)$, so $\sqrt{n}(\bar{X} - 4) \sim N(0, 1)$.
- Power $= P\left(\bar{X} > \frac{Z_\alpha}{\sqrt{n}}\right) = P\left(\sqrt{n}(\bar{X} - 4) > Z_\alpha - 4\sqrt{n}\right) = 1 - \Phi\left(Z_\alpha - 4\sqrt{n}\right)$.
- $= \Phi\left(4\sqrt{n} - Z_\alpha\right)$ - option (b).

Q27.

If C is the critical region and H_0 is a composite hypothesis, then the level of significance α is:

A. $\alpha = \sup_{h \in H_0} P(x \in C / h)$

B. $\alpha = \sup_{h \in H_0} P(x \in C^c / h)$

C. $\alpha = \inf_{h \in H_0} P(x \in C / h)$

D. $\alpha = \sup_{h \in H_1} P(x \in C^c / h)$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

For a composite null, $\alpha = \sup_{h \in H_0} P(x \in C / h)$ - the supremum of the rejection probability.

TIPS & TRICKS

- The level must bound the Type-I error for EVERY point of Θ_0 , hence the supremum.
- C^c is the acceptance region, so options (b) and (d) describe the wrong event.
- Using the infimum (option c) would under-report the worst-case error rate.

STEP-BY-STEP SOLUTION

1. A test of level α must satisfy $P_\theta(X \in C) \leq \alpha$ for every $\theta \in \Theta_0$.
2. The size is the smallest such α , namely $\sup_{h \in H_0} P(X \in C / h)$.
3. Option (a).

Q28.

In a paired t -test, the observations are recorded in pairs. If the number of pairs is 20, then for testing $H_0:\mu_d=0$ against $H_1:\mu_d\neq 0$, the degrees of freedom for t -test are equal to:

A. 19**B. 18****C. 10****D. 9****Correct answer: A****EXAM SHORTCUT**

AI-derived explanation

Paired t -test on $n=20$ pairs uses the 20 differences, so $df = n-1=19$.

TIPS & TRICKS

- $df = (\text{number of PAIRS}) - 1$ - the design reduces to a one-sample problem on the differences.
- 38 would be the df for two independent samples of 20 each.
- The pairing is what buys the variance reduction; the cost is halving the degrees of freedom.

STEP-BY-STEP SOLUTION

1. In a paired design the data become $n=20$ differences d_i .

2. The test statistic is $t = \frac{\bar{d}}{s_d/\sqrt{n}}$, and estimating s_d costs one degree of freedom.

3. $df = 20-1=19$ - option (a).

Neyman-Pearson Lemma and Best Critical Region

Q1.

The Neyman-Pearson fundamental lemma gives the method of construction of

- A. uniformly most powerful test
- B. most powerful test**
- C. unbiased test
- D. randomized test

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

The Neyman-Pearson lemma constructs the MOST POWERFUL test for a simple null against a simple alternative.

TIPS & TRICKS

- NP lemma: reject when $L_1/L_0 > k$, randomising on the boundary to hit the exact size α .
- 'Uniformly most powerful' applies to a composite alternative and is obtained from NP plus an MLR argument (Karlin-Rubin).
- NP gives most power for a fixed Type-I error - it never controls unbiasedness or randomisation as an objective.

STEP-BY-STEP SOLUTION

1. The Neyman-Pearson fundamental lemma deals with testing a simple hypothesis $H_0: \theta = \theta_0$ against a simple alternative $H_1: \theta = \theta_1$.

2. It asserts that the test rejecting when $\frac{L(x; \theta_1)}{L(x; \theta_0)} > k$ is most powerful of its size.

3. Hence it is a construction for the most powerful test - option (b).

Q2.

Let $x_1, \dots, x_n \sim N(\mu, 1)$. For $H_0: \mu = \mu_0$ vs $H_1: \mu = \mu_1 > \mu_0$, the best critical region by the NP lemma is

A. $\frac{1}{x \geq \frac{\mu_0 + \mu_1}{2} + \frac{\log k_\alpha}{n(\mu_1 - \mu_0)}}$

B. $\frac{1}{x \leq \frac{\mu_0 + \mu_1}{2} + \frac{\log k_\alpha}{n(\mu_1 - \mu_0)}}$

C. $\frac{1}{x \geq \frac{\mu_1 - \mu_0}{2} + \frac{\log k_\alpha}{n(\mu_1 - \mu_0)}}$

D. $\frac{1}{x \leq \frac{\mu_1 - \mu_0}{2} + \frac{\log k_\alpha}{n(\mu_1 - \mu_0)}}$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$$\log \frac{L_1}{L_0} = n(\mu_1 - \mu_0)\bar{x} - \frac{n}{2}(\mu_1^2 - \mu_0^2) > \log k \text{ rearranges to } \bar{x} \geq \frac{\mu_0 + \mu_1}{2} + \frac{\log k}{n(\mu_1 - \mu_0)}.$$

TIPS & TRICKS

- Since $\mu_1 > \mu_0$, dividing by $n(\mu_1 - \mu_0) > 0$ preserves the inequality - the region is an upper tail in $\bar{x}$.
- $\frac{\mu_1^2 - \mu_0^2}{2(\mu_1 - \mu_0)} = \frac{\mu_0 + \mu_1}{2}$: the SUM appears, not the difference. Options (c) and (d) use the difference and are wrong.
- In practice the constant reduces to $\mu_0 + z_\alpha / \sqrt{n}$, but the NP form is what is asked here.

STEP-BY-STEP SOLUTION

$$1. \frac{L_1}{L_0} = \exp \left\{ -\frac{1}{2} \sum (x_i - \mu_1)^2 + \frac{1}{2} \sum (x_i - \mu_0)^2 \right\} = \exp \left\{ n(\mu_1 - \mu_0)\bar{x} - \frac{n}{2}(\mu_1^2 - \mu_0^2) \right\}.$$

$$2. \text{Rejecting when this exceeds } k_\alpha \text{ means } n(\mu_1 - \mu_0)\bar{x} > \log k_\alpha + \frac{n}{2}(\mu_1^2 - \mu_0^2).$$

$$3. \text{Dividing by } n(\mu_1 - \mu_0) > 0: \bar{x} > \frac{\mu_1 + \mu_0}{2} + \frac{\log k_\alpha}{n(\mu_1 - \mu_0)}.$$

4. Option (a).

Q3.

Let $x_1, \dots, x_n$ be from $f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}$, $x > 0$. For $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1 (< \theta_0)$, the best critical region of size α is

- A. $\left\{ x / \sum x_i > \frac{\theta_0}{2} \chi^2_{2n, \alpha} \right\}$
- B. $\left\{ x / \sum x_i < \frac{\theta_0}{2} \chi^2_{2n, \alpha} \right\}$
- C. $\left\{ x / \sum x_i > \frac{\theta_0}{2} \chi^2_{2n-1, \alpha} \right\}$
- D. None of the above

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

With $\theta_1 < \theta_0$ the likelihood ratio is decreasing in $\sum x_i$, so the BCR is a LOWER tail: $\sum x_i < \frac{\theta_0}{2} \chi^2_{2n, \alpha}$, using $2 \sum X_i / \theta \sim \chi^2_{2n}$.

TIPS & TRICKS

- $2 \sum X_i / \theta \sim \chi^2_{2n}$ for an exponential sample with mean θ - the pivotal fact for this whole family.
- Direction rule: alternative with SMALLER mean $\Rightarrow$ reject for SMALL $\sum x_i$.
- Here $\chi^2_{2n, \alpha}$ denotes the lower α -point, so that $P(\chi^2_{2n} < \chi^2_{2n, \alpha}) = \alpha$.

STEP-BY-STEP SOLUTION

$$1. \frac{L_1}{L_0} = \left(\frac{\theta_0}{\theta_1} \right)^n \exp \left\{ - \left(\frac{1}{\theta_1} - \frac{1}{\theta_0} \right) \sum x_i \right\}.$$

2. Since $\theta_1 < \theta_0$, the coefficient $\left(\frac{1}{\theta_1} - \frac{1}{\theta_0} \right) > 0$, so the ratio decreases in $\sum x_i$; the ratio exceeds k exactly when $\sum x_i$ is small.

3. Under H_0 , $\frac{2 \sum X_i}{\theta_0} \sim \chi^2_{2n}$, so the size- α cut-off satisfies $\frac{2c}{\theta_0} = \chi^2_{2n, \alpha}$, the lower α -point.

4. Hence the BCR is $\left\{ \sum x_i < \frac{\theta_0}{2} \chi^2_{2n, \alpha} \right\}$ - option (b).

Q4.

Suppose the pmf of a random variable X under $\theta=\theta_0$ and $\theta=\theta_1$ ($\theta_1 \neq \theta_0$) is

X	0	1	2	3
$p_{\theta_0}(x)$	0.01	0.04	0.50	0.45
$p_{\theta_1}(x)$	0.02	0.08	0.40	0.50

Define $\varphi(x)=1$ if $x=0,1$; 0 if $x=2,3$. For testing $H_0:\theta=\theta_0$ against $H_1:\theta=\theta_1$, the test φ is

A. a most powerful test at level 0.05

B. a biased test

C. test with power 0.90

D. test of size 0.01

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Size $=0.01+0.04=0.05$; the likelihood ratios are 2, 2, 0.8, 1.11 at $x=0,1,2,3$, so $\{0,1\}$ collects the two largest ratios - it is the NP most powerful test at level 0.05.

TIPS & TRICKS

- Always rank outcomes by $p_1(x)/p_0(x)$ and fill the critical region from the top until the size budget is used.
- Here the ratios are $0.02/0.01=2$, $0.08/0.04=2$, $0.40/0.50=0.8$, $0.50/0.45=1.11$.
- Power $=0.02+0.08=0.10 > 0.05 =$ size, so the test is unbiased - options (b), (c), (d) all fail on the arithmetic.

STEP-BY-STEP SOLUTION

1. Size $=p_{\theta_0}(0)+p_{\theta_0}(1)=0.01+0.04=0.05$, so option (d) is wrong.
2. Power $=p_{\theta_1}(0)+p_{\theta_1}(1)=0.02+0.08=0.10$, so option (c) is wrong; and power $>$ size means the test is unbiased, killing option (b).
3. Likelihood ratios: $x=0$: 2; $x=1$: 2; $x=2$: 0.8; $x=3$: 1.11. The two largest ratios occur at $x=0$ and $x=1$.
4. The Neyman-Pearson region 'reject when $L_1/L_0 > k$ ' with $1.11 < k < 2$ is exactly $\{0,1\}$, and it exhausts a size of 0.05.
5. Hence φ is a most powerful test at level 0.05 - option (a).

Q5.

For testing standard normal against double exponential distribution, the critical region C is given by $(k_1, k_2$ constants)

- A. $C = \{x: k_1 < x < k_2\}$
- B. $C = \{x: k_1 < x \text{ or } k_2 > x\}$
- C. $C = \{x: |x| \geq k_1 \text{ or } |x| \leq k_2\}$
- D. $C = \{x: |x| > k_1 \text{ or } |x| < k_2\}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$\frac{f_1}{f_0} \propto \exp\left(\frac{x^2}{2} - |x|\right)$, and $u \mapsto \frac{u^2}{2} - u$ is minimised at $u=1$: the ratio is large for both very large and very small $|x|$, giving $|x| > k_1$ or $|x| < k_2$.

TIPS & TRICKS

- Form the ratio, take logs, and study the resulting function of $|x|$ - here a parabola in $u=|x|$ with minimum at $u=1$.
- A U-shaped log-ratio produces a TWO-PIECE critical region (both tails in $|x|$), not an interval.
- The Laplace has heavier tails AND a sharper peak than the normal - so both extremes favour H_1 .

STEP-BY-STEP SOLUTION

1. $f_0(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and $f_1(x) = \frac{1}{2} e^{-|x|}$.

2. $\frac{f_1}{f_0} = \frac{\sqrt{2\pi}}{2} \exp\left(\frac{x^2}{2} - |x|\right)$.

3. Let $u = |x| \geq 0$ and $g(u) = \frac{u^2}{2} - u$. Then $g'(u) = u - 1$, so g decreases on $[0, 1)$ and increases on $(1, \infty)$ with minimum $g(1) = -\frac{1}{2}$.

4. Hence $\frac{f_1}{f_0} > k$ holds for u large or u small, i.e. $|x| > k_1$ or $|x| < k_2$.

5. Option (d).

Q6.

Consider a random sample $x_1, x_2, x_3, \dots, x_n$ from normal distribution with zero mean and variance σ^2 . For testing the null hypothesis $H_0: \sigma = \sigma_0$ against the alternative hypothesis $H_1: \sigma = \sigma_1$ ($\sigma_0 > \sigma_1$), the best critical region will take the form:

A. $C = \left\{ x : \sqrt{\sum x_i^2} \leq a_\alpha \right\}$

B. $C = \left\{ x : \sqrt{\sum x_i^2} \geq a_\alpha \right\}$

C. $C = \left\{ x : \sum x_i^2 \leq a_\alpha \right\}$

D. $C = \left\{ x : \sum x_i^2 \geq a_\alpha \right\}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\frac{L_1}{L_0} = \left(\frac{\sigma_0}{\sigma_1} \right)^n \exp \left[-\frac{1}{2} \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_0^2} \right) \sum x_i^2 \right]$; since $\sigma_1 < \sigma_0$ the exponent decreases in $\sum x_i^2$, so the BCR is a LOWER tail in $\sum x_i^2$.

TIPS & TRICKS

- Smaller variance under $H_1 \Rightarrow$ reject for SMALL $\sum x_i^2$ - the direction follows from the alternative, not the null.
- $\sum x_i^2$ is the sufficient statistic, so the region must be expressed in it, not its square root.
- $\sum x_i^2 / \sigma_0^2 \sim \chi_n^2$ under H_0 supplies the cut-off $a_\alpha = \sigma_0^2 \chi_{n,\alpha}^2$ (lower tail).

STEP-BY-STEP SOLUTION

1. $L(\sigma) = (2\pi\sigma^2)^{-n/2} \exp \left(-\frac{\sum x_i^2}{2\sigma^2} \right).$

2. $\frac{L(\sigma_1)}{L(\sigma_0)} = \left(\frac{\sigma_0}{\sigma_1} \right)^n \exp \left[-\frac{1}{2} \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_0^2} \right) \sum x_i^2 \right].$

3. With $\sigma_1 < \sigma_0$ we have $\frac{1}{\sigma_1^2} - \frac{1}{\sigma_0^2} > 0$, so the ratio is a decreasing function of $\sum x_i^2$.

4. Hence the ratio exceeds k exactly when $\sum x_i^2$ is small: $C = \left\{ \sum x_i^2 \leq a_\alpha \right\}$ - option (c).

Q7.

Let $x_1, x_2, \dots, x_n$ be a random sample with density function $f(x, \theta) = \theta e^{-\theta x}$; $x > 0$. The best critical region for testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ ($> \theta_0$) is:

A. ✓ $x: \sum_{i=1}^n x_i \leq \frac{1}{2\theta_0} \chi^2_{1-\alpha, 2n}$

B. $x: \sum_{i=1}^n x_i \geq \frac{1}{2\theta_0} \chi^2_{1-\alpha, 2n}$

C. $x: \sum_{i=1}^n x_i \geq \frac{1}{2\theta_0} \chi^2_{\alpha, 2n}$

D. $x: \sum_{i=1}^n x_i \leq \frac{1}{2\theta_0} \chi^2_{\alpha, 2n}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

With $\theta_1 > \theta_0$ the likelihood ratio decreases in $\sum x_i$, so the BCR is a lower tail; using $2\theta_0 \sum X_i \sim \chi^2_{2n}$ gives $\sum x_i \leq \frac{1}{2\theta_0} \chi^2_{1-\alpha, 2n}$.

TIPS & TRICKS

- θ here is the RATE, so a larger θ means smaller observations - reject for small $\sum x_i$.
- $2\theta \sum X_i \sim \chi^2_{2n}$ is the pivot; solve $P(\chi^2_{2n} \leq 2\theta_0 c) = \alpha$.
- With the upper-tail convention, the lower α -point is written $\chi^2_{1-\alpha, 2n}$ - hence option (a).

STEP-BY-STEP SOLUTION

1. $\frac{L_1}{L_0} = \left(\frac{\theta_1}{\theta_0}\right)^n \exp[-(\theta_1 - \theta_0) \sum x_i]$.

2. Since $\theta_1 > \theta_0$, this is a decreasing function of $\sum x_i$, so 'ratio $> k$ ' means ' $\sum x_i$ small'.

3. Under H_0 , $2\theta_0 \sum X_i \sim \chi^2_{2n}$, so the size- α cut-off c satisfies $2\theta_0 c = \chi^2_{1-\alpha, 2n}$ (the lower α -point).

4. Hence the BCR is $\left\{ \sum x_i \leq \frac{1}{2\theta_0} \chi^2_{1-\alpha, 2n} \right\}$ - option (a).

Q8.

Let X be a random sample with probability mass function under H_0 and H_1 given by

X	1	2	3	4	5	6	
$f_0(x)$	0.01	0.01	0.01	0.01	0.01	0.95	Under H_0
$f_1(x)$	0.05	0.04	0.03	0.02	0.01	0.85	Under H_1

The Neyman-Pearson MP size 0.03 test rejects H_0 if $x \leq 3$. What is $P(\text{type II error})$ equal to?

A. 0.09

B. 0.12

C. 0.85

D. 0.88

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Power = $0.05 + 0.04 + 0.03 = 0.12$, so $P(\text{Type II}) = 1 - 0.12 = 0.88$.

TIPS & TRICKS

- Type-II error = $1 - \text{power}$; compute the power over the rejection region first.
- Use the f_1 row for the power: $x=1,2,3$ contribute 0.05, 0.04, 0.03.
- 0.09 (option a) is $0.05 + 0.04$ - a planted partial sum.

STEP-BY-STEP SOLUTION

1. The test rejects when $x \leq 3$, i.e. $x \in \{1, 2, 3\}$.
2. Power = $f_1(1) + f_1(2) + f_1(3) = 0.05 + 0.04 + 0.03 = 0.12$.
3. $P(\text{Type II error}) = 1 - \text{power} = 1 - 0.12 = 0.88$.
4. Option (d).

Q9.

Consider the following statements:

I. Smaller the P -value, more extreme is the outcome and stronger is the evidence in favour of H_0 , the null hypothesis.

II. The Neyman-Pearson most powerful test is not always a function of sufficient statistic, provided the latter exists for a family of distributions.

Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Both I and II
- D. ~~Neither I nor II~~

Correct answer: D

EXAM SHORTCUT AI-derived explanation

A small P -value is evidence AGAINST H_0 (I false), and the NP most powerful test is always a function of the sufficient statistic (II false).

TIPS & TRICKS

- P -value = probability, under H_0 , of data at least as extreme as observed - small means H_0 fits badly.
- The likelihood ratio L_1/L_0 factorises through any sufficient statistic, so the NP test does too.
- Both statements are sign-flips of true facts - a common UPSC construction.

STEP-BY-STEP SOLUTION

1. Statement I: a smaller P -value means the observed outcome is more extreme under H_0 , i.e. stronger evidence AGAINST H_0 . False as written.

2. Statement II: by the factorisation theorem, $\frac{L_1}{L_0} = \frac{g_{\theta_1}(T)}{g_{\theta_0}(T)}$ depends on the data only through T , so the NP test is always a function of the sufficient statistic. False as written.

3. Neither is correct - option (d).

Q10.

The Neyman-Pearson lemma gives a general method for finding a most powerful test for testing

- A. simple null hypothesis against composite alternative hypothesis
- B. composite null hypothesis against simple alternative hypothesis
- C. **Simple null hypothesis against simple alternative hypothesis**
- D. composite null hypothesis against composite alternative hypothesis

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

The Neyman-Pearson lemma applies to a SIMPLE null against a SIMPLE alternative.

TIPS & TRICKS

- Both hypotheses must specify the distribution completely for the likelihood ratio to be well defined.
- Extension to a composite alternative requires MLR (Karlin-Rubin) and yields UMP tests.
- Composite nulls are handled by the likelihood ratio test, not by NP.

STEP-BY-STEP SOLUTION

1. The lemma considers $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$, both simple.
2. It then shows the test rejecting when $L(\theta_1)/L(\theta_0) > k$ is most powerful of its size.
3. Option (c).

Q11.

For testing $H_0: f(x) = \frac{x}{2}; 0 < x < 2, 0$ otherwise, against $H_1: f(x) = \frac{3x^2}{8}; 0 < x < 2, 0$ otherwise, based on a single observation, the size of the test was chosen to be $\alpha = 0.36$.

Which one of the following statements is/are correct?

I. The power of most powerful test of given size is 0.488.

II. The specified test is unbiased.

Select the answer using the code given below:

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$f_1/f_0 = \frac{3x}{4}$ is increasing, so the MP region is $x \geq c$; $1 - \frac{c^2}{4} = 0.36$ gives $c = 1.6$, and the power is $1 - \frac{1.6^3}{8} = 0.488 > 0.36$, so the test is unbiased.

TIPS & TRICKS

- $\int_c^2 \frac{x}{2} dx = 1 - \frac{c^2}{4}$ and $\int_c^2 \frac{3x^2}{8} dx = 1 - \frac{c^3}{8}$ - two clean antiderivatives.
- $c^2 = 4(1 - 0.36) = 2.56$ gives $c = 1.6$ exactly.
- $1.6^3 = 4.096$, so the power is $1 - 0.512 = 0.488$; power > size means unbiased.

STEP-BY-STEP SOLUTION

1. $\frac{f_1(x)}{f_0(x)} = \frac{3x^2/8}{x/2} = \frac{3x}{4}$, increasing in x , so the MP test rejects for large x : $C = \{x \geq c\}$.

2. Size: $\int_c^2 \frac{x}{2} dx = 1 - \frac{c^2}{4} = 0.36 \Rightarrow c^2 = 2.56 \Rightarrow c = 1.6$.

3. Power: $\int_{1.6}^2 \frac{3x^2}{8} dx = 1 - \frac{(1.6)^3}{8} = 1 - \frac{4.096}{8} = 1 - 0.512 = 0.488$. Statement I true.

4. Power $0.488 >$ size 0.36 , so the test is unbiased. Statement II true.

5. Option (c).

SI & HT

2025 Q13

Numerical

Medium

Q12.

For testing $H_0: f(x) = 2x; 0 < x < 1, 0$ otherwise, against $H_1: f(x) = 3x^2; 0 < x < 1, 0$ otherwise, on the basis of a single observation, the power of a most powerful test of size $\alpha = 0.19$ is:

A. 0.271

B. 0.275

C. 0.729

D. 0.810

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$f_1/f_0 = \frac{3x}{2}$ increasing, so reject for $x \geq c$; $1 - c^2 = 0.19$ gives $c = 0.9$, and the power is $1 - 0.9^3 = 0.271$.

TIPS & TRICKS

- $\int_c^1 2x \, dx = 1 - c^2$ and $\int_c^1 3x^2 \, dx = 1 - c^3$ - the cleanest pair in this family.
- $c^2 = 0.81 \Rightarrow c = 0.9$; then $c^3 = 0.729$.
- 0.729 is planted as option (c) - that is c^3 , not the power.

STEP-BY-STEP SOLUTION

1. $\frac{f_1(x)}{f_0(x)} = \frac{3x^2}{2x} = \frac{3x}{2}$, increasing, so the MP critical region is $x \geq c$.
2. Size: $P_{H_0}(X \geq c) = \int_c^1 2x \, dx = 1 - c^2 = 0.19 \Rightarrow c^2 = 0.81 \Rightarrow c = 0.9$.
3. Power: $P_{H_1}(X \geq 0.9) = \int_{0.9}^1 3x^2 \, dx = 1 - (0.9)^3 = 1 - 0.729 = 0.271$.
4. Option (a).

Q13.

Let $x_1, x_2, \dots, x_n$ be a random sample from the distribution having p.d.f. $f(x; \theta, \beta) = \beta e^{-\beta(x-\theta)}$; $x \geq \theta$, $\beta > 0$. To test $H_0: \beta = 2, \theta = 3$ against $H_1: \beta = 4, \theta = 2$, the best critical region is given by

where $k > 0$.

A. $\bar{x} \geq \frac{1}{2} \left\{ 8 + \log 2 - \frac{\log k}{n} \right\}$

B. $\bar{x} \geq \left\{ 1 + \frac{\log 2}{2} - \frac{\log k}{n} \right\}$

C. $\bar{x} \leq \left\{ 1 + \log 2 - \frac{\log k}{2n} \right\}$

D. $\bar{x} \leq \left\{ 1 + \frac{\log 2}{2} - \frac{\log k}{2n} \right\}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$$\log \frac{L_1}{L_0} = n \log 2 + 2n - 2 \sum x_i, \text{ so the region } \frac{L_1}{L_0} \geq k \text{ becomes } \bar{x} \leq 1 + \frac{\log 2}{2} - \frac{\log k}{2n}.$$

TIPS & TRICKS

- Both densities are defined only where every $x_i \geq 3$ (the larger of the two thresholds), so the indicator is common and cancels.
- $-4 \sum (x_i - 2) + 2 \sum (x_i - 3) = -2 \sum x_i + 8n - 6n = -2 \sum x_i + 2n$.
- Dividing by $-2n$ flips the inequality, producing an upper-bound (left-tail) region in $\bar{x}$.

STEP-BY-STEP SOLUTION

1. $L_0 = 2^n \exp \left[-2 \sum (x_i - 3) \right]$ and $L_1 = 4^n \exp \left[-4 \sum (x_i - 2) \right]$, both on the common region $x_i \geq 3$.

2. $\log \frac{L_1}{L_0} = n \log 2 - 4 \sum x_i + 8n + 2 \sum x_i - 6n = n \log 2 + 2n - 2 \sum x_i$.

3. $\frac{L_1}{L_0} \geq k \iff n \log 2 + 2n - 2 \sum x_i \geq \log k \iff \sum x_i \leq \frac{n \log 2 + 2n - \log k}{2}$.

4. Dividing by n : $\bar{x} \leq 1 + \frac{\log 2}{2} - \frac{\log k}{2n}$ - option (d).

Q14.

Let $X \sim f(x, \theta)$. The most powerful test of size 0.10 for testing $H_0: f(x, \theta) = 8x^7$, $0 < x < 1$ against $H_1: f(x, \theta) = 4x^3$, $0 < x < 1$ has power

A. $\sqrt{0.10}$

B. $\sqrt{[4]0.10}$

C. $\sqrt{[6]0.10}$

D. $\sqrt{[8]0.10}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$f_1/f_0 = \frac{1}{2x^4}$ is DECREASING, so the MP region is $x \leq c$ with $c^8 = 0.10$; the power is $c^4 = \sqrt{0.10}$.

TIPS & TRICKS

- Decreasing likelihood ratio $\Rightarrow$ LEFT-tail critical region - the direction is the whole trick.
- $P_{H_0}(X \leq c) = \int_0^c 8x^7 dx = c^8$ and $P_{H_1}(X \leq c) = \int_0^c 4x^3 dx = c^4$.
- Power $= c^4 = (c^8)^{1/2} = \sqrt{\alpha}$ - a neat square-root relation.

STEP-BY-STEP SOLUTION

- $\frac{f_1(x)}{f_0(x)} = \frac{4x^3}{8x^7} = \frac{1}{2x^4}$, which decreases in x , so the MP test rejects for SMALL x : $C = \{x \leq c\}$.
- Size: $\int_0^c 8x^7 dx = c^8 = 0.10$, so $c = (0.10)^{1/8}$.
- Power: $\int_0^c 4x^3 dx = c^4 = (0.10)^{4/8} = (0.10)^{1/2}$.
- $= \sqrt{0.10}$ - option (a).

Q15.

Neyman-Pearson fundamental lemma provides the most powerful (MP) test of its size for testing

- A. simple null hypothesis against composite alternative
- B. composite null hypothesis against simple alternative
- C. **Simple null hypothesis against simple alternative**
- D. composite null hypothesis against composite alternative

Correct answer: C

EXAM SHORTCUT[AI-derived explanation](#)

Neyman-Pearson: simple null against simple alternative.

TIPS & TRICKS

- Both hypotheses must be simple for the likelihood ratio L_1/L_0 to be a well-defined number.
- Karlin-Rubin extends this to composite one-sided alternatives via MLR.
- Composite nulls are the domain of the likelihood ratio test.

STEP-BY-STEP SOLUTION

1. The Neyman-Pearson lemma treats $H_0:\theta=\theta_0$ versus $H_1:\theta=\theta_1$ - both simple.
 2. It gives the most powerful test of its size via the likelihood ratio criterion.
 3. Option (c).
-

Q16.

The most powerful test in testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ maximizes

- A. variance
- B. ~~φ~~power
- C. type I error
- D. type II error

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

The most powerful test maximises POWER subject to a fixed size.

TIPS & TRICKS

- Size is CONSTRAINED at α ; power is the objective being maximised.
- Equivalently it minimises the Type-II error probability β .
- Neyman-Pearson solves exactly this constrained optimisation.

STEP-BY-STEP SOLUTION

1. Among all tests of size at most α , the most powerful test is the one with the largest probability of rejecting H_0 when H_1 is true.
 2. That probability is the power, $1-\beta$.
 3. Option (b).
-

Q17.

Consider a random sample of size n from a Bernoulli(p) population. To test $H_0:p=0.4$ against $H_1:p=0.6$, the most powerful test depends on

- A. $\sum_{i=1}^n x_i$
- B. $x_{(1)}$
- C. $\sum_{i=1}^n (x_i - \bar{x})^2$
- D. $\sum_{i=1}^n x_i^2$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\frac{L_1}{L_0} = \left(\frac{0.6}{0.4} \right)^{\sum x_i} \left(\frac{0.4}{0.6} \right)^{n - \sum x_i}, \text{ a monotone function of } \sum x_i.$$

TIPS & TRICKS

- The Bernoulli likelihood ratio always reduces to the total number of successes.
- $p_1 > p_0$ makes the ratio increasing, so the MP test rejects for LARGE $\sum x_i$.
- $\sum x_i^2 = \sum x_i$ for 0-1 data, so option (d) is the same statistic written obscurely - but $\sum x_i$ is the canonical answer.

STEP-BY-STEP SOLUTION

$$1. \frac{L_1}{L_0} = \frac{0.6^{\sum x_i} 0.4^{n - \sum x_i}}{0.4^{\sum x_i} 0.6^{n - \sum x_i}} = \left(\frac{0.6}{0.4} \right)^{\sum x_i - n}, \text{ increasing in } \sum x_i.$$

2. By Neyman-Pearson the most powerful test rejects when this ratio exceeds k , i.e. when $\sum x_i$ is large.

3. So the test depends on the data only through $\sum_{i=1}^n x_i$ - option (a).

Likelihood Ratio Test

Q1.

For testing $\mu = \mu_0$ against $\mu \neq \mu_0$ in $N(\mu, \sigma^2)$, both unknown, the likelihood ratio test is equivalent to

- A. uniformly most powerful test
- B. uniformly most powerful similar test
- C. uniformly most powerful invariant test
- D. uniformly most powerful unbiased test**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

The LRT reduces to the two-sided t -test, which for a two-sided alternative is the uniformly most powerful UNBIASED test (no UMP test exists two-sided).

TIPS & TRICKS

- No UMP test exists for two-sided alternatives in the normal family - restricting to unbiased tests restores optimality.
- The t -test is also similar (its size is α for every nuisance σ^2) and invariant, but the standard optimality statement is UMPU.
- Rule of thumb: one-sided $\Rightarrow$ UMP; two-sided $\Rightarrow$ UMPU.

STEP-BY-STEP SOLUTION

1. For $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$ with σ^2 unknown, the likelihood ratio reduces to a function of

$$|t| = \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right|.$$

2. The test rejects for large $|t|$, i.e. it is the usual two-sided t -test.
3. For two-sided alternatives no uniformly most powerful test exists; within the class of unbiased tests, the t -test is uniformly most powerful.
4. Hence the LRT is the UMPU test - option (d).

Q2.

Let X_i ($i=1, \dots, n$) be $N(\mu_i, 1)$. The LR test for $H_0: \mu_1 = \dots = \mu_n = 0$ against $H_1: \mu_i \neq 0$ for some i , is based on

A. $\sum_{i=1}^n \left(\frac{X_i^2}{\mu_i} \right)$

B. $\sum_{i=1}^n X_i^2$

C. $\sum_{i=1}^n (X_i - \bar{X})^2$

D. $\sum_{i=1}^n X_i$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Unrestricted MLE is $\hat{\mu}_i = X_i$; under H_0 all are 0. The ratio gives $-2 \log \lambda = \sum X_i^2$.

TIPS & TRICKS

- Each μ_i has its own observation, so the unrestricted maximiser is simply $\hat{\mu}_i = X_i$ and the maximised likelihood is $(2\pi)^{-n/2}$.
- Wilks: $-2 \log \lambda = \sum X_i^2 \sim \chi_n^2$ under H_0 , matching the n restrictions imposed.
- $\sum (X_i - \bar{X})^2$ would be the statistic for $H_0: \mu_1 = \dots = \mu_n$ (a common unknown value), not for all being zero.

STEP-BY-STEP SOLUTION

- Under H_1 the parameter space is unrestricted, so $\hat{\mu}_i = X_i$ and $\sup L = (2\pi)^{-n/2}$.
- Under H_0 all $\mu_i = 0$, so $\sup_{H_0} L = (2\pi)^{-n/2} \exp\left(-\frac{1}{2} \sum X_i^2\right)$.
- $\lambda = \exp\left(-\frac{1}{2} \sum X_i^2\right)$, so $-2 \log \lambda = \sum X_i^2$ and the test rejects for large $\sum X_i^2$.
- Option (b).

Q3.

Under regularity conditions, if λ_n is the likelihood ratio, the asymptotic distribution of $-2\log\lambda_n$ as $n \rightarrow \infty$ is

- A. normal distribution
- B. **✓Chi-square distribution**
- C. F -distribution
- D. t -distribution

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Wilks' theorem: $-2\log\lambda_n \rightarrow d\chi^2_r$, where r is the number of independent restrictions imposed by H_0 .

TIPS & TRICKS

- The degrees of freedom equal (dimension of full parameter space) - (dimension under H_0).
- 'Asymptotically normal' is the stock distractor - the LRT statistic is asymptotically chi-square, not normal.
- Regularity conditions matter: they fail when the support depends on θ or H_0 sits on the parameter-space boundary.

STEP-BY-STEP SOLUTION

1. $\lambda_n = \frac{\sup_{\theta_0} L}{\sup_{\theta} L} \in [0, 1]$, so $-2\log\lambda_n \geq 0$.

2. Wilks' theorem states that under H_0 and regularity conditions, $-2\log\lambda_n$ converges in distribution to χ^2_r .

3. Hence the asymptotic distribution is chi-square - option (b).

Q4.

If, for given $\alpha \in [0,1]$, non-randomised NP and LR tests of a simple vs simple hypothesis exist, then

A. They are one and the same

B. They are equivalent

C. They are exactly opposite

D. One cannot say anything about it

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

For simple vs simple, the LR criterion $\lambda = L_0 / \max(L_0, L_1)$ is small exactly when L_1/L_0 is large - the two tests coincide.

TIPS & TRICKS

- $\lambda = \frac{L_0}{\max(L_0, L_1)}$, so $\lambda < c \iff \frac{L_1}{L_0} > \frac{1}{c}$ - ... - a monotone relabelling of the same statistic.
- With randomisation allowed on the boundary they agree exactly; the stem restricts attention to the non-randomised case where both exist.
- For composite hypotheses the two procedures genuinely diverge - the LRT is then generally not most powerful.

STEP-BY-STEP SOLUTION

1. For $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$, $\sup_{\theta_0} L = L_0$ and $\sup_{\theta} L = \max(L_0, L_1)$.
2. Hence $\lambda = \frac{L_0}{\max(L_0, L_1)}$, which is a strictly decreasing function of the likelihood ratio L_1/L_0 whenever $L_1 > L_0$.
3. Rejecting for small λ is therefore identical to rejecting for large L_1/L_0 , which is the Neyman-Pearson rule.
4. When both non-randomised tests exist at level α , they are one and the same test - option (a).

Q5.

Which parameter statement is correct for the Likelihood Ratio Test (LRT)?

- A. In LRT, control is not affected over the probabilities of Type-I error by suitably choosing the cut-off point λ_0
- B. LRT always gives an unbiased test
- C. When the null hypothesis is composite, the LR critical region will always be similar
- D. **Under certain assumptions, an LRT will be consistent**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Under regularity conditions the likelihood ratio test is consistent - the only defensible statement of the four.

TIPS & TRICKS

- Type-I error IS controlled by choosing λ_0 , so (a) is false as written.
- The LRT is not guaranteed unbiased, and its critical region need not be similar when H_0 is composite.
- Consistency: power $\rightarrow 1$ as $n \rightarrow \infty$ for any fixed alternative - a standard asymptotic property of the LRT.

STEP-BY-STEP SOLUTION

1. (a) The cut-off λ_0 is chosen precisely so that $P(\lambda \leq \lambda_0 / H_0) = \alpha$, so Type-I error IS controlled. False.
2. (b) The LRT is not guaranteed to be unbiased in general. False.
3. (c) A similar critical region requires constant size over all of Θ_0 ; the LR region need not have this property. False.
4. (d) Under standard regularity conditions the LRT is consistent: its power tends to 1 at every fixed alternative. True.
5. Option (d).

Q6.

For a likelihood ratio λ , consider:

1. $0 \leq \lambda \leq 1$.
2. $(-2 \log \lambda)$ follows an asymptotic Chi-square distribution.
3. The likelihood ratio test is consistent.

Codes:

- A. 1 and 2 only
B. 1 and 3 only
C. 2 and 3 only
D. 1, 2 and 3

Correct answer: D

EXAM SHORTCUT AI-derived explanation

All three hold: λ is a ratio of a constrained to an unconstrained supremum (hence in $[0,1]$), $-2 \log \lambda$ is asymptotically χ^2 (Wilks), and the LRT is consistent under regularity.

TIPS & TRICKS

- $\lambda \leq 1$ because the numerator maximises over a SUBSET of the denominator's parameter space.
- Wilks' df = number of independent restrictions imposed by H_0 .
- Consistency here means the power tends to 1 at every fixed alternative as $n \rightarrow \infty$.

STEP-BY-STEP SOLUTION

1. Statement 1: $\lambda = \frac{\sup_{\theta_0} L}{\sup_{\theta} L}$ with $\theta_0 \subseteq \theta$, so $0 \leq \lambda \leq 1$. True.
2. Statement 2: by Wilks' theorem, $-2 \log \lambda \rightarrow d\chi^2_r$ under H_0 and regularity. True.
3. Statement 3: under regularity conditions the likelihood ratio test is consistent. True.
4. Option (d).

Q7.

The number of failed satellite launches has a Poisson distribution with parameter λ . Failure counts for the past 12 months are observed to test $H_0: \lambda=2$ against $H_1: \lambda=3$. The Likelihood Ratio is

- A. $1.5e^{-24}$
 B. $1.5e^{-12}$
 C. $(1.5)^{\sum_{i=1}^{12} x_i} e^{-12}$
 D. $(1.5)^{\sum_{i=1}^{24} x_i} e^{-24}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$\frac{L_1}{L_0} = \frac{e^{-36} 3^{\sum x}}{e^{-24} 2^{\sum x}} = (1.5)^{\sum x_i} e^{-12} \text{ with } n=12 \text{ months.}$$

TIPS & TRICKS

- $e^{-12(3)}/e^{-12(2)} = e^{-12}$ - the exponent is $n(\lambda_1 - \lambda_0) = 12$.
- $(3/2)^{\sum x_i} = (1.5)^{\sum x_i}$ - the data enter only through the total, as sufficiency requires.
- $n=12$ throughout; options with 24 double the sample size and are wrong.

STEP-BY-STEP SOLUTION

$$1. \text{ For a Poisson sample of size } n=12, L(\lambda) = \frac{e^{-12\lambda} \lambda^{\sum x_i}}{\prod x_i!}.$$

$$2. \frac{L(\lambda=3)}{L(\lambda=2)} = \frac{e^{-36} 3^{\sum x_i}}{e^{-24} 2^{\sum x_i}}.$$

$$3. = e^{-12} \left(\frac{3}{2} \right)^{\sum x_i} = (1.5)^{\sum_{i=1}^{12} x_i} e^{-12}.$$

4. Option (c).

Q8.

Let $x_1, x_2, \dots, x_n$ be a random sample from a Poisson distribution with parameter $\lambda > 0$. Which of the following statements are correct in respect of the Likelihood Ratio Test (LRT)?

1. The LRT for testing $H_0: \lambda = \lambda_0$ against $H_1: \lambda > \lambda_0$ produces a UMP test identical to the test obtained by NP theory of testing.
2. The LRT for testing $H_0: \lambda = \lambda_0$ against $H_1: \lambda < \lambda_0$ produces a UMP test identical to the test obtained by NP theory of testing.
3. The LRT for testing $H_0: \lambda = \lambda_0$ against $H_1: \lambda \neq \lambda_0$ produces a UMPU test identical to the test obtained by NP theory of testing.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

Correct answer: D

EXAM SHORTCUT**AI-derived explanation**

Poisson has MLR in $\sum x_i$, so both one-sided LRTs reproduce the UMP tests, and the two-sided LRT reproduces the UMPU test. All three statements hold.

TIPS & TRICKS

- MLR family $\Rightarrow$ one-sided UMP tests exist and the LRT coincides with them.
- Two-sided: no UMP test exists, but the LRT gives the UMPU test in exponential families.
- This 'LRT recovers the optimal test' pattern is standard for one-parameter exponential families.

STEP-BY-STEP SOLUTION

1. Poisson is a one-parameter exponential family with MLR in $T = \sum x_i$.
2. Statement 1: for $H_1: \lambda > \lambda_0$, Karlin-Rubin gives the UMP test 'reject if $T > c$ ', and the LRT reduces to exactly this. True.
3. Statement 2: for $H_1: \lambda < \lambda_0$ the same argument gives 'reject if $T < c$ '. True.
4. Statement 3: for $H_1: \lambda \neq \lambda_0$ no UMP test exists, but the two-sided LRT coincides with the UMPU test. True.
5. Option (d).

Q9.

Let $x_1, x_2, \dots, x_9$ be a random sample from Normal distribution $N(\mu, \sigma^2)$, μ and σ are unknown parameters, $\mu \in \mathbb{R}$, $\sigma > 0$. The observed values are $\bar{x} = \frac{\sum_{i=1}^9 x_i}{9} = 9.8$ and $\frac{1}{8} \sum_{i=1}^9 (x_i - \bar{x})^2 = 1.69$. If LRT is used to test $H_0: \mu = 8.8$ versus $H_1: \mu > 8.8$, then what is the calculated value of the statistic?

A. 1.86

B. 2.31

C. 2.90

D. 3.36

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$s^2 = 1.69 \Rightarrow s = 1.3; t = \frac{\sqrt{9} (9.8 - 8.8)}{1.3} = \frac{3}{1.3} = 2.31.$$

TIPS & TRICKS

- The LRT for a normal mean with unknown variance reduces to Student's t statistic.
- $\frac{1}{8} \sum (x_i - \bar{x})^2$ IS s^2 with divisor $n-1=8$ - no further adjustment needed.
- $\sqrt{1.69} = 1.3$ exactly, so $t = 3/1.3 = 2.3077$.

STEP-BY-STEP SOLUTION

- The LRT for $H_0: \mu = \mu_0$ against $\mu > \mu_0$ with σ unknown is a monotone function of $t = \frac{\sqrt{n} (\bar{x} - \mu_0)}{s}$.
- $s^2 = \frac{1}{8} \sum (x_i - \bar{x})^2 = 1.69$, so $s = 1.3$.
- $t = \frac{\sqrt{9} (9.8 - 8.8)}{1.3} = \frac{3 \times 1}{1.3} = 2.3077$.
- ≈ 2.31 - option (b).

Q10.

Consider the following statements about Likelihood Ratio Test (LRT):

1. LRT is generally Uniformly Most Powerful (UMP), if a UMP test exists.
2. Under certain conditions, LRT is consistent.

Which of the statements given above is/are correct?

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

When a UMP test exists (MLR families) the LRT reproduces it, and under regularity the LRT is consistent. Both statements hold.

TIPS & TRICKS

- In one-parameter exponential families the LR statistic is a monotone function of the sufficient statistic, so it gives the same critical region as Karlin-Rubin.
- 'Generally' is doing real work in statement 1 - it is not a universal theorem but holds in the standard families.
- Consistency of the LRT: power $\rightarrow 1$ at every fixed alternative as $n \rightarrow \infty$.

STEP-BY-STEP SOLUTION

1. Statement 1: in families with monotone likelihood ratio, the likelihood ratio statistic is a monotone function of the sufficient statistic, so the LRT critical region coincides with the UMP one whenever a UMP test exists. True.
2. Statement 2: under standard regularity conditions the LRT is consistent. True.
3. Option (c).

Q11.

Consider the following statements regarding likelihood ratio tests:

- I. For a given size α of a test, the non-randomized Neyman-Pearson and likelihood ratio tests of a simple hypothesis against a simple alternative hypothesis are equivalent.
- II. The likelihood ratio λ always lies between 0 and 1.
- III. For testing a null hypothesis against any alternative hypothesis, the likelihood ratio test is a function of every sufficient statistic for θ .
- IV. If λ is the likelihood ratio, then the asymptotic distribution of $-2\ln\lambda$ is normal with mean 0 and variance 1.

Which of the statements given above are correct?

A. I, II and III

B. I, II and IV

C. I, III and IV

D. II, III and IV

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

I, II and III are correct; IV is wrong because $-2\ln\lambda$ is asymptotically CHI-SQUARE, not $N(0,1)$.

TIPS & TRICKS

- Wilks: $-2\ln\lambda \rightarrow d\chi^2_r$ - never normal.
- $\lambda \in [0,1]$ since the numerator maximises over a subset.
- The LR statistic factorises through any sufficient statistic, since the whole likelihood does.

STEP-BY-STEP SOLUTION

1. I: for simple versus simple the two non-randomised tests coincide (the LR criterion is a monotone function of L_1/L_0). True.

2. II: $\lambda = \frac{\sup_{\theta_0} L}{\sup_{\theta} L} \in [0,1]$. True.

3. III: by factorisation both suprema depend on the data only through the sufficient statistic. True.

4. IV: Wilks' theorem gives a chi-square limit, not $N(0,1)$. False.

5. Hence I, II and III - option (a).

Q12.

Consider a random sample $\{x_1, x_2, x_3, x_4\}$ from $N(\mu, \sigma^2)$ distribution. Define $\bar{x} = \frac{\sum_{i=1}^4 x_i}{4}$ and $s^2 = \frac{\sum_{i=1}^4 (x_i - \bar{x})^2}{3}$. The LRT of size $\alpha = 0.05$ rejects $H_0: \mu = 0$ versus $H_1: \mu \neq 0$ if and only if $\frac{|\bar{x}|}{s} > k$. The value of k , when $P(X > t_{(n, \alpha)}) = \alpha$, is

- A. $\frac{1}{2} t_{(3, 0.05)}$
- B. $\frac{1}{2} t_{(3, 0.025)}$
- C. $2t_{(4, 0.025)}$
- D. $2t_{(4, 0.05)}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$t = \frac{\sqrt{4} \bar{x}}{s} = \frac{2\bar{x}}{s}$, and the two-sided size-0.05 test rejects when $|t| > t_{3, 0.025}$, i.e. $\frac{|\bar{x}|}{s} > \frac{1}{2} t_{3, 0.025}$.

TIPS & TRICKS

- $\sqrt{n} = 2$ for $n=4$, so the t statistic is $2\bar{x}/s$ - divide the critical value by 2.
- Degrees of freedom are $n-1=3$, not 4 - options (c) and (d) use the wrong df.
- Two-sided at $\alpha=0.05$ needs the 0.025 upper point, not 0.05.

STEP-BY-STEP SOLUTION

1. With $n=4$, the t statistic is $t = \frac{\sqrt{n} \bar{x}}{s} = \frac{2\bar{x}}{s}$, with $n-1=3$ degrees of freedom.
2. The two-sided size-0.05 LRT rejects when $|t| > t_{3, 0.025}$.
3. $\frac{2|\bar{x}|}{s} > t_{3, 0.025} \iff \frac{|\bar{x}|}{s} > \frac{1}{2} t_{3, 0.025}$.
4. So $k = \frac{1}{2} t_{(3, 0.025)}$ - option (b).

Q13.

In order to test $H_0: \theta = \theta_0$ against the alternative $H_1: \theta \neq \theta_0$, the likelihood ratio to be used as the test statistic is given by

where Θ is the parametric space.

A. ✓ $\frac{L(\theta_0)}{\max(L(\Theta))}$

B. $\frac{\max(L(\Theta))}{L(\theta_0)}$

C. $\min\left(\frac{L(\theta_0)}{L(\Theta)}\right)$

D. $\min\left(\frac{L(\Theta)}{L(\theta_0)}\right)$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\lambda = \frac{L(\theta_0)}{\max_{\Theta} L(\theta)} \text{ - the constrained supremum over the unconstrained supremum.}$$

TIPS & TRICKS

- $\lambda \in [0, 1]$ because the numerator maximises over a subset - options (b) and the reciprocals cannot lie in $[0, 1]$.
- Reject for SMALL λ : the null fits much worse than the best alternative.
- $-2 \log \lambda \rightarrow d\chi^2_r$ under regularity.

STEP-BY-STEP SOLUTION

1. The likelihood ratio criterion is $\lambda = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)}$.

2. With the simple null $\Theta_0 = \{\theta_0\}$ this is $\frac{L(\theta_0)}{\max_{\Theta} L(\theta)}$.

3. Option (a).

Q14.

Consider the following statements:

I. Neyman-Pearson lemma gives a general method for testing simple null hypothesis against composite alternative hypothesis.

II. Likelihood ratio test gives a general method for testing simple or composite null hypothesis against simple or composite alternative hypothesis.

Which of the statements given above is/are correct?

A. I only

B. ~~AI~~ only

C. Both I and II

D. Neither I nor II

Correct answer: B

EXAM SHORTCUT**AI-derived explanation**

Neyman-Pearson covers SIMPLE versus SIMPLE, not simple versus composite (I false); the LRT is the general-purpose method for any combination (II true).

TIPS & TRICKS

- NP needs both likelihoods to be fully specified numbers - a composite alternative breaks that.
- Extension to composite one-sided alternatives requires MLR (Karlin-Rubin), not NP itself.
- The LRT handles composite nulls and alternatives by taking suprema over each region.

STEP-BY-STEP SOLUTION

1. Statement I: the Neyman-Pearson lemma applies to $H_0:\theta=\theta_0$ against $H_1:\theta=\theta_1$, both simple. As stated (composite alternative) it is false.

2. Statement II: $\lambda = \frac{\sup_{\theta_0} L}{\sup_{\theta} L}$ is defined whether θ_0 and θ_1 are single points or whole sets, so the LRT is a general method. True.

3. Hence II only - option (b).

UMP, MLR, Unbiased and Similar Tests

Q1.

For testing H_0 against H_1 , if $\varphi(x)$ is a UMP test, then

- A. $\varphi(x)$ is always unbiased
- B. $\varphi(x)$ is never unbiased
- C. $\varphi(x)$ has power less than its size
- D. None of the above

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Compare a UMP test with the trivial randomised test $\varphi \equiv \alpha$, whose power is α everywhere. Being uniformly most powerful, the UMP test has power $\geq \alpha$, i.e. it is unbiased.

TIPS & TRICKS

- Unbiased test: power $\geq$ size at every point of the alternative.
- The trivial test $\varphi \equiv \alpha$ is the universal comparison device for proving UMP $\Rightarrow$ unbiased.
- The converse fails - an unbiased test need not be UMP.

STEP-BY-STEP SOLUTION

1. A test is unbiased at level α if its power satisfies $\beta_\varphi(\theta) \geq \alpha$ for all $\theta \in \Theta_1$.
2. Consider the randomised test $\varphi_0(x) \equiv \alpha$, which has size α and power exactly α everywhere.
3. If φ is UMP of size α then its power is at least that of φ_0 at every $\theta \in \Theta_1$, i.e. $\beta_\varphi(\theta) \geq \alpha$.
4. Hence a UMP test is always unbiased - option (a).

Q2.

Consider the following statements:

1. A UMP test is unbiased.
2. A consistent test is UMP.
3. A UMP test is biased.
4. An unbiased test need not be UMP.

Codes:

- A. 1 only
B. 2 and 3 only
C. 1 and 2 only
D. 1, 2 and 4

Correct answer: D

EXAM SHORTCUT AI-derived explanation

A UMP test is always unbiased (statement 1) and an unbiased test need not be UMP (statement 4). Statements 2 and 3 are false.

TIPS & TRICKS

- UMP $\Rightarrow$ unbiased: compare with the trivial test $\varphi \equiv \alpha$, whose power is α everywhere.
- Consistency is an asymptotic property (power $\rightarrow 1$ as $n \rightarrow \infty$); it says nothing about optimality at fixed n .
- Statements 1 and 3 are direct contradictions, so at most one of them can be in the key - a useful internal check.

STEP-BY-STEP SOLUTION

1. Statement 1: a UMP test has power at least that of $\varphi \equiv \alpha$, hence power $\geq \alpha$ - it is unbiased. True.
2. Statement 2: consistency means power $\rightarrow 1$ as $n \rightarrow \infty$; it carries no finite-sample optimality, so a consistent test need not be UMP. False.
3. Statement 3: contradicts statement 1. False.
4. Statement 4: many unbiased tests are not most powerful (e.g. any inefficient unbiased test). True.
5. The genuine answer set is $\{1,4\}$; among the printed options only (d) contains both, and it is taken as the key.

Source note: Source anomaly: statements 1 and 4 are true and statements 2 and 3 are false, so the exact truth set $\{1,4\}$ is not among the printed options. Option (d) is the closest offered combination (it contains both true statements) and is taken as the intended key; the source wording and options are preserved unchanged.

Q3.

Let $X_1, X_2 \sim U(0, \theta)$ iid. The power of the UMP test for $H_0: \theta = 2$ vs $H_1: \theta > 2$, and $H_0: \theta = 2$ vs $H_1: \theta \neq 2$, is (with α the size, θ_1 any value under H_1)

A. $1 - \left(\frac{2}{\theta_1}\right)^2(1-\alpha)$

B. α

C. $1 - \left(\frac{\theta_1}{2}\right)^2(1-\alpha)$

D. $1-\alpha$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

UMP region is $\{X_{(2)} > c\}$ with $1 - (c/2)^2 = \alpha$, so $c = 2\sqrt{1-\alpha}$; power at θ_1 is $1 - (c/\theta_1)^2 = 1 - \left(\frac{2}{\theta_1}\right)^2(1-\alpha)$.

TIPS & TRICKS

- $U(0, \theta)$ has monotone likelihood ratio in $X_{(n)}$, so Karlin-Rubin gives a UMP one-sided test based on the maximum.
- $P(X_{(n)} \leq c) = (c/\theta)^n$ - here $n=2$, which is where the square comes from.
- Unusually, the same test is UMP for the two-sided alternative too, because the power function is monotone over the whole admissible range.

STEP-BY-STEP SOLUTION

1. For X_1, X_2 iid $U(0, \theta)$, $P(X_{(2)} \leq c) = \left(\frac{c}{\theta}\right)^2$.

2. The UMP test for $\theta = 2$ vs $\theta > 2$ rejects when $X_{(2)} > c$, with $\alpha = 1 - \left(\frac{c}{2}\right)^2$, i.e. $c^2 = 4(1-\alpha)$.

3. Power at θ_1 : $1 - \left(\frac{c}{\theta_1}\right)^2 = 1 - \frac{4(1-\alpha)}{\theta_1^2} = 1 - \left(\frac{2}{\theta_1}\right)^2(1-\alpha)$.

4. Option (a).

Q4.

A test where $P(\text{reject } H_0 / H_0 \text{ false}) > P(\text{reject } H_0 / H_0 \text{ true})$ is said to be

A. Unbiased

B. Biased

C. Consistent

D. Uniformly most powerful

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Power exceeding size at every alternative is exactly the definition of an unbiased test.

TIPS & TRICKS

- Read the probabilities: LHS is power, RHS is size. Power > size $\Rightarrow$ unbiased.
- A biased test can have power BELOW its size at some alternatives - a genuinely perverse but possible situation.
- UMP $\Rightarrow$ unbiased, but unbiased $\nRightarrow$ UMP.

STEP-BY-STEP SOLUTION

1. $P(\text{reject } H_0 / H_0 \text{ true})$ is the size α ; $P(\text{reject } H_0 / H_0 \text{ false})$ is the power.
2. The stated inequality says power > size for the alternative considered.
3. A test whose power is at least its size throughout the alternative is called unbiased.
4. Option (a).

Q5.

Let $X_1, \dots, X_n$ be iid $N(\theta, 1)$. The UMPU test for $H_0: \theta = \theta_0$ against $H_1: \theta \neq \theta_0$ of size α is

A. ✓ $\varphi = 1$ if $\bar{X} < \theta_0 - \frac{Z_{\alpha/2}}{\sqrt{n}}$ or $\bar{X} > \theta_0 + \frac{Z_{\alpha/2}}{\sqrt{n}}$; 0 otherwise

B. $\varphi = 1$ if $\bar{X} < \theta_0 - \frac{Z_{1-\alpha}}{\sqrt{n}}$ or $\bar{X} > \theta_0 + \frac{Z_{1-\alpha}}{\sqrt{n}}$; 0 otherwise

C. $\varphi = 1$ if $\bar{X} < \theta_0 - \frac{Z_{1-\alpha}}{n}$ or $\bar{X} > \theta_0 + \frac{Z_{1-\alpha}}{n}$; 0 otherwise

D. $\varphi = 1$ if $\bar{X} < \theta_0 - \frac{Z_{\alpha/2}}{n}$ or $\bar{X} > \theta_0 + \frac{Z_{\alpha/2}}{n}$; 0 otherwise

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Reject when $\bar{X}$ lies outside $\theta_0 \pm Z_{\alpha/2}/\sqrt{n}$ - equal tails of $\alpha/2$ each, and the standard error is $1/\sqrt{n}$.

TIPS & TRICKS

- $\bar{X} \sim N(\theta, 1/n)$, so the standard error is $1/\sqrt{n}$ - options dividing by n are dimensionally wrong.
- Two-sided alternative $\Rightarrow$ split α equally, so the multiplier is $Z_{\alpha/2}$, not $Z_{1-\alpha}$.
- Symmetry of the normal makes the equal-tailed test UMPU here.

STEP-BY-STEP SOLUTION

1. Under H_0 , $\bar{X} \sim N\left(\theta_0, \frac{1}{n}\right)$, so $\sqrt{n}(\bar{X} - \theta_0) \sim N(0, 1)$.

2. No UMP test exists two-sided; within unbiased tests the optimum is the symmetric two-tailed test with $\alpha/2$ in each tail.

3. Rejection region: $|\sqrt{n}(\bar{X} - \theta_0)| > Z_{\alpha/2}$, i.e. $\bar{X} < \theta_0 - \frac{Z_{\alpha/2}}{\sqrt{n}}$ or $\bar{X} > \theta_0 + \frac{Z_{\alpha/2}}{\sqrt{n}}$.

4. Option (a).

Q6.

The UMP test for $H_0: X \sim f_0(x, \theta)$ against $H_1: X \sim f_1(x, \theta)$, if it exists,

A. It is unique

B. it is unique but power is greater than size

C. it need not be unique

D. None of the above

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

When a UMP test of size α exists it is unique (almost everywhere) - any other UMP test agrees with it except on a null set.

TIPS & TRICKS

- Uniqueness is 'essentially unique': two UMP tests can differ only where $f_1 = kf_0$, a set of probability zero in the continuous case.
- The NP lemma's uniqueness half is the source of this result.
- Option (b) adds a true but irrelevant clause (power > size is just unbiasedness), so it is not the cleanest answer.

STEP-BY-STEP SOLUTION

1. The Neyman-Pearson lemma has a uniqueness part: any most powerful test of size α must be of the likelihood-ratio form almost everywhere.
2. Hence two UMP tests of the same size can differ only on the boundary set $\{f_1 = kf_0\}$, which has probability zero in the continuous case.
3. So the UMP test, when it exists, is (essentially) unique - option (a).

Q7.

If $\alpha = P(\text{Type-I error})$, $\beta = P(\text{Type-II error})$, a critical region is said to be unbiased if

A. $\alpha + \beta < 1$

B. $\alpha + \beta > 1$

C. $\alpha > \beta$

D. $\alpha < \beta$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Unbiased means power $\geq$ size, i.e. $1 - \beta \geq \alpha$, i.e. $\alpha + \beta \leq 1$.

TIPS & TRICKS

- Rearrange the definition: power $= 1 - \beta \geq \alpha \iff \alpha + \beta \leq 1$.
- A test with $\alpha + \beta > 1$ is worse than tossing a coin - it is biased.
- The strict version $\alpha + \beta < 1$ is the intended answer here.

STEP-BY-STEP SOLUTION

1. A critical region w is unbiased if the probability of rejecting H_0 when H_1 is true is at least the probability of rejecting when H_0 is true.
2. That is, power $\geq$ size: $1 - \beta \geq \alpha$.
3. Rearranging, $\alpha + \beta \leq 1$.
4. Option (a).

Q8.

Every similar test for testing under H_0 has a Neyman structure under which condition?

- A. Sufficient statistic
- B. Consistent statistic
- C. Boundedly complete
- D. Boundedly complete sufficient statistic**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Every similar test has Neyman structure with respect to T when T is a **BOUNDEDLY COMPLETE SUFFICIENT** statistic for the nuisance parameter on the boundary.

TIPS & TRICKS

- Neyman structure: $E[\varphi(X) / T=t] = \alpha$ for (almost) every t - a much stronger condition than mere similarity.
- Bounded completeness is exactly what forces the conditional size to be constant; sufficiency alone is not enough.
- This result is the engine behind UMPU tests in the presence of nuisance parameters (e.g. the t -test).

STEP-BY-STEP SOLUTION

1. A test φ is similar of size α if $E_{\theta}\varphi(X) = \alpha$ for all θ on the boundary of H_0 .
2. φ has Neyman structure with respect to T if $E[\varphi(X) / T=t] = \alpha$ for almost all t .
3. The theorem states: if T is a boundedly complete sufficient statistic for the boundary family, then every similar test has Neyman structure.
4. Sufficiency alone gives the conditional expectation a θ -free meaning; bounded completeness forces it to equal the constant α .
5. Option (d).

Q9.

For testing a simple null against a simple alternative hypothesis, the test is unbiased if (α, β usual meaning)

- A. $\alpha \leq \beta$
- B. ~~$\alpha + \beta < 1$~~
- C. $\alpha + \beta > 1$
- D. $\alpha \geq \beta$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Unbiased means power $\geq$ size: $1 - \beta \geq \alpha$, i.e. $\alpha + \beta \leq 1$.

TIPS & TRICKS

- Write out the definition and rearrange - there is no need to memorise the inequality separately.
- A test with $\alpha + \beta > 1$ would be improved by simply reversing its decisions.
- Comparisons of α with β individually (options (a), (d)) have nothing to do with unbiasedness.

STEP-BY-STEP SOLUTION

1. For simple H_0 against simple H_1 , size $= \alpha$ and power $= 1 - \beta$.
2. Unbiasedness requires power $\geq$ size: $1 - \beta \geq \alpha$.
3. Rearranging, $\alpha + \beta \leq 1$.
4. Option (b).

Q10.

Consider the problem of testing $H_0: \mu = 0$ against $H_1: \mu \neq 0$, based on random sample of size n taken from $N(\mu, \sigma^2)$ normal distribution where σ^2 is unknown. Consider the transformation $Y = bX$, $b \neq 0$. Which one of the following statements is correct?

- A. Sample space will change and hence UMPU similar test will change and hence does not remain invariant under the transformation.
- B. Parametric space will change and hence UMPU similar test does not remain invariant under the transformation.
- C. Sample space will change but parametric space remains unchanged and hence UMPU similar test remains invariant under the transformation.
- D. Sample space as well as parametric space remain unchanged and UMPU similar test becomes invariant under the transformation.**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$Y=bX$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$ and $(\mu, \sigma^2) \mapsto (b\mu, b^2\sigma^2)$ keeps $\mu=0$ equivalent to $b\mu=0$ - both spaces are preserved and the t -based UMPU test is invariant.

TIPS & TRICKS

- The t -statistic $\frac{\sqrt{n} \bar{Y}}{S_Y}$ equals $\pm \frac{\sqrt{n} \bar{X}}{S_X}$, so the two-sided test is literally unchanged.
- Scale changes preserve the null $\mu=0$, so the testing problem itself is invariant.
- Both the sample space ($\mathbb{R}^n$) and the parameter space ($\mathbb{R} \times (0, \infty)$) map onto themselves.

STEP-BY-STEP SOLUTION

1. Under $Y=bX$ with $b \neq 0$: $Y_i \sim N(b\mu, b^2\sigma^2)$, so the parameter space $\{(\mu, \sigma^2): \mu \in \mathbb{R}, \sigma^2 > 0\}$ maps onto itself.
2. The sample space $\mathbb{R}^n$ also maps onto itself, and $H_0: \mu=0$ becomes $H_0: b\mu=0$, the same hypothesis.
3. The UMPU similar test rejects for large $\left| \frac{\sqrt{n} \bar{X}}{S_X} \right|$, and this statistic satisfies $\frac{\sqrt{n} \bar{Y}}{S_Y} = \text{sign}(b) \frac{\sqrt{n} \bar{X}}{S_X}$, so its absolute value is unchanged.
4. Hence the test is invariant under the transformation - option (d).

Q11.

A test of level α is said to be an unbiased test if its power is:

- A. equal to the level of significance
- B. less than the level of significance
- C. greater than probability of type-II error
- D. ~~greater~~ greater than the level of significance**

Correct answer: D

EXAM SHORTCUT[AI-derived explanation](#)

Unbiased means the power is at least the level of significance throughout the alternative.

TIPS & TRICKS

- Power $\geq \alpha$ on Θ_1 is the definition; option (a) states equality, which is similarity, not unbiasedness.
- A UMP test is automatically unbiased (compare with the trivial test $\varphi \equiv \alpha$).
- Comparing the power with the Type-II error probability (option (c)) is meaningless - they sum to 1.

STEP-BY-STEP SOLUTION

1. A level- α test φ is unbiased if $E_{\theta}[\varphi(X)] \geq \alpha$ for every $\theta \in \Theta_1$.
2. In words: the power must be greater than (or at least equal to) the level of significance.
3. Option (d).

Q12.

Consider the following distributions:

1. Bernoulli distribution $b(1, \theta)$
2. Poisson distribution $P(\theta)$
3. Cauchy distribution $C(\theta, 1)$
4. Logistic distribution with location parameter θ

How many of the above distributions possess the Monotone Likelihood Ratio property?

- A. Only one
- B. Only two
- C. **Only three**
- D. All four

Correct answer: C

EXAM SHORTCUT**AI-derived explanation**

Bernoulli, Poisson and the logistic location family have monotone likelihood ratio; the Cauchy does not. So three.

TIPS & TRICKS

- Every one-parameter exponential family with a natural parameter monotone in θ has MLR - Bernoulli and Poisson qualify.
- The logistic location family is not exponential but is still an MLR family (its log-likelihood ratio is monotone).
- The Cauchy's likelihood ratio is non-monotone - the standard counterexample, and the reason no UMP test exists there.

STEP-BY-STEP SOLUTION

1. Bernoulli $b(1, \theta)$: $\frac{L_{\theta_2}}{L_{\theta_1}}$ is monotone in $\sum x_i$ - MLR holds.

2. Poisson $P(\theta)$: likewise monotone in $\sum x_i$ - MLR holds.

3. Cauchy $C(\theta, 1)$: the likelihood ratio is a non-monotone function of the data - MLR fails.

4. Logistic with location θ : the log-likelihood ratio is monotone increasing in x - MLR holds.

5. Three of the four possess MLR - option (c).

Q13.

Let X be a random variable with pdf $f(x, \theta) = 2\theta x + 1 - \theta$, $0 < x < 1$, $-1 \leq \theta \leq 1$. Based on a sample of size one, the Uniformly Most Powerful (UMP) critical region for testing $H_0: \theta = 0$ versus $H_1: \theta > 0$ at level $\alpha = 0.05$ is given by:

- A. $x \geq \frac{1}{20}$
- B. $x \leq \frac{1}{20}$
- C. $x \leq \frac{19}{20}$
- D. ✓ $x \geq \frac{19}{20}$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

The likelihood ratio $1 + \theta(2x - 1)$ is increasing in x , so the UMP region is an upper tail; under $\theta = 0$,

$X \sim U(0, 1)$, so $P(X \geq c) = 1 - c = 0.05$ gives $c = \frac{19}{20}$.

TIPS & TRICKS

- $\frac{f_\theta(x)}{f_0(x)} = 2\theta x + 1 - \theta = 1 + \theta(2x - 1)$, increasing in x for $\theta > 0$ - MLR in x itself.
- Under $H_0: \theta = 0$ the density is $f(x) = 1$, the standard uniform, so tail probabilities are just lengths.
- $\alpha = 0.05$ needs an interval of length 0.05 at the top: $\left[\frac{19}{20}, 1\right]$.

STEP-BY-STEP SOLUTION

1. Under $H_0: \theta = 0$, $f(x, 0) = 1$ on $(0, 1)$, i.e. $X \sim U(0, 1)$.
2. For $\theta > 0$, $\frac{f(x, \theta)}{f(x, 0)} = 2\theta x + 1 - \theta = 1 + \theta(2x - 1)$, which increases in x .
3. By Karlin-Rubin the UMP test rejects for large x : $C = \{x \geq c\}$.
4. Size: $P(X \geq c / \theta = 0) = 1 - c = 0.05 \Rightarrow c = 0.95 = \frac{19}{20}$.
5. Option (d).

Q14.

Let X be a random variable with $N(\theta, 1+a\theta^2)$, $a>0$. The locally Most Powerful Test for testing $H_0:\theta=0$ against $H_1:\theta>0$ is:

where Z_α is the upper $\alpha\%$ value of $N(0,1)$.

A.
$$\varphi(x) = \begin{cases} 1, & \sum_{i=1}^n x_i > \frac{Z_\alpha}{\sqrt{n}} \\ 0, & \text{otherwise} \end{cases}$$

B.
$$\varphi(x) = \begin{cases} 1, & \bar{X} > \frac{Z_{1-\alpha}}{n} \\ 0, & \text{otherwise} \end{cases}$$

C.
$$\varphi(x) = \begin{cases} 1, & \bar{X} > \frac{Z_\alpha}{n} \\ 0, & \text{otherwise} \end{cases}$$

D. ✓
$$\varphi(x) = \begin{cases} 1, & \bar{X} > \frac{Z_\alpha}{\sqrt{n}} \\ 0, & \text{otherwise} \end{cases}$$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

The score at $\theta=0$ reduces to $\sum x_i$, so the LMP test rejects for large $\bar{X}$; under H_0 , $\bar{X} \sim N(0, 1/n)$, giving the cut-off $Z_\alpha/\sqrt{n}$.

TIPS & TRICKS

- Locally most powerful test: base the test on $\left. \frac{\partial \log L}{\partial \theta} \right|_{\theta=\theta_0}$.
- At $\theta=0$ the variance term $1+a\theta^2$ equals 1 and its derivative contributes nothing, so the score is simply $\sum x_i$.
- $V(\bar{X})=1/n$ at $\theta=0$, so the standard error is $1/\sqrt{n}$ - options dividing by n are dimensionally wrong.

STEP-BY-STEP SOLUTION

1. $\log f = -\frac{1}{2} \log [2\pi(1+a\theta^2)] - \frac{(x-\theta)^2}{2(1+a\theta^2)}.$

2. Differentiating and setting $\theta=0$: the terms involving $a\theta$ vanish and $\left. \frac{\partial \log f}{\partial \theta} \right|_{\theta=0} = x.$

3. So the sample score is $\sum x_i$, and the LMP test rejects for large $\sum x_i$, equivalently large $\bar{X}$.

4. Under $H_0:\theta=0$, $\bar{X} \sim N\left(0, \frac{1}{n}\right)$, so the size- α cut-off is $\frac{Z_\alpha}{\sqrt{n}}.$

5. Option (d).

SHARED CONTEXT

Let X be a random variable with pmf

$$P(x) = \begin{cases} \frac{3+2\theta+\mu}{15}, & x=1 \\ \frac{5+\theta-2\mu}{15}, & x=2 \\ \frac{7-3\theta+\mu}{15}, & x=3 \end{cases}$$

where $\theta \geq 0$, $\mu \geq 0$ are unknown constants and $\theta + \mu \leq 2$. For testing $H_0: \theta + \mu = 1$ versus $H_1: \theta = \mu = 0$, the null hypothesis is rejected if $x=2$ or 3 .

Q15.

The test is:

- A. biased with power $\frac{3}{5}$
- B. biased with power $\frac{2}{5}$
- C. ✓ unbiased with power $\frac{4}{5}$
- D. unbiased with power $\frac{13}{15}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Under $H_1: \theta = \mu = 0$ the rejection probability is $\frac{5+7}{15} = \frac{12}{15} = \frac{4}{5}$, which exceeds the size $\frac{11}{15}$ - so the test is unbiased with power $\frac{4}{5}$.

TIPS & TRICKS

- Power = rejection probability under H_1 : set $\theta = \mu = 0$ in the pmf and add the two rejection cells.
- Unbiased iff power $\geq$ size: $\frac{4}{5} = 0.80 > \frac{11}{15} = 0.733$.
- Convert to decimals to compare fractions quickly: $12/15$ against $11/15$ is immediate.

STEP-BY-STEP SOLUTION

- Under $H_1: \theta = \mu = 0$, $P(X=2) = \frac{5}{15}$ and $P(X=3) = \frac{7}{15}$.

2. Power = $\frac{5}{15} + \frac{7}{15} = \frac{12}{15} = \frac{4}{5}$.

3. Size (from the companion item) = $\frac{11}{15}$, and $\frac{4}{5} = \frac{12}{15} > \frac{11}{15}$.

4. Since power exceeds size, the test is unbiased, with power $\frac{4}{5}$ - option (c).

Q16.

Let $X_1, X_2, \dots, X_n$ be iid random variables with $N(\theta, 1)$. The UMPU test of size α for testing $H_0: \theta = \theta_0$ against $H_1: \theta \neq \theta_0$ is given by

$$\varphi(x) = \begin{cases} 1, & \sum_{i=1}^n x_i < c_1 \text{ or } \sum_{i=1}^n x_i > c_2 \\ 0, & \text{otherwise} \end{cases}$$

What are the values c_1 and c_2 ?

- A. $(n\theta_0 - \sqrt{n} Z_{\alpha/2}, n\theta_0 + \sqrt{n} Z_{\alpha/2})$
- B. $(n\theta_0 - \sqrt{n} Z_{\alpha}, n\theta_0 + \sqrt{n} Z_{1-\alpha})$
- C. $(n\theta_0 - \sqrt{n} Z_{1-\alpha}, n\theta_0 + \sqrt{n} Z_{\alpha})$
- D. $(n\theta_0 - \sqrt{n} Z_{1-(\alpha/2)}, n\theta_0 + \sqrt{n} Z_{\alpha/2})$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\sum X_i \sim N(n\theta_0, n)$ under H_0 , so equal tails of $\alpha/2$ give $c_{1,2} = n\theta_0 \mp \sqrt{n} Z_{\alpha/2}$.

TIPS & TRICKS

- $V(\sum X_i) = n$, so the standard deviation is $\sqrt{n}$ - not n .
- Two-sided $\Rightarrow$ split α into two equal tails, giving $Z_{\alpha/2}$.
- Symmetry of the normal makes the equal-tailed test the UMPU one.

STEP-BY-STEP SOLUTION

1. Under $H_0: \theta = \theta_0$, $\sum_{i=1}^n X_i \sim N(n\theta_0, n)$.
2. An equal-tailed size- α region places $\alpha/2$ in each tail: $\left| \frac{\sum X_i - n\theta_0}{\sqrt{n}} \right| > Z_{\alpha/2}$.
3. Hence $c_1 = n\theta_0 - \sqrt{n} Z_{\alpha/2}$ and $c_2 = n\theta_0 + \sqrt{n} Z_{\alpha/2}$.
4. By symmetry this test is UMPU - option (a).

Q17.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from $N(0, \sigma^2)$, $\sigma > 0$. Which one of the following hypotheses has the critical region $\sum_{i=1}^n x_i^2 \geq 2\chi_{(n, \alpha)}^2$ as the UMP critical region of level α , where $P(\chi_{(n)}^2 > \chi_{(n, \alpha)}^2) = \alpha$?

A. $H_0: \sigma^2 = 2$ versus $H_1: \sigma^2 > 2$

B. $H_0: \sigma^2 = 1$ versus $H_1: \sigma^2 > 1$

C. $H_0: \sigma^2 = 4$ versus $H_1: \sigma^2 < 4$

D. $H_0: \sigma^2 = 2$ versus $H_1: \sigma^2 < 2$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Under $H_0: \sigma^2 = \sigma_0^2$, $\frac{\sum x_i^2}{\sigma_0^2} \sim \chi_n^2$, so the size- α upper-tail region is $\sum x_i^2 \geq \sigma_0^2 \chi_{n, \alpha}^2$. Matching the printed $2\chi_{n, \alpha}^2$ gives $\sigma_0^2 = 2$, and an upper tail means $H_1: \sigma^2 > 2$.

TIPS & TRICKS

- Read the multiplier of $\chi_{n, \alpha}^2$: it IS σ_0^2 .
- An UPPER-tail region in $\sum x_i^2$ corresponds to a LARGER variance under H_1 .
- Options (c) and (d) pair an upper tail with a smaller-variance alternative - internally inconsistent.

STEP-BY-STEP SOLUTION

1. With mean zero, $\frac{\sum x_i^2}{\sigma^2} \sim \chi_n^2$.

2. A size- α upper-tail critical region under $H_0: \sigma^2 = \sigma_0^2$ is $\sum x_i^2 \geq \sigma_0^2 \chi_{n, \alpha}^2$.

3. The printed region is $\sum x_i^2 \geq 2\chi_{n, \alpha}^2$, so $\sigma_0^2 = 2$.

4. The likelihood ratio increases with $\sum x_i^2$ when σ^2 increases, so the upper tail is UMP for $H_1: \sigma^2 > 2$ - option (a).

Q18.

What is the distribution having MLR property required for UMP test?

- A. Hypergeometric distribution
- B. Poisson distribution
- C. Cauchy distribution
- D. Normal distribution**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

The normal family (one parameter unknown) is the standard MLR family on which UMP one-sided tests are built; the Cauchy is the classic family WITHOUT MLR.

TIPS & TRICKS

- One-parameter exponential families have MLR in the natural statistic - the normal with known variance is the archetype.
- The Cauchy fails MLR, which is exactly why no UMP test exists for its location parameter.
- Poisson and hypergeometric also possess MLR, so read the question as asking for the standard textbook family used to construct UMP tests.

STEP-BY-STEP SOLUTION

1. A family has monotone likelihood ratio in T if $\frac{L_{\theta_2}}{L_{\theta_1}}$ is a monotone function of T whenever $\theta_2 > \theta_1$.

2. For $N(\theta, \sigma^2)$ with σ^2 known, $\frac{L_{\theta_2}}{L_{\theta_1}} = \exp \left[\frac{(\theta_2 - \theta_1) \sum x_i}{\sigma^2} - \text{const} \right]$, strictly increasing in $\sum x_i$ - MLR holds.

3. Karlin-Rubin then delivers the UMP one-sided test 'reject if $\bar{X} > c$ '.

4. The Cauchy has no such monotone ratio, so it is the non-example. Option (d).

Q19.

Consider the problem of testing $H_0:\theta=\theta_0$ against $H_1:\theta=\theta_1$. The critical region ω and, consequently, the test based on it, is said to be unbiased if:

- A. $1-P(\text{Type-I error})=P(\text{Type-II error})$
- B. $P(\text{Type-I error})>P(\text{Type-II error})$
- C. $1-P(\text{Type-II error})>P(\text{Type-I error})$
- D. $P(\text{Type-I error})<P(\text{Type-II error})$

Correct answer: C

EXAM SHORTCUT[AI-derived explanation](#)

Unbiased means power $>$ size: $1-\beta>\alpha$, i.e. $1-P(\text{Type-II})>P(\text{Type-I})$.

TIPS & TRICKS

- Translate first: power $=1-\beta$, size $=\alpha$. Then the condition reads itself off.
- Equivalently $\alpha+\beta<1$.
- Options comparing α and β directly are irrelevant to unbiasedness.

STEP-BY-STEP SOLUTION

1. A critical region is unbiased if the probability of rejection under H_1 is at least that under H_0 .
2. That is, power $\geq$ size: $1-\beta\geq\alpha$, i.e. $1-P(\text{Type-II error})>P(\text{Type-I error})$ in the strict form.
3. Option (c).

Q20.

Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$ distribution with μ known. Consider the following testing problems for $N(\mu, \sigma^2)$:

I. $H_0: \sigma^2 = \sigma_0^2$ against $H_1: \sigma^2 < \sigma_0^2$

II. $H_0: \sigma^2 = \sigma_0^2$ against $H_1: \sigma^2 > \sigma_0^2$

III. $H_0: \sigma^2 = \sigma_0^2$ against $H_1: \sigma^2 \neq \sigma_0^2$

The UMP level α test exists for

A. I and II

B. III

C. I but not for II

D. II but not for I

Correct answer: A

EXAM SHORTCUT[AI-derived explanation](#)

With μ known the family has MLR in $\sum (x_i - \mu)^2$, so UMP tests exist for both one-sided alternatives but NOT for the two-sided one.

TIPS & TRICKS

- Karlin-Rubin gives UMP tests only for one-sided alternatives in MLR families.
- Two-sided alternatives need the UMPU restriction - no UMP test exists.
- The one-sided regions are $\sum (x_i - \mu)^2 \leq c$ (for $\sigma^2 < \sigma_0^2$) and $\geq c$ (for $\sigma^2 > \sigma_0^2$).

STEP-BY-STEP SOLUTION

1. With μ known, L is a one-parameter exponential family in $T = \sum (x_i - \mu)^2$, with monotone likelihood ratio in T .
2. By Karlin-Rubin, UMP level- α tests exist for $H_1: \sigma^2 > \sigma_0^2$ (reject for large T) and for $H_1: \sigma^2 < \sigma_0^2$ (reject for small T). So I and II admit UMP tests.
3. For the two-sided alternative the two one-sided UMP tests differ, so no single test can be uniformly most powerful - III does not.
4. Option (a).

Q21.

The critical region W and consequently a test for testing $H_0:\theta=\theta_0$ against $H_1:\theta=\theta_1$ is unbiased if

A. $P(x \in W / H_1) \geq P(x \in W / H_0)$

B. $P(x \in W / H_0) \geq P(x \in W / H_1)$

C. $P(x \in W / H_0) = P(x \in W / H_1)$

D. $P(x \in W / H_1) > P(x \in W / H_0)$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Unbiasedness means the rejection probability under H_1 is at least that under H_0 :

$$P(x \in W / H_1) \geq P(x \in W / H_0).$$

TIPS & TRICKS

- Left side is the power; right side is the size.
- Equivalently $1-\beta \geq \alpha$, i.e. $\alpha + \beta \leq 1$.
- Option (b) reverses the inequality, describing a WORSE-than-useless test.

STEP-BY-STEP SOLUTION

1. Power $= P(x \in W / H_1)$ and size $= P(x \in W / H_0)$.
2. A test is unbiased if its power is at least its size throughout the alternative.
3. $P(x \in W / H_1) \geq P(x \in W / H_0)$ - option (a).

Sequential and Decision Theory

27 authentic PYQs in this topic (2018-2026)

Wald's SPRT, OC and ASN Functions

Q1.

In an SPRT, λ_m is the likelihood ratio at stage m ; α, β are Type-I/II error probabilities. Which pairs are correctly matched?

1. $\lambda_m \geq \frac{1-\beta}{\alpha}$: reject H_0

2. $\frac{\beta}{1-\alpha} < \lambda_m < \frac{1-\beta}{\alpha}$: accept H_0

3. $\lambda_m \leq \frac{\beta}{1-\alpha}$: continue sampling

Codes:

A. 1 only

B. 1 and 2

C. 1 and 3

D. 2 and 3

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Reject when $\lambda_m \geq A = \frac{1-\beta}{\alpha}$; ACCEPT when $\lambda_m \leq B = \frac{\beta}{1-\alpha}$; CONTINUE in between. Pairs 2 and 3 are swapped.

TIPS & TRICKS

- SPRT decision triple: $\lambda_m \geq A$ reject, $\lambda_m \leq B$ accept, $B < \lambda_m < A$ continue.
- Mnemonic: a large likelihood ratio favours H_1 (reject), a small one favours H_0 (accept), and the middle band is the undecided zone.
- The statements here deliberately interchange 'accept' and 'continue sampling' - read both columns before matching.

STEP-BY-STEP SOLUTION

1. Wald's SPRT uses boundaries $A \approx \frac{1-\beta}{\alpha}$ and $B \approx \frac{\beta}{1-\alpha}$.
2. Statement 1: $\lambda_m \geq A \Rightarrow$ reject H_0 . Correctly matched.
3. Statement 2: $B < \lambda_m < A$ is the CONTINUE-SAMPLING region, not acceptance. Wrongly matched.
4. Statement 3: $\lambda_m \leq B \Rightarrow$ ACCEPT H_0 , not continue sampling. Wrongly matched.
5. Only statement 1 is correct - option (a).

Q2.

If N is the stopping time for an SPRT, which is correct?

- A. $P[N < \infty] = 1/4$
- B. $P[N < \infty] = 1/2$
- C. $P[N < \infty] = 1$
- D. $P[N < \infty] = 3/4$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Wald proved the SPRT terminates with probability one: $P[N < \infty] = 1$.

TIPS & TRICKS

- The log-likelihood-ratio random walk has non-zero drift (unless H_0 and H_1 coincide), so it must eventually cross a boundary.
- The stopping time N is a random variable with finite moments; its expectation is the ASN.
- This is why the SPRT is well defined as a procedure - the sample size is random but finite almost surely.

STEP-BY-STEP SOLUTION

1. In an SPRT, $\log \lambda_m = \sum_{i=1}^m Z_i$ is a random walk with increments $Z_i = \log \frac{f_1(X_i)}{f_0(X_i)}$.
2. Since $E(Z_i) \neq 0$ under either hypothesis, the walk drifts and crosses one of the two fixed boundaries with probability one.
3. Hence the stopping time satisfies $P[N < \infty] = 1$ - option (c).

Q3.

Let $X \sim f(x, \theta) = \theta^x (1-\theta)^{1-x}$, $x=0,1$. Testing $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$, $Z = \log \left\{ \frac{f(x, \theta_1)}{f(x, \theta_0)} \right\}$, $E(Z)$ equals

A. $\theta \log \left\{ \frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)} \right\}$

B. $\theta \log \left\{ \frac{1-\theta_1}{1-\theta_0} \right\}$

C. $\theta \log \left\{ \frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)} \right\} + \log \left\{ \frac{1-\theta_1}{1-\theta_0} \right\}$

D. $\theta \log \left\{ \frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)} \right\} + \theta \log \left\{ \frac{1-\theta_1}{1-\theta_0} \right\}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$Z = x \log \frac{\theta_1}{\theta_0} + (1-x) \log \frac{1-\theta_1}{1-\theta_0}; \text{ taking } E(X) = \theta \text{ and collecting terms gives } \theta \log \frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)} + \log \frac{1-\theta_1}{1-\theta_0}.$$

TIPS & TRICKS

- Write Z as a linear function of x first: $Z = ax + b$ with $a = \log \frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)}$ and $b = \log \frac{1-\theta_1}{1-\theta_0}$.
- Then $E(Z) = a\theta + b$ - the constant term b carries NO factor θ , which is what distinguishes (c) from (d).
- $E(Z)$ is the drift of the SPRT random walk and appears in the denominator of the ASN formula.

STEP-BY-STEP SOLUTION

1. $\log f(x, \theta) = x \log \theta + (1-x) \log (1-\theta)$, so $Z = x \log \frac{\theta_1}{\theta_0} + (1-x) \log \frac{1-\theta_1}{1-\theta_0}$.

2. Rearranging: $Z = x \left[\log \frac{\theta_1}{\theta_0} - \log \frac{1-\theta_1}{1-\theta_0} \right] + \log \frac{1-\theta_1}{1-\theta_0} = x \log \frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)} + \log \frac{1-\theta_1}{1-\theta_0}$.

3. Taking expectations with $E(X) = \theta$:

4. $E(Z) = \theta \log \frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)} + \log \frac{1-\theta_1}{1-\theta_0}$ - option (c).

Q4.

From sampling from $N(\theta, 1)$ using SPRT for $H_0: \theta = 0$ against $\theta = 2$, the ASN function for $\alpha = 0.05, \beta = 0.1$ is

$$E(n) = \frac{L(\theta) \log B + \{1 - L(\theta)\} \log A}{E(Z)}, \text{ where}$$

A. $A = 18, B = \frac{19}{2}, E(Z) = 4(1 - \theta)$

B. ✓ $A = 18, B = \frac{2}{19}, E(Z) = 4(\theta - 1)$

C. $A = \frac{2}{19}, B = 18, E(Z) = 4(1 - \theta)$

D. $A = \frac{2}{19}, B = 18, E(Z) = 4 - \theta$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$A = \frac{1 - \beta}{\alpha} = \frac{0.9}{0.05} = 18 \text{ and } B = \frac{\beta}{1 - \alpha} = \frac{0.1}{0.95} = \frac{2}{19} - \text{only option (b) has both.}$$

TIPS & TRICKS

- $A > 1 > B$ always. Any option with $A < B$ (here (c) and (d)) is wrong before any arithmetic.
- $\log \frac{f_1}{f_0} = (\theta_1 - \theta_0)x - \frac{\theta_1^2 - \theta_0^2}{2}$, so for $\theta_0 = 0, \theta_1 = 2$: $Z = 2x - 2$ and $E_\theta(Z) = 2(\theta - 1)$.
- $B = \frac{0.1}{0.95} = \frac{2}{19}$, not $\frac{19}{2}$ - option (a) inverts it.

STEP-BY-STEP SOLUTION

1. Wald's boundaries: $A \approx \frac{1 - \beta}{\alpha} = \frac{1 - 0.1}{0.05} = \frac{0.9}{0.05} = 18$.

2. $B \approx \frac{\beta}{1 - \alpha} = \frac{0.1}{1 - 0.05} = \frac{0.1}{0.95} = \frac{2}{19}$.

3. Only option (b) pairs $A = 18$ with $B = \frac{2}{19}$; options (c) and (d) interchange them and (a) inverts B .

4. For completeness, $Z = \log \frac{f_1}{f_0} = -\frac{1}{2}(x - 2)^2 + \frac{1}{2}x^2 = 2x - 2$, so $E_\theta(Z) = 2(\theta - 1)$.

5. Option (b).

Source note: Source note: the Wald boundaries $A = \frac{1 - \beta}{\alpha} = 18$ and $B = \frac{\beta}{1 - \alpha} = \frac{2}{19}$ identify option (b) uniquely. The drift

computed from first principles is $E(Z) = 2(\theta - 1)$, whereas the printed option carries $4(\theta - 1)$; the option set is preserved as printed

and (b) is taken as the key since it is the only one with both boundaries correct.

SI & HT

2021 Q14

Numerical

Medium

Q5.

Consider an SPRT to test $H_0:\theta=\theta_0$ against $H_1:\theta=\theta_1$, continuing as long as $a_m < s_m < r_m$; $m=1,2,\dots$, where $s_m = \sum_{i=1}^m X_i$; $a_m = 0.90 + 0.05m$; $r_m = 2.25 + 0.05m$. On data $\{1,1,0,1,0,0\}$ the decision will be

A. may accept H_1 at 6th stage.

B. reject H_0 at 4th stage.

C. may accept H_1 at 3rd stage.

D. reject H_1 at 3rd stage.

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Partial sums are 1,2,2,3,3,3. At $m=4$, $r_4=2.45$ and $s_4=3 > r_4$, so the upper boundary is crossed: reject H_0 at the 4th stage.

TIPS & TRICKS

- Tabulate s_m , a_m , r_m and stop at the first m where s_m leaves the band.
- $s_m \geq r_m$ (upper boundary) $\Rightarrow$ reject H_0 ; $s_m \leq a_m$ (lower boundary) $\Rightarrow$ accept H_0 .
- Once a boundary is crossed the procedure STOPS - later observations are irrelevant, so options mentioning stage 6 are wrong.

STEP-BY-STEP SOLUTION

1. Cumulative sums of $\{1,1,0,1,0,0\}$: $s_1=1$, $s_2=2$, $s_3=2$, $s_4=3$, $s_5=3$, $s_6=3$.
2. Boundaries: $a_m=0.90+0.05m$ gives 0.95,1.00,1.05,1.10,...; $r_m=2.25+0.05m$ gives 2.30,2.35,2.40,2.45,...
3. $m=1,2,3$: s_m lies strictly between a_m and r_m ($1 \in (0.95, 2.30)$, $2 \in (1.00, 2.35)$, $2 \in (1.05, 2.40)$) - continue.
4. $m=4$: $s_4=3 > 2.45=r_4$, so the upper boundary is crossed and H_0 is rejected at the 4th stage.
5. Option (b).

Q6.

Let $x_1, \dots, x_n$ be from $f(x, \mu) = \frac{1}{2\sqrt{\pi}} \exp\left[-\frac{1}{4}(x-\mu)^2\right]$; $-\infty < x < \infty$. The SPRT for rejecting $H_0: \mu=4$ against $H_1: \mu=6$ for $\alpha=\beta=0.05$ is

A. $\sum_{i=1}^n x_i \geq 5n + \log 19$

B. $\sum_{i=1}^n x_i \geq 5n$

C. $\sum_{i=1}^n x_i \geq n$

D. $\sum_{i=1}^n x_i \geq n + \log 19$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The density is $N(\mu, 2)$, so $Z_i = x_i - 5$ and $\sum Z_i \geq \log A$ with $A = \frac{0.95}{0.05} = 19$ gives $\sum x_i \geq 5n + \log 19$.

TIPS & TRICKS

- Identify the variance from the constant: $\frac{1}{2\sqrt{\pi}} = \frac{1}{\sqrt{2\pi\sigma^2}} \Rightarrow \sigma^2 = 2$.
- $Z_i = \frac{(\mu_1 - \mu_0)x_i}{\sigma^2} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2} = \frac{2x_i}{2} - \frac{36-16}{4} = x_i - 5$.
- $A = \frac{1-\beta}{\alpha} = \frac{0.95}{0.05} = 19$ - the $\log 19$ in the answer is the give-away.

STEP-BY-STEP SOLUTION

1. Matching $\frac{1}{2\sqrt{\pi}} e^{-(x-\mu)^2/4}$ with $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$ gives $\sigma^2 = 2$.

2. $Z_i = \log \frac{f(x_i; 6)}{f(x_i; 4)} = -\frac{(x_i-6)^2}{4} + \frac{(x_i-4)^2}{4} = \frac{4x_i-20}{4} = x_i-5$.

3. SPRT rejects H_0 when $\sum Z_i \geq \log A$ with $A = \frac{1-\beta}{\alpha} = \frac{0.95}{0.05} = 19$.

4. So $\sum x_i - 5n \geq \log 19$, i.e. $\sum x_i \geq 5n + \log 19$ - option (a).

SHARED CONTEXT

The OC function for testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1 (> \theta_0)$ using SPRT from $N(\theta, 1)$ is $L(\theta) = \frac{A^{h(\theta)} - 1}{A^{h(\theta)} - B^{h(\theta)}}$; α, β are Type-I/II error probabilities.

Q7.

What are A and B respectively equal to?

A. $\frac{1-\beta}{\alpha}, \frac{\beta}{1-\alpha}$

B. $\frac{1-\alpha}{\beta}, \frac{\alpha}{1-\beta}$

C. $\frac{\alpha}{1-\beta}, \frac{1-\alpha}{\beta}$

D. $\frac{\beta}{1-\alpha}, \frac{1-\beta}{\alpha}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Wald's boundaries: $A = \frac{1-\beta}{\alpha}$ (upper) and $B = \frac{\beta}{1-\alpha}$ (lower).

TIPS & TRICKS

- $A > 1 > B$: the upper boundary must exceed 1 and the lower must be below 1 - a sign check that eliminates (c) and (d).
- Mnemonic: A carries $(1-\beta)$ = power over α = size.
- $L(\theta_0) = 1-\alpha$ and $L(\theta_1) = \beta$ are the two anchor values of the OC function.

STEP-BY-STEP SOLUTION

1. In Wald's SPRT one continues sampling while $B < \lambda_m < A$, rejects H_0 when $\lambda_m \geq A$ and accepts when $\lambda_m \leq B$.
2. The strength- (α, β) approximations are $A \approx \frac{1-\beta}{\alpha}$ and $B \approx \frac{\beta}{1-\alpha}$.
3. Option (a).

SHARED CONTEXT

The OC function for testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1 (> \theta_0)$ using SPRT from $N(\theta, 1)$ is $L(\theta) = \frac{A^{h(\theta)} - 1}{A^{h(\theta)} - B^{h(\theta)}}$; α, β are Type-I/II error probabilities.

Q8.

What is $h(\theta)$ equal to?

A. $\frac{\theta_1 - \theta_0 - 2\theta}{\theta_1 - \theta_0}$

B. $\frac{\theta_1 + \theta_0 - 2\theta}{\theta_1 - \theta_0}$

C. $\frac{\theta_1 - \theta_0 + 2\theta}{\theta_1 - \theta_0}$

D. $\frac{\theta_1 - \theta_0 + 2\theta}{\theta_0 - \theta_1}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$h(\theta)$ solves $E_\theta[e^{hZ}] = 1$; for $N(\theta, 1)$ this gives $h(\theta) = \frac{\theta_0 + \theta_1 - 2\theta}{\theta_1 - \theta_0}$.

TIPS & TRICKS

- $h(\theta_0) = 1$ and $h(\theta_1) = -1$ - use these two checks to identify the right option instantly.
- Option (a) gives $h(\theta_0) = \frac{\theta_1 - 2\theta_0 + \dots}{\dots}$, which fails the check; only (b) passes both.
- $L(\theta_0) = \frac{A-1}{A-B} = 1 - \alpha$ and $L(\theta_1) = \frac{A^{-1}-1}{A^{-1}-B^{-1}} = \beta$ confirm the construction.

STEP-BY-STEP SOLUTION

1. For $N(\theta, 1)$, $Z = \log \frac{f(x; \theta_1)}{f(x; \theta_0)} = (\theta_1 - \theta_0)x - \frac{\theta_1^2 - \theta_0^2}{2}$.

2. $h = h(\theta)$ is the non-zero solution of $E_\theta[e^{hZ}] = 1$. Using the normal mgf, this reduces to

$$h(\theta_1 - \theta_0)\theta - \frac{h(\theta_1^2 - \theta_0^2)}{2} + \frac{h^2(\theta_1 - \theta_0)^2}{2} = 0.$$

3. Dividing by $h(\theta_1 - \theta_0)/2 \neq 0$: $2\theta - (\theta_1 + \theta_0) + h(\theta_1 - \theta_0) = 0$.

4. Hence $h(\theta) = \frac{\theta_1 + \theta_0 - 2\theta}{\theta_1 - \theta_0}$, and indeed $h(\theta_0) = 1$, $h(\theta_1) = -1$ - option (b).

Q9.

Let a random variable X have the following probability distribution under $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$:

X	-1	1	2
$P_{\theta_0}(x)$	1/3	1/3	1/3
$P_{\theta_1}(x)$	1/4	1/4	1/2

To test sequentially the hypothesis H_0 against H_1 , it is decided to continue the procedure as long as

$-\frac{n+1}{2} < S_n < \frac{n+2}{2}$, where $S_n = \sum_{i=1}^n X_i$. Then the probability under H_1 that the procedure will terminate with the second observation, is:

- A. $\frac{1}{9}$
- B. $\frac{1}{8}$
- C. $\frac{1}{3}$
- D. $\frac{1}{2}$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Under H_1 the walk survives the first stage only if $X_1=1$ (probability $\frac{1}{4}$); it then terminates at the second stage with probability $\frac{1}{2}$, giving $\frac{1}{8}$.

TIPS & TRICKS

- Tabulate S_n against the two boundaries stage by stage - the whole question is bookkeeping.
- Stage 1 boundaries: $-1 < S_1 < 1.5$, so $X_1=-1$ and $X_1=2$ both stop the procedure immediately.
- Stage 2 boundaries: $-1.5 < S_2 < 2$ with $S_2=1+X_2$.

STEP-BY-STEP SOLUTION

1. Stage 1: continue while $-1 < S_1 < 1.5$. With $S_1 = X_1 \in \{-1, 1, 2\}$, only $X_1=1$ keeps the procedure alive;

$$P_{H_1}(X_1=1) = \frac{1}{4}.$$

2. Stage 2: $S_2=1+X_2$, and the continuation band is $-1.5 < S_2 < 2$.

3. $X_2 = -1 \Rightarrow S_2 = 0$ (continue); $X_2 = 1 \Rightarrow S_2 = 2$ (boundary); $X_2 = 2 \Rightarrow S_2 = 3$ (terminate).

4. Taking the upper boundary as non-strict, termination at the second observation requires $X_2 = 2$, with

$$P_{H_1} = \frac{1}{2}.$$

5. Hence $P = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$ - option (b).

Source note: Source note: with strict inequalities the procedure continues past $n=1$ only when $X_1=1$ (probability $\frac{1}{4}$ under

H_1), and at $n=2$ the boundary $S_2 < \frac{2+2}{2} = 2$ is violated by $X_2 \in \{1, 2\}$, giving $\frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$. Reading the upper boundary as

non-strict ($S_n \leq \frac{n+2}{2}$ continues) makes only $X_2=2$ terminate, giving $\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$, which is option (b) and is taken as the

intended key. Options and wording are preserved as printed.

Q10.

In SPRT, to test $H_0 = X \sim f(x, \theta_0)$ versus $H_1 = X \sim f(x, \theta_1)$, Z is defined as $Z = \log \left\{ \frac{f(x, \theta_1)}{f(x, \theta_0)} \right\}$ and $h(\theta)$ is the non-zero solution of the equation $E(e^{Zh(\theta)}) = 1$. Under H_0 , the OC function $L(\theta)$ becomes:

- A. α
- B. β
- C. $1-\alpha$
- D. $1-\beta$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

The OC function is the probability of ACCEPTING H_0 ; under H_0 that probability is $1-\alpha$.

TIPS & TRICKS

- $L(\theta) = P_\theta(\text{accept } H_0)$; so $L(\theta_0) = 1-\alpha$ and $L(\theta_1) = \beta$.
- These two anchor values are what calibrate $h(\theta)$ and the boundaries A, B .
- Do not confuse the OC function with the power function - they are complements.

STEP-BY-STEP SOLUTION

1. The operating characteristic function of an SPRT is $L(\theta) = P_\theta(\text{accept } H_0)$.
2. Under H_0 the probability of wrongly rejecting is α , so the probability of accepting is $1-\alpha$.
3. Hence $L(\theta_0) = 1-\alpha$ - option (c).

Q11.

Let $X_1, X_2, X_3, \dots, X_n$ be a sequence of iid observations from an exponential distribution with mean $\frac{1}{\theta}$, $\theta > 0$. To test $H_0: \theta = 1$ versus $H_1: \theta = 2$, SPRT is used. What is the probability that the procedure has no decision at the first stage when $\alpha = \beta = 0.1$ and H_0 is true?

- A. $\frac{1}{18}$
- B. $\frac{2}{9}$
- C. $\frac{8}{9}$
- D. $\frac{17}{18}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$\lambda_1 = 2e^{-x}$ and the continuation band is $\frac{1}{9} < 2e^{-x} < 9$, i.e. $x < \ln 18$; under H_0 this has probability

$$1 - e^{-\ln 18} = \frac{17}{18}.$$

TIPS & TRICKS

- $A = \frac{1-\beta}{\alpha} = 9$ and $B = \frac{\beta}{1-\alpha} = \frac{1}{9}$ when $\alpha = \beta = 0.1$.
- $\lambda_1 = \frac{2e^{-2x}}{e^{-x}} = 2e^{-x} < 2 < 9$ always, so only the lower boundary can bind.
- $2e^{-x} > \frac{1}{9} \iff e^{-x} > \frac{1}{18} \iff x < \ln 18$, and $P(X < \ln 18 / \theta = 1) = 1 - \frac{1}{18}$.

STEP-BY-STEP SOLUTION

1. With $\alpha = \beta = 0.1$: $A = \frac{1-\beta}{\alpha} = 9$ and $B = \frac{\beta}{1-\alpha} = \frac{1}{9}$.

2. $\lambda_1 = \frac{f_1(x_1)}{f_0(x_1)} = \frac{2e^{-2x_1}}{e^{-x_1}} = 2e^{-x_1}$.

3. No decision means $B < \lambda_1 < A$: the upper condition $2e^{-x_1} < 9$ always holds, and $2e^{-x_1} > \frac{1}{9}$ gives $x_1 < \ln 18$.

4. Under H_0 ($\theta = 1$), $P(X_1 < \ln 18) = 1 - e^{-\ln 18} = 1 - \frac{1}{18} = \frac{17}{18}$.

5. Option (d).

Q12.

Consider a sequential probability ratio test for testing $H_0: \mu = \mu_0$ against $H_1: \mu = \mu_1$, where μ is the mean of normal distribution with variance unity. For this test, which of the following statements is/are correct?

- I. The OC function $L(\mu)$ may be obtained for a given μ using the relation $h(\mu) = \frac{\mu_1 - \mu_0 - 2\mu}{\mu_1 + \mu_0}$ in $L(\mu)$.
- II. If probabilities of errors α and β involved in the test procedure are equal, then the average sample numbers under H_0 and H_1 are equal.

Select the answer using the code given below:

- A. I only
- B. AI only**
- C. Both I and II
- D. Neither I nor II

Correct answer: B

EXAM SHORTCUT AI-derived explanation

The correct OC exponent is $h(\mu) = \frac{\mu_0 + \mu_1 - 2\mu}{\mu_1 - \mu_0}$, so statement I is mis-stated. Statement II is true for the normal because $|E_{\mu_0}(Z)| = E_{\mu_1}(Z)$.

TIPS & TRICKS

- Check h at the endpoints: the correct formula gives $h(\mu_0) = 1$ and $h(\mu_1) = -1$; the printed version does not.
- For $N(\mu, 1)$, $E_\mu(Z) = (\mu_1 - \mu_0) \left[\mu - \frac{\mu_0 + \mu_1}{2} \right]$, so the drifts under H_0 and H_1 are equal and opposite.
- With $\alpha = \beta$ we get $B = 1/A$, so $\log B = -\log A$ and the two ASN expressions coincide.

STEP-BY-STEP SOLUTION

1. Statement I: solving $E_\mu[e^{hZ}] = 1$ for $N(\mu, 1)$ gives $h(\mu) = \frac{\mu_0 + \mu_1 - 2\mu}{\mu_1 - \mu_0}$, with $h(\mu_0) = 1$ and $h(\mu_1) = -1$. The printed expression has the sum and difference interchanged, so it is incorrect. False.
2. Statement II: $E_\mu(Z) = (\mu_1 - \mu_0) \left[\mu - \frac{\mu_0 + \mu_1}{2} \right]$, so $E_{\mu_0}(Z) = -\frac{(\mu_1 - \mu_0)^2}{2}$ and $E_{\mu_1}(Z) = +\frac{(\mu_1 - \mu_0)^2}{2}$.
3. With $\alpha = \beta$, $A = \frac{1 - \alpha}{\alpha}$ and $B = \frac{\alpha}{1 - \alpha} = \frac{1}{A}$, so $\log B = -\log A$. Substituting into $E(N) = \frac{L \log B + (1 - L) \log A}{E(Z)}$ with $L(\mu_0) = 1 - \alpha$, $L(\mu_1) = \beta = \alpha$ gives the same value under both hypotheses. True.

4. Hence II only - option (b).

SI & HT

2025 Q49

Conceptual

Medium

SHARED CONTEXT

Let the SPRT of strength (α, β) and the boundary points (A, B) terminate with probability 1.

Q13.

Which one of the following is correct?

A. $A \geq \frac{1-\beta}{\alpha}$

B. ✓ $A \leq \frac{1-\beta}{\alpha}$

C. $A \geq \frac{\alpha}{1-\beta}$

D. $A \leq \frac{\alpha}{1-\beta}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Wald's exact inequality is $A \leq \frac{1-\beta}{\alpha}$ (the approximation replaces $\leq$ by $=$).

TIPS & TRICKS

- The exact relations are $A \leq \frac{1-\beta}{\alpha}$ and $B \geq \frac{\beta}{1-\alpha}$ - note the opposite directions.
- Derivation: integrate the likelihood ratio over the rejection region to get $1-\beta \geq A\alpha$.
- Using the equalities as approximations makes the achieved error probabilities slightly smaller than nominal.

STEP-BY-STEP SOLUTION

1. Let R be the event of rejecting H_0 . On R the terminal likelihood ratio is at least A , so $P_1(R) \geq A P_0(R)$.
2. $P_1(R) = 1-\beta$ and $P_0(R) = \alpha$, giving $1-\beta \geq A\alpha$.
3. Hence $A \leq \frac{1-\beta}{\alpha}$ - option (b).

SHARED CONTEXT

Let the SPRT of strength (α, β) and the boundary points (A, B) terminate with probability 1.

Q14.

Which one of the following is correct?

A. $B \leq \frac{\beta}{1-\alpha}$

B. ✓ $B \geq \frac{\beta}{1-\alpha}$

C. $B \leq \frac{1-\alpha}{\beta}$

D. $B \geq \frac{1-\alpha}{\beta}$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

The companion inequality is $B \geq \frac{\beta}{1-\alpha}$.

TIPS & TRICKS

- Apply the same argument on the ACCEPTANCE event, where the ratio is at most B .
- $\beta \leq B(1-\alpha)$ rearranges to $B \geq \frac{\beta}{1-\alpha}$.
- $A \leq \cdot$ but $B \geq \cdot$ - the directions are opposite and are frequently swapped in distractors.

STEP-BY-STEP SOLUTION

1. Let A_c be the event of accepting H_0 . On A_c the terminal likelihood ratio is at most B , so $P_1(A_c) \leq B P_0(A_c)$.
2. $P_1(A_c) = \beta$ and $P_0(A_c) = 1-\alpha$, giving $\beta \leq B(1-\alpha)$.
3. Hence $B \geq \frac{\beta}{1-\alpha}$ - option (b).

Q15.

If the SPRT of strength (α, β) and boundary points (A, B) terminates with probability 1, then which of the following is/are correct?

I. $A \leq \frac{1-\beta}{\alpha}$

II. $B \geq \frac{\beta}{1-\alpha}$

Select the answer using the code given below.

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Wald's exact inequalities: $A \leq \frac{1-\beta}{\alpha}$ and $B \geq \frac{\beta}{1-\alpha}$. Both hold.

TIPS & TRICKS

- Opposite directions: upper boundary bounded ABOVE, lower boundary bounded BELOW.
- Derived by integrating the likelihood ratio over the rejection and acceptance events respectively.
- Using equalities as approximations makes the realised α, β no larger than nominal.

STEP-BY-STEP SOLUTION

1. On the rejection event the terminal ratio is at least A , giving $1-\beta \geq A\alpha$, i.e. $A \leq \frac{1-\beta}{\alpha}$. Statement I true.
2. On the acceptance event the ratio is at most B , giving $\beta \leq B(1-\alpha)$, i.e. $B \geq \frac{\beta}{1-\alpha}$. Statement II true.
3. Option (c).

Q16.

In an SPRT with n , α and β , which one of the following is correct?

where the symbols have their usual meanings.

- A. $n \rightarrow$ fixed constant, $\alpha \rightarrow$ fixed constant, $\beta \rightarrow$ fixed constant
- B. $n \rightarrow$ random variable, $\alpha \rightarrow$ fixed constant, $\beta \rightarrow$ fixed constant**
- C. $n \rightarrow$ fixed constant, $\alpha \rightarrow$ random variable, $\beta \rightarrow$ random variable
- D. $n \rightarrow$ random variable, $\alpha \rightarrow$ random variable, $\beta \rightarrow$ fixed constant

Correct answer: B

EXAM SHORTCUT[AI-derived explanation](#)

In an SPRT the sample size n is a RANDOM VARIABLE while α and β are fixed in advance.

TIPS & TRICKS

- This is the defining inversion versus fixed-sample testing, where n is fixed and the error rates follow.
- $E(N)$ is the average sample number (ASN), typically smaller than the fixed-sample size for the same (α, β) .
- $P[N < \infty] = 1$, so the random n is finite with probability one.

STEP-BY-STEP SOLUTION

1. The SPRT continues sampling until the likelihood ratio leaves the band (B, A) , so the stopping time N is determined by the data - a random variable.
2. The strength (α, β) is specified in advance and fixes the boundaries A and B .
3. Hence n is random while α and β are fixed constants - option (b).

Q17.

Let $f(x; \theta) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}}$; $-\infty < x < \infty$, $-\infty < \theta < \infty$. The sequential probability ratio test is used for testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1 (> \theta_0)$. Then we reject H_0 if

- A. $\sum_{i=1}^m x_i \leq \frac{1}{\theta_1 - \theta_0} \log \left(\frac{\beta}{1-\alpha} \right) + \frac{m(\theta_0 + \theta_1)}{2}$
- B. ✓ $\sum_{i=1}^m x_i \geq \frac{1}{\theta_1 - \theta_0} \log \left(\frac{1-\beta}{\alpha} \right) + \frac{m(\theta_0 + \theta_1)}{2}$
- C. $\sum_{i=1}^m x_i = \frac{1}{\theta_1 - \theta_0} \log \left(\frac{\beta}{1-\alpha} \right) - \frac{m(\theta_0 + \theta_1)}{2}$
- D. $\sum_{i=1}^m x_i \geq \frac{1}{\theta_1 - \theta_0} \log \left(\frac{1-\beta}{\alpha} \right) - \frac{m(\theta_0 + \theta_1)}{2}$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\sum Z_i = (\theta_1 - \theta_0) \sum x_i - \frac{m(\theta_1^2 - \theta_0^2)}{2} \geq \log A \text{ rearranges to } \sum x_i \geq \frac{\log \frac{1-\beta}{\alpha}}{\theta_1 - \theta_0} + \frac{m(\theta_0 + \theta_1)}{2}.$$

TIPS & TRICKS

- $\frac{\theta_1^2 - \theta_0^2}{2(\theta_1 - \theta_0)} = \frac{\theta_0 + \theta_1}{2}$ - the SUM appears, with a PLUS sign.
- Rejection uses the upper boundary $A = \frac{1-\beta}{\alpha}$; acceptance uses $B = \frac{\beta}{1-\alpha}$.
- Since $\theta_1 > \theta_0$, dividing preserves the inequality, so rejection is an upper tail in $\sum x_i$.

STEP-BY-STEP SOLUTION

$$1. Z_i = \log \frac{f(x_i; \theta_1)}{f(x_i; \theta_0)} = (\theta_1 - \theta_0)x_i - \frac{\theta_1^2 - \theta_0^2}{2}.$$

$$2. \text{Reject } H_0 \text{ when } \sum_{i=1}^m Z_i \geq \log A \text{ with } A = \frac{1-\beta}{\alpha}:$$

$$3. (\theta_1 - \theta_0) \sum x_i - \frac{m(\theta_1^2 - \theta_0^2)}{2} \geq \log \frac{1-\beta}{\alpha}.$$

$$4. \text{Dividing by } \theta_1 - \theta_0 > 0: \sum x_i \geq \frac{1}{\theta_1 - \theta_0} \log \left(\frac{1-\beta}{\alpha} \right) + \frac{m(\theta_0 + \theta_1)}{2}.$$

5. Option (b).

Decision Theory, Loss, Risk and Bayes Estimators

SI & HT

2018 Q48

Definition

Easy

Q1.

Let $\delta(x)$ estimate θ with loss $L[\theta, \delta(x)]$. The risk $R(\theta, \delta) = E[L(\theta, \delta(x))]$ is mean squared error if the loss function is

A. $L = (\theta - a)^3$

B. $L = (\theta - a)^2$

C. $L = |\theta - a|$

D. $L = \theta(\theta - a)^2$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Squared error loss $L = (\theta - a)^2$ makes the risk $E[(\theta - \delta)^2]$, which is the mean squared error.

TIPS & TRICKS

- Squared error loss $\rightarrow$ risk is MSE $\rightarrow$ Bayes rule is the posterior mean.
- Absolute error loss $\rightarrow$ posterior median; 0-1 loss $\rightarrow$ posterior mode.
- A cubic loss is not even non-negative, so it cannot be a loss function at all.

STEP-BY-STEP SOLUTION

1. Risk is $R(\theta, \delta) = E_{\theta}[L(\theta, \delta(X))]$.
2. With $L(\theta, a) = (\theta - a)^2$, $R(\theta, \delta) = E_{\theta}[(\theta - \delta(X))^2]$, which is precisely the mean squared error of δ .
3. Hence option (b).

Q2.

Let $(X_1, \dots, X_n) \sim N(\mu, 1)$ with prior $\mu \sim N(0, 1)$, $L(\mu, \delta) = [\mu - \delta(x)]^2$. The Bayes' risk associated with δ is

A. $\frac{\sum_{i=1}^n x_i}{n+1}$

B. $\frac{2}{n+2}$

C. ✓ $\frac{1}{n+1}$

D. $\frac{\sum_{i=1}^n x_i^2}{n}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Normal-normal conjugacy gives posterior $N\left(\frac{\sum x_i}{n+1}, \frac{1}{n+1}\right)$; under squared-error loss the Bayes risk is the posterior variance $\frac{1}{n+1}$.

TIPS & TRICKS

- Precisions add: prior precision 1, data precision n , posterior precision $n+1$, hence posterior variance $\frac{1}{n+1}$.
- A risk is a number, not a statistic - options containing x_i cannot be the Bayes risk. That screen alone leaves (b) and (c).
- Bayes risk under squared error loss equals the expected posterior variance; here that is constant, so it is just $\frac{1}{n+1}$.

STEP-BY-STEP SOLUTION

1. Likelihood: $X_i/\mu \sim N(\mu, 1)$ iid; prior $\mu \sim N(0, 1)$.

2. Posterior precision $= 1+n$, so posterior variance $= \frac{1}{n+1}$, and posterior mean $= \frac{\bar{nx}}{n+1} = \frac{\sum x_i}{n+1}$.

3. Under squared error loss the Bayes estimator is the posterior mean, and the resulting posterior expected loss is the posterior variance $\frac{1}{n+1}$.

4. Since this does not depend on the data, the Bayes risk is also $\frac{1}{n+1}$ - option (c).

SI & HT

2018 Q50

Conceptual

Easy

Q3.

The Bayes' estimator of a parameter under squared error loss is

A. posterior mean

B. posterior median

C. posterior mode

D. posterior variance

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Minimising $E[(\theta-a)^2 / x]$ over a gives $a=E(\theta / x)$, the posterior mean.

TIPS & TRICKS

- The three-line table: squared error $\rightarrow$ mean, absolute error $\rightarrow$ median, 0-1 loss $\rightarrow$ mode.
- Proof: $E[(\theta-a)^2 / x] = V(\theta / x) + [E(\theta / x) - a]^2$ is minimised when the bracket vanishes.
- Posterior variance is the resulting minimum risk, never the estimator itself.

STEP-BY-STEP SOLUTION

1. The Bayes estimator minimises the posterior expected loss $E[(\theta-a)^2 / x]$.
2. Expanding: $E[(\theta-a)^2 / x] = V(\theta / x) + [E(\theta / x) - a]^2$.
3. This is minimised by taking $a=E(\theta / x)$, the posterior mean.
4. Option (a).

Q4.

Which statement is **not** correct?

- A. If any Bayes' solution is unique, it must be admissible.
- B. **A uniformly best decision rule always exists.**
- C. Posterior median is the Bayes' estimator under absolute error loss.
- D. A Bayes' decision rule whose risk is constant is also a minimax rule.

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

A uniformly best decision rule (best for every θ) generally does NOT exist - that is why admissibility, minimaxity and Bayes criteria were invented.

TIPS & TRICKS

- If a uniformly best rule existed, the whole apparatus of decision theory would be unnecessary.
- A unique Bayes rule is admissible; a Bayes rule with constant risk is minimax; the posterior median is Bayes under absolute error loss - all true.
- Counterexample to (b): the constant rule $\delta \equiv \theta_0$ has zero risk at θ_0 , so no rule can beat every rule everywhere.

STEP-BY-STEP SOLUTION

1. (a) A unique Bayes solution is admissible - standard theorem. Correct.
2. (b) A uniformly best decision rule would have risk no larger than every other rule at every θ . The constant rule $\delta \equiv \theta_0$ has zero risk at $\theta = \theta_0$, so a uniformly best rule would have to have zero risk there too, for every θ - impossible. Hence this statement is NOT correct.
3. (c) Under $L = |\theta - a|$ the posterior median minimises posterior expected loss. Correct.
4. (d) A Bayes rule with constant risk is minimax - standard theorem. Correct.
5. The incorrect statement is option (b).

Q5.

The Bayes estimator of θ , $X \sim f_{\theta}(x)$, prior $\pi(\theta)$, squared error loss, is

- A. Mean of the distribution, $f_{\theta}(x)$
- B. Mean of the prior distribution, $\pi(\theta)$
- C. **Mean of the posterior distribution, $\pi(\theta / x)$**
- D. Median of the posterior distribution, $\pi(\theta / x)$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Under squared error loss the Bayes estimator is the posterior mean, $E(\theta / x)$.

TIPS & TRICKS

- Squared error $\rightarrow$ posterior mean; absolute error $\rightarrow$ posterior median; 0-1 loss $\rightarrow$ posterior mode.
- The prior mean is the Bayes estimator only when there are no data - it ignores the likelihood.
- Posterior mean minimises $E[(\theta - a)^2 / x]$ by the variance-decomposition identity.

STEP-BY-STEP SOLUTION

1. The Bayes rule minimises the posterior expected loss $E[(\theta - a)^2 / x]$.
2. $E[(\theta - a)^2 / x] = V(\theta / x) + [E(\theta / x) - a]^2$, minimised at $a = E(\theta / x)$.
3. So the Bayes estimator is the mean of the posterior distribution $\pi(\theta / x)$ - option (c).

Q6.

For estimating θ under absolute error loss $L(\theta, d) = |\theta - d|$, the Bayes rule is given by

- A. Mean of the posterior distribution of θ given X
- B. Mode of the posterior distribution of θ given X
- C. **Median of the posterior distribution of θ given X**
- D. MLE of θ in the posterior distribution of θ given X

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Absolute error loss is minimised by the posterior MEDIAN.

TIPS & TRICKS

- $E[|\theta - d| \mid X]$ is minimised at the median of the conditional distribution - a general fact about L_1 loss.
- The table to memorise: squared error $\rightarrow$ mean; absolute error $\rightarrow$ median; 0-1 loss $\rightarrow$ mode.
- Median-based rules are more robust to heavy-tailed posteriors than mean-based ones.

STEP-BY-STEP SOLUTION

1. The Bayes rule minimises the posterior expected loss $E[|\theta - d| \mid X = x]$.
2. For any random variable θ with conditional distribution $\pi(\theta \mid x)$, the function $d \mapsto E[|\theta - d| \mid x]$ is minimised at any median of that conditional distribution.
3. Hence the Bayes rule under absolute error loss is the posterior median - option (c).

Q7.

What is the expectation of the loss function?

A. The risk function

B. The power function

C. The error function

D. The convex loss function

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

The expected loss is the risk function $R(\theta, \delta) = E_{\theta}[L(\theta, \delta(X))]$.

TIPS & TRICKS

- Risk is loss averaged over the sampling distribution, for a FIXED θ .
- Bayes risk averages the risk further over the prior; do not confuse the two.
- The power function belongs to testing, not to decision-theoretic estimation.

STEP-BY-STEP SOLUTION

1. In decision theory, a rule δ incurs loss $L(\theta, \delta(x))$ when x is observed.
2. Averaging this over the sampling distribution of X for a fixed θ gives $R(\theta, \delta) = E_{\theta}[L(\theta, \delta(X))]$.
3. This is called the risk function - option (a).

SHARED CONTEXT

Let X be binomial(n, θ). The prior of θ is beta (first kind) with parameters α, β . The posterior of $\theta \mid X=x$ is beta with parameters p_1, p_2 .

Q8.

What is p_1 equal to?

A. $n-\alpha-\beta$

B. $x+\alpha$

C. $n-x+\beta$

D. $x-\alpha$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Beta-binomial conjugacy: posterior is Beta($x+\alpha, n-x+\beta$), so $p_1 = x+\alpha$.

TIPS & TRICKS

- Posterior $\propto \theta^{x+\alpha-1}(1-\theta)^{n-x+\beta-1}$ - read the parameters off the exponents plus one.
- Mnemonic: add the successes to α and the failures to β .
- Posterior mean $= \frac{x+\alpha}{n+\alpha+\beta}$, a weighted average of x/n and the prior mean.

STEP-BY-STEP SOLUTION

- Likelihood $\propto \theta^x(1-\theta)^{n-x}$; prior $\propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$.
- Posterior $\propto \theta^{x+\alpha-1}(1-\theta)^{n-x+\beta-1}$, which is Beta($x+\alpha, n-x+\beta$).
- Hence $p_1 = x+\alpha$ - option (b).

SHARED CONTEXT

Let X be binomial(n, θ). The prior of θ is beta (first kind) with parameters α, β . The posterior of $\theta / X=x$ is beta with parameters p_1, p_2 .

Q9.

What is p_2 equal to?

A. $n+\alpha+\beta$

B. $x-\alpha$

C. $n+x-\beta$

D. $x+\alpha$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Posterior is Beta($x+\alpha, n-x+\beta$), so the second parameter counts the FAILURES $n-x$ added to β .

TIPS & TRICKS

- p_1 collects successes ($x+\alpha$); p_2 collects failures ($n-x+\beta$).
- Check: $p_1+p_2=n+\alpha+\beta$, which is why option (a) is planted - it is the SUM, not p_2 .
- Posterior mean $\frac{p_1}{p_1+p_2} = \frac{x+\alpha}{n+\alpha+\beta}$ shrinks the sample proportion toward the prior mean.

STEP-BY-STEP SOLUTION

1. Likelihood $\alpha\theta^x(1-\theta)^{n-x}$; prior $\alpha\theta^{\alpha-1}(1-\theta)^{\beta-1}$.
2. Posterior $\alpha\theta^{x+\alpha-1}(1-\theta)^{(n-x+\beta)-1}$.
3. So the posterior is Beta($x+\alpha, n-x+\beta$) and the second parameter is $p_2=n-x+\beta$.
4. Among the printed options only (c) is formed from n, x and β ; it is taken as the intended key (see the source flag).

Source note: Source anomaly: the beta-binomial posterior is Beta($x+\alpha, n-x+\beta$), so the exact value is $p_2=n-x+\beta$. The printed option (c) reads $n+x-\beta$, which appears to carry two sign slips; it is nevertheless the only option built from n, x and β and is taken as the intended key. Options and wording are preserved as printed.

Q10.

Consider the following statements:

I. Bayes estimator with a constant risk is a minimax estimator.

II. If a minimax estimator of a parametric function is unique, then it must be admissible.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

A Bayes rule with constant risk is minimax (I true), and a unique minimax estimator is admissible (II true).

TIPS & TRICKS

- Constant risk $\Rightarrow$ the Bayes risk equals the maximum risk, which no rule can beat - hence minimax.
- If a unique minimax rule were inadmissible, the dominating rule would also be minimax, contradicting uniqueness.
- These two results are the standard bridge between Bayes, minimax and admissibility.

STEP-BY-STEP SOLUTION

1. Statement I: let δ be Bayes with constant risk c . For any rule δ' , $\sup_{\theta} R(\theta, \delta') \geq \text{Bayes risk of } \delta' \geq \text{Bayes risk of } \delta = c = \sup_{\theta} R(\theta, \delta)$. So δ is minimax. True.
2. Statement II: if a unique minimax δ were inadmissible, some δ' would dominate it, making $\sup_{\theta} R(\theta, \delta') \leq \sup_{\theta} R(\theta, \delta)$ - so δ' would also be minimax, contradicting uniqueness. True.
3. Option (c).